Introduction to Smooth Manifolds

March 6, 2026

Contents

1 Smooth Manifolds **1**
- 1.1 Topological Manifolds . . . 1
  - 1.1.1 Coordinate Chart . . . 2
  - 1.1.2 Examples . . . 3
  - 1.1.3 Topological Properties of Manifolds . . . 3
- 1.2 Smooth Structures . . . 5
  - 1.2.1 Local Coordinate Representation . . . 7
- 1.3 Examples of Smooth Manifolds . . . 8
  - 1.3.1 Einstein Summation Convention . . . 8
- 1.4 Manifolds with Boundary . . . 11
  - 1.4.1 Smooth Structure on Manifolds with Boundary . . . 13

2 Smooth Maps **15**
- 2.1 Smooth Functions and Smooth Maps . . . 15
  - 2.1.1 Smooth Functions on Manifolds . . . 15
  - 2.1.2 Smooth Maps Between Manifolds . . . 16
  - 2.1.3 Diffeomorphisms . . . 18
- 2.2 Partitions of Unity . . . 18

3 Tangent Vectors **21**
- 3.1 Tangent Vectors . . . 21
  - 3.1.1 Geometric Tangent Vectors . . . 21
  - 3.1.2 Tangent Vectors on Manifolds . . . 23
- 3.2 The Differential of a Smooth Map . . . 23
- 3.3 Computation in Coordinates . . . 27
  - 3.3.1 The Differential in Coordinates . . . 28
  - 3.3.2 Change of Coordinates . . . 28
- 3.4 The Tangent Bundle . . . 29
- 3.5 Velocity Vectors of Curves . . . 31
- 3.6 Alternative Definition of Tangent Vectors . . . 33
  - 3.6.1 Derivations of the Space of Germs . . . 33
  - 3.6.2 Equivalent Class of Curves . . . 33
- 3.7 Categories and Functors . . . 34

4 Submersions, Immersions, and Embeddings **35**
- 4.1 Maps of Constant Rank . . . 35
  - 4.1.1 Local Diffeomorphisms . . . 36

4.1.2 The Rank Theorem . . . 37
4.1.3 The Rank Theorem with Boundary . . . 38
4.2 Embeddings . . . 39
4.3 Submersions . . . 41
4.4 Smooth Covering Maps . . . 42

5 Submanifolds **45**
5.1 Embedded Submanifolds . . . 45
5.1.1 Slice Charts for Embedded Submanifolds . . . 47
5.1.2 Level Sets . . . 48
5.2 Immersed Submanifolds . . . 50
5.3 Restricting Maps to Submanifolds . . . 52
5.3.1 Uniqueness of Smooth Structures on Submanifolds . . . 53
5.3.2 Extending Functions from Submanifolds . . . 53
5.4 The Tangent Space to a Submanifold . . . 53
5.5 Submanifolds with Boundary . . . 55

6 Sard's Theorem **59**
6.1 The Sard's Theorem . . . 59
6.2 The Whitney Embedding Theorem . . . 59
6.3 The Whitney Approximation Theorem . . . 60
6.3.1 Tabular Neighborhoods . . . 60
6.3.2 Smooth Approximation between Manifolds . . . 61
6.4 Transversality . . . 62

7 Lie Groups **65**
7.1 Definition and Examples . . . 65
7.2 Lie Group Homomorphisms . . . 66
7.2.1 Universal Covering Groups . . . 67
7.3 Lie Subgroups . . . 67
7.4 Group Actions and Equivariant Maps . . . 69
7.4.1 Equivariant Maps . . . 70
7.4.2 Semidirect Products . . . 72
7.4.3 Representations of Lie Groups . . . 73

8 Vector Fields **75**
8.1 Vector Fields on Manifolds . . . 75
8.1.1 Local and Global Frames . . . 77
8.1.2 Vector Fields as Derivations . . . 78
8.2 Vector Fields and Smooth Maps . . . 79
8.2.1 Vector Fields and Submanifolds . . . 80
8.3 Lie Brackets . . . 81
8.4 The Lie Algebra of Lie Groups . . . 83
8.4.1 Induced Lie Algebra Homomorphisms . . . 86
8.4.2 Lie Algebra of Lie Subgroups . . . 87

9 Integral Curves and Flows **89**
9.1 Integral Curves . . . 89
9.2 Flows . . . 90
9.2.1 The Fundamental Theorem of Flows . . . 91
9.2.2 Complete Vector Fields . . . 92
9.3 Flowouts . . . 93
9.3.1 Regular Points and Singular Points . . . 94
9.4 Flows and Flowouts on Manifolds with Boundary . . . 95
9.5 Lie Derivatives . . . 97
9.6 Commuting Vector Fields . . . 100
9.6.1 Commuting Frames . . . 101
9.7 Time-Dependent Vector Fields . . . 103
9.8 First-Order PDEs . . . 104
9.8.1 Linear Equations . . . 104

10 Vector Bundles **107**
10.1 Vector Bundles . . . 107
10.2 Local and Global Sections . . . 110
10.2.1 Local and Global Frames . . . 111
10.3 Bundle Homomorphisms . . . 113
10.4 Subbundles . . . 114
10.5 Fiber Bundles . . . 115

11 The Cotangent Bundle **117**
11.1 Covectors . . . 117
11.1.1 Covector Fields . . . 118
11.1.2 Coframes . . . 119
11.2 The Differential of a Smooth Function . . . 120
11.3 Pullback of Covector Fields . . . 122
11.3.1 Restricting Covector Fields to Submanifolds . . . 123
11.4 Line Integrals . . . 123
11.5 Conservative Covector Fields . . . 126

12 Tensors **129**
12.1 Multilinear Algebra . . . 129
12.1.1 Tensors Products of Vector Spaces . . . 130
12.1.2 Covariant and Contravariant Tensors . . . 132
12.2 Symmetric and Alternating Tensors . . . 132
12.2.1 Symmetric Tensors . . . 133
12.2.2 Alternating Tensors . . . 134
12.3 Tensors and Tensor Fields on Manifolds . . . 134
12.3.1 Pullbacks of Tensor Fields . . . 135
12.3.2 Lie Derivatives of Tensor Fields . . . 136

13 Riemannian Metrics **139**
13.1 Riemannian Manifolds . . . 139
13.1.1 Pullback Metrics . . . 140
13.1.2 Riemannian Submanifolds . . . 141
13.1.3 The Normal Bundle . . . 143
13.2 The Riemannian Distance Function . . . 144
13.3 The Tangent-Cotangent Isomorphism . . . 145
13.4 Pseudo-Riemannian Metrics . . . 146

14 Differential Forms **149**
14.1 Alternating Tensors . . . 149
14.1.1 Elementary Alternating Tensors . . . 150
14.1.2 Wedge Product . . . 151
14.1.3 Interior Multiplication . . . 153
14.2 Differential Forms on Manifolds . . . 154
14.3 Exterior Derivatives . . . 156
14.3.1 Vector Calculus on $\mathbb{R}^3$ . . . 160
14.3.2 An Invariant Formulation of Exterior Derivative . . . 161
14.3.3 Lie Derivative of Differential Forms . . . 162

15 Orientations **165**
15.1 Orientation of Vector Spaces . . . 165
15.2 Orientation of Manifolds . . . 166
15.2.1 Orientation of Hypersurfaces . . . 169
15.2.2 Boundary Orientation . . . 170
15.3 The Riemannian Volume Form . . . 171
15.3.1 Hypersurfaces in Riemannian Manifolds . . . 172
15.4 Orientations and Covering Maps . . . 173
15.4.1 The Orientation Covering . . . 174

16 Integration on Manifolds **177**
16.1 The Geometry of Volume Measurement . . . 177
16.2 Integration of Differential Forms . . . 177
16.2.1 Integration on Manifolds . . . 178
16.2.2 Integration on Lie Groups . . . 182
16.3 Stokes' Theorem . . . 183
16.4 Manifolds with Corners . . . 185
16.5 Integration on Riemannian Manifolds . . . 188
16.5.1 Integration of Functions on Riemannian Manifolds . . . 188
16.5.2 The Hodge Star Operator . . . 188
16.5.3 The Divergence Theorem . . . 190
16.5.4 Surface Integrals . . . 192
16.6 Densities . . . 193
16.6.1 The Riemannian Density . . . 196

17 Symplectic Manifolds **199**
17.1 Symplectic Tensors 199
17.2 Symplectic Structures on Manifolds 202
17.2.1 The Canonical Symplectic Form on Cotangent Bundles 203
17.3 The Darboux Theorem 204
17.4 Hamiltonian Vector Fields 204
17.4.1 Poisson Bracket 206

Chapter 1

Smooth Manifolds

In simple terms, smooth manifolds are spaces that locally look like $\mathbb{R}^n$, and on which we can do calculus. We can visualize them like smooth plane curves like circles and parabolas.

The simplest manifold and topological manifolds, which encode just the properties of what we mean by "locally look like $\mathbb{R}^n$ ". However, to do calculus (volume, curvature, etc.), we need a stronger restriction – the notion of smoothness. Intuitively, we can describe smoothness by having a tangent structure that moves continuously from point to point. For more sophisticated applications we can restrict it to be embedded in some ambient Euclidean vector space. The structure of this ambient space is superfluous that is not guaranteed by the internal structure of the manifold itself.

Also, it is evidently that we cannot define smoothness solely by topological structure. A circle and a square are homeomorphic topological space, but we all agree that square is not smooth but circle is. Therefore, we should think a smooth manifold has two layers of structure: topological manifolds and smoothness.

1.1 Topological Manifolds

Definition 1.1.1: Topological Manifolds

Suppose $(M, \mathcal{T})$ is a topological space, we say that M is a topological manifold of dimension n if it has the following property:

- M is a Hausdorff space. $\forall p \neq q$ in M, there are disjoint open sets $U, V \subseteq M$ such that $p \in U, q \in V$.
- M is second-countable. There exists a countable basis for the topology of M.
- M is locally Euclidean of dimension n : Each point of M has a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$, in the Euclidean topology. We call the n here the dimension of the topological manifold, denoted $\dim M$.

The last property can be expressed explicitly as: $\forall p \in M, \exists$ open set $U \subseteq M, p \in U$ and $\hat{U} \subseteq \mathbb{R}^n$ such that $U \cong \hat{U}$.

Remark:
We can change the definition to letting U to be homeomorphic to some open balls in $\mathbb{R}^n$. This is equivalent to the original definition.

Proof. If we have a neighborhood that is homeomorphic to a open subspace of $\mathbb{R}^n$, then we have an open ball subspace that would do. □

We also abbreviate M being a topological manifold of dimension n by M^n. It is worth mentioning that we do not consider spaces with mixed dimensions, like a disjoint union of a plane and a line. The dimension here is global to all the point in the space.

Theorem 1.1.1: Topological Invariant of Dimension

A nonempty n-dimensional topological manifold cannot be homeomorphic to an m-dimensional manifold unless $m = n$.

Remark:
The empty set satisfies the definition of a topological manifold of dimension n for every n. But in most circumstances we shall just ignore the trivial case.

A basic example of an n-dimensional topological space is $\mathbb{R}^n$ itself. As every metrizable space is Hausdorff and $\{B(a, r) \mid a \in \mathbb{Q}^n, r \in \mathbb{Q}\}$ is a countable basis.

1.1.1 Coordinate Chart

Definition 1.1.2: Coordinate Chart

Let M be a topological manifold of dimension n, a coordinate chart on M is a pair (U, φ), where U is an open set of M and $\varphi : U \to \hat{U}$ is a homeomorphism from U to an open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.
An *atlas* on M is a collection of coordinate charts $\{(U_\alpha, \varphi_\alpha) \mid \alpha \in A\}$ such that $\bigcup_{\alpha \in A} U_\alpha = M$.

By the definition of a topological manifold, $\forall p \in M$, we can find a neighborhood where we can define a (U, φ).

- If $\varphi(p) = 0$, we say that the chart is centered at p. (We can always find a chart centered at p by subtracting $\varphi(p)$.
- Given a (U, φ), we say U a coordinate domain. If $\varphi(U)$ is a ball, we say U a coordinate ball.
- φ is called a (local) coordinate map. And the component functions $(x^1, \ldots, x^n)$ of φ are called local coordinates on U. We have $\varphi(p) = (x^1(p), \ldots, x^n(p))$.

1.1.2 Examples

Example: **Graphs of Continuous Functions**

Let $U \subseteq \mathbb{R}^n$ be an open set. And $f : U \to \mathbb{R}^k$ be a continuous function. The graph of f is the subset of $\mathbb{R}^n \times \mathbb{R}^k$ defined by

$$\Gamma(f) = \big\{(x, y) \in \mathbb{R}^n \times \mathbb{R}^k : x \in U \wedge y = f(x)\big\} \tag{1.1}$$

with the subspace topology. Let $\pi : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ be the projection map, and let $\varphi : \Gamma(f) \to U$ be the restriction of π to $\Gamma(f)$.

$$\varphi(x, y) = x, (x, y) \in \Gamma(f)$$

Then $\Gamma(f)$ is a topological manifold of dimension n. $(\Gamma(f), \varphi)$ is a global coordinate chart.

Example: **Spheres**

For each $n \in \mathbb{N}$, the uni sphere $\mathbb{S}^n$ is a subspace of $\mathbb{R}^{n+1}$, and a local part (hemisphere would do) is the graph of a continuous mapping.

Example: **Projective Spaces**

The n-dimensional real projective space $\mathbb{RP}^n$, is defined as $(X, \mathcal{T})$, where

- X is the 1-dimensional linear subspaces of $\mathbb{R}^n$. (The lines that cross the origin)
- $\mathcal{T}$ is the quotient topology.

Example: **Product Manifold**

Suppose $M_1, \ldots, M_k$ are topological manifolds of dimension $n_1, \ldots, n_k$ respectively. Then the product space $M_1 \times \ldots \times M_k$ is a topological manifold of dimension $n_1 + \ldots + n_k$.

Proof. The Hausdorff and second-countable properties follows from the product topology itself. Given any point $p = (p_1, \ldots, p_k) \in M_1 \times \ldots \times M_k$, we can find a neighborhood U_i of p_i such that $U_i \cong \hat{U}_i \subseteq \mathbb{R}^{n_i}$. Then $U = U_1 \times \ldots \times U_k$ is a neighborhood of p and homeomorphic to $\hat{U}_1 \times \ldots \times \hat{U}_k \subseteq \mathbb{R}^{n_1 + \ldots + n_k}$. □

1.1.3 Topological Properties of Manifolds

We shall see that manifolds have a well-behaved topological structure, thanks to the Hausdorff and second-countable properties.

Lemma 1.1.1: Precompact Coordinate Balls

Every topological manifold has a countable basis of precompact coordinate balls. (Precompact means its closure is compact)

First we shall show that every second countable space is Lindelöf(every open cover has a countable subcover).

Proof. First, let $\mathcal{B} = \{B_i \mid i \in \mathbb{N}\}$ be a countable basis of the topology of M. Given any open cover $\{U_\alpha \mid \alpha \in A\}$ of M, for each B_i, we can find a U_{α_i} such that $B_i \subseteq U_{\alpha_i}$. Then $\{U_{\alpha_i} \mid i \in \mathbb{N}\}$ is a countable subcover of M. □

Now we prove the lemma. For any chart (U, φ), as $\varphi(U)$ is an open subset of $\mathbb{R}^n$, we can find a countable basis of precompact balls $\{B_i \mid i \in \mathbb{N}\}$ of $\varphi(U)$. Then $\{\varphi^{-1}(B_i) \mid i \in \mathbb{N}\}$ is a countable basis of precompact coordinate balls of U. As M is Lindelöf, we can find a countable collection of charts that cover M. The union of the countable bases of precompact coordinate balls of these charts is a countable basis of precompact coordinate balls of M.

Connectedness Topological manifolds also have nice connectedness properties.

Proposition: **Connectedness Properties of Manifolds**

Let M be a topological manifold, then

- M is locally path-connected.
- M is connected iff it is path-connected.
- The components of M are the same as its path components.
- M has countably many components, each is an open subset of M and a topological manifold itself.

Proof. Since each coordinate ball is path-connected, M has a basis of path-connected neighborhoods, so it is locally path-connected. The second and third properties follows from general topology. The openness of components follows from local path-connectedness. The countability of components follows from second-countability and the disjointness of components.(The components are an open cover of M, so we can find a countable subcover. As the components are disjoint, the only subcover is itself.) □

Local Compactness and Paracompactness Topological manifolds are also locally compact and paracompact.

Definition 1.1.3: Exhaustion

Let X be a topological space, an exhaustion of X is a sequence of compact sets $\{K_j\}_{j\in\mathbb{Z}}$ such that

- $K_j \subseteq \operatorname{Int} K_{j+1}$ for all $j \in \mathbb{Z}$.

- $\bigcup_{j\in\mathbb{Z}} K_j = X$.

We say that X is *exhausted* by $\{K_j\}_{j\in\mathbb{Z}}$.

We can see that for a second-countable locally compact Hausdorff space, we can find a countable exhaustion. This is because we can find a countable basis of compact sets, and we can take the union of these compact sets to form an exhaustion.

Proposition: **Local Compactness and Paracompactness of Manifolds**

Let M be a topological manifold, then

- M is locally compact.
- M is paracompact. In fact, given any open cover $\mathcal{X}$ of M and any basis $\mathcal{B}$, there is a countable, locally finite open refinement of $\mathcal{X}$ by elements of $\mathcal{B}$.

Proof. Local compactness follows from the fact that each point has a precompact coordinate ball neighborhood. As second-countable Hausdorff spaces are normal, and every regular Lindelöf space is paracompact, M is paracompact. For a construction, let $\{K_j\}_{j\in\mathbb{Z}}$ be an exhaustion of M by compact sets. For each j, let $V_j = K_{j+1} - \operatorname{Int} K_j$ and $W_j = \operatorname{Int} K_{j+2} - K_{j-1}$. Then □

Fundamental Groups of Manifolds The topological restrictions on manifolds also limit their fundamental groups, which is of great importance when we study covering spaces of manifolds.

Theorem 1.1.2: Fundamental Groups of Manifolds

The fundamental group of a topological manifold is at most countable.

Proof. SORRY, but fairly obvious due to the countability of coordinate balls. □

1.2 Smooth Structures

If we only have the topological structure of a manifold, we cannot do calculus on it. One may try to define derivatives of functions on the manifold by using coordinate charts, but the problem is that this definition is not invariant under homeomorphisms.

For example, the map given by

$$\varphi : \mathbb{R}^2 \to \mathbb{R}^2, \qquad \varphi(x, y) = (x^{1/3}, y^{1/3})$$

is a homeomorphism, but it is easy to construct a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that f is differentiable at 0, but $f \circ \varphi$ is not differentiable at 0.

The smooth structure allows us to formalize the idea of smooth transition between different coordinate charts, so that we can define derivatives of functions on the manifold in an invariant way. Let $U \in \mathbb{R}^n$, and $V \in \mathbb{R}^m$ be two open sets, a map $F : U \to V$ is said to be *smooth* (or C^∞, infinitely differentiable) if all its component functions have continuous partial derivatives of all orders. If F is bijective and both F and F^{-1} are smooth, then F is called a *diffeomorphism*.

Let M be a n-dimensional topological manifold, and for $p \in M$, take a coordinate chart (U, φ) with $p \in U$. We would think that a function $f : U \to \mathbb{R}$ is smooth if $f \circ \varphi^{-1} : \hat{U} \to \mathbb{R}$ is smooth (here $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$). But this would only make sense if this is independent of the choice of the chart (U, φ). Therefore, we need to impose some restrictions on the charts, called *smooth charts*. As this is not preserved by arbitrary homeomorphisms, we should thought this as a new structure on the manifold, called *smooth structure*.

Definition 1.2.1: Transition Map

For an n-dimensional topological manifold M, let (U, φ) and (V, ψ) be two coordinate charts such that $U \cap V \neq \varnothing$. Then the map

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V) \tag{1.2}$$

is called a *transition map* from (U, φ) to (V, ψ). It is a composition of homeomorphisms, so it is a homeomorphism itself.
Two coordinate charts (U, φ) and (V, ψ) are said to be *smoothly compatible* if either $U \cap V = \varnothing$, or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism. A smooth atlas on M is an atlas whose charts are pairwise smoothly compatible.

However, there may be many different smooth atlases that gave the same set of smooth functions on M. We could define an equivalence relation on the set of smooth atlases, but a more straightforward way is to define a maximal smooth atlas: A smooth atlas $\mathcal{A}$ is said to be *maximal* or *complete* if any coordinate chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

Definition 1.2.2: Smooth Structure

Let M be a topological manifold. A *smooth structure* on M is a maximal smooth atlas $\mathcal{A}$ on M. A smooth manifold is a pair $(M, \mathcal{A})$.

It is not convenient to work with maximal smooth atlases directly, so we have the following theorem that allows us to work with arbitrary smooth atlases.

Proposition: **Existence of Maximal Smooth Atlas**

Let M be a topological manifold,

- Every smooth atlas $\mathcal{A}$ on M is contained in a unique maximal smooth atlas, called the maximal smooth atlas determined by $\mathcal{A}$.
- Two smooth atlases $\mathcal{A}$ and $\mathcal{A}'$ on M determine the same smooth structure if and only if their union $\mathcal{A} \cup \mathcal{A}'$ is a smooth atlas.

Remark:
Intuitively, this means that we can define an equivalence relation on the set of smooth atlases, where two atlases are equivalent if they can be combined to form a larger smooth atlas. Each equivalence class has a unique maximal element, and all elements in the equivalence class are

just the sub-atlases of this maximal element.

Proof. Let $\mathcal{A}$ be a smooth atlas on M. Let $\overline{\mathcal{A}}$ be the set of all coordinate charts that are smoothly compatible with every chart in $\mathcal{A}$. We claim that $\overline{\mathcal{A}}$ is a maximal smooth atlas containing $\mathcal{A}$.

First, let $(U, \varphi), (V, \psi) \in \overline{\mathcal{A}}$, for $x = \varphi(p) \in \varphi(U \cap V)$, we have some chart $(W, \theta) \in \mathcal{A}$ with $p \in W$. Therefore, we have

$$\psi \circ \varphi^{-1} = (\psi \circ \theta^{-1}) \circ (\theta \circ \varphi^{-1})$$

is smooth in a neighborhood of x, so we have $\overline{\mathcal{A}}$ is a smooth atlas. Moreover, every chart that is smoothly compatible with every chart in $\overline{\mathcal{A}}$ is also smoothly compatible with every chart in $\mathcal{A}$, so it is already in $\overline{\mathcal{A}}$. Therefore, $\overline{\mathcal{A}}$ is maximal.

For the second part, if $\mathcal{A}$ and $\mathcal{A}'$ determine the same smooth structure, then they are both contained in the same maximal smooth atlas, so their union is a smooth atlas. Conversely, if their union is a smooth atlas, then every chart in $\mathcal{A}'$ is smoothly compatible with every chart in $\mathcal{A}$, so $\mathcal{A}' \subseteq \overline{\mathcal{A}}$. Then both $\mathcal{A}$ and $\mathcal{A}'$ are contained in $\overline{\mathcal{A}}$, so they determine the same smooth structure. □

Remark:
There exists topological manifolds that do not admit any smooth structure. For example, the E8 manifold in dimension 4. The first such example was constructed by Kervaire in 1960. On the other hand, there are also topological manifolds that admit more than one smooth structure. The first such example is the 7-sphere, discovered by Milnor in 1956. In fact, it is known that for every $n \geqslant 7$, there exist topological manifolds of dimension n that admit more than one smooth structure.
NOTE that different smooth manifold can be diffeomorphic, which we shall justify later.

We can produce various kinds of structures by changing the requirements on the transition maps:

- If we require the transition maps to be homeomorphisms, we get the notion of a topological manifold.
- If we require the transition maps to be diffeomorphisms of class C^k (i.e., having continuous derivatives up to order k), we get the notion of a C^k-manifold.
- If we require the transition maps to be real-analytic (can be expanded as a convergent power series around each point) diffeomorphisms, we get the notion of a real-analytic manifold.
- If we have even dimension, we can identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$, and require the transition maps to be holomorphic (analytic) diffeomorphisms, we get the notion of a complex manifold.

1.2.1 Local Coordinate Representation

If M is a smooth manifold, any chart (U, φ) in the smooth structure is called a smooth chart, and the coordinate map φ is called a smooth coordinate map.

We say a set $B \subseteq M$ is *Regular coordinate ball* if there is a larger coordinate ball $B' \subseteq M$ such that $\overline{B} \subseteq B'$ and a smooth coordinate map $\varphi : B' \to \mathbb{R}^n$ such that for some positive number $r < r'$, we have

$$\varphi(B) = B_r(0), \qquad \varphi(\overline{B}) = \overline{B_r(0)}, \qquad \varphi(B') = B_{r'}(0) \tag{1.3}$$

Therefore, the regular coordinate ball is precompact.

Remark:
This is not true for arbitrary coordinate balls, take $M = \mathbb{R} - \{0\}$, and $B = B_{(1)}(1)$, there is no larger coordinate ball that contains the closure of B, and it is not precompact.

Proposition: **Countabe Basis of Regular Coordinate Balls**

Every smooth manifold has a countable basis of regular coordinate balls.

Proof. This is a slight improvement of lemma 1.1.1. Let $\{(U_\alpha, \varphi_\alpha) \mid \alpha \in A\}$ be a countable atlas of smooth charts that cover M. For each $\alpha \in A$, as $\varphi_\alpha(U_\alpha)$ is an open subset of $\mathbb{R}^n$, we can find a countable basis of regular balls $\{B_{\alpha,i} \mid i \in \mathbb{N}\}$ of $\varphi_\alpha(U_\alpha)$. Then $\{\varphi_\alpha^{-1}(B_{\alpha,i}) \mid \alpha \in A, i \in \mathbb{N}\}$ is a countable basis of regular coordinate balls of M. □

If we have a chart (U, φ), we can simply identify U with $\varphi(U) \subseteq \mathbb{R}^n$. Therefore, for simplicity we shall say that a point $p \in M$ has coordinates $(x^1(p), \ldots, x^n(p))$ instead of writing $\varphi(p) = (x^1(p), \ldots, x^n(p))$.

A simple example is the polar coordinate on an open set of $\mathbb{R}^2$.

1.3 Examples of Smooth Manifolds

0-dimensional Smooth Manifolds 0-dimensional topological manifolds are just countable discrete spaces. Therefore, the only smooth structure on a 0-dimensional topological manifold is the trivial one, where every chart is a homeomorphism to an open subset of $\mathbb{R}^0 = \{0\}$.

Euclidean Spaces For each $n \in \mathbb{N}$, the space $\mathbb{R}^n$ is a smooth manifold of dimension n with the smooth structure given by the *standard smooth atlas*, which consists of the single global chart $(\mathbb{R}^n, _{\mathbb{R}^n})$.

There are other smooth structures on $\mathbb{R}^n$. For example, consider $\psi(x) = x^3$, then $(\mathbb{R}, \psi)$ determines a smooth structure on $\mathbb{R}$ that is different from the standard one. However, it can be shown that there are diffeomorphic.

Finite-Dimensional Vector Spaces Let V be a finite-dimensional real vector space of dimension n. Then V is isomorphic to $\mathbb{R}^n$ as a vector space if we take a basis. It is fairly obvious that all basis give the same smooth structure on V, making it a smooth manifold of dimension n, called the *standard smooth structure* on V.

1.3.1 Einstein Summation Convention

In differential geometry, we often deal with objects that have multiple components, such as vectors and tensors. To simplify the notation, we use the Einstein summation convention, which states that when an index appears twice in a single term, once as a subscript and once as a superscript, it implies summation over all possible values of that index. For example,

$$E(x) = x^i e_i = \sum_{i=1}^{n} x^i e_i$$

To be consistent, we shall use superscripts for components of vectors and subscripts for basis vectors.

Space of Matrices Let $m, n \in \mathbb{N}$, the space of all $m \times n$ real matrices, denoted by $\mathbb{R}^{m\times n}$, is a finite-dimensional vector space of dimension mn. Therefore, it has a standard smooth structure, making it a smooth manifold of dimension mn.

Open Submanifolds Let M be a smooth manifold of dimension n, and let $U \subseteq M$ be an open subset. Then U is a topological manifold of dimension n with the subspace topology. Define a smooth structure on U by

$$\mathcal{A}_U = \{(V, \varphi) \in \mathcal{A}_M, V \subseteq U\} \tag{1.4}$$

Then $\mathcal{A}_U$ is a smooth atlas on U, making it a smooth manifold of dimension n, called an *open submanifold* of M.

Remark:
As $\mathcal{A}$ is maximal, for any chart, its subchart is also in $\mathcal{A}$. Therefore, our requirement is sufficient for a mere inclusion.

The General Linear Group Let $n \in \mathbb{N}$, the general linear group $(n, \mathbb{R})$ is the set of all invertible $n \times n$ real matrices, which is an open subset of $\mathbb{R}^{n\times n}$ (the determinant function is continuous, and $(n, \mathbb{R})$ is the preimage of $\mathbb{R} - \{0\}$). Therefore, it is a smooth manifold of dimension n^2, with the smooth structure induced from the standard smooth structure on $\mathbb{R}^{n\times n}$.

Full Rank Matrices Let $m < n$ be two natural numbers, the set of all $m \times n$ real matrices of rank m, denoted by $M_m(m \times n, \mathbb{R})$, is an open subset of $\mathbb{R}^{m\times n}$ (the map that sends a matrix to the maximum absolute value of its $m \times m$ minors is continuous, and $M_m(m \times n, \mathbb{R})$ is the preimage of $(0, \infty)$). Therefore, it is a smooth manifold of dimension mn, with the smooth structure induced from the standard smooth structure on $\mathbb{R}^{m\times n}$.

For $m = n$, we have $M_n(n \times n, \mathbb{R}) = (n, \mathbb{R})$.

Linear Map Spaces Let V and W be finite-dimensional real vector spaces of dimension m and n respectively. The set of all linear maps from V to W, denoted by $\mathcal{L}(V, W)$, is a finite-dimensional vector space of dimension mn. Therefore, it has a standard smooth structure, making it a smooth manifold of dimension mn.

Graphs of Smooth Functions Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}^k$ be a smooth function. The graph of f is a n-dimensional smooth manifold, by the projection map as a global smooth chart.

Example: **Spheres**

For each $n \in \mathbb{N}$, the unit sphere $\mathbb{S}^n$ is a topological n-manifold. Each hemisphere is the graph of a smooth mapping, and it is fairly easy to check that the transition maps are all smooth. Therefore, $\mathbb{S}^n$ is a smooth manifold of dimension n, called the *standard smooth structure* on $\mathbb{S}^n$.

Level Sets of Smooth Functions Suppose $U \in \mathbb{R}^n$ is an open set, and $\Phi : U \to \mathbb{R}$ is a smooth function. For any $c \in \mathbb{R}$, the set

$$M_c = \Phi^{-1}(c) = \{x \in U \mid \Phi(x) = c\} \tag{1.5}$$

is called a *level set* of Φ. Suppose $M_c \neq \varnothing$, and for every $a \in M_c$, the derivative $D\Phi(a) : \mathbb{R}^n \to \mathbb{R}$ is non zero. Then by the implicit function theorem, take $\partial\Phi/\partial x^i(a) \neq 0$, we can find a neighborhood U_a of a such that $M_c \cap U_a$ is the graph of a smooth function from an open subset of $\mathbb{R}^{n-1}$ to $\mathbb{R}$. Therefore, M_c is a topological manifold of dimension $n-1$. By checking the transition maps, we can see that M_c is a smooth manifold of dimension $n-1$.

Projective Spaces The n-dimensional real projective space $\mathbb{RP}^n$ can be given a smooth structure by using standard charts.

Proposition: **Smooth Product Manifolds**

Suppose $M_1, \ldots, M_k$ are smooth manifolds of dimension $n_1, \ldots, n_k$ respectively. Then the product space $M_1 \times \ldots \times M_k$ is a smooth manifold of dimension $n_1 + \ldots + n_k$, with the smooth structure determined by charts of the form

$$(U_1 \times \ldots \times U_k, \varphi_1 \times \ldots \times \varphi_k)$$

where (U_i, φ_i) is a smooth chart on M_i.

Up to now, we construct smooth manifolds from topological manifolds. By the following lemma, we can construct smooth manifolds directly from smooth atlases.

Lemma 1.3.1: The Smooth Manifold Chart Lemma

Let M be a set, and let $\{U_\alpha\}_{\alpha \in A}$ be a collection of subsets of M and $\varphi_\alpha : U_\alpha \to \hat{U}_\alpha \subseteq \mathbb{R}^n$, such that

- For each $\alpha \in A$, φ_α is a bijection from U_α to an open subset $\hat{U}_\alpha$ of $\mathbb{R}^n$.
- For each $\alpha, \beta \in A$, the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$, and the map

  $$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

  is a diffeomorphism.
- Countable many U_α cover M.
- For any two distinct points $p, q \in M$, either there is an $\alpha \in A$ such that $p, q \in U_\alpha$, or there are $\alpha, \beta \in A$ such that $p \in U_\alpha, q \in U_\beta$ and $U_\alpha \cap U_\beta = \varnothing$.

Then there is a unique topology and smooth structure on M such that $\{(U_\alpha, \varphi_\alpha) \mid \alpha \in A\}$ is a smooth atlas on M, making M a smooth manifold of dimension n.

Remark:
The sets U_α gives local properties of every point in M, so we can define a topology on M by declaring open sets of M by inverses of open sets in $\mathbb{R}^n$ via the maps φ_α. The second requirement ensures that the charts are smoothly compatible, so we can define a smooth structure on M. The third requirement ensures that M is second-countable, and the fourth requirement ensures that M is Hausdorff.

Example: **Grassmann Manifolds**

Let V be a finite-dimensional real vector space of dimension n. For each $k \leqslant n$, the Grassmannian $G_k(V)$ is the set of all k-dimensional linear subspaces of V. We can give $G_k(V)$ a smooth structure.

Proof. SORRY □

1.4 Manifolds with Boundary

Many spaces, like closed balls and half-spaces, are not manifolds in the usual sense, because they have "edges". However, we can generalize the notion of manifolds to include such spaces because they still locally resemble Euclidean spaces, except at the boundary points.

Definition 1.4.1: Manifold with Boundary

An n-dimensional topological manifold with boundary is a Hausdorff, second-countable topological space M such that for every point $p \in M$, there exists a neighborhood U of p that is either homeomorphic to an open subset of $\mathbb{R}^n$ or to an (relative) open subset of the closed half-space $\mathbb{H}^n$, where

$$\mathbb{H}^n = \left\{(x^1, \ldots, x^n) \in \mathbb{R}^n \mid x^n \geqslant 0\right\}$$

We call a chart (U, φ) a *boundary chart* if $\varphi(U)$ is an open subset of $\mathbb{H}^n$ with $\varphi(U) \cap \partial\mathbb{H}^n \neq \varnothing$, and an *interior chart* if $\varphi(U)$ is an open subset of $\mathbb{R}^n$.

A point $p \in M$ is called an *interior point* if there is an interior chart (U, φ) with $p \in U$. p is a *boundary point* if there is a boundary chart (U, φ) with $p \in U$ and $\varphi(p) \in \partial\mathbb{H}^n$.

The set of all boundary points of M is called the *boundary* of M, denoted by ∂M. The set of all interior points of M is called the *interior* of M, denoted by $\operatorname{Int} M$.

Remark:
A point must be either a boundary point or an interior point. If p is not a boundary point, then either it is in the domain of an interior chart, or it is in the domain of a boundary chart but mapped to the interior of $\mathbb{H}^n$. In the latter case, we can shrink the domain to get an interior chart containing p.

The following theorem shows that a point cannot be both a boundary point and an interior point.

Theorem 1.4.1: Topological Invariance of Boundary

Let M be a topological manifold with boundary, then each point $p \in M$ is either a boundary point or an interior point, but not both. Thus

$$M = \partial M \cup \operatorname{Int} M, \qquad \partial M \cap \operatorname{Int} M = \varnothing \tag{1.6}$$

Remark:
NOTE that here the concept of boundary is not the same as the boundary of a subspace in topology. When in confusion, we shall call the former the *manifold boundary* and the latter the *topological boundary*.
Manifold boundary is a local, absolute concept, while topological boundary is a global, relative concept. For example, consider the closed unit disk $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leqslant 1\}$ as a manifold with boundary. The manifold boundary of D is the unit circle $\mathbb{S}^1$, while the topological boundary of D in $\mathbb{R}^2$ is also $\mathbb{S}^1$. However, if we consider D as a subspace of itself, then its topological boundary is empty, since D has no points outside itself.

Proposition: **Manifold Structure on Interior and Boundary**

Let M be a topological manifold with boundary of dimension n. Then

- The interior $\operatorname{Int} M$ is an n-dimensional topological manifold (without boundary), with the subspace topology.
- The boundary ∂M is an $(n-1)$-dimensional topological manifold (without boundary), with the subspace topology.
- M is a topological manifold (without boundary) iff $\partial M = \varnothing$.
- If $n = 0$, then $\partial M = \varnothing$ and M is a 0-dimensional topological manifold (without boundary).

Proof. SORRY □

Proposition: **Topological Properties of Manifolds with Boundary**

Let M be a topological manifold with boundary, then

- M has countable basis of precompact coordinate balls and half-balls.
- M is locally compact.
- M is paracompact.
- M is locally path-connected.
- M has countablely many components, each is an open subset of M and a connectd topological manifold with boundary itself.

- The fundamental group of M is at most countable.

1.4.1 Smooth Structure on Manifolds with Boundary

First we shall define smooth functions on arbitrary subset of $\mathbb{R}^n$:

Definition 1.4.2: Smooth Maps on subset of $\mathbb{R}^n$

Let $A \subseteq \mathbb{R}^n$ be an arbitrary subset, a map $f : A \to \mathbb{R}^k$ is said to be *smooth* if for every point $p \in A$, there is an open neighborhood U of p in $\mathbb{R}^n$ and a smooth map $\tilde{f} : U \to \mathbb{R}^k$ such that $\tilde{f}|_{U \cap A} = f|_{U \cap A}$.

The definition of smooth atlases and smooth structures on manifolds with boundary are similar to those on manifolds without boundary, except that we now allow charts to be homeomorphisms to open subsets of $\mathbb{H}^n$, and tweak the definition of smooth compatibility accordingly.

Proposition: **Properties of Smooth Manifolds with Boundary**

Let M be a smooth manifold with boundary of dimension n. Then

- The interior $\operatorname{Int} M$ is an n-dimensional smooth manifold (without boundary), with the subspace topology and the smooth structure induced from M.
- The boundary ∂M is an $(n-1)$-dimensional smooth manifold (without boundary), with the subspace topology and the smooth structure induced from M.
- Every smooth manifold with boundary has a countable basis of regular coordinate balls and half-balls.
- The smooth manifold chart lemma 1.3.1 also holds for smooth manifolds with boundary. Just replace $\mathbb{R}^n$ by $\mathbb{R}^n$ or $\mathbb{H}^n$ accordingly.

As a product of $\mathbb{H}^m$ and $\mathbb{H}^n$ is not a half space, the product of two manifolds with boundary is not a manifold with boundary in general. (It is a smooth manifold with corners, which we shall not discuss here.)

Proposition: **Products of Smooth Manifold with Boundary**

Suppose $M_1, M_2, \ldots, M_k$ are smooth manifolds and N is a smooth manifold with boundary. Then the product space $M_1 \times M_2 \times \ldots \times M_k \times N$ is a smooth manifold with boundary, with the boundary

$$\partial(M_1 \times M_2 \times \ldots \times M_k \times N) = M_1 \times M_2 \times \ldots \times M_k \times \partial N$$

Chapter 2

Smooth Maps

We shall do calculus on smooth manifolds via smooth maps between them.

2.1 Smooth Functions and Smooth Maps

Although formally maps and functions are the same thing, we shall technically denote functions as maps from a manifold to $\mathbb{R}^n$ and maps as maps between manifolds.

2.1.1 Smooth Functions on Manifolds

Definition 2.1.1: Smooth Functions on Manifolds

Let M be a smooth n-manifold and $k \in \mathbb{N}$. A function $f : M \to \mathbb{R}^k$ is a **smooth function** if for every $p \in M$, there exists a smooth chart (U, φ) containing p the corresponding coordinate representation $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^k$ is a smooth function (in the usual sense) on the open subset $\varphi(U) \subseteq \mathbb{R}^n$.
The definition for manifolds with boundary is similar.
We denote all smooth functions from M to $\mathbb{R}^k$ by $C^\infty(M, \mathbb{R}^k)$ or simply $C^\infty(M)$ when $k = 1$. It is a vector space over $\mathbb{R}$.

Remark:
If $M \subseteq \mathbb{R}^n$, the definition coincide with the usual definition of smooth functions on subsets of $\mathbb{R}^n$, obviously.

We shall see that the definition holds for all charts containing p if it holds for one chart containing p, thanks to the smoothness of transition maps.

Proposition: **Smoothness is Chart-Independent**

Let M be a smooth manifold, with or without boundary, and let $f : M \to \mathbb{R}^k$ be a function. Then f is a smooth function if and only if for every $p \in M$ and every smooth chart (U, φ) containing p, the coordinate representation $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^k$ is a smooth function (in the usual sense) on the open subset $\varphi(U) \subseteq \mathbb{R}^n$.

Given a function $f : M \to \mathbb{R}^k$ and a chart (U, φ) on M, the function

$$\hat{f} : \varphi(U) \to \mathbb{R}^k, \quad \hat{f} = f \circ \varphi^{-1} \tag{2.1}$$

is called the **coordinate representation** of f with respect to the chart (U, φ). Then the definition just says that f is smooth if and only if for every $p \in M$, there exists a chart (U, φ) containing p such that the coordinate representation $\hat{f}$ is smooth in the usual sense.

2.1.2 Smooth Maps Between Manifolds

Definition 2.1.2: Smooth Maps Between Manifolds

Let M and N be smooth manifolds. A map $F : M \to N$ is a **smooth map** if for every $p \in M$, there exist smooth charts (U, φ) on M containing p and (V, ψ) on N containing $F(p)$ such that $F(U) \subseteq V$ and the coordinate representation

$$\hat{F} : \varphi(U) \to \psi(V), \quad \hat{F} = \psi \circ F \circ \varphi^{-1} \tag{2.2}$$

is a smooth map (in the usual sense) between the open subsets $\varphi(U) \subseteq \mathbb{R}^m$ and $\psi(V) \subseteq \mathbb{R}^n$.
The definition for manifolds with boundary is similar.
We denote all smooth maps from M to N by $C^\infty(M, N)$.

Our previous definition of smooth functions is a special case of this definition when $N = \mathbb{R}^k$.

Remark:
The requirement that $F(U) \subseteq V$ is crucial, as we need to make F completely in control when we express it in coordinates. So we can identify F with its coordinate representation $\hat{F}$ on U.

Proposition: **Smooth Maps are Continuous**

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a map. If F is smooth, then it is continuous.

Proof. As F is smooth, for every $p \in M$, there exist smooth charts (U, φ) on M containing p and (V, ψ) on N containing $F(p)$ such that the coordinate representation $\hat{F} = \psi \circ F \circ \varphi^{-1}$ is smooth in the usual sense.

$$F_U = \psi^{-1} \circ \hat{F} \circ \varphi : U \to V$$

is continuous as a composition of continuous maps. Since F agrees with F_U on U, F is continuous at p. As p is arbitrary, F is continuous. □

Proposition: **Characterization of Smooth Maps**

Suppose M and N are smooth manifolds, with or without boundary, and let $F : M \to N$ be a map. Then F is smooth if and only if one of the following equivalent conditions holds:

- For every $p \in M$ there exist smooth charts (U, φ) on M containing p and (V, ψ) on N

containing $F(p)$ such that $U \cap F^{-1}(V)$ is open in M and the composite map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth in the usual sense.

- F is continuous and there exists smooth atlases $\{(U_\alpha, \varphi_\alpha)\}$ on M and $\{(V_\beta, \psi_\beta)\}$ on N such that for every α and β, the composite map $\psi_\beta \circ F \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap F^{-1}(V_\beta)) \to \psi_\beta(V_\beta)$ is smooth in the usual sense.

It is also obvious that smooth maps does not depend on the choice of charts, thanks to the smoothness of transition maps.

Proposition: **Smoothness is Local**

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a map. Then F is smooth if and only if for every $p \in M$ and every smooth chart (U, φ) on M containing p and every smooth chart (V, ψ) on N containing $F(p)$ such that $F(U) \subseteq V$, the coordinate representation $\hat{F} = \psi \circ F \circ \varphi^{-1}$ is smooth in the usual sense.

Proposition: **Algebra of Smooth Maps**

Let M, N, P be smooth manifolds, with or without boundary.

- The constant map $C : M \to N$ defined by $C(p) = q$ for some fixed $q \in N$ is smooth.
- The identity map $\mathrm{Id}_M : M \to M$ is smooth.
- If $U \subseteq M$ is an open submanifold, then the inclusion map $\iota : U \hookrightarrow M$ is smooth.
- If $F : M \to N$ and $G : N \to P$ are smooth maps, then the composition $G \circ F : M \to P$ is smooth.

Proposition: **Smooth Maps by Components**

Suppose $M_1, \ldots, M_k$ and N are smooth manifolds, with or without boundary (at most one of $M_1, \ldots, M_k$ has nonempty boundary), and let $\pi_i : M_1 \times \cdots \times M_k \to M_i$ be the projection map onto the i-th factor. A map $F : N \to M_1 \times \cdots \times M_k$ is smooth if and only if each component map $F_i = \pi_i \circ F : N \to M_i$ is smooth for $i = 1, \ldots, k$.

Example: **Smooth Maps**

- If M is a 0-manifold, then every map $F : M \to N$ is smooth.
- The wrapping map $\epsilon : \mathbb{R} \to S^1$ defined by $\epsilon(t) = \exp(2\pi i t)$ is smooth. So is $\epsilon^n : \mathbb{R}^n \to T^n$ defined by $\epsilon^n(t_1, \ldots, t_n) = (\exp(2\pi i t_1), \ldots, \exp(2\pi i t_n))$.
- The inclusion map $\iota : S^n \hookrightarrow \mathbb{R}^{n+1}$ is smooth.
- The quotient map $\pi : \mathbb{R}^{n+1} \backslash \{0\} \to \mathbb{R}P^n$ defined by $\pi(x) = [x]$ is smooth.

- If $M_1, \ldots, M_k$ are smooth manifolds, then each projection map $\pi_i : M_1 \times \cdots \times M_k \to M_i$ is smooth.

2.1.3 Diffeomorphisms

Definition 2.1.3: Diffeomorphisms

A **diffeomorphism** is a smooth map $F : M \to N$ that is a bijection and whose inverse $F^{-1} : N \to M$ is also smooth. If such a map exists, we say that M and N are **diffeomorphic**, denoted by $M \cong N$.

Remark:
Diffeomorphisms are isomorphisms in the category of smooth manifolds, so diffeomorphic manifolds are "the same" from the smooth manifold point of view.

Diffeomorphisms give an equivalence relation on the class of smooth manifolds. And it is fairly interesting to ask whether a given manifold has multiple smooth structures that are not diffeomorphic to each other. As it turns out, for $n \neq 4$, $\mathbb{R}^n$ has a unique smooth structure up to diffeomorphism, while for $n = 4$, there are uncountably many non-diffeomorphic smooth structures on $\mathbb{R}^4$!

2.2 Partitions of Unity

The Gluing lemma in topology states that

> Let X, Y be topological spaces, and if one of the following holds:
>
> - X is the union of finitely many closed subsets $A_1, \ldots, A_n$.
> - X is the union of open subsets $\{U_\alpha\}_{\alpha \in A}$.
>
> If we are given continuous maps $f_i : A_i \to Y$ (or $f_\alpha : U_\alpha \to Y$) that agree on the overlaps, then there exists a unique continuous map $f : X \to Y$ such that $f|_{A_i} = f_i$ (or $f|_{U_\alpha} = f_\alpha$).

We can glue smooth maps for the open cover case, but not for the closed cover case. This is fairly obvious, Take $f(x) = |x|$ on $\mathbb{R}$, and cover $\mathbb{R}$ by the two closed sets $(-\infty, 0]$ and $[0, \infty)$. The restrictions $f|_{(-\infty,0]}$ and $f|_{[0,\infty)}$ are both smooth, but f is not smooth at 0.

A slight disadvantage of gluing smooth maps over open covers is that we need to make sure the maps agree on the overlaps. To get around this, we introduce partitions of unity, which allow us to glue local smooth properties into global smooth properties without worrying about the overlaps.

Our discussion is based on the existence of smooth bump functions that are positive in a specific part and vanish outside a slightly larger part. Take the function

$$f(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leqslant 0. \end{cases}$$

on $\mathbb{R}$ for example.

Lemma 2.2.1: Smooth Bump on $\mathbb{R}^n$

Given any $0 < r_1 < r_2$, there exists a smooth function $H : \mathbb{R}^n \to \mathbb{R}$ such that $H(x) = 1$ for $||x|| \leqslant r_1$, $H(x) = 0$ for $||x|| \geqslant r_2$, and $0 < H(x) < 1$ for $r_1 < ||x|| < r_2$.

Proof. Using f to patch the two regions together would do. □

Definition 2.2.1: Partition of Unity

Suppose M is a topological space and $\mathcal{X} = \{X_\alpha\}_{\alpha \in A}$ is an open cover of M. A **partition of unity subordinate to $\mathcal{X}$** is a collection of continuous functions $\{\varphi_\alpha : M \to \mathbb{R}\}_{\alpha \in A}$ such that

- For each $\alpha \in A$, $0 \leqslant \varphi_\alpha(p) \leqslant 1$ for all $p \in M$.
- $\operatorname{supp} \varphi_\alpha \subseteq X_\alpha$ for each $\alpha \in A$.
- The family of supports $\{\operatorname{supp} \varphi_\alpha\}_{\alpha \in A}$ is locally finite.
- For every $p \in M$, $\sum_{\alpha \in A} \varphi_\alpha(p) = 1$ (only finitely many terms are nonzero by local finiteness).

If each φ_α is smooth, we say it is a **smooth partition of unity**.

Theorem 2.2.1: Existence of Smooth Partitions of Unity

Let M be a smooth manifold, with or without boundary, and let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be any open cover of M. Then there exists a smooth partition of unity $\{\varphi_\alpha : M \to \mathbb{R}\}_{\alpha \in A}$ subordinate to $\mathcal{U}$.

Proof. SORRY □

As you can see, we can use smooth partitions of unity to glue local smooth properties into global smooth properties. This is extremely useful in differential geometry.

Definition 2.2.2: Bump functions

Let M be a smooth manifold, with or without boundary, and let $A \subseteq M$ be closed and $A \subseteq U \subseteq M$ for some open set U. A **bump function** for A supported in U is a continuous function $\psi : M \to \mathbb{R}$ such that

- $0 \leqslant \psi(p) \leqslant 1$ for all $p \in M$.
- $\psi(p) = 1$ for all $p \in A$.
- $\operatorname{supp} \psi \subseteq U$.

Proposition: **Existence of Smooth Bump Functions**

Let M be a smooth manifold, with or without boundary, and let $A \subseteq M$ be closed and $A \subseteq U \subseteq M$ for some open set U. Then there exists a smooth bump function for A supported in U.

Proof. Use the existence of smooth partitions of unity. Let $U_0 = U, U_1 = M - A$. □

Now we deal with smooth maps on arbitrary subsets of manifolds. Suppose M, N are smooth manifolds, with or without boundary, and $A \subseteq M$ is arbitrary. A map $F : A \to N$ is **smooth** if for every $p \in A$, there exists an open neighborhood U of p in M and a smooth map $\tilde{F} : U \to N$ such that $\tilde{F}|_{U \cap A} = F|_{U \cap A}$.

Lemma 2.2.2: Extension Lemma for Smooth Functions

Let M be a smooth manifold, with or without boundary, and let $A \subseteq M$ be closed and $f : A \to \mathbb{R}^k$ be a smooth function. Then for any open set U containing A, there exists a smooth function $\tilde{f} : M \to \mathbb{R}^k$ such that $\tilde{f}|_A = f$ and $\operatorname{supp} \tilde{f} \subseteq U$.

Proof. For each $p \in A$, take an open neighborhood $W_p \subseteq U$ and a smooth function $\tilde{f}_p : W_p \to \mathbb{R}^k$ such that $\tilde{f}_p|_{W_p \cap A} = f|_{W_p \cap A}$. Then the family $\{W_p\}_{p \in A} \cup \{M - A\}$ is an open cover of M. Let $\{\varphi_p : M \to \mathbb{R}\}_{p \in A} \cup \{\varphi_0\}$ be a smooth partition of unity subordinate to this cover. Define

$$\tilde{f}(x) = \sum_{p \in A} \varphi_p(x) \tilde{f}_p(x).$$

From local finiteness, the sum is well-defined and smooth. Also, $\operatorname{supp} \tilde{f} \subseteq U$ and for any $x \in A$,

$$\tilde{f}(x) = \sum_{p \in A} \varphi_p(x) \tilde{f}_p(x) = \sum_{p \in A} \varphi_p(x) f(x) = f(x).$$

Remark:
Note that the codomain is $\mathbb{R}^k$ here, this lemma would fail for arbitrary manifolds.

□

Definition 2.2.3: Exhaustion Functions

If M is a topological space, a continuous function $f : M \to \mathbb{R}$ is an **exhaustion function** if for every $c \in \mathbb{R}$, the sublevel set $M_c = f^{-1}((-\infty, c])$ is compact.

Well, as $n \in \mathbb{Z}_+$, the sets M_n forms an exhaustion of M by compact sets, hence the name.

Proposition: **Existence of Smooth Exhaustion Functions**

Every smooth manifold M without boundary admits a smooth positive exhaustion function.

Proof. SORRY □

Theorem 2.2.2: Level Sets of Smooth Functions

Let M be a smooth manifold. If K is a closed subset of M, then there exists a smooth nonnegative function $f : M \to \mathbb{R}$ such that $f^{-1}(0) = K$.

Proof. SORRY □

Chapter 3

Tangent Vectors

The basic idea of calculus is linear approximation.

In analysis, we come across the idea of geometric tangent vectors in $\mathbb{R}^n$, which are used for "directional derivatives" of multivariable functions. We shall follow this path initially, and then move to a more abstract definition of tangent vectors as derivations.

3.1 Tangent Vectors

Take $S^{n-1} \subseteq \mathbb{R}^n$ for example. For a point $x \in S^{n-1}$, usually we think it as a location, expressed by coordinates $(x^1, x^2, \ldots, x^n)$. But when doing calculus, we sometimes need to think of it as a vector. Geometrically, we can think of a vector as an arrow which has arbitrary start point. We imagine tangent vectors as arrows starting from the point x. That is to say, they live in a copy of $\mathbb{R}^n$ that is "attached" to the point x.

3.1.1 Geometric Tangent Vectors

Given $a \in \mathbb{R}^n$, define the geometric tangent space to $\mathbb{R}^n$ at a as the vector space

$$\mathbb{R}^n_a = \{a\} \times \mathbb{R}^n = \{(a, v) : v \in \mathbb{R}^n\}, \quad (a, v) + (a, w) = (a, v + w), \quad c(a, v) = (a, cv). \tag{3.1}$$

A geometric tangent vector in $\mathbb{R}^n$ is an element of $\mathbb{R}^n_a$ for some $a \in \mathbb{R}^n$. We shall denote $v_a = (a, v)$

From this perspective, we can think of the tangent space of S^{n-1} at $a \in S^{n-1}$ as a subspace of $\mathbb{R}^n_a$: As all vectors in $\mathbb{R}^n_a$ that are perpendicular to the radius vector from the origin to a. To do this, we must have an inner product inherited from $\mathbb{R}^n$ via the natural isomorphism between $\mathbb{R}^n_a$ and $\mathbb{R}^n$.

This cannot be generalized to arbitrary manifolds, since there is no ambient Euclidean space to provide such a notion of perpendicularity. We shall use smooth structures to define tangent vectors in a more abstract way.

We turn to directional derivatives to motivate our definition. Every geometric tangent vector $v_a \in \mathbb{R}^n_a$ defines a map

$$D_v|_a : C^\infty(\mathbb{R}^n) \to \mathbb{R}, \quad D_v|_a f = D_v f(a) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} f(a + tv). \tag{3.2}$$

This map is linear and satisfies the Leibniz product rule:

$$D_v|_a(fg) = f(a)D_v|_a g + g(a)D_v|_a f.$$

If $v_a = v^i e_{i,a}$ in the standard basis, then we have

$$D_v|_a(f) = v^i \frac{\partial f}{\partial x^i}(a).$$

We now reverse the process.

Definition 3.1.1: Derivation

A **derivation** at $a \in \mathbb{R}^n$ is a linear map

$$w : C^\infty(\mathbb{R}^n) \to \mathbb{R}$$

that satisfies the Leibniz product rule:

$$w(fg) = f(a)w(g) + g(a)w(f)$$

for all $f, g \in C^\infty(\mathbb{R}^n)$. The set of all derivations at a is denoted by $T_a\mathbb{R}^n$. Then $T_a\mathbb{R}^n$ is a vector space under the operations

$$(w_1 + w_2)(f) = w_1(f) + w_2(f), \quad (cw)(f) = cw(f).$$

It is fairly surprising that $T_a\mathbb{R}^n$ is isomorphic to the geometric tangent space $\mathbb{R}^n_a$.

Lemma 3.1.1: Property of Derivations

Suppose $a \in \mathbb{R}^n$ and $w \in T_a\mathbb{R}^n$, $f, g \in C^\infty(\mathbb{R}^n)$.

- If f is constant, then $w(f) = 0$.
- If $f(a) = g(a) = 0$, then $w(fg) = 0$.

Proof. If $f(x) = 1$, then $wf = w(ff) = f(a)wf + f(a)wf = 2wf$, so $wf = 0$. If $f(x) = c$, then $wf = w(cf_1) = cw(f_1) = 0$. □

Proposition: **The Structure of $T_a\mathbb{R}^n$**

Let $a \in \mathbb{R}^n$. Then

- For each geometric tangent vector $v_a \in \mathbb{R}^n_a$, the map $D_v|_a$ defined above is a derivation at a.
- The map $v_a \mapsto D_v|_a$ is a vector space isomorphism from $\mathbb{R}^n_a$ to $T_a\mathbb{R}^n$.

Proof. To prove isomorphism:

- Linearity: we have

$$D_{c_1v+c_2w}|_a f = (c_1v + c_2w)^i \frac{\partial f}{\partial x^i}(a) = c_1 v^i \frac{\partial f}{\partial x^i}(a) + c_2 w^i \frac{\partial f}{\partial x^i}(a) = c_1 D_v|_a f + c_2 D_w|_a f.$$

- Injectivity: if $D_v|_a = 0$, then for all $f \in C^\infty(\mathbb{R}^n)$, $D_v|_a f = v^i \dfrac{\partial f}{\partial x^i}(a) = 0$. Taking $f(x) = x^j$, we have $v^j = 0$ for all j, so $v = 0$.

- Surjectivity: let $w \in T_a\mathbb{R}^n$. Define $v^i = w(x^i)$, and let $v_a = v^i e_i|_a$. For any $f \in C^\infty(\mathbb{R}^n)$, by Taylor's theorem, we have

$$f(x) = f(a) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a)(x^i - a^i) + R(x),$$

$$R(x) = \sum_{i,j=1}^{n} (x^i - a^i)(x^j - a^j) \int_0^1 (1-t) \frac{\partial^2 f}{\partial x^i \partial x^j}(a + t(x-a))\, \mathrm{d}t.$$

As $R(x)$ is the sum of products of functions vanishing at a, by the previous lemma we have $w(R) = 0$. Thus,

$$w(f) = w\left(f(a) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a)(x^i - a^i)\right) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a) w(x^i) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(a) v^i = D_v|_a f.$$

□

We have thus established the equivalence. And this definition can be generalized to arbitrary smooth manifolds.

3.1.2 Tangent Vectors on Manifolds

Definition 3.1.2: Tangent Vectors on Manifolds

Let M be a smooth manifold, with or without boundary, and let $p \in M$. A linear map $v : C^\infty(M) \to \mathbb{R}$ is called a **derivation at** p if it satisfies the Leibniz product rule:

$$v(fg) = f(p)v(g) + g(p)v(f), \quad \forall f, g \in C^\infty(M).$$

The set of all derivations at p is denoted by T_pM and called the **tangent space** of M at p. Its elements are called **tangent vectors** to M at p.

Proposition: **Property of Tangent Vectors on Manifolds**

Let M be a smooth manifold, with or without boundary, and let $p \in M$. If $v \in T_pM$ and $f, g \in C^\infty(M)$, then

- If f is constant, then $v(f) = 0$.
- If $f(p) = g(p) = 0$, then $v(fg) = 0$.

3.2 The Differential of a Smooth Map

We talk about the differential in analysis as linear approximations of functions at a given point. In the manifold case, there makes no sense to talk about linear transformations between manifolds, so we do it in terms of tangent spaces.

Definition 3.2.1: Differential on Manifolds

If M, N are smooth manifolds, with or without boundary, and $F : M \to N$ is a smooth map, then for each $p \in M$, we define a map

$$\mathrm{d}F_p : T_pM \to T_{F(p)}N \tag{3.3}$$

to be the **differential** of F at p, defined by

$$(\mathrm{d}F_p(v))(f) = v(f \circ F), \quad \forall f \in C^\infty(N), v \in T_pM. \tag{3.4}$$

Remark:
This is quite natural. To give a geometric intuition, take a curve γ in M with $\gamma(0) = p$ and $\gamma'(0) = v$. Then $F \circ \gamma$ is a curve in N with $(F \circ \gamma)(0) = F(p)$, and the tangent vector of $F \circ \gamma$ at 0 is $\mathrm{d}F_p(v)$.
Here v is a directional derivative operator acting on functions on M. which is given: $v(g) = \frac{\mathrm{d}}{\mathrm{d}t}\big|_{t=0} g(\gamma(t))$ for $g \in C^\infty(M)$. Then $\mathrm{d}F_p(v)$ is also a directional derivative operator acting on functions on N: for $f \in C^\infty(N)$,

$$(\mathrm{d}F_p(v))(f) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} f((F \circ \gamma)(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} (f \circ F)(\gamma(t)) = v(f \circ F).$$

The operator $\mathrm{d}F_p$ is linear, as for $v, w \in T_pM$, $c \in \mathbb{R}$, we have

$$(\mathrm{d}F_p(cv + w))(f) = (cv + w)(f \circ F) = cv(f \circ F) + w(f \circ F) = c(\mathrm{d}F_p(v))(f) + (\mathrm{d}F_p(w))(f).$$

Is also follows the Leibniz product rule:

$$\begin{aligned}(\mathrm{d}F_p(v))(fg) &= v((fg) \circ F) = v((f \circ F)(g \circ F)) \\ &= (f \circ F)(p)v(g \circ F) + (g \circ F)(p)v(f \circ F) \\ &= f(F(p))(\mathrm{d}F_p(v))(g) + g(F(p))(\mathrm{d}F_p(v))(f).\end{aligned}$$

Proposition: **Properties of Differential**

Let M, N, P be smooth manifolds, with or without boundary, and let $F : M \to N$ and $G : N \to P$ be smooth maps. Then for each $p \in M$,

- $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is a linear map.
- (Chain Rule) $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$.
- If $\mathrm{id}_M : M \to M$ is the identity map, then $\mathrm{d}(\mathrm{id}_M)_p$ is the identity map on T_pM.
- If F is a diffeomorphism, then $\mathrm{d}F_p$ is an isomorphism, and $(\mathrm{d}F_p)^{-1} = \mathrm{d}(F^{-1})_{F(p)}$.

Our first application of differentials is to relate tangent spaces of manifoldes to those of Euclidean spaces via charts. But first we shall prove that tangent vectors are local-behaved, for charts only give local information.

Proposition: **Locality of Tangent Vectors**

Let M be a smooth manifold, with or without boundary, and let $p \in M$. If $v \in T_pM$ and $f, g \in C^\infty(M)$ agree on an open neighborhood of p, then $v(f) = v(g)$.

Proof. Let $f, g \in C^\infty(M)$ agree on an open neighborhood U of p. Then $h = f - g$ vanishes on U. Let $\psi \in C^\infty(M)$ be a smooth bump function that is 1 on $\operatorname{supp} h$ and $\operatorname{supp} \psi \subseteq M - \{p\}$. Then $h = h\psi$, so by the previous proposition, we have

$$v(h) = v(h\psi) = h(p)v(\psi) + \psi(p)v(h) = 0.$$

Thus, $v(f) = v(g)$. □

Proposition: **Tangent Space to Open Subsets**

Let M be a smooth manifold, with or without boundary, and let $U \subseteq M$ be an open subset. Let $\iota : U \hookrightarrow M$ be the inclusion map. Then for $p \in U$, the differential

$$\mathrm{d}\iota_p : T_pU \to T_pM$$

is an isomorphism.

Proof. Via the extension lemma, every $f \in C^\infty(U)$ can be extended to a function $\tilde{f} \in C^\infty(M)$ such that $\tilde{f}|_U = f$. Thus, with the locality of tangent vectors, we can easily see the result. □

Therefore, it is safe to identify T_pU with T_pM via the inclusion map.

Theorem 3.2.1: Dimension of Tangent Space

Let M be a smooth manifold of dimension n, and let $p \in M$. Then T_pM is an n-dimensional real vector space.

Proof. Take a chart (U, φ) containing p. Then as φ is a diffeomorphism from U to an open subset $\hat{U} \subseteq \mathbb{R}^n$, by the previous proposition, we have an isomorphism

$$T_pM \cong T_pU \cong T_{\varphi(p)}\hat{U} \cong T_{\varphi(p)}\mathbb{R}^n.$$

Thus, by the earlier result on $\mathbb{R}^n$, we have $\dim T_pM = n$. □

Now we address points on the boundary of manifolds with boundary. The situation is similar. First, we shall relate the tangent spaces of $T_a\mathbb{H}^n$ to those of $\mathbb{R}^n$ when $a \in \partial\mathbb{H}^n$. As $\mathbb{H}^n$ is not an open subset of $\mathbb{R}^n$, we cannot use the previous proposition 3.2 directly. However, we have the following result.

Lemma 3.2.1: Inclusion of $\mathbb{H}^n$

Let $\iota : \mathbb{H}^n \hookrightarrow \mathbb{R}^n$ be the inclusion map. Then for each $a \in \partial\mathbb{H}^n$, the differential $\mathrm{d}\iota_a : T_a\mathbb{H}^n \to T_a\mathbb{R}^n$ is a linear isomorphism.

Proof. Assume $\mathrm{d}\iota_a(v) = 0$, then for all $f \in C^\infty(\mathbb{H}^n)$, we let $\tilde{f} \in C^\infty(\mathbb{R}^n)$ be an extension of f. Thus, $\tilde{f} \circ \iota = f$, and we have

$$v(f) = v(\tilde{f} \circ \iota) = (\mathrm{d}\iota_a(v))(\tilde{f}) = 0.$$

So $v = 0$ and $\mathrm{d}\iota_a$ is injective.

For surjectivity, let $w \in T_a\mathbb{R}^n$. Define $v : C^\infty(\mathbb{H}^n) \to \mathbb{R}$ by $v(f) = w(\tilde{f})$, where $\tilde{f} \in C^\infty(\mathbb{R}^n)$ is any extension of f. Thus

$$v(f) = w^i \frac{\partial \tilde{f}}{\partial x^i}(a).$$

From continuity, this does not depend on the choice of extension $\tilde{f}$, as we can get the result by limiting process from points in the interior of $\mathbb{H}^n$. So we have $\mathrm{d}\iota_a(v) = w$. □

Therefore, it is safe to identify $T_a\mathbb{H}^n$ with $T_a\mathbb{R}^n$ via the inclusion map, even for $a \in \partial\mathbb{H}^n$.

Proposition: **Dimension of Tangent Space with Boundary**

Let M be a smooth manifold of dimension n with boundary, and let $p \in M$. Then T_pM is an n-dimensional real vector space.

Next, as we know that for a finite-dimensional vector space, there exists a natural smooth structure on it. We shall see that the tangent space to a vector space at any point is naturally isomorphic to the vector space itself.

Proposition: **Tangent Space to a Vector Space**

Let V be a finite-dimensional real vector space with the standard smooth structure, and let $v \in V$. Then there is a natural isomorphism $V \cong T_vV$, defined by

$$v \mapsto D_v|_a, \quad D_v|_a(f) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} f(a + tv), \quad \forall f \in C^\infty(V).$$

For any liner transformation $T : V \to W$ between finite-dimensional real vector spaces, we have

$$L \cong \mathrm{d}L_a, \qquad \mathrm{d}L_a(D_v|_a) = D_{L(v)}|_{L(a)}. \tag{3.5}$$

Therefore, we can identify T_vV with V itself via the above isomorphism. For example, since $GL(n, \mathbb{R})$ is an open subset of the vector space $M_{n\times n}(\mathbb{R})$, we can identify $T_AGL(n, \mathbb{R})$ with $M_{n\times n}(\mathbb{R})$ for each $A \in GL(n, \mathbb{R})$.

For products, we have the following result.

Theorem 3.2.2: Tangent Space to Product Manifolds

Let $M_1, \ldots, M_k$ be smooth manifolds, at most one have boundary. Let $\pi_j : M_1 \times \cdots \times M_k \to M_j$ be the projection map onto the j-th factor. Then for each $p = (p_1, \ldots, p_k) \in M_1 \times \cdots \times M_k$, the map

$$\alpha_p : T_p(M_1 \times \cdots \times M_k) \to T_{p_1}M_1 \oplus \cdots \oplus T_{p_k}M_k, \quad \alpha_p(v) = (\mathrm{d}\pi_1)_p(v), \ldots, (\mathrm{d}\pi_k)_p(v)$$

Is an isomorphism.

Therefore, we can identify $T_p(M_1 \times \cdots \times M_k)$ with $T_{p_1}M_1 \oplus \cdots \oplus T_{p_k}M_k$ via the above isomorphism.

3.3 Computation in Coordinates

We shall use charts to compute tangent vectors and differentials in coordinates.

Suppose M is a smooth manifold of dimension n (without boundary for simplicity), and (U, φ) is a chart containing $p \in M$. Then $\varphi : U \to \hat{U} \subseteq \mathbb{R}^n$ is a diffeomorphism, thus $\mathrm{d}\varphi_p : T_pM \to T_{\varphi(p)}\mathbb{R}^n$ is an isomorphism.

In $\mathbb{R}^n$, we have the standard basis

$$\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)} : f \mapsto \frac{\partial f}{\partial x^i}(\varphi(p)), \quad i = 1, \ldots, n.$$

Therefore, the preimages of these basis vectors under $\mathrm{d}\varphi_p$ form a basis of T_pM, denoted by

$$\frac{\partial}{\partial x^i}\bigg|_p = (\mathrm{d}\varphi_p)^{-1}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right) = \mathrm{d}(\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right), \quad i = 1, \ldots, n. \tag{3.6}$$

Acting on $f \in C^\infty(M)$, we have

$$\frac{\partial}{\partial x^i}\bigg|_p (f) = \frac{\partial(f \circ \varphi^{-1})}{\partial x^i}(\varphi(p)) = \frac{\partial \hat{f}}{\partial x^i}(\hat{p}), \quad \hat{f} = f \circ \varphi^{-1}, \quad \hat{p} = \varphi(p).$$

which is the coordinate expression of f and p in $\mathbb{R}^n$. We call $\left\{\frac{\partial}{\partial x^i}\bigg|_p : i = 1, \ldots, n\right\}$ the **coordinate basis** of T_pM induced by the chart (U, φ).

Remark:
In $\mathbb{R}^n$, the coordinate basis vectors are just the partial derivative operators along the coordinate axes.

For points on the boundary of manifolds with boundary, the situation is similar, just replacing $\mathbb{R}^n$ with $\mathbb{H}^n$, and using the inclusion isomorphism between $T_a\mathbb{H}^n$ and $T_a\mathbb{R}^n$ for $a \in \partial\mathbb{H}^n$.

Theorem 3.3.1: The Coordinate Basis

Let M be a smooth manifold of dimension n, with or without boundary, and let $p \in M$. Then take any chart (U, φ) containing p. Then the coordinate vectors

$$\frac{\partial}{\partial x^i}\bigg|_p = \mathrm{d}(\varphi^{-1})_{\varphi(p)}\left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)}\right), \quad i = 1, \ldots, n$$

form a basis of T_pM.

This a tangent vector $v \in T_pM$ can be expressed in coordinates as

$$v = v^i \frac{\partial}{\partial x^i}\bigg|_p, \quad v^i = v(x^i), \tag{3.7}$$

where $x^i = \pi_i \circ \varphi$ are the coordinate functions on U. The numbers v^i are called the **components** of v with respect to the coordinate basis induced by the chart (U, φ).

3.3.1 The Differential in Coordinates

Now, we shall do computations of differentials of smooth maps in coordinates form. First, for simplicity consider $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open subsets, and let $F : U \to V$ be a smooth map. For $p \in U$, we have $\mathrm{d}F_p : T_p\mathbb{R}^n \to T_{F(p)}\mathbb{R}^m$ being a linear map. In the standard coordinate bases, we have

$$\mathrm{d}F_p\left(\frac{\partial}{\partial x^i}\Big|_p\right) f = \frac{\partial}{\partial x^i}\Big|_p (f \circ F) = \frac{\partial f}{\partial y^j}(F(p))\frac{\partial F^j}{\partial x^i}(p) = \left(\frac{\partial F^j}{\partial x^i}(p)\frac{\partial}{\partial y^j}\Big|_{F(p)}\right) f.$$

Thus, we have

$$\mathrm{d}F_p\left(\frac{\partial}{\partial x^i}\Big|_p\right) = \frac{\partial F^j}{\partial x^i}(p)\frac{\partial}{\partial y^j}\Big|_{F(p)}. \tag{3.8}$$

Writing in matrix form, we have

$$\mathrm{d}F_p = \begin{pmatrix} \frac{\partial F^1}{\partial x^1}(p) & \frac{\partial F^1}{\partial x^2}(p) & \cdots & \frac{\partial F^1}{\partial x^n}(p) \\ \frac{\partial F^2}{\partial x^1}(p) & \frac{\partial F^2}{\partial x^2}(p) & \cdots & \frac{\partial F^2}{\partial x^n}(p) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F^m}{\partial x^1}(p) & \frac{\partial F^m}{\partial x^2}(p) & \cdots & \frac{\partial F^m}{\partial x^n}(p) \end{pmatrix} \tag{3.9}$$

which is just the Jacobian matrix of F at p. The same can be said if U is an open subset of $\mathbb{H}^n$, so do V.

For a more general case, let M, N be smooth manifolds of dimension n, m respectively, with or without boundary, and let $F : M \to N$ be a smooth map. Take charts (U, φ) and (V, ψ) containing $p \in M$ and $F(p) \in N$ respectively. Then we have $\mathrm{d}\hat{F}_{\hat{p}} : T_{\hat{p}}\mathbb{R}^n \to T_{\hat{F}(\hat{p})}\mathbb{R}^m$ being the differential of the smooth map $\hat{F} = \psi \circ F \circ \varphi^{-1} : \hat{U} \to \hat{V}$ at $\hat{p} = \varphi(p)$. In the coordinate bases, we have

$$\begin{aligned} \mathrm{d}F_p\left(\frac{\partial}{\partial x^i}\Big|_p\right) &= \mathrm{d}(\psi^{-1})_{\hat{F}(\hat{p})} \circ \mathrm{d}\hat{F}_{\hat{p}} \circ \mathrm{d}\varphi_p\left(\frac{\partial}{\partial x^i}\Big|_p\right) \\ &= \mathrm{d}(\psi^{-1})_{\hat{F}(\hat{p})}\left(\mathrm{d}\hat{F}_{\hat{p}}\left(\frac{\partial}{\partial x^i}\Big|_{\hat{p}}\right)\right) \\ &= \mathrm{d}(\psi^{-1})_{\hat{F}(\hat{p})}\left(\frac{\partial \hat{F}^j}{\partial x^i}(\hat{p})\frac{\partial}{\partial y^j}\Big|_{\hat{F}(\hat{p})}\right) \\ &= \frac{\partial \hat{F}^j}{\partial x^i}(\hat{p})\frac{\partial}{\partial y^j}\Big|_{F(p)}. \end{aligned} \tag{3.10}$$

Which is just the pushforward of the Jacobian matrix of $\hat{F}$ at $\hat{p}$ via the charts.

3.3.2 Change of Coordinates

Suppose (U, φ) and (V, ψ) are two smooth charts on M and $p \in U \cap V$. Denote the coordinate functions of φ by $x^i = \pi_i \circ \varphi$ and those of ψ by $\tilde{x}^i = \pi_i \circ \psi$. Therefore, any tangent vector

$v \in T_pM$ can be expressed in both coordinate bases, and we want to find the relation between the components.

To do it, consider the transition map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$, and we write its coordinate functions by

$$\varphi \circ \psi^{-1}(x) = (\tilde{x}^1(x), \ldots, \tilde{x}^n(x)).$$

We have identified V with $\psi(V) \subseteq \mathbb{R}^n$ via the chart ψ, so we use the same notation $\tilde{x}^i$ for the coordinate functions on V for simplicity. Then we have the differential

$$\mathrm{d}(\psi \circ \varphi^{-1})_{\varphi(p)} : T_{\varphi(p)}\mathbb{R}^n \to T_{\psi(p)}\mathbb{R}^n.$$

by the previous result, we have

$$\mathrm{d}(\psi \circ \varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)} \right) = \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j}\bigg|_{\psi(p)}. \tag{3.11}$$

So we have pull back to T_pM via the charts:

$$\begin{aligned} \frac{\partial}{\partial x^i}\bigg|_p &= \mathrm{d}(\varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)} \right) \\ &= \mathrm{d}(\psi^{-1})_{\psi(p)} \circ \mathrm{d}(\psi \circ \varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i}\bigg|_{\varphi(p)} \right) \\ &= \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j}\bigg|_p. \end{aligned} \tag{3.12}$$

Therefore, the components of v in the two coordinate bases are related by

$$\tilde{v}^j = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p}) v^i. \tag{3.13}$$

3.4 The Tangent Bundle

Definition 3.4.1: The Tangent Bundle

Let M be a smooth manifold, with or without boundary. The **tangent bundle** of M is the disjoint union

$$TM = \bigsqcup_{p \in M} T_pM = \{(p, v) : p \in M, v \in T_pM\}.$$

The map $\pi : TM \to M$ defined by $\pi(p, v) = p$ is called the **bundle projection**.

For example, the tangent bundle of $\mathbb{R}^n$ is naturally isomorphic to $\mathbb{R}^n \times \mathbb{R}^n$ via the isomorphism

$$\mathbb{R}^n \times \mathbb{R}^n \to T\mathbb{R}^n, \quad (a, v) \mapsto (a, D_v|_a).$$

But for general manifolds, we cannot identify TM with $M \times \mathbb{R}^n$ globally because we cannot have a natural way to identify each tangent space T_pM with each other.

Theorem 3.4.1: Structure of the Tangent Bundle

Let M be a smooth manifold of dimension n. Then TM has a natural topology and smooth structure such that TM is a smooth manifold of dimension $2n$. With this structure, the bundle projection $\pi : TM \to M$ is a smooth map.

Proof. The ultimate intuition is to do it locally via charts. For each smooth chart (U, φ) on M, note that $\pi^{-1}(U) = \bigsqcup_{p \in U} T_pM$. Define a map $\tilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2n}$ by

$$\tilde{\varphi}(p, v) = \tilde{\varphi}\left(v^i \frac{\partial}{\partial x^i}\bigg|_p\right) = \left(x^1(p), \ldots, x^n(p), v^1, \ldots, v^n\right), \tag{3.14}$$

So the image set is $\hat{U} \times \mathbb{R}^n$, being an open subset of $\mathbb{R}^{2n}$. It is also a bijection from $\pi^{-1}(U)$ to $\hat{U} \times \mathbb{R}^n$, because

$$\tilde{\varphi}^{-1}(x^1, \ldots, x^n, v^1, \ldots, v^n) = v_i \frac{\partial}{\partial x^i}\bigg|_{\varphi^{-1}(x^1, \ldots, x^n)}.$$

Now suppose we have two smooth charts (U, φ) and (V, ψ) on M and let $(\pi^{-1}(U), \tilde{\varphi})$ and $(\pi^{-1}(V), \tilde{\psi})$ be the corresponding charts on TM. Then the sets

$$\tilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \varphi(U \cap V) \times \mathbb{R}^n, \quad \tilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \psi(U \cap V) \times \mathbb{R}^n$$

are both open subsets of $\mathbb{R}^{2n}$. The transition map is given by

$$\tilde{\psi} \circ \tilde{\varphi}^{-1}(x^1, \ldots, x^n, v^1, \ldots, v^n) = \left(\tilde{x}^1(x), \ldots, \tilde{x}^n(x), \frac{\partial \tilde{x}^j}{\partial x^1}(x) v^i, \ldots, \frac{\partial \tilde{x}^j}{\partial x^n}(x) v^i\right),$$

which is smooth.

Finally, choose a countable cover of charts $\{(U_\alpha, \varphi_\alpha)\}$ of M, then the corresponding charts $\{(\pi^{-1}(U_\alpha), \tilde{\varphi}_\alpha)\}$ form an atlas of TM. The conditions of 1.3.1 are easily verified. □

Remark:
For smooth manifolds with boundary, the construction is similar, just replacing $\mathbb{R}^n$ with $\mathbb{H}^n$ in the above proof. We note that the only "half-ness" happens in the base manifold M, while each tangent space T_pM is a full n-dimensional vector space, so no harm is done to the tangent bundle structure.

Proposition: **Single-Chart Tangent Bundle**

If M is a smooth manifold of dimension n (with or without boundary) that can be covered by a single chart (M, φ), then the tangent bundle TM is diffeomorphic to $M \times \mathbb{R}^n$.

Proof. Obvious. □

Remark:
NOTE that although we can locally view TM as $U \times \mathbb{R}^n$ via charts, there is no natural way to identify TM with $M \times \mathbb{R}^n$ globally in general. In fact, this may not be true in many cases.

Putting all pointwise differentials together, we have a map

$$\mathrm{d}F : TM \to TN, \quad \mathrm{d}F(p, v) = (F(p), \mathrm{d}F_p(v)),$$

called the global differential of F.

Theorem 3.4.2: Global Differential is Smooth

Let M, N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. Then the global differential

$$\mathrm{d}F : TM \to TN$$

is a smooth map.

Proof. From the coordinate expression, we have

$$\mathrm{d}F(p, v) = \left(F(p), \frac{\partial F^j}{\partial x^i}(p)v^i\right),$$

which is smooth for F is. □

Proposition: **Properties of Global Differential**

Let M, N, P be smooth manifolds, with or without boundary, and let $F : M \to N$ and $G : N \to P$ be smooth maps. Then for each $(p, v) \in TM$,

- (Chain Rule) $\mathrm{d}(G \circ F) = \mathrm{d}G \circ \mathrm{d}F$.
- If $\mathrm{id}_M : M \to M$ is the identity map, then $\mathrm{d}(\mathrm{id}_M) = \mathrm{id}_{TM}$.
- If F is a diffeomorphism, then $\mathrm{d}F$ is a diffeomorphism, and $(\mathrm{d}F)^{-1} = \mathrm{d}(F^{-1})$.

Just using proposition 11.2 would do. From now we may denote $\mathrm{d}F^{-1}$ for either $\mathrm{d}(F^{-1})$ or $(\mathrm{d}F)^{-1}$, when F is a diffeomorphism.

3.5 Velocity Vectors of Curves

Definition 3.5.1: Curves

Let M be a manifold, with or without boundary. A **curve** in M is a continuous map $\gamma : J \to M$, where $J \subseteq \mathbb{R}$ is an open interval. Sometimes we may want J to have one or both endpoints, in which case slight modifications are needed.

Definition 3.5.2: Velocity

Let M be a smooth manifold, with or without boundary, and let $\gamma : J \to M$ be a smooth curve. The **velocity** of γ at $t_0 \in J$ is the tangent vector

$$\gamma'(t_0) = \mathrm{d}\gamma_{t_0}\left(\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t_0}\right) \in T_{\gamma(t_0)}M. \tag{3.15}$$

Other notations include
$$\dot{\gamma}(t_0) = \frac{\mathrm{d}\gamma}{\mathrm{d}t}(t_0) = \frac{\mathrm{d}\gamma}{\mathrm{d}t}\Big|_{t=t_0}$$

The tangent vector $\gamma'(t_0)$ acts on functions by

$$\gamma'(t_0)(f) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=t_0} f(\gamma(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=t_0} (f \circ \gamma)(t).$$

which is the rate of change of f along the curve γ at t_0. For a smooth chart (U, φ) containing $\gamma(t_0)$, we can express the velocity in coordinates as

$$\gamma'(t_0) = \frac{\mathrm{d}\gamma^i}{\mathrm{d}t}(t_0)\frac{\partial}{\partial x^i}\Big|_{\gamma(t_0)} = \left(\frac{\mathrm{d}\gamma^1}{\mathrm{d}t}(t_0), \ldots, \frac{\mathrm{d}\gamma^n}{\mathrm{d}t}(t_0)\right),$$

which is familiar in Euclidean space.

Next, we shall see that every tangent vector can be expressed as the velocity of some curve, which will lead us to an equivalent definition of tangent vectors.

Proposition: **Tangent Vector as Velocity**

Let M be a smooth manifold, with or without boundary, and let $p \in M$. Then for any tangent vector $v \in T_pM$, there exists a smooth curve $\gamma : J \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.

Proof. First suppose $p \in \operatorname{Int} M$, then let (U, φ) be a smooth chart centering p. Then we write $v = v^i \partial/\partial x^i|_p$. For sufficiently small ϵ, we have a smooth curve $\gamma : (-\epsilon, \epsilon) \to U$ defined by

$$\gamma(t) = \varphi^{-1}\left(tv^1, \ldots, tv^n\right)$$

which is smooth because φ^{-1} is smooth.

Now if $p \in \partial M$, then let (U, φ) be a smooth boundary chart centering p. We can similarly define a smooth curve $\gamma : [0, \epsilon) \to U$ or $(-\epsilon, 0] \to U$ by the same formula for sufficiently small $\epsilon > 0$, depending on the sign of the first component of v. □

For composition, we have the following result.

Proposition: **Velocity under Composition**

Let M, N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. If $\gamma : J \to M$ is a smooth curve, then for each $t_0 \in J$,

$$(F \circ \gamma)'(t_0) = \mathrm{d}F_{\gamma(t_0)}\left(\gamma'(t_0)\right).$$

Proof. Just the chain rule:

$$(F \circ \gamma)'(t_0)(f) = \mathrm{d}(F \circ \gamma)_{t_0}\left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t_0}\right)(f) = \mathrm{d}F \circ \mathrm{d}\gamma_{t_0}\left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t_0}\right)(f) = \mathrm{d}F_{\gamma(t_0)}\left(\gamma'(t_0)\right)(f).$$

□

We can also use curve velocity to compute differentials: Suppose M, N are smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map, then to compute $\mathrm{d}F_p(v)$ for $p \in M$ and $v \in T_pM$, we can first find a smooth curve $\gamma : J \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$, then we have

$$\mathrm{d}F_p(v) = \mathrm{d}F_p(\gamma'(0)) = (F \circ \gamma)'(0).$$

3.6 Alternative Definition of Tangent Vectors

3.6.1 Derivations of the Space of Germs

A smooth function element on M is an ordered pair (f, U), where $U \subseteq M$ is an open set and $f \in C^\infty(U)$. Two smooth function elements (f, U) and (g, V) are said to be equivalent at $p \in U \cap V$ if there exists an open neighborhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) at p is called the **germ** of f at p, and the set if all germs of smooth functions at p is denoted by $C_p^\infty(M)$.

Remark:
Intuitively, $C_p^\infty(M)$ of a smooth function at p contains all distinguishable smooth functions locally around p.

We notice that $C_p^\infty(M)$ is a real vector space and an associative algebra under operations defined by

- Addition: $[(f, U)] + [(g, V)] = [(f + g, U \cap V)]$.
- Scalar Multiplication: $c[(f, U)] = [(cf, U)]$.
- Multiplication: $[(f, U)] \cdot [(g, V)] = [(fg, U \cap V)]$.

Now we denote the germ of f at p simply by $[f]_p$ when there is no confusion.

A derivation of $C_p^\infty(M)$ is a linear map $v : C_p^\infty(M) \to \mathbb{R}$ such that for all $[f]_p, [g]_p \in C_p^\infty(M)$,

$$v([f]_p \cdot [g]_p) = f(p)v([g]_p) + g(p)v([f]_p). \tag{3.16}$$

The set of all derivations of $C_p^\infty(M)$ is denoted by $\mathcal{D}_p(M)$. And it is simple to verify that $\mathcal{D}_p(M)$ is naturally isomorphic to T_pM.

3.6.2 Equivalent Class of Curves

This definition captures the intuitive idea of tangent vectors as "directions" at a point. Suppose p is a point of M, and consider all smooth curves $\gamma : J \to M$ such that $\gamma(0) = p$. We say two such curves γ_1 and γ_2 are equivalent at p if for any smooth function $f : M \to \mathbb{R}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}(f \circ \gamma_1)(t) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}(f \circ \gamma_2)(t). \tag{3.17}$$

The equivalence classes are denoted by $[\gamma]$, and all such equivalence classes form a set denoted by $\mathcal{V}_p(M)$, which is naturally isomorphic to T_pM.

3.7 Categories and Functors

A category $\mathcal{C}$ consists of

- A class $\mathrm{Ob}(\mathcal{C})$, whose elements are called objects of $\mathcal{C}$.
- A class $\mathrm{Hom}(\mathcal{C})$, whose elements are called morphisms of $\mathcal{C}$.
- For each morphism $f \in \mathrm{Hom}(\mathcal{C})$, there are two objects $X, Y \in \mathrm{Ob}(\mathcal{C})$ called the source and target of f, denoted by $f : X \to Y$.
- For each triplet of objects $X, Y, Z \in \mathrm{Ob}(\mathcal{C})$, there is a mapping called composition

$$\circ : \mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z), \quad (g, f) \mapsto g \circ f.$$

  where $\mathrm{Hom}(A, B)$ is the class of all morphisms from A to B.

The morphisms and objects must satisfy the following axioms:

- (Associativity) For each $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- (Identity) For each object $X \in \mathrm{Ob}(\mathcal{C})$, there exists an identity morphism $\mathrm{id}_X : X \to X$ such that for each $f : X \to Y$.

$$\mathrm{id}_Y \circ f = f, \quad f \circ \mathrm{id}_X = f.$$

A morphism $f : X \to Y$ is called an isomorphism if there exists a morphism $g : Y \to X$ such that

$$g \circ f = \mathrm{id}_X, \quad f \circ g = \mathrm{id}_Y.$$

Chapter 4

Submersions, Immersions, and Embeddings

We shall study the geometric properties of smooth maps by there differential.

4.1 Maps of Constant Rank

Suppose M and N are smooth manifolds, with or without boundary, and $F : M \to N$ is a smooth map. For each point $p \in M$, we define the **rank** of F at p to be the rank of the linear map $\mathrm{d}F_p : T_pM \to T_{F(p)}N$, which is just the rank of the Jacobian matrix of F in any coordinate chart containing p and $F(p)$, or just $\dim \operatorname{range} \mathrm{d}F_p$. If the rank of F is the same at every point of M, we say that F is a **map of constant rank**.

The maximum possible rank of F at any point is $\min\{\dim M, \dim N\}$. If the rank of F at p is equal to this maximum value, we say that F has **full rank** at p. If F has full rank at every point of M, we say that F has **constant full rank**.

Definition 4.1.1: smooth Submersions and Immersions

A smooth map $F : M \to N$ between smooth manifolds is a **smooth submersion** if $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is surjective for every point $p \in M$.
A smooth map $F : M \to N$ between smooth manifolds is a **smooth immersion** if $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is injective for every point $p \in M$.

From the continuity of $\mathrm{d}F$, we have the following result.

Proposition: **Local Surjectivity and Local Injectivity**

If $F : M \to N$ is a smooth map and $p \in M$. If $\mathrm{d}F_p$ is surjective, then there exists an open neighborhood U of p such that $F|_U : U \to N$ is a submersion. If $\mathrm{d}F_p$ is injective, then there exists an open neighborhood U of p such that $F|_U : U \to N$ is an immersion.

Proof. Choose any smooth coordinate chart (U, φ) on M containing p and any smooth coordinate chart (V, ψ) on N containing $F(p)$. Then the Jacobian matrix of F has full rank at p. As $\mathrm{d}F$ is continuous, there exists an open neighborhood $U' \subset U$ of p such that the Jacobian matrix of F has full rank at every point of U'. $\square$

Example: **Submersion and Immersion**

- Suppose $M_1, \ldots, M_k$ are smooth manifolds, then the projection map

$$\pi_i : M_1 \times \cdots \times M_k \to M_i, \quad (p_1, \ldots, p_k) \mapsto p_i$$

  is a smooth submersion for each $1 \leqslant i \leqslant k$.

- If $\gamma : J \to M$ is a smooth curve on a smooth manifold M, with or without boundary, then γ is a smooth immersion if and only if $\gamma'(t) \neq 0$ for all $t \in J$.

- If M is a smooth manifold then the tangent bundle projection $\pi : TM \to M$ is a smooth submersion.

Proposition: **Properties of Submersions and Immersions**

- If $F : M \to N$ and $G : N \to P$ are smooth submersions, then $G \circ F : M \to P$ is a smooth submersion.

- If $F : M \to N$ and $G : N \to P$ are smooth immersions, then $G \circ F : M \to P$ is a smooth immersion.

- The composition of maps of constant rank need not have constant rank.

Proof. For the third claim, take

$$f : \mathbb{R} \to \mathbb{R}^2, \quad t \mapsto (t, t^2)$$

and

$$g : \mathbb{R}^2 \to \mathbb{R}, \quad (x, y) \mapsto y.$$

Then both f and g have constant rank 1, but the composition

$$g \circ f : \mathbb{R} \to \mathbb{R}, \quad t \mapsto t^2$$

does not have constant rank. □

4.1.1 Local Diffeomorphisms

If M, N are smooth manifolds with or without boundary, a smooth map $F : M \to N$ is a **local diffeomorphism** if for each point $p \in M$, there exists an open neighborhood U of p such that $F(U)$ is open in N and $F|_U : U \to F(U)$ is a diffeomorphism.

Theorem 4.1.1: The Inverse Function Theorem

Suppose M and N are smooth manifolds, and $F : M \to N$ is a smooth map. If $p \in M$ and $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ is invertible, then there exist connected open neighborhoods U_0 of p in M and V_0 of $F(p)$ in N such that $F|_{U_0} : U_0 \to V_0$ is a diffeomorphism.

Proof. Firstly, this implies M, N have the same dimension n, then choose smooth charts (U, φ) centering at p and (V, ψ) centering at $F(p)$ with $F(U) \subseteq V$. Then $\hat{F} = \psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$

is a smooth map between open subsets of $\mathbb{R}^n$ with invertible Jacobian matrix at $\varphi(p)$. By the Euclidean version of the Inverse Function Theorem, there exist open neighborhoods U' of $\varphi(p)$ and V' of $\psi(F(p))$ such that $\hat{F}|_{U'} : U' \to V'$ is a diffeomorphism. Let $U_0 = \varphi^{-1}(U')$ and $V_0 = \psi^{-1}(V')$, then $F|_{U_0} : U_0 \to V_0$ is a diffeomorphism. □

Remark:
NOTE that this is true only for manifolds without boundary.

Proposition: **Properties of Local Diffeomorphisms**

- Compositions of local diffeomorphisms are local diffeomorphisms.
- Finite products of local diffeomorphisms are local diffeomorphisms.
- The restriction of a local diffeomorphism to an open submanifold (with or without boundary) is a local diffeomorphism.
- Every diffeomorphism is a local diffeomorphism. Every bijective local diffeomorphism is a diffeomorphism.
- A map between smooth manifolds, with or without boundary, is a local diffeomorphism if and only if it has a local diffeomorphism coordinate representation at each point.

Proposition: **Local Diffeomorphisms, Submersions, and Immersions**

A smooth map between smooth manifolds (without boundary), is a local diffeomorphism if and only if it is both a smooth immersion and a smooth submersion.
Moreover, if $\dim M = \dim N$, then a smooth map $F : M \to N$ is a smooth submersion if and only if it is a smooth immersion, and in either case it is a local diffeomorphism.

Example: **Local Diffeomorphisms**

The map $\mathbb{R} \to S^1$ defined by $t \mapsto (\cos t, \sin t)$ is a local diffeomorphism, but not a diffeomorphism.

4.1.2 The Rank Theorem

Theorem 4.1.2: The Rank Theorem

Suppose M and N are smooth manifolds of dimensions m and n, respectively, and $F : M \to N$ is a smooth map of constant rank k. Then for each point $p \in M$, there exist smooth coordinate charts (U, φ) on M centered at p and (V, ψ) on N centered at $F(p)$ such that

$$\hat{F} = \psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$$

is given by

$$\hat{F}(x^1, \ldots, x^m) = (x^1, \ldots, x^k, 0, \ldots, 0).$$

The linear version of the Rank Theorem is that under certain choice of basis, any linear map can be represented by a matrix of the form

$$\begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

Proof. From locality, just replace M, N by open subsets $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$ containing p and $F(p)$, respectively. We also assume $p = 0$ and $F(p) = 0$.

SORRY □

The following corollary is an immediate consequence and also can be viewed as a restatement of the Rank Theorem.

Corollary 4.1.1: Local Linearity

Let M and N be smooth manifolds of dimensions m and n, respectively, and let $F : M \to N$ be a smooth map. If M is connected, then F has constant rank k if and only if for each point $p \in M$, there exist smooth coordinate charts (U, φ) on M centered at p and (V, ψ) on N centered at $F(p)$ such that the coordinate representation

$$\hat{F} = \psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$$

is linear.

Theorem 4.1.3: Global Rank Theorem

Let M and N be smooth manifolds and let $F : M \to N$ be a smooth map of constant rank k. Then

- If F is surjective, then it is a smooth submersion.
- If F is injective, then it is a smooth immersion.
- If F is bijective, then it is a diffeomorphism.

Proof. SORRY □

4.1.3 The Rank Theorem with Boundary

The rank theorem does not generalize to manifolds with boundary in full generality. However, we do have the following partial result.

Theorem 4.1.4: The Local Immersion Theorem with Boundary

Suppose M is a smooth manifold with boundary of dimension m, N is a smooth manifold of dimension n, and $F : M \to N$ is a smooth immersion. Then for each point $p \in \partial M$, there exist smooth boundary charts (U, φ) on M centered at p and smooth chart (V, ψ) on N centered at $F(p)$ such that $F(U) \subseteq V$ and the coordinate representation

$$\hat{F}(x^1, \ldots, x^m) = (x^1, \ldots, x^m, 0, \ldots, 0).$$

Proof. SORRY □

4.2 Embeddings

Definition 4.2.1: Smooth Embeddings

Let M and N be smooth manifolds, with or without boundary. A **smooth embedding** is a smooth immersion $F : M \to N$ that is also a topological embedding; that is, F is a homeomorphism onto its image $F(M)$, where $F(M)$ is given the subspace topology inherited from N.

Proposition: **Compositions of Embeddings**

If $F : M \to N$ and $G : N \to P$ are smooth embeddings, then $G \circ F : M \to P$ is a smooth embedding.

Example: **Smooth Embeddings**

- Let M be a smooth manifold with or without boundary and $U \subseteq M$ be an open submanifold. Then the inclusion map $\iota : U \hookrightarrow M$ is a smooth embedding.
- If $M_1, \ldots, M_k$ are smooth manifolds and $p_i \in M_i$ for each $1 \leqslant i \leqslant k$, then each of

$$\iota_i : M_i \hookrightarrow M_1 \times \cdots \times M_k, \quad q \mapsto (p_1, \ldots, p_{i-1}, q, p_{i+1}, \ldots, p_k)$$

  is a smooth embedding. Indeed, $\mathbb{R}^n \iota \mathbb{R}^{n+k}$ defined by $x \mapsto (x, 0)$ is a smooth embedding.

Here are some counterexamples that illustrate the definition.

- The map $\mathbb{R} \to \mathbb{R}^2$ defined by $t \mapsto (t^3, 0)$ is a smooth map and a topological embedding, but not a smooth immersion at $t = 0$, so it is not a smooth embedding.
- The figure-eight curve $\mathbb{R} \to \mathbb{R}^2$ defined by $t \mapsto (\sin t, \sin 2t)$ is a smooth immersion, but not a topological embedding because it is not a homeomorphism onto its image. (Compactness fails)
- Now is an interesting example that shows mere injectivity is not enough for a smooth embedding. Consider $\gamma : \mathbb{R} \to \mathbb{T}^2$ by

$$\gamma(t) = (e^{it}, e^{i\alpha t}),$$

where α is an irrational number. Then γ is a smooth immersion, injective (because α is irrational), but not a smooth embedding because it is not a homeomorphism onto its image. Because $\gamma(\mathbb{Z})$ is dense in $\mathbb{T}^2$, but $\mathbb{Z}$ is not dense in $\mathbb{R}$.

Proposition: **Criterion for Smooth Embeddings**

Suppose M and N are smooth manifolds, with or without boundary, and $F : M \to N$ is an injective smooth immersion. Then F is a smooth embedding if one of the following equivalent conditions holds:

- F is an open or closed map.
- F is a proper map, that is, for every compact subset $K \subseteq M$, the image $F(K)$ is compact in N.
- M is compact.
- M has empty boundary and $\dim M = \dim N$.

Proof. The first claim is just the definition of a topological embedding. The second claim follows from the fact that a proper map is a closed map, and the third claim follows from the fact that a compact set is closed. The fourth claim follows from the fact that if M has empty boundary, then it is an open submanifold of itself, and if $\dim M = \dim N$, then F is a local diffeomorphism, hence an open map. □

Theorem 4.2.1: Local Embedding Theorem

Suppose M and N are smooth manifolds, with or without boundary, and $F : M \to N$ is a smooth map. Then F is a smooth immersion iff every point $p \in M$ has a neighborhood U such that $F|_U : U \to N$ is a smooth embedding.

Proof. If F is a local smooth embedding on every point, then it has full rank everywhere, hence it is a smooth immersion. Conversely, if F is a smooth immersion, then for each point $p \in M$:

If $F(p) \notin \partial N$, then by the Rank Theorem, there exist coordinate charts (U, φ) on M centered at p and (V, ψ) on N centered at $F(p)$ such that the coordinate representation

$$\hat{F}(x^1, \ldots, x^m) = (x^1, \ldots, x^m, 0, \ldots, 0).$$

Then restricting U if necessary, we have that $F|_U : U \to N$ is injective. If $F(p) \in \partial N$, then the same argument applies slight adjustment by $\mathbb{H}^n$ coordinate charts.

Next, let U_1 be a precompact neighborhood of p in U such that $F|_{\overline{U_1}}$ is injective and has compact domain. Then $F|_{\overline{U_1}} : \overline{U_1} \to N$ is a topological embedding. □

Remark:
This gives a direct notion for topological immersions: a continuous map between topological spaces that is a local topological embedding at every point.

4.3 Submersions

Definition 4.3.1: Section

Let M and N be topological spaces, and let $\pi : M \to N$ be a continuous map. A **section** of π is a continuous map $\sigma : N \to M$ such that $\pi \circ \sigma = \mathrm{id}_N$.
A local section of π over an open set $U \subseteq N$ is a continuous map $\sigma : U \to M$ such that $\pi \circ \sigma = \mathrm{id}_U$.

Really, sections are right inverses of the map π, so it is injective.

Note that global sections need not exist, for example, consider the S^1 projection onto RP^1. However, local sections always exist for submersions, as the following result shows.

Theorem 4.3.1: Local Section Theorem

Let M and N be smooth manifolds, and $\pi : M \to N$ be a smooth map. Then π is a smooth submersion if and only if for each point $p \in M$, it is in an image of a smooth local section of π.

Proof. Suppose π is a smooth submersion, and given $p \in M$. By the Rank Theorem, there exist smooth coordinate charts (U, φ) on M centered at p and (V, ψ) on N centered at $\pi(p)$ such that the coordinate representation $\pi(x^1, \ldots, x^m) = (x^1, \ldots, x^n)$. For sufficiently small ϵ, the coordinate cube

$$C_\epsilon = \{(x^1, \ldots, x^n) \in \mathbb{R}^n : |x^i| < \epsilon, 1 \leqslant i \leqslant m\}$$

Then $\pi(C_\epsilon)$ is also a coordinate cube in $\mathbb{R}^n$. The coordinate map by

$$\sigma(x^1, \ldots, x^n) = (x^1, \ldots, x^n, 0, \ldots, 0)$$

is a smooth local section.

Conversely, from $\pi \circ \sigma = \mathrm{id}_N$, we have that $\mathrm{d}\pi_{\sigma(q)} \circ \mathrm{d}\sigma_q = \mathrm{id}_{T_qN}$ for each $q \in N$. Hence, $\mathrm{d}\pi_{\sigma(q)}$ is surjective for each $q \in N$, so π is a smooth submersion. □

Remark:
This theorem motivates the definition of a topological submersion: a continuous map between topological spaces such that every point in the domain is in the image of a local section.

Proposition: **Properties of Smooth Submersions**

Let M and N be smooth manifolds, and let $\pi : M \to N$ be a smooth submersion. Then

- π is an open map.
- If π is surjective, then it is a quotient map.

Proof. For the first claim, let $W \subseteq M$ be open and $q \in \pi(W)$, take any $p \in W$ such that $\pi(p) = q$. By the Local Section Theorem, there exists a smooth local section $\sigma : U \to M$ of π such that $\sigma(q) = p$. Then $\sigma^{-1}(W)$ is an open neighborhood of q contained in $\pi(W)$, so $\pi(W)$ is open in N.

The second claim follows from the first claim and the definition of a quotient map. □

We can see that smooth submersions plays a similar role to that of quotient maps in topology.

Theorem 4.3.2: Passing Smoothly to Quotient

Suppose M and N are smooth manifolds, and $\pi : M \to N$ is a surjective smooth submersion. If P is a smooth manifold, with or without boundary, then a map $F : N \to P$ is smooth if and only if $F \circ \pi : M \to P$ is smooth.
Moreover, if $G : M \to P$ is a smooth map that is constant on each fiber of π, then there exists a unique smooth map $\tilde{G} : N \to P$ such that $G = \tilde{G} \circ \pi$.

Proof. If $F : N \to P$ is smooth, then $F \circ \pi : M \to P$ is smooth by composition of smooth maps. Conversely, suppose that $F \circ \pi : M \to P$ is smooth. Given any point $q \in N$, take any $p \in M$ such that $\pi(p) = q$. By the Local Section Theorem, there exists a smooth local section $\sigma : U \to M$ of π such that $\sigma(q) = p$. Then the restriction $F|_U = (F \circ \pi) \circ \sigma : U \to P$ is smooth. Since q was arbitrary, F is smooth. □

Theorem 4.3.3: Uniqueness of Smooth Quotients

Suppose M and N_1, N_2 are smooth manifolds, and $\pi_1 : M \to N_1$ and $\pi_2 : M \to N_2$ are surjective smooth submersions that are constant on each other's fibers. Then there exists a unique diffeomorphism $F : N_1 \to N_2$ such that $F \circ \pi_1 = \pi_2$.

4.4 Smooth Covering Maps

In general topology, a **covering map** is a continuous surjective map $\pi : X \to Y$ such that for each point $y \in Y$, there exists an open neighborhood U of y such that $\pi^{-1}(U)$ is a disjoint union of open sets in X, each of which is homeomorphic to U via π.

In the context of smooth manifolds, we have the following definition.

Definition 4.4.1: Smooth Covering Map

Let E and M be connected smooth manifolds, with or without boundary. A **smooth covering map** is a smooth surjective map $\pi : E \to M$ such that for each point $p \in M$, there exists an open neighborhood U of p such that $\pi^{-1}(U)$ is a disjoint union of open sets in E, each of which is diffeomorphic to U via π.
We say that M is the **base space**, E is the **covering space**, and if E is simply connected, we say that E is the **universal covering space** of M.

Proposition: **Properties of Smooth Covering Maps**

Let E and M be connected smooth manifolds, with or without boundary, and let $\pi : E \to M$ be a smooth covering map. Then

- π is a local diffeomorphism, a smooth submersion, an open map and a quotient map.
- An injective smooth covering map is a diffeomorphism.
- A topological covering map between smooth manifolds is a smooth covering map if and only if it is a local diffeomorphism.

- Every local section of π is a smooth local section.

Example: **Smooth Covering Maps**

- The map $\mathbb{R} \to S^1$ defined by $t \mapsto (\cos t, \sin t)$ is a smooth covering map. Its universal covering space is $\mathbb{R}$ itself.
- For $n \geqslant 1$, the map $q : S^n \to \mathbb{R}P^n$ defined by $q(x) = [x]$ is a two-to-one smooth covering map.

For smooth covering maps, we also have the local section theorem strengthened.

Theorem 4.4.1: Local Section Theorem for Smooth Covering Maps

Let E and M be connected smooth manifolds, with or without boundary, and let $\pi : E \to M$ be a smooth covering map. Then for each evenly covered open set $U \subseteq M$, a point $q \in U$, and $p \in \pi^{-1}(q)$, there exists a unique smooth local section $\sigma : U \to E$ of π such that $\sigma(q) = p$.

Proof. This is quite obvious from the definition of smooth covering maps. □

Proposition: **Products of Covering Maps**

Let $E_1, \ldots, E_k$ and $M_1, \ldots, M_k$ be smooth manifolds without boundary, and let $\pi_i : E_i \to M_i$ be smooth covering maps for each $1 \leqslant i \leqslant k$. Then the product map

$$\pi_1 \times \cdots \times \pi_k : E_1 \times \cdots \times E_k \to M_1 \times \cdots \times M_k$$

is a smooth covering map.

Theorem 4.4.2: Covering Space of Smooth manifolds

Suppose M is a connected smooth n-manifold, and $\pi : E \to M$ is a topological covering map, then E is a topological n-manifold and has a unique smooth structure such that $\pi : E \to M$ is a smooth covering map.
Moreover, if M is smooth manifold with boundary, then E is a topological manifold with boundary such that $\partial E = \pi^{-1}(\partial M)$, and has a unique smooth structure with boundary such that $\pi : E \to M$ is a smooth covering map.

Proof. SORRY. □

Corollary 4.4.1: Existence of Universal Covering Manifold

If M is a connected smooth manifold, there exists a simply connected smooth manifold $\tilde{M}$ called the **universal covering manifold** of M, and a smooth covering map $\pi : \tilde{M} \to M$. The universal covering manifold is unique up to diffeomorphism. That is, if $\tilde{M}'$ is another simply connected smooth manifold with a smooth covering map $\pi' : \tilde{M}' \to M$, then there exists a diffeomorphism $F : \tilde{M} \to \tilde{M}'$ such that $\pi' \circ F = \pi$.

There are not many simple criterion for a smooth map to be a smooth covering map, but we do have the following sufficient (not necessary) condition.

Proposition: **Local Proper Diffeomorphism is a Smooth Covering Map**

Let E and M be connected smooth manifolds, with or without boundary, and let $\pi : E \to M$ be a smooth map. If π is a local diffeomorphism and a proper map, then it is a smooth covering map.

Chapter 5

Submanifolds

We have already seen that open subsets of manifolds are themselves manifolds. But the range of possible submanifolds is much broader.

5.1 Embedded Submanifolds

Definition 5.1.1: Embedded Submanifold

Suppose M is a smooth manifold, with or without boundary. An embedded submanifold of M is a subset $S \subseteq M$ equipped with the subspace topology and a smooth structure such that the inclusion map $\iota_S : S \hookrightarrow M$ is a smooth embedding.

If S is an embedded submanifold of M, then the difference $\dim M - \dim S$ is called the **codimension** of S in M. M is called the ambient manifold of S. An embedded hypersurface is an embedded submanifold of codimension 1.

Proposition: **Open Submanifolds**

Suppose M is a smooth manifold. The embedded submanifolds of M of codimension 0 are precisely the open subsets of M.

Proof. Suppose $U \subseteq M$ is open. Then it has the same dimension as M, and the inclusion map $\iota_U : U \hookrightarrow M$ is a smooth embedding, so U is an embedded submanifold of codimension 0.

Conversely, suppose U is an embedded submanifold of codimension 0. Then the inclusion map $\iota_U : U \hookrightarrow M$ is a smooth embedding, so it is a local diffeomorphism and an open map. Thus, U is open in M. □

We can produce embedded submanifolds using images of embeddings.

Proposition: **Images of Embeddings**

Suppose M is a smooth manifold, with or without boundary, and N is a smooth manifold without boundary. If $F : N \to M$ is a smooth embedding, then $S = F(N)$ with the subspace topology has a unique smooth structure making it into an embedded submanifold of M such that $F : N \to S$ is a diffeomorphism.

Remark:
As by definition, embedded submanifolds are images of embeddings, this proposition shows that embedded submanifolds are exactly images of embeddings.

Proposition: **Slices of Products**

Suppose M and N are smooth manifolds. For each $p \in N$, the subset $M \times \{p\} \subseteq M \times N$ is an embedded submanifold that is diffeomorphic to M.

Proposition: **Graphs as Submanifolds**

Suppose M is a smooth m-manifold without boundary, and N is a smooth n-manifold with or without boundary. If $U \subseteq M$ be open, and $f : U \to N$ is a smooth map, then let $\Gamma(f)$ denote the **graph** of f,

$$\Gamma(f) = \{(p, f(p)) \in M \times N : p \in U\}.$$

Then $\Gamma(f)$ is an embedded m-submanifold of $M \times N$ that is diffeomorphic to U.

Sometimes, merely being an embedded submanifold is not enough. An embedded submanifold S of M is said to be properly embedded if the inclusion map $\iota_S : S \hookrightarrow M$ is a proper map.

Proposition: **Criterion for Properly Embedded Submanifolds**

Suppose M is a smooth manifold, with or without boundary, and S is an embedded submanifold of M. Then S is properly embedded if and only if S is a closed subset of M.
Therefore, we have every compact embedded submanifold is properly embedded.

Proof. Suppose S is properly embedded. Then the inclusion map $\iota_S : S \hookrightarrow M$ is a proper map, so the preimage of every compact set in M is compact in S. In particular, the preimage of every closed set in M is closed in S. Since S has the subspace topology, this implies that S is closed in M.

Conversely, suppose S is closed in M. Then for any compact set $K \subseteq M$, the intersection $K \cap S$ is closed in K, and since K is compact, $K \cap S$ is also compact. Thus, the preimage of every compact set in M under the inclusion map ι_S is compact in S, so ι_S is a proper map. Therefore, S is properly embedded. □

Proposition: **Global Graphs are Properly Embedded**

Suppose M is a smooth m-manifold without boundary, and N is a smooth n-manifold with or without boundary. If $f : M \to N$ is a smooth map, then the graph $\Gamma(f)$ is a properly embedded m-submanifold of $M \times N$ that is diffeomorphic to M.

5.1.1 Slice Charts for Embedded Submanifolds

We will show that embedded submanifolds are modeled locally by the standard embedding R^k into $\mathbb{R}^n$ as the first k-coordinates:

$$\mathbb{R}^k \hookrightarrow \mathbb{R}^n, \quad (x^1, \ldots, x^k) \mapsto (x^1, \ldots, x^k, 0, \ldots, 0).$$

More generally, a k-slice of $U \subseteq \mathbb{R}^n$ is any subset of the form

$$S = \{(x^1, \ldots, x^n) \in U : x^{k+1} = c^{k+1}, \ldots, x^n = c^n\}$$

for some constants $c^{k+1}, \ldots, c^n \in \mathbb{R}$. Note that every k-slice is diffeomorphic to an open subset of $\mathbb{R}^k$ via the projection map onto the first k-coordinates.

Now, let M be a smooth manifold and (U, φ) be a smooth chart on M. If $S \subseteq U$ that $\varphi(S)$ is a k-slice of $\varphi(U)$, then we say that S is a k-**slice of** U. In general, we allow slices have any constant values in their last $n-k$ coordinates.

Given a subset $S \subseteq M$, and $k \geqslant 0$, we say S satisfies the local k-slice condition if each point $p \in S$ has a smooth chart (U, φ) for M such that $p \in U$ and $S \cap U$ is a k-slice of U. Any such chart is called a **slice chart** for S in M and the corresponding coordinates $(x^1, \ldots, x^n)$ are called **slice coordinates** for S in M.

Theorem 5.1.1: Local Slice Criterion for Embedded Submanifolds

Let M be a smooth n-manifold, and if $S \subseteq M$ is an embedded k-submanifold, then S satisfies the local k-slice condition.
Conversely, if $S \subseteq M$ is a subset that satisfies the local k-slice condition, then with the subspace topology S is a topological manifold of dimension k, and there is a smooth structure making S into an embedded k-submanifold of M.

Proof. SORRY □

Later we shall see that the smooth structure constructed in the theorem is the unique one in which S is an embedded submanifold of M.

Also, if M is a smooth manifold with boundary, and $S \subseteq M$ is an embedded submanifold, then S might interment ∂M in complicated ways. However, if $S = \partial M$ itself, then the boundary charts for M is just slice charts for S in M, and we do have the following proposition.

Theorem 5.1.2: Boundary as Embedded Submanifold

If M is a smooth n-manifold with boundary, then ∂M with the subspace topology is a topological $(n-1)$-manifold without boundary, and there is a smooth structure making ∂M a properly embedded $(n-1)$-submanifold of M.

Later, we shall see that this smooth structure is the unique one in which ∂M is an embedded submanifold of M.

Remark:
In order to study submanifolds of manifolds with boundary in greater generality, a typical approach is to find an embedding of M into a larger smooth manifold $\tilde{M}$ without boundary.

5.1.2 Level Sets

In practice, many embedded submanifolds arise as solution sets of systems of equations. If $\Phi : M \to N$ be any map and $c \in N$, we call the set

$$\Phi^{-1}(c) = \{p \in M : \Phi(p) = c\}$$

the **level set** of Φ at c. In the special case where $N = \mathbb{R}^k$ and $c = 0$, we call $\Phi^{-1}(0)$ the **zero set** of Φ.

It is easy to find examples where level sets of smooth functions that are not smooth submanifold. As we previously saw, all closed subset of M can be expressed as the zero set of some smooth function $M \to \mathbb{R}$. However, we have

Theorem 5.1.3: Constant Rank Level Set Theorem

Suppose M, N are smooth manifolds, and $\Phi : M \to N$ is a smooth map with constant rank r. Then each level set of Φ is a properly embedded submanifold of M with codimension r. Specifically, if Φ is a submersion, then each level set is a properly embedded submanifold of codimension $\dim N$.

Proof. Let $\dim M = m$ and $\dim N = n$, and $k = m - r$ be the codimension. $\forall c$, let $S = \Phi^{-1}(c)$. From the rank theorem, $\forall p \in S$, there exist smooth charts (U, φ) around p in M and (V, ψ) around c in N such that Φ has the local representation

$$\Phi : (x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^r, 0, \ldots, 0).$$

So we have

$$S \cap U = \{(x^1, \ldots, x^m) \in U : x^1 = 0, \ldots, x^r = 0\},$$

Hence, S satisfies the local k-slice condition. By the Local Slice Criterion for Embedded Submanifolds, S is an embedded k-submanifold of M.

Finally, to see that S is properly embedded, note that Φ is continuous, so $S = \Phi^{-1}(c)$ is closed in M. By the Criterion for Properly Embedded Submanifolds, S is properly embedded. □

Remark:
This corresponds to the familiar rank-nullity theorem from linear algebra.

Definition 5.1.2: Regular and Critical Point

Suppose M, N are smooth manifolds, and $\Phi : M \to N$ is a smooth map. A point $p \in M$ is called a **regular point** of Φ if the differential $d\Phi_p : T_pM \to T_{\Phi(p)}N$ is surjective. If p is not a regular point, then it is called a **critical point** of Φ. The point $c \in N$ is called a **regular value** of Φ if every point in the preimage $\Phi^{-1}(c)$ is a regular point (we include $\Phi^{-1}(c) = \varnothing$). If c is not a regular value, then it is called a **critical value** of Φ.

Now, we weakens the hypothesis of the Constant Rank Level Set Theorem to only require that c is a regular value.

Corollary 5.1.1: Regular Level Set Theorem

Suppose M, N are smooth manifolds, and $\Phi : M \to N$ is a smooth map. If $c \in N$ is a regular value of Φ, then the level set $\Phi^{-1}(c)$ is a properly embedded submanifold of M with codimension $\dim N$.

Proof. From proposition 4.1, The set

$$U = \{p \in M : d\Phi_p \text{ is surjective}\}$$

is open in M and contains $\Phi^{-1}(c)$. The restriction $\Phi|_U : U \to N$ is a submersion, so by the Constant Rank Level Set Theorem, $\Phi^{-1}(c)$ is a properly embedded submanifold of U with codimension $\dim N$. Then take the composition $\Phi^{-1}(c) \hookrightarrow U \hookrightarrow M$ would would do. □

Not all embedded submanifolds arise as level sets of smooth maps. However, we do know that they at least locally do.

Proposition: **Local Level Set Representation of Embedded Submanifolds**

Suppose M is a smooth m-manifold, and $S \subseteq M$, then S is a embedded k-submanifold of M if and only if $\forall p \in S$, there exist an open neighborhood U of p in M and a smooth submersion $\Phi : U \to \mathbb{R}^{m-k}$ such that $S \cap U = \Phi^{-1}(c)$ for some $c \in \mathbb{R}^{m-k}$.

Definition 5.1.3: Defining Map

If $S \subseteq M$ is an embedded submanifold, then a smooth map $\Phi : M \to N$ that has S as a regular level set is called a **defining map** for S in M. If $N = \mathbb{R}^{m-k}$, we say Φ is a **defining function** for S in M. For example, $f(x) = |x|^2$ is a defining function for the sphere in $\mathbb{R}^n$. Generally, if $U \subseteq M$ is open and $\Phi : U \to N$ is a smooth map that has $S \cap U$ as a regular level set, then we say Φ is a **local defining map** for S in M.

The last proposition shows that every embedded submanifold has local defining function.

Example: **Surface of Revolution**

Let H be the half plane $\{(r, z) \in \mathbb{R}^2 : r > 0\}$, and $C \subseteq H$ be a one-dimensional embedded submanifold. Then the **surface of revolution** generated by C is the subset

$$S_C = \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2}, z) \in C\}. \tag{5.1}$$

If $\varphi : U \to \mathbb{R}$ is any locally defining function for C in H, then the map

$$\Phi : \tilde{U} \to \mathbb{R}, \quad \Phi(x, y, z) = \varphi(\sqrt{x^2 + y^2}, z) \tag{5.2}$$

where $\tilde{U} = \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2}, z) \in U\}$ is a local defining function for S_C in $\mathbb{R}^3$. Thus, by the Local Level Set Representation of Embedded Submanifolds, S_C is an embedded submanifold of $\mathbb{R}^3$.

5.2 Immersed Submanifolds

Definition 5.2.1: Immersed Submanifold

Suppose M is a smooth manifold, with or without boundary. An immersed submanifold of M is a subset $S \subseteq M$ equipped with a topology (not necessarily the subspace topology) making it a topological manifold without boundary, and a smooth structure such that the inclusion map $\iota_S : S \hookrightarrow M$ is a smooth immersion.

Remark:
This is a rather larger class of submanifolds, as every embedded submanifold is an immersed submanifold, but not conversely. We shall simply denote "smooth submanifold" to mean "immersed submanifold" unless otherwise specified. A smooth hypersurface is an immersed submanifold of codimension 1.
We can also define a immersed topological submanifold similarly, as a subset $S \subseteq M$ equipped with a topology making it a topological manifold (not necessarily the subspace topology) such that the inclusion map $\iota_S : S \hookrightarrow M$ is a topological immersion.

Usually, immersed submanifolds arise as images of immersions.

Proposition: **Images of Immersions as Submanifolds**

Suppose M is a smooth manifold, with or without boundary, and N is a smooth manifold without boundary. If $F : N \to M$ is an injective smooth immersion, then $S = F(N)$ has a unique topology and smooth structure making it into an immersed submanifold of M such that $F : N \to S$ is a diffeomorphism.

Proof. We shall define the topology on S to be $\{U \cap S : F^{-1}(U) \text{ is open in } N\}$. And the smooth structure is defined by the charts $\{F(U), \varphi \circ F^{-1}\}$ where (U, φ) are charts of N. □

Example: **Immersed Submanifolds**

The figure eight curve and the dense curve on the torus are examples of immersed submanifolds that are not embedded submanifolds.

Remark:
In fact, suppose M is a smooth manifold and $S \subseteq M$ is an immersed submanifold. Then every subset of S that is open in the subspace topology is open in the topology of S, but the converse is not necessarily true. The converse holds if and only if S is an embedded submanifold of M.

Proposition: **From Immersed to Embedded Submanifolds**

Suppose M is a smooth manifold, with or without boundary, and S is an immersed submanifold of M. If one of the following conditions holds, then S is an embedded submanifold of

M:

- S has codimension 0 in M.
- The inclusion map $\iota_S : S \hookrightarrow M$ is a proper map.
- S is compact.

Proposition: **Locally Embeddedness of Immersed Submanifolds**

Suppose M is a smooth manifold, with or without boundary, and S is an immersed submanifold of M. Then $\forall p \in S$, there exists an open neighborhood U of p in S that U is an embedded submanifold of U.

Remark:
This does NOT mean that we can find an open neighborhood W of p in M such that $S \cap W$ is an embedded submanifold of W.

Definition 5.2.2: Local Parametrization

Suppose M is a smooth manifold, with or without boundary, and S is an immersed k-submanifold of M. A **local parametrization** for S in M is a continuous map $X : U \to M$ such that

- U is an open subset of $\mathbb{R}^k$,
- $X(U)$ is an open subset of S (in the topology of S),
- $X : U \to X(U)$ is a homeomorphism (in the topology of S),

It is called a **smooth local parametrization** if $X : U \to X(U)$ is a diffeomorphism onto its image (with the smooth structure of S). If $X(U) = S$, then X is called a **global parametrization** of S in M.

Proposition: **Criterion for Local Parametrization**

Suppose M is a smooth manifold, with or without boundary, and S is an immersed k-submanifold of M. Let $\iota : S \hookrightarrow M$ be the inclusion map. A map $X : U \to M$ is a local parametrization for S in M if and only if there is a smooth coordinate chart (V, φ) for S that $X = \iota \circ \varphi^{-1}$.
Therefore, every point in S is in the image of a local parametrization.

Example: **Parametrizations**

- Graph parametrizations: Suppose $U \subseteq \mathbb{R}^n$ is an open subset and $f : U \to \mathbb{R}^k$ is a

smooth function. Then the map

$$\gamma_f : U \to \mathbb{R}^{n+k}, \quad \gamma_f(x) = (x, f(x))$$

is a global parametrization for the graph $\Gamma(f)$ of f in $\mathbb{R}^{n+k}$.

- Figure-eight curve: Let $S \subseteq \mathbb{R}^2$ be the figure-eight curve, Then the map

$$\beta : (-\pi, \pi) \to \mathbb{R}^2, \quad \beta(t) = (\sin t, \sin(2t))$$

is a global parametrization for S in $\mathbb{R}^2$.

5.3 Restricting Maps to Submanifolds

Theorem 5.3.1: Restricting the Domain of a Smooth Map

Suppose M, N are smooth manifolds, with or without boundary, and S is an immersed submanifold of M. If $F : M \to N$ is a smooth map, then the restriction $F|_S : S \to N$ is a smooth map.

Proof. The inclusion map $\iota_S : S \hookrightarrow M$ is smooth. And $F_S = F \circ \iota_S$, so F_S is smooth. □

But we cannot generally restrict the codomain of a smooth map to an immersed submanifold. For example, take

$$G : \mathbb{R} \to \mathbb{R}^2, \quad G(t) = (\sin t, \sin 2t),$$

it is a smooth map whose image is the figure-eight curve S. However, if we consider G as a map from $\mathbb{R}$ to S, then it is not even continuous at π.

Theorem 5.3.2: Restricting the Codomain of a Smooth Map

Suppose M, N are smooth manifolds without boundary, and S is an immersed submanifold of M, and $F : N \to M$ is a smooth map such that $F(N) \subseteq S$. If $F : N \to S$ is continuous in the topology of S, then it is smooth.
The result also holds when M has nonempty boundary. If S is an embedded submanifold of M, then the continuity hypothesis is automatically satisfied.

However, there are certain immersed but not embedded submanifolds that the result automatically holds without the continuity hypothesis. To distinguish them, we introduce the following definition.

Definition 5.3.1: Weakly Embedded

Suppose M is a smooth manifold, and S is an immersed submanifold of M. Then S is said to be **weakly embedded** if for every smooth manifold N and every smooth map $F : N \to M$ such that $F(N) \subseteq S$, the induced map $F : N \to S$ is smooth (without any additional continuity hypothesis).

5.3.1 Uniqueness of Smooth Structures on Submanifolds

SORRY

5.3.2 Extending Functions from Submanifolds

Lemma 5.3.1: Extension Lemma For Submanifolds

Suppose M is a smooth manifold, and $S \subseteq M$ is an immersed submanifold. If $f : S \to \mathbb{R}$ is a smooth function on the submanifold structure, denote $f \in C^\infty(S)$. Then

- If S is embedded, then there exist a neighborhood U of S in M and a smooth function $\tilde{f} \in C^\infty(U)$ such that $\tilde{f}|_S = f$.
- If S is properly embedded, then U can be taken to be all of M.

5.4 The Tangent Space to a Submanifold

Suppose M is a smooth manifold, with or without boundary, and S is an immersed submanifold of M. Since the inclusion map $\iota : S \hookrightarrow M$ is a smooth immersion, for each $p \in S$, we can identify the tangent space T_pS as a subspace of T_pM $\mathrm{d}\iota_p$:

$$\mathrm{d}\iota_p(v)f = v(f \circ \iota) = v(f|_S), \quad \forall v \in T_pS, \forall f \in C^\infty(M).$$

Proposition: **Identify Submanifold Tangent Space**

Suppose M is a smooth manifold, with or without boundary, and S is an immersed submanifold of M, and $p \in S$. Then a vector $v \in T_pM$ is in the subspace $T_pS \subseteq T_pM$ if and only if there exists a smooth curve $\gamma : J \to M$ such that

- $\gamma(J) \subseteq S$,
- γ is smooth as a map into S,
- $0 \in J$ and $\gamma(0) = p$,
- $\gamma'(0) = v$.

If S is an embedded submanifold, then we have

$$T_pS = \{v \in T_pM : \forall f \in C^\infty(M), f|_S = 0, vf = 0\}. \tag{5.3}$$

Proof. Quite clear from the local embeddedness of immersed submanifolds. □

We can also characterize tangent spaces via defining maps.

Proposition: **Tangent Space via Defining Maps**

Suppose M is a smooth manifold, and S is an embedded submanifold of M. If $\Phi : U \to N$ is any local defining map for S in M then

$$T_pS = \ker \mathrm{d}\Phi_p : T_pM \to T_{\Phi(p)}N, \tag{5.4}$$

for each $p \in S \cap U$.
Specifically, if S is a level set of a smooth submersion $\Phi : M \to \mathbb{R}^k$, then

$$T_pS = \{v \in T_pM : v\Phi^i = 0, i = 1, \dots, k\}, \tag{5.5}$$

If M is a smooth manifold with boundary, and $p \in \partial M$, intuitively, we expect that we can classify the tangent vectors to three categories: those that point inward to M, those that point outward to M, and those that are tangent to the boundary ∂M itself. This is indeed the case. We interpret the boundary as an embedded submanifold of M from theorem 5.1.2.

Definition 5.4.1: Tangent Vectors On the Boundary

If $p \in \partial M$, then a vector $v \in T_pM - T_p(\partial M)$ is said to be **inward-pointing** if $\exists \epsilon > 0$, there is a smooth curve $\gamma : [0, \epsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$. It is said to be **outward-pointing** if the same holds for a smooth curve $\gamma : (-\epsilon, 0] \to M$.

Proposition: **Boundary Tangent Vectors from Coordinates**

Let M be a smooth n-manifold with boundary, and $p \in \partial M$. If (U, φ) is a boundary chart for M around p, with coordinates $(x^1, \dots, x^n)$, then a vector $v \in T_pM$ is inward-pointing if and only if it has positive x^n-component, outward-pointing if and only if it has negative x^n-component, and tangent to the boundary if and only if its x^n-component is zero.
This gives a partition of T_pM by

$$T_pM = \{\text{inward-pointing}\} \sqcup T_p(\partial M) \sqcup \{\text{outward-pointing}\}. \tag{5.6}$$

Definition 5.4.2: Boundary Defining Function

Suppose M is a smooth manifold with boundary. A **boundary defining function** for M is a smooth function $f : M \to [0, \infty)$ such that $\partial M = f^{-1}(0)$ and $\mathrm{d}f_p \neq 0$ for all $p \in \partial M$.

For example, the defining function for a closed unit ball in $\mathbb{R}^n$ is $f(x) = 1 - |x|^2$.

Proposition: **Existence of Boundary Defining Functions**

Suppose M is a smooth manifold with boundary. Then there exists a boundary defining function for M.

Proof. Let $\{U_\alpha, \varphi_\alpha\}$ be a collection of smooth charts covering M, define $f_\alpha : U_\alpha \to [0, \infty)$ by

- If U_α is an interior chart then $f_\alpha = 1$.
- If U_α is a boundary chart with coordinates $(x^1, \dots, x^n)$, then $f_\alpha = x^n$.

Thus $f_\alpha > 0$ in the interior and $f_\alpha = 0$ on the boundary. Take any partition of unity $\{\psi_\alpha\}$ subordinate to the cover and taking $f = \sum_\alpha \psi_\alpha f_\alpha$ would do. To see $\mathrm{d}f_p \neq 0$ for all $p \in \partial M$, we have

$$\mathrm{d}f_p(v) = \sum_\alpha (f_\alpha \mathrm{d}\psi_\alpha|_p(v) + \psi_\alpha(p)\mathrm{d}f_\alpha|_p(v)) = \sum_\alpha \psi_\alpha(p)\mathrm{d}f_\alpha|_p(v),$$

□

Remark:
Usually, it is fairly easy to say that if a subset of M is an embedded submanifold for they are exactly those satisfying the local slice condition. However, it is often much more difficult to determine whether a subset is an immersed submanifold. A common technique is to first assume it is, then derive a contradiction from some phenomenon:

- $\forall p \in S$, the tangent space T_pS is a linear subspace of T_pM with constant dimension.
- $\forall p \in S$, it is in the image of a local parametrization.
- Each vector tangent to S at p is the velocity vector of a smooth curve in S through p.
- Each vector tangent to S at p annihilates all smooth functions on M that vanish on S.

Example: **Proving Smooth Submanifolds**

Consider

$$S = \{(x, y) : y = |x|, x \in \mathbb{R}\} \subseteq \mathbb{R}^2.$$

It is easy that $S - \{(0, 0)\}$ is an embedded submanifold of $\mathbb{R}^2$. If S is a smooth submanifold of $\mathbb{R}^2$, then it must be one-dimensional from local embeddedness. Then $T_{(0,0)}S$ must be a one-dimensional linear subspace of $T_{(0,0)}\mathbb{R}^2 \cong \mathbb{R}^2$. This means that there is a smooth curve $\gamma : (-\epsilon, \epsilon) \to S$ such that $\gamma(0) = (0, 0)$ and $\gamma'(0) \neq 0$. However, the only such curve is $\gamma(t) = (0, 0)$ for all t, which contradicts $\gamma'(0) \neq 0$. Hence, S is not a smooth submanifold of $\mathbb{R}^2$.

5.5 Submanifolds with Boundary

The definition is very similar to the case without boundary.

Definition 5.5.1: Submanifold with Boundary

Suppose M is a smooth manifold with or without boundary. A submanifold with boundary of M is a subset $S \subseteq M$ equipped with a topology making it a topological manifold with boundary, and a smooth structure such that the inclusion map $\iota_S : S \hookrightarrow M$ is a smooth immersion.
If the inclusion map is a smooth embedding, then S is called an **embedded submanifold with boundary** of M, and in the general case, it is called an **immersed submanifold with boundary** of M.
A regular domain of M is a properly embedded submanifold with boundary of codimension 0.

Proposition: **Topological Boundary and Manifold Boundary**

Suppose M is a smooth manifold without boundary, and $D \subseteq M$ is a regular domain. Then the topological boundary and interior of D in M coincide with the manifold boundary and interior of D as a manifold with boundary, respectively.

Proof. Simply due to D having the subspace topology from M. □

Proposition: **Generating Regular Domains**

Suppose M is a smooth manifold without boundary, and $f \in C^\infty(M)$, then

- For each regular value $b \in \mathbb{R}$ of f, the set $f^{-1}((-\infty, b])$ is a regular domain in M. It is called a **sublevel set** of f. And if D is a regular domain that $D = f^{-1}((-\infty, b])$ for some f and b, then f is called a **defining function** for D in M.
- For each regular value $a < b$ in f, then the set $f^{-1}([a, b])$ is a regular domain in M.

Theorem 5.5.1: Existence of Sublevel Defining Functions

If M is a smooth manifold without boundary, and $D \subseteq M$ is a regular domain, then there exists a smooth function $f \in C^\infty(M)$ being a defining function for D in M. If D is compact, then f can be chosen to be a smooth exhaustion function on M.

Proposition: **Properties of Submanifolds with Boundary**

Suppose M is a smooth manifold with or without boundary, then

- Every open subset of M is an embedded submanifold with or without boundary of codimension 0.
- If N is a smooth manifold with boundary, and $F : N \to M$ is a smooth embedding, then $F(N)$ is an embedded submanifold with boundary of M, with the subspace topology and smooth structure.
- If $S \subseteq M$ is an immersed submanifold with boundary of M, then for each $p \in S$ there exists a neighborhood U of p in S such that U is an embedded submanifold with boundary of M.

A k-dimensional half-slice of U is a subset

$$\{(x^1, \ldots, x^n) \in U : x^{k+1} = c^{k+1}, \ldots, x^n = c^n, x^n \geqslant 0\},$$

We say that $S \subseteq M$ satisfies the **local half-slice condition** if $\forall p \in S$, there exist a chart (U, φ) for M around p such that $S \cap U$ is a k-dimensional usual slice or half-slice of U. In the former case it is called the interior slice chart of S in M, and in the latter case it is called the boundary slice chart of S in M.

Theorem 5.5.2: Local Half-Slice Criterion for Embedded Submanifolds with Boundary

Suppose M is a smooth manifold with or without boundary, and $S \subseteq M$. Then S is an embedded submanifold with boundary of M if and only if it satisfies the local half-slice condition.

Theorem 5.5.3: Restricting the Domain of a Smooth Map to a Submanifold with Boundary

Suppose M, N are smooth manifolds with boundary and $S \subseteq M$ is an embedded submanifold with boundary.

- Restricting the domain: If $F : M \to N$ is a smooth map, then the restriction $F|_S : S \to N$ is a smooth map.
- Restricting the codomain: If $\partial M \neq \varnothing$, and $F : N \to M$ is a smooth map such that $F(N) \subseteq S$, then the induced map $F : N \to S$ is smooth.

Remark:
Actually, the requirement $\partial M \neq \varnothing$ is not necessary.

Chapter 6

Sard's Theorem

We study the behavior of critical values of smooth maps between manifolds. Sard's Theorem states that the set of critical values has measure zero in the target manifold.

6.1 The Sard's Theorem

Theorem 6.1.1: The Sard's Theorem

Suppose M, N are smooth manifolds, with or without boundary, and $F : M \to N$ is a smooth map. Then the set of critical values of F has measure zero in N.

This shows that if $\dim M < \dim N$, then $F(M)$ has measure zero in N. Because each point of M is critical for F.

6.2 The Whitney Embedding Theorem

Now we formalize out intuition that smooth manifolds are smoooth "surfaces" in Euclidean space.

Firstly, we show that an injective immersion of an n-dimensional manifold into $\mathbb{R}^N$ can be turned into a lower dimensional immersion if $N > 2n + 1$.

Lemma 6.2.1: Lower the Immersion Dimension

Suppose $M \subseteq \mathbb{R}^N$ is a smooth n-dimensional submanifold with or without boundary. Let $\mathbb{R}^{N-1}$ be the subspace of $\mathbb{R}^N$ with last coordinate zero. For any $v \in \mathbb{R}^N - \mathbb{R}^{N-1}$, let $\pi_v : \mathbb{R}^N \to \mathbb{R}^{N-1}$ be the projection with kernel $\mathbb{R}v$. If $N > 2n+1$, then there exists a dense set of $v \in \mathbb{R}^N - \mathbb{R}^{N-1}$ such that $\pi_v|_M : M \to \mathbb{R}^{N-1}$ is an injective immersion.

Lemma 6.2.2: Lowering to $2n + 1$

Let M be a smooth n-dimensional manifold with or without boundary. If M has a smooth embedding into $\mathbb{R}^N$ for some $N > 2n + 1$, then it has a smooth embedding into $\mathbb{R}^{2n+1}$.

Theorem 6.2.1: Whitney Embedding Theorem

Every smooth n-dimensional manifold with or without boundary admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.

Theorem 6.2.2: Whitney Immersion Theorem

Every smooth n-dimensional manifold with or without boundary admits a smooth immersion into $\mathbb{R}^{2n}$.

Theorem 6.2.3: Strong Whitney Embedding Theorem

Every smooth n-dimensional manifold with or without boundary admits a smooth embedding into $\mathbb{R}^{2n}$.

Theorem 6.2.4: Strong Whitney Immersion Theorem

Every smooth n-dimensional manifold with or without boundary admits a smooth immersion into $\mathbb{R}^{2n-1}$.

6.3 The Whitney Approximation Theorem

If $\delta : M \to \mathbb{R}$ is a positive continuous function, then we say two functiona $F_1, F_2 : M \to \mathbb{R}^k$ are **δ-close** if for all $p \in M$, we have

$$\|F_1(p) - F_2(p)\| < \delta(p).$$

Theorem 6.3.1: Whitney Approximation Theorem for Functions

Let M be a smooth manifold with or without boundary, and let $F : M \to \mathbb{R}^k$ be a continuous map. Given a positive continuous function $\delta : M \to \mathbb{R}$, there exists a smooth map $G : M \to \mathbb{R}^k$ that is δ-close to F. If F is smooth on a closed subset $A \subseteq M$, then we can choose G so that $G|_A = F|_A$.

6.3.1 Tabular Neighborhoods

We need to generalized the Whitney Approximation Theorem to maps between manifolds. If $F : N \to M$ is smooth, then by the Whitney Embedding Theorem, we can embed M into some $\mathbb{R}^n$, and approximate F by a smooth map into $\mathbb{R}^n$. However, the image may not lie in M. To fix this, we use **tabular neighborhoods**.

Definition 6.3.1: Normal Space

Suppose $M \subseteq \mathbb{R}^n$ is an embedded m-dimensional submanifold. For each $p \in M$, the **normal space** to M at p is the subspace $N_pM \subseteq T_x\mathbb{R}^n$ that are orthogonal to T_pM via the inherited Euclidean inner product on $\mathbb{R}^n$ itself.
The **normal bundle** of M is the set

$$NM = \{(p, v) \in \mathbb{R}^n \times \mathbb{R}^n : p \in M, v \in N_pM\}. \tag{6.1}$$

with a natural projection map $\pi_{NM} : NM \to M$ defined as the restriction of the projection map $\pi : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ to NM.

Theorem 6.3.2: Structure of Normal Bundle

If $M \subseteq \mathbb{R}^n$ is an embedded m-dimensional submanifold, then the normal bundle NM is an embedded n-dimensional submanifold of $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$.

Definition 6.3.2: Tabular Neighborhoods

Think NM as a submanifold in $\mathbb{R}^n \times \mathbb{R}^n$. Define

$$E : NM \to \mathbb{R}^n, \quad E(p, v) = p + v. \tag{6.2}$$

Then E is smooth. A **tabular neighborhood** of M is a neighborhood $U \subseteq \mathbb{R}^n$ of M such that it is diffeomorphic image under E of an open subset $V \subseteq NM$ that

$$V = \{(p, v) \in NM : \|v\| < \delta(p)\} \tag{6.3}$$

for some positive continuous function $\delta : M \to \mathbb{R}$.

Theorem 6.3.3: Tabular Neighborhood Theorem

Every embedded submanifold $M \subseteq \mathbb{R}^n$ has a tabular neighborhood.

Definition 6.3.3: Retraction

A **retraction** of a topological space X onto a subspace $A \subseteq X$ is a continuous map $r : X \to A$ such that $r(a) = a$ for all $a \in A$.

Proposition: **Tabular Neighborhood to Retraction**

If $M \subseteq \mathbb{R}^n$ is an embedded submanifold with a tabular neighborhood U, then there exists a smooth retraction $r : U \to M$ that is also a smooth submersion.

6.3.2 Smooth Approximation between Manifolds

Theorem 6.3.4: Whitney Approximation Theorem

Let N be smooth manifolds with or without boundary, M be a smooth manifold without boundary, and let $F : N \to M$ be a continuous map, then F is homotopic to a smooth map $G : N \to M$. Furthermore, if F is a smooth map on a closed subset $A \subseteq N$, then the homotopy can be taken to be relative to A.

Corollary 6.3.1: Extension Lemma for Smooth Maps

Suppose N is a smooth manifold with or without boundary, and M is a smooth manifold without boundary. If $A \subseteq N$ is closed and $F : A \to M$ is a smooth map, then f has a smooth extension to N if and only if it has a continuous extension to N.

Definition 6.3.4: Smooth Homotopy

A **smooth homotopy** between smooth maps $F_0, F_1 : M \to N$ is a smooth map $H : M \times [0,1] \to N$ such that $H(p,0) = F_0(p)$ and $H(p,1) = F_1(p)$ for all $p \in M$.
If N, M are smooth manifolds with or without boundary, it is easy to see that smooth homotopy is an equivalence relation on the set of smooth maps from N to M.

Theorem 6.3.5: Homotopy and Smooth Homotopy

Suppose N is a smooth manifold with or without boundary, and M is a smooth manifold without boundary. $F, G : N \to M$ are smooth maps. Then if F is homotopic to G, then they are smoothly homotopic. If F, G are homotopic relative to a closed subset $A \subseteq N$, then they are smoothly homotopic relative to A.

6.4 Transversality

Vector space intersects nicely: If $V_1, V_2 \subseteq W$ are subspaces of a vector space W, then $V_1 \cap V_2$ is also a subspace of W. For manifolds, it is not always true: There are smooth submanifolds whose intersection is not a submanifold.

Theorem 6.4.1: Transversality

Suppose M is a smooth manifold. Two embedded submanifolds $S_1, S_2 \subseteq M$ intersect **transversely** if for every $p \in S_1 \cap S_2$, we have

$$T_pS_1 + T_pS_2 = T_pM.$$

in linear algebraic sense, where we consider T_pS_1, T_pS_2 as subspaces of T_pM.
If $F : N \to M$ is a smooth map and $S \subseteq M$ is an embedded submanifold, then we say F is **transverse** to S if for every $p \in F^{-1}(S)$, we have

$$T_{F(p)}S + \mathrm{d}F_p(T_pN) = T_{F(p)}M. \tag{6.4}$$

Remark:
The first definition is easy to understand: At each intersection point, the two submanifolds' tangent spaces cross nicely to span the whole tangent space of the ambient manifold. The second definition generalizes the first: It means that the image of N and the submanifold S intersect nicely in M.
Specially, if F is a submersion, then it is transverse to every embedded submanifold of M. And two embedded submanifolds $S_1, S_2 \subseteq M$ intersect transversely if and only if the inclusion

map $i : S_1 \hookrightarrow M$ is transverse to S_2.

Theorem 6.4.2: Generalized Level Set Theorem

Suppose M, N are smooth manifolds and $S \subseteq M$ is an embedded submanifold.

- If $F : N \to M$ is a smooth map that is transverse to S, then $F^{-1}(S)$ is an embedded submanifold of N with codimension equal to that of S in M.
- If $S' \subseteq M$ is another embedded submanifold that intersects S transversely, then $S \cap S'$ is an embedded submanifold of M with codimension equal to the sum of the codimensions of S and S' in M.
- If $F : N \to M$ is a smooth submersion, then for any embedded submanifold S with codimension k in M, $F^{-1}(S)$ is an embedded submanifold of N with codimension k.

This shows in $\mathbb{R}^3$, a smooth surface and a smooth curve intersecting transversely will yield a discrete set of points, and two smooth surfaces intersecting transversely will yield a smooth curve.

Theorem 6.4.3: Global Characterization of Graphs

Suppose M, N are smooth manifolds and $S \subseteq M \times N$ is an immersed submanifold. Let π_M and π_N be the projection maps from $M \times N$ to M and N respectively. Then the following are equivalent:

- S is the graph of a smooth map $f : M \to N$.
- $\pi_M|_S$ is a diffeomorphism from S to M.
- For each $p \in M$, the submanifolds S and $\pi_M^{-1}(p)$ intersect transversely in $M \times N$ at exactly one point.
- S is the graph of $f = \pi_N \circ (\pi_M|_S)^{-1}$.

Corollary 6.4.1: Local Characterization of Graphs

Suppose M, N are smooth manifolds and $S \subseteq M \times N$ is an immersed submanifold. If $(p, q) \in S$, S intersects $\pi_M^{-1}(p)$ transversely at (p, q), then there exists an open neighborhood U of p in M and V of (p, q) in S such that V is the graph of a smooth map $f : U \to N$.

Now, we generalize the concept of smooth homotopy:

Suppose N, M, S are smooth manifolds and $\forall s \in S$, we have a smooth map $F_s : N \to M$. If the map $F : N \times S \to M$ defined as $F(p, s) = F_s(p)$ is smooth, then we say $\{F_s\}_{s \in S}$ is a **smooth family** of smooth maps from N to M. This is just a higher dimensional generalization of smooth homotopy.

Proposition: **Smooth Family and Homotopy**

If $\{F_s\}_{s\in S}$ is a smooth family of smooth maps from N to M, if S is connected, then for any $s_1, s_2 \in S$, F_{s_1} is smoothly homotopic to F_{s_2}.

Proof. S is connected so it is path connected. Let $\gamma : [0,1] \to S$ be a smooth path with $\gamma(0) = s_1$ and $\gamma(1) = s_2$. Define $H(p,s) = F(p,\gamma(s))$ is a smooth homotopy between F_{s_1} and F_{s_2}. □

Theorem 6.4.4: Parametric Transversality Theorem

Suppose N, M are smooth manifolds, $X \subseteq M$ is an embedded submanifold, and $\{F_s\}_{s\in S}$ is a smooth family of smooth maps from N to M. If the map $F : N \times S \to M$ defined as $F(p,s) = F_s(p)$ is transverse to X, then for almost every $s \in S$, the map $F_s : N \to M$ is transverse to X.

Theorem 6.4.5: Transversality Homotopy Theorem

Suppose N, M are smooth manifolds and $X \subseteq M$ is an embedded submanifold. Every smooth map $F : N \to M$ is homotopic to a smooth map $g : N \to M$ that is transverse to X.

Chapter 7

Lie Groups

7.1 Definition and Examples

Definition 7.1.1: Lie Group

A **Lie group** is a smooth manifold G without boundary that is also a group, such that the group operations (multiplication and inversion) are smooth maps.

Proposition: **Identify Lie Groups**

If G is a smooth manifold and a group such that the map

$$f : G \times G \to G, \quad (g, h) \mapsto gh^{-1} \tag{7.1}$$

is smooth, then G is a Lie group.

Proof. We have

$$g \cdot h = f(g, h^{-1}), \quad h^{-1} = f(e, h).$$

□

If G is a Lie group, $\forall g \in G$, define

$$L_g : G \to G, \quad h \mapsto gh, \quad R_g : G \to G, \quad h \mapsto hg.$$

These are called **left** and **right translations** by g, respectively. Both are diffeomorphisms with inverses $L_{g^{-1}}$ and $R_{g^{-1}}$, obviously.

Example: **Lie Groups**

- The general linear group $GL(n, \mathbb{R})$ is a Lie group under matrix multiplication. It is an open submanifold of $M(n, \mathbb{R}) \cong \mathbb{R}^{n^2}$.

  Some subsets of $GL(n, \mathbb{R})$: $GL^+(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det(A) > 0\}$, $SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det(A) = 1\}$, $O(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I\}$, $SO(n) = O(n) \cap SL(n, \mathbb{R})$.

  Similarly, we have complex versions $GL(n, \mathbb{C})$, $SL(n, \mathbb{C})$, $U(n)$, $SU(n)$.

- Generally, for any vector space V over $\mathbb{R}$ or $\mathbb{C}$ with finite dimension, the group of all automorphisms $Aut(V)$ is a Lie group isomorphic to $GL(n, \mathbb{R})$ or $GL(n, \mathbb{C})$.
- If G is a Lie group, and $H \subseteq G$ be an open subgroup, then H is also a Lie group.
- The $\mathbb{R}$ and $\mathbb{C}$ under addition are Lie groups. The $\mathbb{R}^* = \mathbb{R}\backslash\{0\}$ and $\mathbb{C}^* = \mathbb{C}\backslash\{0\}$ under multiplication are also Lie groups.
- The S^1 under complex multiplication is a Lie group, called the **circle group**.
- Given $G_1, \ldots, G_k$ Lie groups, their product $G_1 \times \cdots \times G_k$ is also a Lie group under component-wise multiplication:

$$(g_1, \ldots, g_k) \cdot (h_1, \ldots, h_k) = (g_1 h_1, \ldots, g_k h_k). \tag{7.2}$$

  So the n-torus $T^n = S^1 \times \cdots \times S^1$ is a Lie group.
- Any group with discrete topology is a 0-dimensional Lie group.

7.2 Lie Group Homomorphisms

Definition 7.2.1: Lie Group Homomorphism

A **Lie group homomorphism** is a smooth map $F : G \to H$ between Lie groups that is also a group homomorphism, i.e., $F(gh) = F(g)F(h)$ for all $g, h \in G$. It is called a **Lie group isomorphism** if it is a diffeomorphism as well.

Example: **Lie Group Homomorphisms**

- The inclusion map $S^1 \hookrightarrow \mathbb{C}^*$ is a Lie group homomorphism.
- The map $\exp : (\mathbb{R}, +) \to (\mathbb{R}^*, \cdot)$ is a Lie group homomorphism. And $\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \cdot)$ is a Lie group isomorphism. The same goes for $\mathbb{C}$.
- The map $\epsilon : \mathbb{R} \to S^1$ defined by $\epsilon(t) = e^{2\pi i t}$ is a Lie group homomorphism with kernel $\mathbb{Z}$. Same goes for $\epsilon^n : \mathbb{R}^n \to T^n$ defined by $\epsilon^n(t_1, \ldots, t_n) = (e^{2\pi i t_1}, \ldots, e^{2\pi i t_n})$ with kernel $\mathbb{Z}^n$.
- The determinant map $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$ is a Lie group homomorphism.
- For any Lie group G, the conjugation map $C_g : G \to G$ defined by $C_g(h) = ghg^{-1}$ is a Lie group isomorphism for each fixed $g \in G$.

Theorem 7.2.1: Constant Rank for Lie Group Homomorphisms

Every Lie group homomorphism has constant rank. A Lie group homomorphism is an isomorphism if and only if it is a bijection.

Proof. Let $F : G \to H$ be a Lie group homomorphism, and e_g and e_h be the identity elements. $\forall g_0 \in G$, we have

$$F(L_{g_0}(g)) = F(g_0 g) = F(g_0)F(g) = L_{F(g_0)}(F(g)), \qquad F \circ L_{g_0} = L_{F(g_0)} \circ F.$$

Taking the differential at e_g, we get

$$dF_{g_0} \circ dL_{g_0}|_{e_g} = dL_{F(g_0)}|_{e_h} \circ dF_{e_g}.$$

As L_{g_0} and $L_{F(g_0)}$ are diffeomorphisms, $dL_{g_0}|_{e_g}$ and $dL_{F(g_0)}|_{e_h}$ are isomorphisms. Thus, dF_{g_0} has the same rank as dF_{e_g} for all $g_0 \in G$.

The second claim follows from the global rank theorem. □

7.2.1 Universal Covering Groups

Theorem 7.2.2: Existence of Universal Covering Groups

Every connected Lie group G has a simply connected Lie group $\tilde{G}$ called the **universal covering group** of G, that has a smooth covering map $\pi : \tilde{G} \to G$ which is also a Lie group homomorphism.
Also, the universal covering group is unique up to isomorphism.

Example: **Universal Covering Group**

- $\epsilon^n : \mathbb{R}^n \to T^n$ by
$$\epsilon^n(t_1, \ldots, t_n) = (e^{2\pi i t_1}, \ldots, e^{2\pi i t_n})$$
is a Lie group homomorphism and a smooth covering map. So $\mathbb{R}^n$ is the universal covering group of T^n.
- $\exp : \mathbb{C} \to \mathbb{C}^*$ is a Lie group homomorphism and a smooth covering map. So $\mathbb{C}$ is the universal covering group of $\mathbb{C}^*$.

7.3 Lie Subgroups

Definition 7.3.1: Lie Subgroup

A **Lie subgroup** of a Lie group G is a subgroup H of G with a topology and smooth structure such that H is a Lie group and an immersed submanifold of G.

The following shows that embedded subgroups are automatically Lie subgroups.

Proposition: **Embedded Subgroup is Lie Subgroup**

If H is an embedded submanifold of a Lie group G and a subgroup of G, then H is a Lie subgroup.

Proof. We only need to show the multiplication and inversion maps on H are smooth. □

The possibility of open subgroups as a candidate is limited:

Lemma 7.3.1: Open Subgroups as Lie Subgroups

Suppose G is a Lie group and H is an open subgroup of G. Then H is an embedded Lie subgroup of G. In addition H is also closed, and thus a union of connected components of G.

Proof. If H is open, then it is embedded from proposition 5.1. Then every left coset gH is also open. As $G - H$ is a union of left cosets, it is also open. Thus, H is closed. □

If G is a group and $S \subseteq G$, then the subgroup generated by S is the intersection of all subgroups of G containing S.

Proposition: **Generating Lie Subgroups**

Suppose G is a Lie group and $W \subseteq G$ is any neighborhood of the identity element e.

- W generates an open subgroup of G.
- If W is connected, then the subgroup generated by W is also connected.
- If G is connected, then W generates G.

Proposition: **The Identity Component**

Let G be a Lie group, and let G_0 be the connected component of the identity element $e \in G$, called the **identity component** of G. Then G_0 is a normal subgroup of G, and it is the only connected open subgroup of G. Every connected component of G is a coset of G_0, thus diffeomorphic to G_0.

Now lets move on to Lie subgroups that are not open subgroups.

Proposition: **Kernel as Lie Subgroup**

If $F : G \to H$ is a Lie group homomorphism, then $\ker F$ is a properly embedded Lie subgroup of G, with codimension equal to the rank of F.

Proposition: **Image as Lie Subgroup**

If $F : G \to H$ is an injective Lie group homomorphism, then $F(G)$ has a unique smooth structure making it a Lie subgroup of H, such that $F : G \to F(G)$ is a Lie group isomorphism.

Example: **Embedded Lie Subgroups**

- $GL^+(n, \mathbb{R})$ is an open subgroup of $GL(n, \mathbb{R})$, thus an embedded Lie subgroup.
- S^1 is an embedded Lie subgroup of $\mathbb{C}^*$.
- $SL(n, \mathbb{R})$ is the kernel of the determinant map $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$, thus a properly embedded Lie subgroup of $GL(n, \mathbb{R})$ with codimension 1. As determinant is a smooth submersion, $SL(n, \mathbb{R})$ has dimension $n^2 - 1$.

On the torus, let $\gamma : \mathbb{R} \to T^2$ be defined by $\gamma(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$ for some irrational number α. Then γ is a Lie group homomorphism from $\mathbb{R}$ to T^2. Its image is a Lie subgroup of T^2 that is not an embedded submanifold. In fact, its image is dense in T^2.

More interesting, in general smooth submanifolds can be closed without being embedded submanifolds, like figure-eight curve in $\mathbb{R}^2$, and can be embedded without being closed. However, for Lie subgroups, we have the following result:

Theorem 7.3.1: Embeddedness and Closeness of Lie Groups

Suppose G is a Lie group and H is a Lie subgroup of G. Then H is closed in G if and only if H is an embedded submanifold of G.

7.4 Group Actions and Equivariant Maps

Lie groups often act on smooth manifolds in a smooth way. Generally, if G is a group and M is a set, a **left group action** of G on M is a map $G \times M \to M$, $(g, x) \mapsto g \cdot x$ such that $\forall g, h \in G$ and $x \in M$, $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$. A **right group action** is defined similarly.

Now, let G be a Lie group and M be a smooth manifold. A **smooth left group action** of G on M is a left group action such that the map $\theta : G \times M \to M$, $(g, x) \mapsto g \cdot x$ is smooth.

We denote the left action by $\theta_g : M \to M$, $x \mapsto g \cdot x$. For a smooth left group action, each θ_g is a diffeomorphism with inverse $\theta_{g^{-1}}$. Some frequently used notions are listed below.

- Orbit: For each $p \in M$, the **orbit** of p is
$$G \cdot p = \{g \cdot p : g \in G\}.$$
- Isotropy Group: The **isotropy group** or **stabilizer** of p is
$$G_p = \{g \in G : g \cdot p = p\}.$$
The stabilizer is a subgroup of G.
- Transitive Action: If for every $p, q \in M$, there exists $g \in G$ such that $g \cdot p = q$, then the action is called **transitive**. Equivalently, the only orbit is M itself.
- Free Action: If for every $p \in M$, the only $g \in G$ such that $g \cdot p = p$ is $g = e$. Equivalently, the isotropy group G_p is trivial for all $p \in M$.

Example: **Lie Group Action**

- Trivial Action: G is any Lie group, and M is any smooth manifold. The trivial action is defined by $g \cdot p = p$ for all $g \in G$ and $p \in M$. This is a smooth left group action. Every orbit is a single point, and the isotropy group is the whole group G.
- Natural action of $GL(n, \mathbb{R})$ on $\mathbb{R}^n$: For $A \in GL(n, \mathbb{R})$ and $v \in \mathbb{R}^n$, the action is defined by $A \cdot v = Av$. This is a smooth left group action. The orbit of v is the line spanned by v, and the isotropy group is the set of all matrices that scale v. There are two orbits: $\{0\}$ and $\mathbb{R}^n \setminus \{0\}$.
- Every Lie Group acts smoothly on itself by left multiplication. The action is both free and transitive. Also, every Lie group acts smoothly on itself by conjugation, i.e., $g \cdot h = ghg^{-1}$.
- A discrete group G acts smoothly on a smooth manifold M if and only if for each $g \in G$, the map $\theta_g : M \to M$ is a smooth map from M to itself. For example, $\mathbb{Z}^n$ acts on $\mathbb{R}^n$ by translation, i.e., $(m_1, \ldots, m_n) \cdot (x_1, \ldots, x_n) = (x_1 + m_1, \ldots, x_n + m_n)$.

Another important situation is covering maps. Suppose E, M be topological spaces and $\pi : E \to M$ is a topological covering map. An automorphism of π (or **covering transformation** or **deck transformation**) is a homeomorphism $\varphi : E \to E$ such that $\pi \circ \varphi = \pi$. The set of all automorphisms of π forms a group under composition, denoted by $\mathrm{Aut}_\pi(E)$, left acting on E.

Proposition: **Automorphisms as Lie Group**

Suppose E, M are smooth manifolds, with or without boundary, and $\pi : E \to M$ is a smooth covering map. With the discrete topology, $\mathrm{Aut}_\pi(E)$ is a zero-dimensional Lie group that acts smoothly and freely on E.

7.4.1 Equivariant Maps

Definition 7.4.1: Equivariant Maps

Suppose G is a Lie group acting smoothly on smooth manifolds M and N. A map $F : M \to N$ is called **G-equivariant** if $\forall g \in G$ and $p \in M$,

$$F(g \cdot p) = g \cdot F(p).$$

We also say that F intertwines the actions of G on M and N.

Theorem 7.4.1: Equivariant Rank Theorem

Let M, N be smooth manifolds, and G be a Lie group. Suppose $F : M \to N$ is a G-equivariant smooth map. If G acts transitively on M, then F has constant rank.

Proof. Let θ and φ be the actions of G on M and N, respectively. $\forall p, q \in M$, choose $g \in G, \theta_g(p) =$

q. Then from $\varphi_g \circ F = F \circ \theta_g$, we have

$$\mathrm{d}\varphi_g|_{F(p)} \circ \mathrm{d}F_p = \mathrm{d}F_q \circ \mathrm{d}\theta_g|_p.$$

So $\mathrm{d}F_q$ has the same rank as $\mathrm{d}F_p$. □

Definition 7.4.2: Orbit Map

Suppose G is a Lie group acting smoothly on a smooth manifold M. For each $p \in M$, the **orbit map** at p is the map $\theta^p : G \to M$ defined by $\theta^p(g) = g \cdot p$. The image of θ^p is the orbit $G \cdot p$.

Proposition: **Properties of Orbit Maps**

Suppose θ is a smooth left group action of a Lie group G on a smooth manifold M. For each $p \in M$, the orbit map $\theta^{(p)} : G \to M$ is smooth and has constant rank. So the isotropy group $G_p = (\theta^{(p)})^{-1}(p)$ is a properly embedded Lie subgroup of G. If $G_p = e$, then $\theta^{(p)}$ is an injective smooth immersion, so the orbit $G \cdot p$ is an immersed submanifold of M.

Remark:
In fact, every orbit is an immersed submanifold of M. But we shall postpone the proof until we develop more tools.

Proof. The orbit map $\theta^{(p)}$ is smooth as

$$G \cong G \times \{p\} \hookrightarrow G \times M \xrightarrow{\theta} M.$$

As G acts transitively on itself by left multiplication, by the equivariant rank theorem, $\theta^{(p)}$ has constant rank. Thus, G_p is a properly embedded Lie subgroup of G.

Also, when $G_p = e$, $\mathrm{d}\theta^{(p)}|_e$ is injective, so $\theta^{(p)}$ is an injective smooth immersion. Thus the orbit $G \cdot p$ is an immersed submanifold of M. □

Example: **The Orthogonal Group**

$O(n)$ is naturally a subgroup of $GL(n, \mathbb{R})$. Define a smooth map $\Phi : GL(n, \mathbb{R}) \to M(n, \mathbb{R})$ by $\Phi(A) = A^T A$. Then $O(n) = \Phi^{-1}(I)$.
Define an action of $GL(n, \mathbb{R})$ on $M(n, \mathbb{R})$ by $A \cdot X = A^T X A$. This is a smooth left group action. And $GL(n, \mathbb{R})$ acts on itself by left multiplication. The map Φ is $GL(n, \mathbb{R})$-equivariant, since

$$\Phi(A \cdot B) = \Phi(AB) = (AB)^T(AB) = B^T(A^T A)B = A \cdot \Phi(B).$$

So by the equivariant rank theorem, Φ has constant rank. Thus, $O(n)$ is a properly embedded Lie subgroup of $GL(n, \mathbb{R})$. It is compact for it is closed and bounded in $M(n, \mathbb{R}) \cong \mathbb{R}^{n^2}$.
To compute the dimension of $O(n)$, take $B \in T_I GL(n, \mathbb{R}) \cong M(n, \mathbb{R})$. Then take $\gamma(t) = I + tB$, we have

$$\mathrm{d}\Phi_I(B) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \Phi(I + tB) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} (I + tB)^T(I + tB) = B^T + B.$$

So $\mathrm{d}\Phi_I(T_I GL(n, \mathbb{R}))$ is the space of symmetric matrices, which has dimension $\frac{n(n+1)}{2}$. Then,

$O(n)$ is a properly embedded Lie subgroup of $GL(n, \mathbb{R})$ with dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$. The special orthogonal group $SO(n)$ is the intersection of $O(n)$ and $SL(n, \mathbb{R})$. It is also a properly embedded Lie subgroup of $GL(n, \mathbb{R})$ with dimension $\frac{n(n-1)}{2}$.

Example: **Unitary Group**

The unitary group $U(n)$ is defined as

$$U(n) = \{A \in GL(n, \mathbb{C}) : A^* A = I\},$$

where A^* is the conjugate transpose of A. Similar to the orthogonal group, the unitary group is a properly embedded Lie subgroup of $GL(n, \mathbb{C})$ with dimension n^2. The special unitary group $SU(n) = U(n) \cap SL(n, \mathbb{C})$ is also a properly embedded Lie subgroup of $GL(n, \mathbb{C})$ with dimension $n^2 - 1$.

7.4.2 Semidirect Products

Definition 7.4.3: Semidirect Product

Suppose H, N are Lie groups, and $\theta : H \times N \to N$ is a smooth left group action of H on N. It is said to be by **automorphisms** if $\forall h \in H$, the map $\theta_h : N \to N$, $n \mapsto h \cdot n$ is a Lie group automorphism of N.
Now we define a new Lie group $G = N \rtimes_\theta H$, called the **semidirect product** of N and H with respect to θ. As a smooth manifold, $G = N \times H$. The group operation is defined by

$$(n_1, h_1) \cdot (n_2, h_2) = (n_1(h_1 \cdot n_2), h_1 h_2). \tag{7.3}$$

Remark:
Intuitively, a semidirect product is a generalization of a direct product. If the action θ is trivial, i.e., $h \cdot n = n$ for all $h \in H$ and $n \in N$, then the semidirect product reduces to the direct product $N \times H$. Its like first twisting N by the action of H, then taking the product.

Example: **The Euclidean Group**

Consider $\mathbb{R}^n$ as a Lie group under addition, and $O(n)$ as a Lie group under matrix multiplication. Then the natural action of $O(n)$ on $\mathbb{R}^n$ by matrix multiplication is by automorphisms. The semidirect product $E(n) = \mathbb{R}^n \rtimes O(n)$ is called the **Euclidean group**, which is the group of all isometries of $\mathbb{R}^n$ that preserve distances. The group acting on $\mathbb{R}^n$ is given by

$$(b, A) \cdot x = Ax + b, \quad (b, A)(b', A') = (b + Ab', AA'),$$

Proposition: **Properties of Semidirect Products**

Suppose H, N are Lie groups, and $\theta : H \times N \to N$ is a smooth left group action of H on N by automorphisms. Let $G = N \rtimes_\theta H$ be the semidirect product of N and H with respect to θ. Then

- The subsets $\tilde{N} = N \times \{e_H\}$ and $\tilde{H} = \{e_N\} \times H$ are closed Lie subgroups of G that are isomorphic to N and H, respectively.
- $\tilde{N}$ is a normal subgroup of G.
- $\tilde{N} \cap \tilde{H} = \{e_G\}$, where $e_G = (e_N, e_H)$ is the identity element of G.
- $\tilde{N}\tilde{H} = G$.

Theorem 7.4.2: Characterization of Semidirect Products

Suppose G is a Lie group, and $N, H \subseteq G$ are closed Lie subgroups such that N is normal. Also, suppose $N \cap H = \{e_G\}$ and $NH = G$. Then the map $(n, h) \mapsto nh$ is a Lie group isomorphism from the semidirect product $N \rtimes_\theta H$ to G, where $\theta : H \times N \to N$ is the action by conjugation, i.e., $\theta(h, n) = hnh^{-1}$.

7.4.3 Representations of Lie Groups

SORRY

Chapter 8

Vector Fields

8.1 Vector Fields on Manifolds

Definition 8.1.1: Vector Fields

If M is a smooth manifold, with or without boundary, a **vector field** on M is a smooth section of the map $\pi : TM \to M$. In other words, a vector field is a continuous map $X : M \to TM$ such that $\pi \circ X = \mathrm{id}_M$. This means that for each point $p \in M$, the vector field X assigns a tangent vector $X(p) \in T_pM$.
Usually, we are interested in smooth vector fields, meaning that the map X is smooth. If X is not even necessarily continuous, we say that X is a **rough vector field**.

We shall denote $X(p)$ by X_p for each $p \in M$, to be more readable. For any vector field X on M, we can write its component functions

$$X_p = X^i(p) \left.\frac{\partial}{\partial x^i}\right|_p$$

according to some local smooth chart $(U, \varphi = (x^1, \ldots, x^n))$ around p.

Proposition: **Smoothness Criterion for Vector Fields**

Let M be a smooth manifold, with or without boundary, and let $X : M \to TM$ be a rough vector field on M. If $(U, \varphi = (x^1, \ldots, x^n))$ is a smooth chart on M, then the restriction of X to U is smooth if and only if the component functions $X^i : U \to \mathbb{R}$ is smooth for each i.

Example: **Vector Fields**

- Now if $(U, \varphi = (x^i))$ is any smooth chart on M, then we can define the **coordinate vector fields** on U by
$$p \mapsto \left.\frac{\partial}{\partial x^i}\right|_p . \tag{8.1}$$
called the i-th coordinate vector field on U.

- Euler vector field: The vector field V on $\mathbb{R}^n$ by

$$V_x = x^i \left.\frac{\partial}{\partial x^i}\right|_x = x^1 \left.\frac{\partial}{\partial x^1}\right|_x + \cdots + x^n \left.\frac{\partial}{\partial x^n}\right|_x$$

  is called the **Euler vector field** on $\mathbb{R}^n$.

- The angle vector field: Let θ be any angle coordinate on a proper open subset $U \subseteq S^1$, then let $\mathrm{d}/\mathrm{d}\theta$ be the corresponding coordinate vector field on U. Any other angle coordinate only differs from θ by an additive constant, so the vector field $\mathrm{d}/\mathrm{d}\theta$ is independent of the choice of angle coordinate. Thus, we can define a vector field on all of S^1 by defining it to be $\mathrm{d}/\mathrm{d}\theta$ on U at each proper open subset $U \subseteq S^1$. This vector field is called the **angle vector field** on S^1.

  The same goes for tori T^n.

We can see that tangent spaces behave locally. So we can identify T_pU with T_pM without ambiguity.

Definition 8.1.2: Vector Field Along a Subset

Let M be a smooth manifold, with or without boundary, and let $A \subseteq M$ be any subset. A **vector field along** A is a continuous map $X : A \to TM$ such that $\pi \circ X = \mathrm{id}_A$. We call it a smooth vector field along A if for each $p \in A$, there is an open neighborhood V of p in M and a smooth vector field $\tilde{X}$ on V such that $\tilde{X}|_{V \cap A} = X|_{V \cap A}$.

Lemma 8.1.1: Extension Lemma for Vector Fields

Let M be a smooth manifold, with or without boundary, and let $A \subseteq M$ be any closed subset. If X is a smooth vector field along A, then for any open neighborhood U of A in M, there is a smooth vector field $\tilde{X}$ on M such that $\tilde{X}|_A = X$ and $\mathrm{supp}(\tilde{X}) \subseteq U$.

Specifically, any vector at a point $p \in M$ can be extended to a smooth vector field on M that vanishes outside a small neighborhood of p.

Definition 8.1.3: Vector Field Spaces

If M is a smooth manifold, with or without boundary, we denote the space of all smooth vector fields on M by $\mathfrak{X}(M)$. It is a vector space over $\mathbb{R}$ under pointwise addition and scalar multiplication.

In addition, smooth vector fields can be multiplied by smooth functions to produce new smooth vector fields. If $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$, then we define a new vector field $fX \in \mathfrak{X}(M)$ by

$$(fX)_p = f(p)X_p \tag{8.2}$$

Proposition: **Properties of $\mathfrak{X}(M)$**

Let M be a smooth manifold, with or without boundary.

- If $X, Y \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$, then $fX + gY \in \mathfrak{X}(M)$.
- $\mathfrak{X}(M)$ is a module over the ring $C^\infty(M)$.

8.1.1 Local and Global Frames

Definition 8.1.4: Local and Global Frames

Let M be a smooth manifold, with or without boundary, and an ordered k-tuple $(X_1, \ldots, X_k)$ of vector fields on some subset $A \subseteq M$. We say that they are linearly independent if $\forall p \in A$, the vectors $(X_1)_p, \ldots, (X_k)_p$ are linearly independent in T_pM, and it spans the tangent bundle over A if $\forall p \in A$, the vectors $(X_1)_p, \ldots, (X_k)_p$ span T_pM.
A **local frame** on M is an ordered n-tuple of vector fields $(E_1, \ldots, E_n)$ on some open subset $U \subseteq M$ that is linearly independent and spans the tangent bundle over U. The vectors $(E_1)_p, \ldots, (E_n)_p$ then form a basis for T_pM for each $p \in U$. It is called a **global frame** if $U = M$, and a **smooth frame** if each E_i is a smooth vector field.
A smooth manifold M is called **parallelizable** if it admits a smooth global frame.

Example: **Local and Global Frames**

- The standard coordinate vector fields $(\partial/\partial x^1, \ldots, \partial/\partial x^n)$ form a smooth global frame on $\mathbb{R}^n$.
- For any smooth chart $(U, \varphi = (x^1, \ldots, x^n))$ on a smooth manifold M, the coordinate vector fields $(\partial/\partial x^1, \ldots, \partial/\partial x^n)$ form a smooth local frame on U.
- The angle vector field $\mathrm{d}/\mathrm{d}\theta$ on S^1 is a smooth global frame. And the n angle vector fields $\partial/\partial\theta^1, \ldots, \partial/\partial\theta^n$ form a smooth global frame on the torus T^n.

Proposition: **Completion of Local Frames**

Let M be a n-smooth manifold, with or without boundary,

- Let $U \subseteq M$ be any open subset, and let $(X_1, \ldots, X_k)$ be a linearly independent k-tuple of smooth vector fields on U, where $k < n$. Then there exist smooth vector fields $X_{k+1}, \ldots, X_n$ on U such that $(X_1, \ldots, X_n)$ is a smooth local frame on U.
- If $(v_1, \ldots, v_k)$ is any linearly independent k-tuple of vectors in T_pM at some point $p \in M$, where $k \leqslant n$, then there exist a smooth local frame (X_i) on some open neighborhood U of p such that $X_i|_p = v_i$ for each $1 \leqslant i \leqslant k$.
- If $X_1, \ldots, X_n$ is a linearly independent n-tuple of smooth vector fields on some closed subset $A \subseteq M$, then there exist smooth local frame $(\tilde{X}_i)$ on some open neighborhood U of A such that $\tilde{X}_i|_A = X_i$ for each $1 \leqslant i \leqslant n$.

Specially for subsets of $\mathbb{R}^n$, we can use the Euclidean inner product to define orthonormal frames. For example, the standard coordinate vector fields on $\mathbb{R}^n$ or the polar coordinate vector fields on $\mathbb{R}^2 \backslash \{0\}$.

Lemma 8.1.2: Gram-Schmidt Algorithm for Frames

Suppose (X_j) is a smooth local frame for $T\mathbb{R}^n$ over some open subset $U \subseteq \mathbb{R}^n$. Then there exists a smooth orthonormal local frame (E_j) for $T\mathbb{R}^n$ over U such that

$$\operatorname{span}\{X_1, \dots, X_k\} = \operatorname{span}\{E_1, \dots, E_k\} \quad \text{for each } 1 \leqslant k \leqslant n.$$

Generally speaking, parallelizable manifolds are rare. For example, spheres S^n are only parallelizable for $n = 1, 3, 7$. We shall later see that all Lie groups are parallelizable.

8.1.2 Vector Fields as Derivations

Vector fields define operators on smooth functions. If $X \in \mathfrak{X}(M)$ and f is a smooth function on some open subset $U \subseteq M$, then we can define a new smooth function Xf on U by

$$(Xf)(p) = X_p(f) \quad \forall p \in U. \tag{8.3}$$

Remark:
Note the difference between Xf and fX. The former is a smooth function on U, while the latter is a smooth vector field on M.

It is quite direct that Xf is defined locally, for any open subset $V \subseteq U$, we have

$$(Xf)|_V = X|_V(f|_V).$$

Proposition: **Properties of Vector Field Derivations**

Let M be a smooth manifold, with or without boundary, and let $X : M \to TM$ be a rough vector field on M. Then the following are equivalent:

- X is smooth.
- For every $f \in C^\infty(M)$, the function $Xf : M \to \mathbb{R}$ is smooth.
- For every open subset $U \subseteq M$ and every $f \in C^\infty(U)$, the function $Xf : U \to \mathbb{R}$ is smooth.

Proof. Quite obvious, just taking a local chart around each point. □

Recall the definition of a derivation at a point 3.1.1. Now we present the global version.

Definition 8.1.5: Derivations

Let M be a smooth manifold, with or without boundary. A **derivation** on M is a linear map $D : C^\infty(M) \to C^\infty(M)$ such that for all $f, g \in C^\infty(M)$,

$$D(fg) = fD(g) + gD(f).$$

The derivation at a point $p \in M$ is just the composition of D with the evaluation map at p:

$$D_p : C^\infty(M) \to \mathbb{R}, \quad D_p(f) = D(f)(p).$$

We can see that for a vector field $X \in \mathfrak{X}(M)$, we have

$$X(fg)(p) = X_p(fg) = f(p)X_p(g) + g(p)X_p(f) = f(p)(Xg)(p) + g(p)(Xf)(p) = (fXg + gXf)(p).$$

which matches our definition.

Theorem 8.1.1: Vector Fields and Derivations

Let M be a smooth manifold, with or without boundary. A map $D : C^\infty(M) \to C^\infty(M)$ is a derivation on M if and only if there exists a smooth vector field $X \in \mathfrak{X}(M)$ such that $D(f) = Xf$ for all $f \in C^\infty(M)$.
Thus, we can identify the space of derivations on M with the space $\mathfrak{X}(M)$ of smooth vector fields on M.

Proof. We have already shown the "if" part. For the "only if" part, for each $p \in M$, define $X_p : C^\infty(M) \to \mathbb{R}$ by $X_p(f) = D(f)(p)$ would do. Smoothness follows from the fact that $D(f)$ is smooth for each $f \in C^\infty(M)$. □

8.2 Vector Fields and Smooth Maps

Given $F : M \to N$ a smooth map and X a vector field on M, we can try to "push forward" X to a vector field on N by $\mathrm{d}F_p(X_p) \in T_{F(p)}N$. However, this does not necessarily define a vector field on N. If F is not surjective, then there may be points in N that are not in the image of F, and if F is not injective, then there may be points in N that have multiple preimages in M with different pushed-forward vectors.

Definition 8.2.1: F-related Vector Fields

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. If X is a vector field on M and Y is a vector field on N, we say that X and Y are **F-related** if for every $p \in M$,

$$\mathrm{d}F_p(X_p) = Y_{F(p)}.$$

Proposition: **F-related Vector Fields on Smooth Functions**

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. If $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$, then they are F-related if and only if for every $g \in C^\infty(N)$,

$$X(g \circ F) = (Yg) \circ F. \tag{8.4}$$

Proof. We have

$$X(g \circ F)(p) = X_p(g \circ F) = \mathrm{d}F_p(X_p)(g) = Y_{F(p)}(g) = (Yg)(F(p)) = ((Yg) \circ F)(p).$$

□

Example: **F-related Vector Fields**

Let $F : \mathbb{R} \to \mathbb{R}^2, t \mapsto (\cos t, \sin t)$ be the standard embedding of the unit circle. Then the vector field $X = \mathrm{d}/\mathrm{d}t$ on $\mathbb{R}$ is F-related to the angle vector field

$$Y_{(x,y)} = -y \left.\frac{\partial}{\partial x}\right|_{(x,y)} + x \left.\frac{\partial}{\partial y}\right|_{(x,y)}$$

For an arbitrary smooth map $F : M \to N$, there may not exist any nontrivial F-related vector fields. But for diffeomorphisms, we have the following result.

Proposition: **Existence of Related Vector Fields for Diffeomorphisms**

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a diffeomorphism. Then for every vector field $X \in \mathfrak{X}(M)$, there exists a unique vector field $Y \in \mathfrak{X}(N)$ that is F-related to X, and vice versa.
We often denote Y by F_*X, called the **pushforward** of X by F. Explicitly, for each $q \in N$,

$$(F_*X)_q = \mathrm{d}F_{F^{-1}(q)}(X_{F^{-1}(q)}). \tag{8.5}$$

and we have

$$((F_*X)g) \circ F = X(g \circ F) \quad \forall g \in C^\infty(N). \tag{8.6}$$

8.2.1 Vector Fields and Submanifolds

If $S \subseteq M$ is an immersed or embedded submanifold, with or without boundary, then in general, a vector field on M does not restrict to a vector field on S, because the vectors may not lie in the tangent spaces of S.

Definition 8.2.2: Tangent to a Submanifold

Let M be a smooth manifold, with or without boundary, and let $S \subseteq M$ be an immersed or embedded submanifold, with or without boundary. A vector field $X \in \mathfrak{X}(M)$ is said to be **tangent to** S if for every $p \in S$, $X_p \in T_pS \subseteq T_pM$.

Proposition: **Criterion for Tangency to a Submanifold**

Let M be a smooth manifold, with or without boundary, and let $S \subseteq M$ be an embedded submanifold, with or without boundary. A vector field $X \in \mathfrak{X}(M)$ is tangent to S if and only if for every smooth function $f \in C^\infty(M)$ that vanishes on S, the function Xf also vanishes on S.

If $S \subseteq M$ is an immersed submanifold, with or without boundary, and $Y \in \mathfrak{X}(S)$, then if there is a vector field $X \in \mathfrak{X}(S)$ that is $\iota : S \hookrightarrow M$-related to Y, then clearly Y is tangent to S. Because for each $p \in S$, $Y_p = \mathrm{d}\iota_p(X_p) = X_p \in T_pS$. We shall see that the converse is also true.

Proposition: **Restricting Vector Fields to Submanifolds**

Let M be a smooth manifold, and $S \subseteq M$ be an immersed submanifold, with or without boundary. Let $\iota : S \hookrightarrow M$ be the inclusion map. A vector field $Y \in \mathfrak{X}(M)$ is tangent to S if and only if there exists a vector field $X \in \mathfrak{X}(S)$ that is ι-related to Y. In this case, X is unique, and we often denote it by $Y|_S$, called the **restriction** of Y to S.

8.3 Lie Brackets

Now, we introduce an important operation on vector fields, joining two vector fields to produce a new vector field, called the Lie bracket.

Let $X, Y \in \mathfrak{X}(M)$ be two smooth vector fields on a smooth manifold M. Given a $f \in C^\infty(M)$, we can successively apply X and Y to f to get a new smooth function $Y(Xf)$. However, this operatio $f \mapsto YXf$n does not satisfy the Leibniz rule, so is not a vector field. To fix this, we introduce the Lie bracket.

Definition 8.3.1: Lie Bracket

Let M be a smooth manifold, with or without boundary, and let $X, Y \in \mathfrak{X}(M)$ be two smooth vector fields on M. The **Lie bracket** of X and Y is the map $[X, Y] : C^\infty(M) \to C^\infty(M)$ defined by

$$[X, Y]f = X(Yf) - Y(Xf) \quad \forall f \in C^\infty(M). \tag{8.7}$$

Then $[X, Y]$ is a smooth vector field on M.

Proof. We shall show that $[X, Y]$ is a derivation on M. For any $f, g \in C^\infty(M)$, we have

$$\begin{aligned}
[X, Y](fg) &= X(Y(fg)) - Y(X(fg)) \\
&= X(fYg + gYf) - Y(fXg + gXf) \\
&= fX(Yg) + (Xf)(Yg) + gX(Yf) + (Xg)(Yf) \\
&\quad - fY(Xg) - (Yf)(Xg) - gY(Xf) - (Yg)(Xf) \\
&= f(X(Yg) - Y(Xg)) + g(X(Yf) - Y(Xf)) \\
&= f[X, Y]g + g[X, Y]f.
\end{aligned}$$

□

Theorem 8.3.1: Coordinate Formula for Lie Bracket

Let M be a smooth manifold, with or without boundary, and let $(U, \varphi = (x^1, \ldots, x^n))$ be a smooth chart on M. If $X, Y \in \mathfrak{X}(M)$ are two smooth vector fields on M, then on U, we can write

$$X = X^i \frac{\partial}{\partial x^i}, \quad Y = Y^j \frac{\partial}{\partial x^j},$$

where $X^i, Y^j \in C^\infty(U)$ are the component functions of X and Y with respect to the coordinate vector fields. Then the Lie bracket $[X, Y]$ on U is given by

$$[X, Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j} = \left(X(Y^j) - Y(X^j) \right) \frac{\partial}{\partial x^j}.$$

Proof. Mere computation: take $f \in C^\infty(U)$,

$$\begin{aligned}
[X,Y]f &= X(Yf) - Y(Xf) \\
&= X^i \frac{\partial}{\partial x^i}\left(Y^j \frac{\partial f}{\partial x^j}\right) - Y^i \frac{\partial}{\partial x^i}\left(X^j \frac{\partial f}{\partial x^j}\right) \\
&= X^i \left(\frac{\partial Y^j}{\partial x^i}\frac{\partial f}{\partial x^j} + Y^j \frac{\partial^2 f}{\partial x^i \partial x^j}\right) - Y^i \left(\frac{\partial X^j}{\partial x^i}\frac{\partial f}{\partial x^j} + X^j \frac{\partial^2 f}{\partial x^i \partial x^j}\right) \\
&= \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i}\right)\frac{\partial f}{\partial x^j} + \left(X^i Y^j - Y^i X^j\right)\frac{\partial^2 f}{\partial x^i \partial x^j} \\
&= \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i}\right)\frac{\partial f}{\partial x^j}.
\end{aligned}$$

□

A trivial example is that

$$\left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right] = 0, \quad \forall 1 \leqslant i, j \leqslant n.$$

This is only that mixed partial derivatives commute for smooth functions.

Proposition: **Properties of Lie Bracket**

Let M be a smooth manifold, with or without boundary, and let $X, Y, Z \in \mathfrak{X}(M)$ be three smooth vector fields on M, and let $f, g \in C^\infty(M)$ be two smooth functions on M. Then the Lie bracket satisfies the following properties:

- Bilinearity: $[aX + bY, Z] = a[X, Z] + b[Y, Z]$ and $[Z, aX + bY] = a[Z, X] + b[Z, Y]$ for all $a, b \in \mathbb{R}$.
- Antisymmetry: $[X, Y] = -[Y, X]$.
- Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.
- Leibniz rule: $[fX, gY] = fg[X, Y] + f(Xg)Y - g(Yf)X$.

Theorem 8.3.2: Naturality of Lie Bracket

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. If $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \mathfrak{X}(N)$ are vector fields such that X_i is F-related to Y_i for $i = 1, 2$, then the Lie bracket $[X_1, X_2]$ is F-related to the Lie bracket $[Y_1, Y_2]$.

Proof. We have

$$X_1 X_2 (f \circ F) = X_1\left((Y_2 f) \circ F\right) = (Y_1(Y_2 f)) \circ F, \qquad X_2 X_1 (f \circ F) = (Y_2(Y_1 f)) \circ F.$$

So putting them together,

$$[X_1, X_2](f \circ F) = (Y_1(Y_2 f) - Y_2(Y_1 f)) \circ F = ([Y_1, Y_2]f) \circ F.$$

□

Corollary 8.3.1: Pushforward of Lie Bracket

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a diffeomorphism. If $X_1, X_2 \in \mathfrak{X}(M)$ then

$$F_*[X_1, X_2] = [F_*X_1, F_*X_2]. \tag{8.8}$$

Corollary 8.3.2: Brackets Tangent to Submanifolds

Let M be a smooth manifold and S be an immersed submanifold, with or without boundary. If $X, Y \in \mathfrak{X}(M)$ are vector fields that are tangent to S, then the Lie bracket $[X, Y]$ is also tangent to S.

8.4 The Lie Algebra of Lie Groups

Definition 8.4.1: Left-Invariant

Let G be a Lie group, and let $X \in \mathfrak{X}(G)$ be a smooth vector field on G. We say that X is **left-invariant** if for every $g \in G$, X is L_g-related to itself. In other words

$$\forall g, h \in G, \quad \mathrm{d}L_g(X_h) = X_{gh}. \tag{8.9}$$

Sine L_g is a diffeomorphism for each $g \in G$, this means $(L_g)_*X = X$ for all $g \in G$.
From linearity, the set of all left-invariant vector fields on G forms a vector subspace of $\mathfrak{X}(G)$, and it is closed under the Lie bracket.

Proof. We have from $(L_g)_*X = X$ and $(L_g)_*Y = Y$ that

$$(L_g)_*[X, Y] = [(L_g)_*X, (L_g)_*Y] = [X, Y].$$

□

Definition 8.4.2: Lie Algebra

A Lie algebra over $\mathbb{R}$ is a real vector space $\mathfrak{g}$ equipped with a map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ called the bracket, such that for all $X, Y, Z \in \mathfrak{g}$ and $a, b \in \mathbb{R}$, the following properties hold:

- Bilinearity: $[aX + bY, Z] = a[X, Z] + b[Y, Z]$ and $[Z, aX + bY] = a[Z, X] + b[Z, Y]$.
- Antisymmetry: $[X, Y] = -[Y, X]$.
- Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

If $\mathfrak{g}$ is a Lie algebra, a linear subspace $\mathfrak{h} \subseteq \mathfrak{g}$ is called a **Lie subalgebra** if it is closed under the bracket operation.
If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, a linear map $\varphi : \mathfrak{g} \to \mathfrak{h}$ is called a **Lie algebra homomorphism** if for all $X, Y \in \mathfrak{g}$, $\varphi([X, Y]) = [\varphi(X), \varphi(Y)]$. It is called a **Lie algebra isomorphism** if it is bijective, and we say that $\mathfrak{g}$ and $\mathfrak{h}$ are **isomorphic** Lie algebras.

Example: **Lie Algebras**

- The space $\mathfrak{X}(M)$ of all smooth vector fields on a smooth manifold M, with or without boundary, is a Lie algebra under the Lie bracket.
- If G is a Lie group, then the set of all left-invariant vector fields on G forms a Lie subalgebra of $\mathfrak{X}(G)$, denoted by $\mathrm{Lie}(G)$.
- The vector space $M(n, \mathbb{R})$ of all $n \times n$ real matrices is a Lie algebra under the bracket operation defined by the commutator:

$$[A, B] = AB - BA \quad \forall A, B \in M(n, \mathbb{R}).$$

  denoted by $\mathfrak{gl}(n, \mathbb{R})$. Similarly, $\mathfrak{gl}(n, \mathbb{C})$ is a $2n^2$-dimensional real Lie algebra.
- Generally, if V is a vector space over $\mathbb{R}$, the space $\mathfrak{gl}(V)$ of all linear operators on V is a Lie algebra under the commutator bracket.
- Any vector space V becomes a Lie algebra under the trivial bracket operation $[X, Y] = 0$ for all $X, Y \in V$. Such a Lie algebra is called an **abelian** Lie algebra. (This name reflects that most cases Lie algebras are just commutators like above.)

For $\mathrm{Lie}(G)$, intuitively, if we know the vector at any point on the manifold, then we can determine the whole vector field by left translations, so the dimension of $\mathrm{Lie}(G)$ should be the same as that of G.

Theorem 8.4.1: Structure of $\mathrm{Lie}(G)$

Let G be a Lie group of dimension n. Then the evaluation map $\epsilon : \mathrm{Lie}(G) \to T_eG$ defined by $\epsilon(X) = X_e$ is a vector space isomorphism. In particular, $\dim \mathrm{Lie}(G) = \dim G = n$.
The inversion map is given by

$$T_eG \to \mathrm{Lie}(G), \quad v \mapsto v^L|_g = \mathrm{d}(L_g)_e(v). \tag{8.10}$$

Proof. It is clear that ϵ is linear. To show it is injective, suppose $X \in \mathrm{Lie}(G)$ such that $X_e = 0$. Then for any $g \in G$, $X_g = \mathrm{d}(L_g)_e(X_e) = \mathrm{d}(L_g)_e(0) = 0$. So X is the zero vector field, and hence ϵ is injective. Surjectivity follows from the construction in the second part of the theorem, and smoothness is clear: take any smooth curve $\gamma : (-\delta, \delta) \to G$ with $\gamma(0) = e$ and $\gamma'(0) = v$, then

$$(v^L f)(g) = v^L|_g(f) = \mathrm{d}(L_g)_e(v)(f) = v(f \circ L_g) = \gamma'(0)(f \circ L_g) = \frac{\mathrm{d}}{\mathrm{d}t}(f \circ L_g \circ \gamma)(0).$$

If we define $\varphi : (-\delta, \delta) \times G \to \mathbb{R}$ by $\varphi(t, g) = f \circ L_g \circ \gamma(t) = f(g\gamma(t))$, then φ is smooth, so $(v^L f)(g) = \partial\varphi/\partial t(0, g)$ is smooth in g. Thus v^L is a smooth vector field, and hence $v^L \in \mathrm{Lie}(G)$. $\square$

Therefore, given any vector $v \in T_eG$, there exists a unique left-invariant vector field $X \in \mathrm{Lie}(G)$ such that $X_e = v$, denoted by X^L.

We shall see that the smoothness condition in the definition of left-invariant vector fields is actually superfluous.

Corollary 8.4.1: Left-Invariant Rough Field

Let G be a Lie group. For any rough vector field $X : G \to TG$ that is left-invariant, X is smooth, and hence $X \in \mathrm{Lie}(G)$.

Corollary 8.4.2: Parallelizability of Lie Groups

Every Lie group admits a left-invariant smooth global frame, and is therefore parallelizable.

Example: **Lie Algebras of Lie Groups**

- $\mathbb{R}^n$: as a Lie group under addition, so a vector field X is left-invariant if and only if $X^i\partial/\partial x^i$ has constant component functions X^i. Thus $\mathrm{Lie}(\mathbb{R}^n) \cong T_0\mathbb{R}^n \cong \mathbb{R}^n$.
- S^1: as a Lie group under multiplication, the basis is just $\mathrm{d}/\mathrm{d}\theta$, so $\mathrm{Lie}(S^1) \cong T_1S^1 \cong \mathbb{R}$. The same goes for the torus T^n, the basis is $\partial/\partial\theta^1, \ldots, \partial/\partial\theta^n$, so $\mathrm{Lie}(T^n) \cong T_eT^n \cong \mathbb{R}^n$.

We notice that the group $\mathbb{R}^n, S^1$ are abelian, and their Lie algebras are also abelian. This is not a coincidence. Every abelian Lie group has an abelian Lie algebra.

Proof. If G is abelian, then for any $X, Y \in \mathrm{Lie}(G)$, we have SORRY □

We shall also see that the converse holds when G is connected.

Now we come to inspect the Lie algebra of $GL(n,\mathbb{R})$. Consider $GL(n,\mathbb{R})$ is an open subset of $\mathfrak{gl}(n,\mathbb{R}) \cong \mathbb{R}^{n^2}$, so its tangent space is naturally isomorphic to $\mathfrak{gl}(n,\mathbb{R})$ itself. Also, from the structure theorem of $\mathrm{Lie}(G)$ 8.4.1, we have a vector space isomorphism $\mathrm{Lie}(GL(n,\mathbb{R})) \cong T_IGL(n,\mathbb{R}) \cong \mathfrak{gl}(n,\mathbb{R})$. However, note that $\mathrm{Lie}(GL(n,\mathbb{R}))$ and $\mathfrak{gl}(n,\mathbb{R})$ have independently defined brackets, the former is defined by Lie brackets of left-invariant vector fields, while the latter is defined by commutators of matrices. We shall see that these two brackets actually agree under the above isomorphism.

Theorem 8.4.2: Lie Algebra of $GL(n,\mathbb{R})$

The composition of natural vector space isomorphisms

$$\mathrm{Lie}(GL(n,\mathbb{R})) \xrightarrow{\epsilon} T_IGL(n,\mathbb{R}) \xrightarrow{\cong} \mathfrak{gl}(n,\mathbb{R}) \tag{8.11}$$

is a Lie algebra isomorphism between the left-invariant Lie algebra $\mathrm{Lie}(GL(n,\mathbb{R}))$ and the matrix Lie algebra $\mathfrak{gl}(n,\mathbb{R})$. This can also be generalized to any finite-dimensional real vector space V to give a Lie algebra isomorphism between $\mathrm{Lie}(GL(V))$ and $\mathfrak{gl}(V)$.

Proof. Using the matrix entries X^i_j as global coordinates on $GL(n,\mathbb{R}) \subseteq \mathfrak{gl}(n,\mathbb{R})$, the isomorphism

$$T_IGL(n,\mathbb{R}) \xrightarrow{\cong} \mathfrak{gl}(n,\mathbb{R}), \quad A^i_j \frac{\partial}{\partial X^i_j}\bigg|_I \mapsto A = (A^i_j)$$

So the isomorphism from $\mathfrak{gl}(n,\mathbb{R})$ to $\mathrm{Lie}(GL(n,\mathbb{R}))$ is given by, take any $A \in \mathfrak{gl}(n,\mathbb{R})$, then the corresponding left-invariant vector field $X^A \in \mathrm{Lie}(GL(n,\mathbb{R}))$ is given by

$$A^L|_X = \mathrm{d}(L_X)_I\left(A^i_j \frac{\partial}{\partial X^i_j}\bigg|_I\right) = A^i_j \mathrm{d}(L_X)_I\left(\frac{\partial}{\partial X^i_j}\bigg|_I\right) = A^i_j X^k_i \frac{\partial}{\partial X^k_j}\bigg|_X = X^i_j A^j_k \frac{\partial}{\partial X^i_k}\bigg|_X.$$

Next is pure computation: Take any $A, B \in \mathfrak{gl}(n, \mathbb{R})$, then

$$\begin{aligned}
[A^L, B^L] &= \left[X^i_j A^j_k \frac{\partial}{\partial X^i_k}, X^p_q B^q_r \frac{\partial}{\partial X^p_r} \right] \\
&= X^i_j A^j_k \frac{\partial}{\partial X^i_k} \left(X^p_q B^q_r \right) \frac{\partial}{\partial X^p_r} - X^p_q B^q_r \frac{\partial}{\partial X^p_r} \left(X^i_j A^j_k \right) \frac{\partial}{\partial X^i_k} \\
&= X^i_j A^j_k B^k_r \frac{\partial}{\partial X^i_r} - X^p_q B^q_r A^r_k \frac{\partial}{\partial X^p_k} \\
&= X^i_j (A^j_k B^k_r - B^j_k A^k_r) \frac{\partial}{\partial X^i_r} \\
&= [A, B]^L.
\end{aligned}$$

□

8.4.1 Induced Lie Algebra Homomorphisms

We shall see that Lie group homomorphisms induce Lie algebra homomorphisms.

Theorem 8.4.3: Induced Lie Algebra Homomorphisms

Let G and H be Lie groups, and let $\mathfrak{g}, \mathfrak{h}$ be their respective Lie algebras. Suppose $F : G \to H$ is a Lie group homomorphism. For every $X \in \mathfrak{g}$, there exists a unique vector field $Y \in \mathfrak{h}$ that is F-related to X, denoted by $F_* X$. The map $F_* : \mathfrak{g} \to \mathfrak{h}$ defined by $X \mapsto F_* X$ is a Lie algebra homomorphism.

Proof. By the spirit of one point generates all, the only choice for Y is given by

$$Y = (\mathrm{d}F_e(X_e))^L.$$

And we have

$$\begin{aligned}
\mathrm{d}F_g(X_g) &= \mathrm{d}F_g\left(\mathrm{d}(L_g)_e(X_e)\right) = \mathrm{d}(F \circ L_g)_e(X_e) = \mathrm{d}(L_{F(g)} \circ F)_e(X_e) \\
&= \mathrm{d}(L_{F(g)})_{F(e)}\left(\mathrm{d}F_e(X_e)\right) = (\mathrm{d}F_e(X_e))^L|_{F(g)} = Y_{F(g)}.
\end{aligned}$$

where as $F(L_g g') = F(gg') = F(g)F(g') = L_{F(g)}(F(g'))$ for all $g' \in G$, so we have $F \circ L_g = L_{F(g)} \circ F$. So precisely, Y is F-related to X. Next from naturality of Lie bracket, F_* is a Lie algebra homomorphism. □

Remark:
Note that here we only require F to be a Lie group homomorphism, not necessarily a diffeomorphism. So the induced map F_* may not be an isomorphism.

Proposition: **Properties of Induced Lie Algebras**

- The identity homomorphism $\mathrm{id}_G : G \to G$ induces the identity Lie algebra homomorphism $\mathrm{id}_{\mathfrak{g}} : \mathfrak{g} \to \mathfrak{g}$.
- Transitive property: if $F : G \to H$ and $H \to K$ are Lie group homomorphisms, then

the composition $K \circ F : G \to K$ induces the composition of Lie algebra homomorphisms $K_* \circ F_* : \mathfrak{g} \to \mathfrak{k}$.

- Isomorphic Lie groups have isomorphic Lie algebras.

8.4.2 Lie Algebra of Lie Subgroups

Intuitively, if H is a subgroup of a Lie group G, then the Lie algebra of H should be a Lie subalgebra of that of G. There is a slight confusion however, for elements of $\mathrm{Lie}(H)$ are vector fields on H, not on G. We propose a small patch here nevertheless.

Theorem 8.4.4: Lie Algebra of Lie Subgroups

Suppose $H \subseteq G$ is a Lie subgroup of a Lie group G, and $\iota : H \hookrightarrow G$ is the inclusion map. There is a Lie algebra $\mathfrak{h} \subseteq \mathrm{Lie}(G)$ isomorphic to $\mathrm{Lie}(H)$, given by

$$\mathfrak{h} = \iota_*(\mathrm{Lie}(H)) = \{X \in \mathrm{Lie}(G) : X \in T_eH \subseteq T_eG\}. \tag{8.12}$$

This is quite natural, as both can be generated by the tangent space at the identity. This gives a nice way to identify the Lie algebra of a Lie subgroup as a Lie subalgebra of that of the bigger Lie group.

Example: **Lie Algebra of $O(n)$**

The orthogonal group $O(n)$ is a Lie subgroup of $GL(n, \mathbb{R})$. For $\Phi(A) = A^TA$, it is equal to the level set $\Phi^{-1}(I)$. We have

$$T_IO(n) = \{B \in \mathfrak{gl}(n, \mathbb{R}) : B^T + B = 0\},$$

consisting of all skew-symmetric matrices. It is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$ under the commutator bracket, denoted by $\mathfrak{o}(n)$.

We can do the same for $GL(n, \mathbb{C})$ viewed as a real Lie group.

Definition 8.4.3: Representation of Lie Algebra

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{R}$. A **representation** of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\varphi : \mathfrak{g} \to \mathfrak{gl}(V), \tag{8.13}$$

for some finite-dimensional real vector space V. If such a representation is injective, we say that $\mathfrak{g}$ is **faithfully represented** on V, in this case it is isomorphic to a Lie subalgebra of $\mathfrak{gl}(V)$.

Theorem 8.4.5: Ado's Theorem

Every finite-dimensional Lie algebra over $\mathbb{R}$ admits a faithful finite-dimensional representation, so it is isomorphic to a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$ for some n with the commutator bracket.

Chapter 9

Integral Curves and Flows

9.1 Integral Curves

Suppose M is a smooth manifold, with or without boundary, and $\gamma : J \to M$ is a smooth curve, then $\forall t \in J$, the velocity vector $\gamma'(t) \in T_{\gamma(t)}M$. Now we want to work in the reverse direction.

Definition 9.1.1: Integral Curve

If V is a vector field on M, then a differentiable curve $\gamma : J \to M$ is called an **integral curve** of V if $\forall t \in J$,

$$\gamma'(t) = V_{\gamma(t)}.$$

Usually if $0 \in J$, we say γ is an integral curve of V **starting at** $\gamma(0)$.

Suppose V is a smooth vector field on M. For a smooth chart $U \subseteq M$, in local coordinates, we can write $\gamma(t) = (\gamma^i(t))$, and $V = (V^i)$, then the integral curve equation becomes

$$\frac{d\gamma^i}{dt}(t) = V^i(\gamma(t)), \quad i = 1, \ldots, n. \tag{9.1}$$

From ODE theory, we have

Proposition: **Existence of Integral Curve**

Let V be a smooth vector field on a smooth manifold M. For each point $p \in M$, there exists $\epsilon > 0$ and an integral curve $\gamma : (-\epsilon, \epsilon) \to M$ of V such that $\gamma(0) = p$.

Next, we investigate some reparametrization properties of integral curves.

Proposition: **Reparametrization of Integral Curves**

Let V be a smooth vector field on a smooth manifold M, and let $\gamma : J \to M$ be an integral curve of V.

- Rescaling: For any $a \in \mathbb{R}$, the curve $\tilde{\gamma} : \tilde{J} \to M$ defined by $\tilde{\gamma}(s) = \gamma(as)$, where $\tilde{J} = \{s \in \mathbb{R} : as \in J\}$, is an integral curve of the vector field aV.
- Translation: For any $b \in \mathbb{R}$, the curve $\hat{\gamma} : \hat{J} \to M$ defined by $\hat{\gamma}(u) = \gamma(u + b)$, where

$\hat{J} = \{u \in \mathbb{R} : u + b \in J\}$, is an integral curve of the vector field V.

Proposition: **Naturality of Integral Curves**

Let M and N be smooth manifolds, and let $F : M \to N$ be a smooth map. Then $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are F-related if and only if for every integral curve γ of X, the curve $F \circ \gamma$ is an integral curve of Y.

Proof. First suppose X and Y are F-related, and let $\gamma : J \to M$ be an integral curve of X. Then define $\sigma = F \circ \gamma : J \to N$. For any $t \in J$,

$$\sigma'(t) = \mathrm{d}F_{\gamma(t)}(\gamma'(t)) = \mathrm{d}F_{\gamma(t)}(X_{\gamma(t)}) = Y_{F(\gamma(t))} = Y_{\sigma(t)},$$

so σ is an integral curve of Y.

Conversely, suppose that for every integral curve γ of X, the curve $F \circ \gamma$ is an integral curve of Y. Let $p \in M$, and let $\gamma : (-\epsilon, \epsilon) \to M$ be an integral curve of X with $\gamma(0) = p$. Then $\sigma = F \circ \gamma : (-\epsilon, \epsilon) \to N$ is an integral curve of Y. Evaluating at $t = 0$, we have

$$Y_{F(p)} = (F \circ \gamma)'(0) = \mathrm{d}F_{\gamma(0)}(\gamma'(0)) = \mathrm{d}F_p(X_p),$$

so X and Y are F-related. □

9.2 Flows

We come by another way to look at integral curves. Suppose for each point $p \in M$, we have a unique integral curve $\theta^{(p)} : \mathbb{R} \to M$ (Actually, it may always be defined on the whole $\mathbb{R}$, but for simplicity we assume this here) starting at p. Then we can define a map

$$\theta_t : M \to M, \quad \theta_t(p) = \theta^{(p)}(t).$$

which sends each point p to the point that slides along the integral curve starting at p for time t. It is easy to see that

$$\theta_t \circ \theta_s = \theta_{t+s}, \quad \theta_0 = \mathrm{id}_M.$$

well, this is just the property of group action. So the map $\theta : \mathbb{R} \times M \to M$ defined by $\theta(t, p) = \theta_t(p)$ is an action of the group $\mathbb{R}$ on M.

Definition 9.2.1: Global Flow

Let M be a smooth manifold. A **global flow** on M is a continuous left action $\theta : \mathbb{R} \times M \to M$ of the group $\mathbb{R}$ on M.

The geometric meaning of a global flow is find where each point goes as time passes.

- For each fixed $t \in \mathbb{R}$, the map $\theta_t : M \to M$ defined by $\theta_t(p) = \theta(t, p)$ is a homeomorphism, and if the flow is smooth, then θ_t is a diffeomorphism.
- For each fixed $p \in M$, the map $\theta^{(p)} : \mathbb{R} \to M$ defined by $\theta^{(p)}(t) = \theta(t, p)$ is the orbit of p under the action.

Every smooth global flow can be associated with integral curves of a smooth vector field. For any $\theta : \mathbb{R} \times M \to M$ smooth global flow, we can define a vector field V on M by

$$V_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \theta_t(p).$$

called the **infinitesimal generator** of the flow.

Proposition: **Infinitesimal Generator**

Let $\theta : \mathbb{R} \times M \to M$ be a smooth global flow on a smooth manifold M, and let V be its infinitesimal generator. Then V is a smooth vector field on M, and for each $p \in M$, the orbit $\theta^{(p)} : \mathbb{R} \to M$ is the unique integral curve of V starting at p.

Proof. Take any $f \in C^\infty(M)$ on some open neighborhood of p. Then

$$Vf(p) = V_p f = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (f \circ \theta_t)(p) = \left.\frac{\partial}{\partial t}\right|_{(0,p)} (f \circ \theta)(t, p),$$

So it is smooth. The second part follows according to the definition of integral curves. □

9.2.1 The Fundamental Theorem of Flows

We cannot say that every smooth vector field on a smooth manifold generates a global flow because integral curves may not be defined for all $\mathbb{R}$. So we introduce a small patch here.

Definition 9.2.2: Flow

Let M be a manifold. A **flow domain** in M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ such that for each $p \in M$, the set $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$ is an open interval containing 0. A **flow** or **local flow** on M is a continuous map $\theta : \mathcal{D} \to M$ such that

- $\forall p \in M$, we have $\theta(0, p) = p$;
- For any $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta(s,p))}$, such that $s + t \in \mathcal{D}^{(p)}$, we have

$$\theta(t, \theta(s, p)) = \theta(s + t, p). \tag{9.2}$$

If θ is a flow, we define $\theta_t : M_t \to M$ by $\theta_t(p) = \theta(t, p)$, where $M_t = \{p \in M : (t, p) \in \mathcal{D}\}$. Then if θis smooth, the infinitesimal generator of θ is defined as before.

Proposition: **Infinitesimal Generator of a Flow**

Let $\theta : \mathcal{D} \to M$ be a smooth flow on a smooth manifold M, and let V be its infinitesimal generator. Then V is a smooth vector field on M, and for each $p \in M$, the orbit $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ is an integral curve of V starting at p.

Definition 9.2.3: Maximal Integral Curve

Let V be a smooth vector field on a smooth manifold M. An integral curve $\gamma : J \to M$ of V is called a **maximal integral curve** of V if there is no integral curve $\tilde{\gamma} : \tilde{J} \to M$ of V such that $J \subsetneq \tilde{J}$ and $\tilde{\gamma}|_J = \gamma$.
A maximal flow of V is a flow $\theta : \mathcal{D} \to M$ that cannot be extended to a larger flow domain.

Theorem 9.2.1: The Fundamental Theorem of Flows

Let V be a smooth vector field on a smooth manifold M. Then there exists a unique maximal flow $\theta : \mathcal{D} \to M$ on M whose infinitesimal generator is V. And θ has the following properties:

- For each $p \in M$, the map $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ defined by $\theta^{(p)}(t) = \theta(t, p)$ is the unique maximal integral curve of V starting at p.
- If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))} = \mathcal{D}^{(p)} - s = \{t - s : t \in \mathcal{D}^{(p)}\}$.
- For each $t \in \mathbb{R}$, the set $M_t = \{p \in M : (t, p) \in \mathcal{D}\}$ is open in M, and the map $\theta_t : M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

Proposition: **Naturality of Flows**

Let M and N be smooth manifolds, and let $F : M \to N$ be a smooth map. Let $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ be F-related smooth vector fields, and let θ and η be their respective maximal flows. Then for all $t \in \mathbb{R}$, $F(M_t) \subseteq N_t$, and the following diagram commutes:

$$\begin{tikzcd} M_t \arrow[r, "\theta_t"] \arrow[d, "F"] & M_{-t} \arrow[d, "F"] \\ N_t \arrow[r, "\eta_t"] & N_{-t} \end{tikzcd}$$

Proof. For any $p \in M$, the curve $F \circ \theta^{(p)}$ is an integral curve of Y starting at $F(p)$. So by the uniqueness of maximal integral curves, $\eta^{(F(p))}$ extends $F \circ \theta^{(p)}$ and must be defined at least on $\mathcal{D}^{(p)}$. Thus, $F(M_t) \subseteq N_t$ for all $t \in \mathbb{R}$. Moreover,

$$F(\theta^{(p)}(t)) = F \circ \theta^{(p)}(t) = \eta^{(F(p))}(t) = \eta_t(F(p)),$$

thus $\eta_t \circ F = F \circ \theta_t$ on M_t. □

9.2.2 Complete Vector Fields

As we have seen, not every smooth vector field generates a global flow.

Definition 9.2.4: Complete Vector Field

A smooth vector field V on a smooth manifold M is called **complete** if its maximal flow $\theta : \mathcal{D} \to M$ has flow domain $\mathcal{D} = \mathbb{R} \times M$; that is, it generates a global flow on M.

We will show that compactly supported vector fields are complete.

Lemma 9.2.1: Uniform Time Lemma

Let V be a smooth vector field on a smooth manifold M, and let $\theta : \mathcal{D} \to M$ be its maximal flow. If there exists $\epsilon > 0$ such that for each $p \in M$, the interval $(-\epsilon, \epsilon) \subseteq \mathcal{D}^{(p)}$, then V is complete.

Proof. Suppose for some $p \in M$, the domain $\mathcal{D}^{(p)}$ is bounded above, let $b = \sup \mathcal{D}^{(p)}$ and let $b - \epsilon < t_0 < b$, and $q = \theta(t_0, p)$. Then by the hypothesis, $\theta^{(q)}$ is defined on $(-\epsilon, \epsilon)$, so $\gamma : (-\epsilon, t_0 + \epsilon) \to M$ defined by

$$\gamma(t) = \begin{cases} \theta^{(p)}(t), & t \in (-\epsilon, b) \\ \theta^{(q)}(t - t_0), & t \in (t_0 - \epsilon, t_0 + \epsilon) \end{cases}$$

is an integral curve of V extending $\theta^{(p)}$, contradicting the maximality of $\theta^{(p)}$. Thus, $\mathcal{D}^{(p)}$ is unbounded above. A similar argument shows that $\mathcal{D}^{(p)}$ is unbounded below, so $\mathcal{D}^{(p)} = \mathbb{R}$ for all $p \in M$, and hence V is complete. □

Theorem 9.2.2: Completeness of Compactly Supported Vector Fields

Let M be a smooth manifold, and let V be a smooth vector field on M with compact support. Then V is complete.
Therefore, on a compact smooth manifold, every smooth vector field is complete.

Proof. Obvious. □

Theorem 9.2.3: Completeness of Left-Invariant Vector Fields

Let G be a Lie group, and let V be a left-invariant vector field on G. Then V is complete.

Proof. There is some ϵ that $\theta^{(e)}$ is defined on $(-\epsilon, \epsilon)$, where e is the identity element of G. For any $g \in G$, the integral curve $\theta^{(g)}$ starting at g is given by $L_g \circ \theta^{(e)}$, which is defined on $(-\epsilon, \epsilon)$ as well. Thus, by the Uniform Time Lemma, V is complete. □

Lemma 9.2.2: Escape Lemma

Suppose M is a smooth manifold, and $V \in \mathfrak{X}(M)$. If $\gamma : J \to M$ is a maximal integral curve of V and J is bounded above, let $b = \sup J$. Then for any $t_0 \in J$, $\gamma([t_0, b))$ is not contained in any compact subset of M.

9.3 Flowouts

Flows provide some technique for geometric constructions on manifolds.

Theorem 9.3.1: Flowout Theorem

Let M be a smooth manifold, let $S \subseteq M$ be an embedded k-dimensional submanifold, and let V be a smooth vector field on M that is nowhere tangent to S. Let $\theta : \mathcal{D} \to M$ be the maximal flow of V. Then let $\mathcal{O} = (\mathbb{R} \times S) \cap \mathcal{D}$ be the submanifold part of the flow domain. Let $\Phi = \theta|_{\mathcal{O}} : \mathcal{O} \to M$ be the restriction of the flow to $\mathcal{O}$.

- $\Phi : \mathcal{O} \to M$ is an immersion;
- $\partial/\partial t \in \mathfrak{X}(\mathcal{O})$ is Φ-related to $V \in \mathfrak{X}(M)$;
- There exists a smooth positive function $\delta : S \to \mathbb{R}$ such that the restriction $\Phi|_{\mathcal{O}_\delta} : \mathcal{O}_\delta \to M$ is injective, where
$$\mathcal{O}_\delta = \{(t, p) \in \mathbb{R} \times S : |t| < \delta(p)\}.$$
Thus $\Phi(\mathcal{O}_\delta)$ is an immersed submanifold of M containing S, and V is tangent to $\Phi(\mathcal{O}_\delta)$.
- If S has codimension 1, then $\Phi|_{\mathcal{O}_\delta} : \mathcal{O}_\delta \to \Phi(\mathcal{O}_\delta)$ is a diffeomorphism onto an open submanifold of M.

The submanifold $\Phi(\mathcal{O}_\delta)$ is called the **flowout** of S along V.

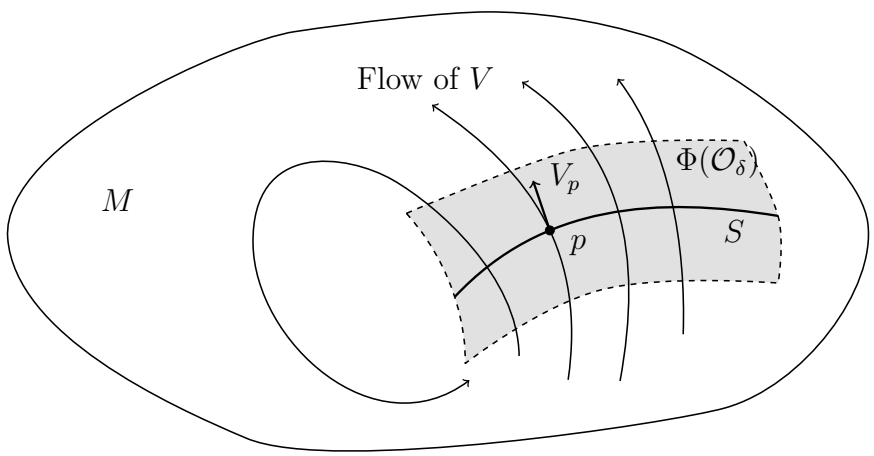

Figure 9.1: Flowout

Proof. SORRY □

9.3.1 Regular Points and Singular Points

Definition 9.3.1: Regular Point and Singular Point

Let M be a smooth manifold, and let V be a vector field on M. A point $p \in M$ is called a **regular point** of V if $V_p \neq 0$. Otherwise, it is called a **singular point** of V.

Proposition: **Regular and Singular Points**

Let V be a smooth vector field on a smooth manifold M. Let $\theta : \mathcal{D} \to M$ be the maximal flow of V. Then if $p \in M$ is a singular point of V, then $\mathcal{D}^{(p)} = \mathbb{R}$, and $\theta^{(p)}(t) = p$ for all $t \in \mathbb{R}$ is a constant curve. If p is a regular point of V, then $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ is a smooth immersion.

This shows that equilibrium points (The points where $\theta^{(p)}$ is constant) of a flow are exactly the singular points of its infinitesimal generator.

Now we give a complete local structure of a vector field around a regular point.

Theorem 9.3.2: Canonical Form near a Regular Point

Let M be a smooth manifold, and let V be a smooth vector field on M. If $p \in M$ is a regular point of V, then there exists a smooth chart (U, φ) near p, with coordinates $(s^1, \ldots, s^n)$, such that on U,

$$V = \frac{\partial}{\partial s^1}.$$

If $S \subseteq M$ is any embedded hypersurface (codimension 1 submanifold) with $p \in S$ and $V_p \notin T_pS$, then the chart (U, φ) can be chosen so that s^1 is the local defining function of S.

9.4 Flows and Flowouts on Manifolds with Boundary

Well from definition point we see that vector fields on manifolds with boundary need not generate flows when the point is on the boundary, which only permits half-open intervals as domains of integral curves. However, there are some flowout results that are important.

Theorem 9.4.1: Boundary Flowout Theorem

Let M be a smooth manifold with nonempty boundary, let N be a smooth vector field on M that is inward points on every $p \in \partial M$. There exists a smooth function $\delta : \partial M \to \mathbb{R}^+$ and a smooth embedding $\Phi : \mathcal{P}_\delta \to M$, where $\mathcal{P}_\delta = \{(t, p) : p \in \partial M, 0 \leqslant t < \delta(p)\} \subseteq \mathbb{R} \times \partial M$, such that $\Phi(\mathcal{P}_\delta)$ is an open neighborhood of ∂M in M, and for each $p \in \partial M$, the curve $\Phi^{(p)} : [0, \delta(p)) \to M$ defined by $\Phi^{(p)}(t) = \Phi(t, p)$ is the integral curve of N starting at p.

Lemma 9.4.1: Existence of Inward Vector Fields

Let M be a smooth manifold with nonempty boundary. There exists a smooth global vector field on M that inward points at every point of ∂M.

Theorem 9.4.2: Collar Neighborhood Theorem

Let M be a smooth manifold with nonempty boundary. A neighborhood of ∂M is called a **collar neighborhood** if it is the image of a smooth embedding $[0, 1) \times \partial M \to M$ that restricts to the identity on $\{0\} \times \partial M$.
Then every smooth manifold with nonempty boundary has a collar neighborhood.

Theorem 9.4.3: Homotopy to Interior

Let M be a smooth manifold with nonempty boundary. And let $\iota : \operatorname{Int} M \to M$ be the inclusion map. Then there exists a proper smooth embedding $R : M \to \operatorname{Int} M$ such that both $\iota \circ R : M \to M$ and $R \circ \iota : \operatorname{Int} M \to \operatorname{Int} M$ are smoothly homotopic to the respective identity maps. Therefore, ι is a homotopy equivalence between $\operatorname{Int} M$ and M.

Proof. Quite natural, just shrinking the boundary along the collar neighborhood a little bit to get R would do. □

Theorem 9.4.4: Whitney Approximation Theorem for Manifolds with Boundary

Let M, N be smooth manifolds with boundary, then every continuous map $F : M \to N$ is homotopic to a smooth map.

We next generalize theorem 6.3.5, we have

Theorem 9.4.5: Homotopy and Smooth Homotopy for Manifolds with Boundary

Let M, N be smooth manifolds with boundary, and let $F, G : M \to N$ be smooth maps that are homotopic. Then F and G are smoothly homotopic.

The following theorem shows how to attach manifolds along their boundaries.

Theorem 9.4.6: Attaching Manifolds along their Boundaries

Let M, N be smooth n-manifolds with nonempty boundaries, and suppose $h : \partial N \to \partial M$ is a diffeomorphism. Let

$$M \cup_h N = (M \sqcup N)/\sim, \qquad \text{where } x \sim h(x) \text{ for all } x \in \partial N.$$

Then $M \cup_h N$ is a topological manifold without boundary, and it admits a smooth structure such that there are regular domains $M', N' \subseteq M \cup_h N$ that are diffeomorphic to M and N, respectively, and satisfies

$$M' \cup N' = M \cup_h N, \quad M' \cap N' = \partial M' = \partial N'. \tag{9.3}$$

If M, N are both compact, then $M \cup_h N$ is also compact. If they are both connected then $M \cup_h N$ is also connected.

Proof. SORRY □

Example: **Connected Sums**

Let M_1, M_2 be connected smooth n-manifolds, for $i = 1, 2$ let U_i denote a regular coordinate ball centered at some point $p_i \in M_i$, and let $M_i' = M_i \backslash U_i$. Then M_i' is a smooth manifold with boundary diffeomorphic to $\mathbb{S}^{n-1}$. A smooth **connected sum** of M_1 and M_2, denoted $M_1 \# M_2$, is the smooth manifold obtained by attaching M_1' and M_2' along their boundaries via some diffeomorphism $h : \partial M_2' \to \partial M_1'$. The resulting manifold is independent of the

choice of coordinate balls and diffeomorphism, up to diffeomorphism.

Theorem 9.4.5 shows a way of embedding a smooth manifold with boundary into another one without boundary, namely just $\operatorname{Int} M$. We can also have another way.

Example: **The Double of a Manifold with Boundary**

Let M be a smooth manifold with nonempty boundary. The **double** of M, denoted $D(M)$, is the smooth manifold without boundary obtained by attaching two copies of M along their boundaries via the identity map of ∂M, namely, $M \cup_{\mathrm{id}} M$.
Then $D(M)$ is compact if and only if M is compact, connected if and only if M is connected.

Although vector fields on manifolds with boundary may not generate flows, there are circumstances that they do.

Lemma 9.4.2: Tangent Fields Generate Flows

Let M be a smooth manifold, and $D \subseteq M$ is a regular domain. Let V be a smooth vector field on M that is tangent to ∂D at every point of ∂D. Then every integral curve of V that starts in D remains in D for all time.

Theorem 9.4.7: Flows on Manifolds with Boundary

Let M be a smooth manifold with nonempty boundary, and let V be a smooth vector field on M that is tangent to ∂M at every point of ∂M. Then the fundamental theorem of flows 9.2.1 holds for V on M.

Proof. Consider M to be the regular domain in its double $D(M)$ would do. □

Theorem 9.4.8: Canonical Form near a Regular Point on Manifolds with Boundary

Let M be a smooth manifold with boundary, and let V be a smooth vector field on M that is tangent to ∂M. If $p \in \partial M$ is a regular point of V, then there exists a smooth chart (U, φ) near p, with coordinates $(s^1, \ldots, s^n)$, such that on U, $V = \partial/\partial s^1$.

9.5 Lie Derivatives

We know how to interpret a directional derivative of a real-valued function on a manifold. Indeed a tangent vector $v \in T_pM$ can be viewed as this. So what about the directional derivative of a vector field? In Euclidean space $\mathbb{R}^n$, the directional derivative of a smooth vector field W in the direction of a vector $v \in T_p\mathbb{R}^n$ is defined as

$$D_vW(p) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} W(p+tv) = \lim_{t\to 0} \frac{W(p+tv) - W(p)}{t}.$$

however, this definition depends on the vector space structure of $\mathbb{R}^n$ because we need to do substraction of vectors at different points.

There is no natural way to define the directional derivative of a vector field on a general manifold. However, if we have a vector field V instead of a single tangent vector v, then we can use the flow to push forward and pull back vectors.

Definition 9.5.1: Lie Derivative

Suppose M is a smooth manifold, V is a smooth vector field on M with maximal flow θ. Let W be another smooth vector field on M. The **Lie derivative** of W with respect to V is the rough vector field defined by

$$(\mathcal{L}_V W)_p = \lim_{t\to 0} \frac{(\mathrm{d}\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t} = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (\mathrm{d}\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}). \tag{9.4}$$

If M is a smooth manifold with boundary, and V is tangent to ∂M, then the above definition still makes sense using the flow theorem for manifolds with boundary 9.4.7.

Lemma 9.5.1: Smoothness of Lie Derivative

Let M be a smooth manifold, with or without boundary. Let $V, W \in \mathfrak{X}(M)$. If $\partial M \neq \varnothing$, assume that V is tangent to ∂M. Then $\mathcal{L}_V W$ exists and is a smooth vector field on M.

Proof. Let θ be the flow of V. For any $p \in M$, let $(U, (x^i))$ be a smooth chart about p. Choose a small open interval J_0 containing 0 and an open subset $U_0 \subseteq U$ containing p such that θ maps $J_0 \times U_0$ into U. Write $\theta(t,x) = (\theta^1(t,x), \dots, \theta^n(t,x))$ when $(t,x) \in J_0 \times U_0$. Then for any $(t,x) \in J_0 \times U_0$, the matrix of $(\mathrm{d}\theta_{-t})_{\theta_t(x)}$ with respect to the basis $\{\partial/\partial x^i|_{\theta_t(x)}\}$ is given by

$$\left(\frac{\partial \theta^i}{\partial x^j}(-t, \theta_t(x))\right)_{i,j=1}^n.$$

So we have

$$\mathrm{d}(\theta_{-t})_{\theta_t(x)}(W_{\theta_t(x)}) = \frac{\partial \theta^i}{\partial x^j}(-t, \theta(t,x)) W^j(\theta_t(x)) \frac{\partial}{\partial x^i}\bigg|_x,$$

is smooth in both t and x. Thus, taking the derivative with respect to t at $t = 0$ gives a smooth vector field on U_0 that agrees with $\mathcal{L}_V W$ on U_0. Since p was arbitrary, $\mathcal{L}_V W$ is smooth on M. □

The definition is not quite computation friendly. We have the following more useful property that links Lie derivatives with Lie brackets.

Theorem 9.5.1: Structure of Lie Derivative

Let M be a smooth manifold and $V, W \in \mathfrak{X}(M)$. then

$$\mathcal{L}_V W = [V, W]. \tag{9.5}$$

Proof. Let $\mathcal{R}(V) \subseteq M$ be the set of regular points of V (points where $V_p \neq 0$). Since $\mathcal{R}(V)$ is open by continuity, its closure is the support of V. Take $p \in M$.

- Case 1, $p \in \mathcal{R}(V)$: By the Canonical Form near a Regular Point 9.3.2, there exists a smooth chart $(U, (u^1, \dots, u^n))$ about p such that on U, $V = \partial/\partial u^1$. Then the flow of V is $\theta_t(u) = (u^1 + t, u^2, \dots, u^n)$. For each t, then $\mathrm{d}(\theta_{-t})_{\theta_t(x)}$ is the identity map on $T_{\theta_t(x)}M$. Thus for any

$u \in U$, we have

$$\mathrm{d}(\theta_{-t})_{\theta_t(u)}(W_{\theta_t(u)}) = W^j(u^1 + t, u^2, \ldots, u^n)\frac{\partial}{\partial u^j}\bigg|_u,$$

So we have

$$(\mathcal{L}_V W)_u = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} W^j(u^1 + t, u^2, \ldots, u^n)\frac{\partial}{\partial u^j}\bigg|_u = \frac{\partial W^j}{\partial u^1}(u)\frac{\partial}{\partial u^j}\bigg|_u.$$

On the other hand,

$$[V, W]_u = \left[\frac{\partial}{\partial u^1}, W^j\frac{\partial}{\partial u^j}\right]_u = \frac{\partial W^j}{\partial u^1}(u)\frac{\partial}{\partial u^j}\bigg|_u.$$

- Case 2, $p \in \operatorname{supp} V$. By continuity.
- Case 3, $p \notin \operatorname{supp} V$. Then $V = 0$ on some open neighborhood U of p. Then θ_t is the identity map on U for all t sufficiently small. Thus, for any $u \in U$,

$$\mathrm{d}(\theta_{-t})_{\theta_t(u)}(W_{\theta_t(u)}) = W_u,$$

so $(\mathcal{L}_V W)_u = 0$. On the other hand, since $V = 0$ on U, we have $[V, W]_u = 0$ as well.

□

Remark:
This is the geometric interpretation of Lie brackets we mentioned in the chapter on Lie groups. Lie brackets measure the change of one vector field along the flow generated by another vector field.

Proposition: **Properties of Lie Derivatives**

Suppose M is a smooth manifold, with or without boundary. Let $V, W, X \in \mathfrak{X}(M)$,

- $\mathcal{L}_V W = -\mathcal{L}_W V$.
- $\mathcal{L}_V[W, X] = [\mathcal{L}_V W, X] + [W, \mathcal{L}_V X]$. (The Jacobi Identity)
- $\mathcal{L}_{[V,W]}X = \mathcal{L}_V\mathcal{L}_W X - \mathcal{L}_W\mathcal{L}_V X$. (Again the Jacobi Identity)
- $\mathcal{L}_V(gW) = (Vg)W + g\mathcal{L}_V W$, for any $g \in C^\infty(M)$. (Note this gives a geometric interpretation of the product rule for Lie brackets.)
- If $F : M \to N$ is a diffeomorphism, then $F_*(\mathcal{L}_V W) = \mathcal{L}_{F_*V}(F_*W)$.

To compute derivative at other time than 0, we have

Proposition: **Derivatives at Other Times**

Let M be a smooth manifold, with or without boundary. Let $V, W \in \mathfrak{X}(M)$, and if $\partial M \neq \varnothing$,

assume that V is tangent to ∂M. Let θ be the flow of V. Then for any $(t_0, p) \in \mathcal{D}$,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=t_0} \mathrm{d}(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = \mathrm{d}(\theta_{-t_0})_{\theta_{t_0}(p)}((\mathcal{L}_V W)_{\theta_{t_0}(p)}). \tag{9.6}$$

This can be seen as conputing the Lie derivative at time t_0 AT the position $t = 0$, it pulls back the Lie derivative at time t_0 and all t back to time 0 for comparison. So the answer is just the pullback of the Lie derivative at time t_0.

9.6 Commuting Vector Fields

Let M be a smooth manifold, and let $V, W \in \mathfrak{X}(M)$. We say that V, W commute if their Lie bracket vanishes, i.e. $[V, W] = 0$. A vector field W is said to be invariant under some flow θ if W is θ_t-related to itself for all t in the flow domain. (More precisely, $W|_{M_t}$ is θ_t-related to $W|_{M_{-t}}$ for all $t \in \mathbb{R}$.) This means that $\mathrm{d}(\theta_t)_p(W_p) = W_{\theta_t(p)}$ for all $(t, p) \in \mathcal{D}$.

Theorem 9.6.1: Structure of Invariant Fields under Flows

Let M be a smooth manifold, and let $V, W \in \mathfrak{X}(M)$. Then the following are equivalent:

- V and W commute, i.e. $[V, W] = 0$;
- W is invariant under the flow of V;
- V is invariant under the flow of W.

Specially, every smooth vector field is invariant under its own flow.

Proof. If W is invariant under the flow of V, then for any $(t, p) \in \mathcal{D}$, $W_{\theta_t(p)} = \mathrm{d}(\theta_t)_p(W_p)$. Thus,

$$\mathrm{d}(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) = \mathrm{d}(\theta_{-t})_{\theta_t(p)} \circ \mathrm{d}(\theta_t)_p(W_p) = W_p.$$

So $\mathcal{L}_V W = 0$, i.e. $[V, W] = 0$.

Conversely, if $[V, W] = 0$, then reverse the argument above would do. □

Remark:
We can use $[V, V] = 0$ to see that every smooth vector field is invariant under its own flow. But this is quite obvious from the naturality of flows proposition 9.2.1, since V is always θ_t-related to itself, because θ_t action gives the same flow.

As infinitesimal generators of flows, we can visualize commuting vector fields as infinitesimal symmetries of each other. Moving along V a little bit and then moving along W a little bit is the same as moving along W first and then moving along V, which is exactly why we need W not to change too much along the flow of V (and vice versa). We may naturally think that this is also equivalent to the commutivity of the flows generated by V and W.

To be more precise,

Definition 9.6.1: Commuting Flows

Suppose M is a smooth manifold, and let θ, ψ be two flows on M, then we say that θ and ψ **commute** if

$$\begin{aligned}&\forall p \in M, \forall J, K \subseteq \mathbb{R} \text{ open intervals containing } 0\\&[(\forall (s,t) \in J \times K, \theta_s \circ \psi_t \text{ is defined }) \vee (\forall (s,t) \in J \times K, \psi_t \circ \theta_s \text{ is defined })\\&\to (\forall (s,t) \in J \times K, \theta_s \circ \psi_t(p) = \psi_t \circ \theta_s(p)) \text{ are both defined }].\end{aligned} \tag{9.7}$$

For global flows, this means that $\theta_s \circ \psi_t = \psi_t \circ \theta_s$ for all $s, t \in \mathbb{R}$.

Theorem 9.6.2: Equivalence of Field Commuting and Flow Commuting

Smooth vector fields commute if and only if their flows commute.

Proof. SORRY □

Remark:
Note the condition of commutativity of flows is a bit complicated due to the possible non-completeness of the vector fields. It is easily mistaken to be

$$\begin{aligned}&\forall p \in M, \forall J, K \subseteq \mathbb{R} \text{ open intervals containing } 0\\&\forall (s,t) \in J \times K, [\theta_s \circ \psi_t \text{ is defined } \wedge \psi_t \circ \theta_s \text{ is defined } \to \theta_s \circ \psi_t(p) = \psi_t \circ \theta_s(p)].\end{aligned}$$

But this is not correct. There exists certain commuting vector fields and for some certain s, t that both $\theta_s \circ \psi_t$ and $\psi_t \circ \theta_s$ are defined, but they are not equal.

Proof. SORRY □

9.6.1 Commuting Frames

Definition 9.6.2: Commuting Frame

Let M be a smooth n-manifold. A smooth local frame $(U, (E_1, \ldots, E_n))$ on M is called a **commuting frame** if $[E_i, E_j] = 0$ for all $1 \leqslant i, j \leqslant n$.

Example: **Commuting Frames**

- The coordinate frame $(U, (\partial/\partial x^1, \ldots, \partial/\partial x^n))$ associated to any smooth chart $(U, (x^1, \ldots, x^n))$ is a commuting frame.
- On $\mathbb{R}\backslash\{0\}$, define

$$E_1 = \frac{x}{r}\frac{\partial}{\partial x} + \frac{y}{r}\frac{\partial}{\partial y}, \quad E_2 = -\frac{y}{r}\frac{\partial}{\partial x} + \frac{x}{r}\frac{\partial}{\partial y}, \quad r = \sqrt{x^2 + y^2}.$$

  Then we have

$$[E_1, E_2] = \frac{y}{r^2}\frac{\partial}{\partial x} - \frac{x}{r^2}\frac{\partial}{\partial y}0.$$

So it it not a commuting frame. To see why, in angular coordinates (r, θ), we have

$$E_1 = \frac{\partial}{\partial r}, \quad E_2 = \frac{1}{r}\frac{\partial}{\partial \theta},$$

which has a normalizing factor $1/r$ that varies along the flow of E_1.

Therefore, this frame cannot be expressed as a coordinate frame of any smooth chart on $\mathbb{R}^2 \backslash \{0\}$.

Next, we shall show that commuting is also a sufficient condition for a local frame to be a coordinate frame.

Theorem 9.6.3: Canonical Form of Commuting Vector Fields

Let M be a smooth n-manifold, and let $(V_1, \ldots, V_k)$ be a collection of linearly independent commuting smooth vector fields on an open set $W \subseteq M$, then for each point $p \in W$, there exists a smooth chart $(U, (s^i))$ about p such that $V_i = \partial/\partial s^i$ on U for $i = 1, \ldots, k$.
If $p \in S \subseteq W$ is an embedded submanifold of M with codimension k such that T_pS is complementary to the span of $\{V_1|_p, \ldots, V_k|_p\}$ in T_pM, then the coordinates $(s^1, \ldots, s^n)$ can be chosen so that $S \cap U$ is the slice by $s^1 = \cdots = s^k = 0$.

This gives a way to define the local coordinates by commuting vector fields.

- First start with an $n-k$ dimensional submanifold S that is complementary to the span of the k commuting vector fields at a point p;
- For any point p, take it as origin, and trace along the flows of the k vector fields to get k coordinates; As the vector fields commute, the order of tracing does not matter;

Example: **Constructing Local Frames from Commuting Fields**

Take $\mathbb{R}^2$, define

$$V = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}, \quad W = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}.$$

We have $[V, W] = 0$. So they commute. The flow for V and W are

$$\theta_t(x, y) = (x\cos t - y\sin t, x\sin t + y\cos t), \quad \eta_t(x, y) = (e^t x, e^t y).$$

For the new coordinate (s, t), start at point $(1, 0)$, trace along the flow of V for time s then along the flow of W for time t, we get

$$\Phi(s, t) = \eta_t(\theta_s(1, 0)) = (e^t \cos s, e^t \sin s).$$

So we have

$$(s, t) = (\tan^{-1}(y/x), \log\sqrt{x^2 + y^2}).$$

for a local chart around $(1, 0)$.

9.7 Time-Dependent Vector Fields

In many physical applications, the vector fields involved depend on time explicitly. We shall generalize our discussion to cover these cases.

Definition 9.7.1: Time-Dependent Vector Field

Let M be a smooth manifold. A **time-dependent vector field** on M is a continuous map $V : J \times M \to TM$, where $J \subseteq \mathbb{R}$ is an interval. For each $t \in J$, the map $V_t : M \to TM$ defined by $V_t(p) = V(t, p)$ is a vector field on M.
An integral curve of V is a differentiable curve $\gamma : J_0 \to M$, where $J_0 \subseteq J$ is an interval, such that

$$\gamma'(t) = V(t, \gamma(t)) \quad \forall t \in J_0. \tag{9.8}$$

Note that time-dependent vector fields may not generate flows in the usual sense, since the integral curves starting at the same point at different times may not agree.

Theorem 9.7.1: Fundamental Theorem on Time-Dependent Flows

Let M be a smooth manifold, and let $J \subseteq \mathbb{R}$ be an open interval. Let $V : J \times M \to TM$ be a time-dependent smooth vector field on M. There exists an open subset $\mathcal{E} \subseteq J \times J \times M$ and a smooth map $\psi : \mathcal{E} \to M$ called the **time-dependent flow** of V, such that

- For each $(t_0, p) \in J \times M$, the set $\mathcal{E}^{(t_0,p)} = \{t \in J : (t_0, t, p) \in \mathcal{E}\}$ is an open interval containing t_0, and the smooth curve $\psi^{(t_0,p)} : \mathcal{E}^{(t_0,p)} \to M$ defined by $\psi^{(t_0,p)}(t) = \psi(t, t_0, p)$ is the unique maximal integral curve of V satisfying $\psi^{(t_0,p)}(t_0) = p$.
- If $t_1 \in \mathcal{E}^{(t_0,p)}, q = \psi^{(t_0,p)}(t_1)$, then $\mathcal{E}^{(t_1,q)} = \mathcal{E}^{(t_0,p)}$ and $\psi^{(t_1,q)} = \psi^{(t_0,p)}$.
- For any $(t_1, t_0) \in J \times J$, the set $M_{(t_1,t_0)} = \{p \in M : (t_0, t_1, p) \in \mathcal{E}\}$ is an open subset of M, and the map $\psi_{(t_1,t_0)} : M_{(t_1,t_0)} \to M$ defined by $\psi_{(t_1,t_0)}(p) = \psi(t_1, t_0, p)$ is a diffeomorphism from $M_{(t_1,t_0)}$ onto $M_{(t_0,t_1)}$, with inverse $\psi_{(t_0,t_1)}$.
- If $p \in M_{t_1,t_0}$ and $\psi_{(t_1,t_0)}(p) \in M_{t_2,t_1}$, then $p \in M_{t_2,t_0}$ and
$$\psi_{(t_2,t_0)}(p) = \psi_{(t_2,t_1)}(\psi_{(t_1,t_0)}(p)).$$

We can reduce time-dependent vector fields to time-independent ones by considering an augmented manifold $J \times M$. Consider the vector field $\tilde{V}$ on $J \times M$ defined by

$$\tilde{V}_{(s,p)} = \left(\left.\frac{\partial}{\partial t}\right|_s, V(s, p) \right) \in T_s J \oplus T_p M \cong T_{(s,p)}(J \times M). \tag{9.9}$$

where s is the standard coordinate on J. Let $\tilde{\theta} : \tilde{\mathcal{D}} \to J \times M$ be the flow of $\tilde{V}$. Then we can write

$$\tilde{\theta}(t, (s, p)) = (\alpha(t, (s, p)), \beta(t, (s, p))),$$

Then $\alpha : \tilde{\mathcal{D}} \to J$ and $\beta : \tilde{\mathcal{D}} \to M$ have

$$\begin{aligned} \frac{\partial \alpha}{\partial t}(t, (s, p)) &= 1, & \alpha(0, (s, p)) &= s; \\ \frac{\partial \beta}{\partial t}(t, (s, p)) &= V(\alpha(t, (s, p)), \beta(t, (s, p))), & \beta(0, (s, p)) &= p. \end{aligned}$$

So we have $\alpha(t,(s,p)) = t+s$, so

$$\frac{\partial \beta}{\partial t}(t,(s,p)) = V(t+s, \beta(t,(s,p))).$$

Let $\mathcal{E} \subseteq \mathbb{R} \times J \times M$ defined by

$$\mathcal{E} = \{(t,t_0,p) : (t-t_0,(t_0,p)) \in \tilde{\mathcal{D}}\}.$$

If $(t,t_0,p) \in \mathcal{E}$, then $t = t - t_0 + t_0 = \alpha(t-t_0,(t_0,p)) \in J$, so $\mathcal{E} \subseteq J \times J \times M$.

Now $\psi : \mathcal{E} \to M$ defined by

$$\psi(t,t_0,p) = \beta(t-t_0,(t_0,p))$$

also corresponds to the time-dependent flow of V. All the above results can be obtained from the properties of $\tilde{\theta}$.

9.8 First-Order PDEs

In coordinated, any first order PDE for a single unknown function $u : \mathbb{R}^n \to \mathbb{R}$ can be written as

$$F(x^1,\dots,x^n,u(x),\frac{\partial u}{\partial x^1}(x),\dots,\frac{\partial u}{\partial x^n}(x)) = 0, \tag{9.10}$$

where F is a smooth function of $2n+1$ variables. We also add a initial / boundary condition: given a smooth hypersurface $S \subseteq \mathbb{R}^n$ and a smooth function $\varphi : S \to \mathbb{R}$, we require that

$$u|_S = \varphi. \tag{9.11}$$

The problem of finding a solution to the PDE in a neighborhood of S that satisfies the initial condition is called the **Cauchy problem**.

We point out that not all Cauchy problems have solutions, we need to impose some conditions between F and S to ensure the existence of solutions, called some **non-characteristic** conditions.

9.8.1 Linear Equations

$$a^1(x)\frac{\partial u}{\partial x^1} + \cdots + a^n(x)\frac{\partial u}{\partial x^n} + b(x)u(x) = f(x), \tag{9.12}$$

where a^i, b, f are smooth functions on some $\Omega \subseteq \mathbb{R}^n$. Take a smooth vector field $A \in \mathfrak{X}(\Omega)$ defined by

$$A_x = a^i(x)\frac{\partial}{\partial x^i}\bigg|_x.$$

then the PDE can be written as

$$Au + bu = f, \tag{9.13}$$

with initial hypersurface S being non-characteristic if A is nowhere tangent to S at every point of S.

Proposition: **The Linear First-Order Cauchy Problem**

Let M be a smooth n-manifold, let $S \subseteq M$ be an embedded hypersurface, let $A \in \mathfrak{X}(M)$ be a smooth vector field that is nowhere tangent to S, and let $b, f \in C^\infty(M)$, and $\varphi \in C^\infty(S)$. Then for some neighborhood U of S in M, there exists a unique smooth function $u \in C^\infty(U)$ such that

$$Au + bu = f \qquad u|_S = \varphi. \tag{9.14}$$

Chapter 10

Vector Bundles

10.1 Vector Bundles

We have encountered the tangent bundle TM of a smooth manifold M. Locally it looks like a product $U \times \mathbb{R}^n$, but globally it may be twisted in a nontrivial way. This motivates the following definition.

Definition 10.1.1: Vector Bundles

Let M be a topological space. A (real) **vector bundle** of rank k over M is a topological space E together with a surjective continuous map $\pi : E \to M$ such that

- For each $p \in M$, the fiber $E_p := \pi^{-1}(p)$ is equipped with the structure of a real vector space of dimension k.
- For each $p \in M$, there exists an open neighborhood U of p in M and a homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ called a **local trivialization** such that
  - $\pi_U \circ \Phi = \pi$, where $\pi_U : U \times \mathbb{R}^k \to U$ is the projection onto the first factor.
  - For each $q \in U$, the restriction $\Phi|_{E_q} : E_q \to \{q\} \times \mathbb{R}^k$ is a vector space isomorphism.

If M, E are smooth manifolds, with or without boundary, and π is a smooth map, and the local trivializations Φ are diffeomorphisms, then we say that E is a **smooth vector bundle** over M.
The space E is called the **total space** of the vector bundle, M is called the **base space**, and the map π is called the **projection map**.

If there exists a local trivialization of E over all of M, then we say that it is a **global trivialization**, and E is called a **trivial vector bundle**. In this case, E is homeomorphic (or diffeomorphic, in the smooth case) to the product manifold $M \times \mathbb{R}^k$.

Example: **Vector Bundles**

- Product bundles: For any topological space (or smooth manifold) M and integer $k \geqslant 0$, the product space $M \times \mathbb{R}^k$ with the projection map $\pi : M \times \mathbb{R}^k \to M$ defined by

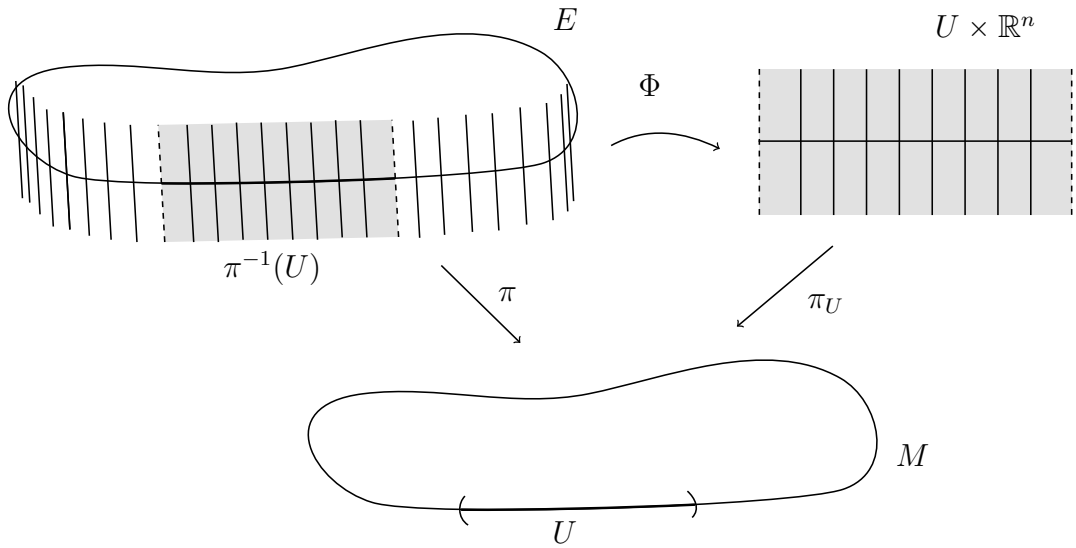

Figure 10.1: Vector Bundles

$\pi(p, v) = p$ is a trivial vector bundle of rank k over M.

- The Möbius Bundle: Define an equivalence relation on $\mathbb{R}^2$ generated by

$$(x, y) \sim (x + 1, -y).$$

Let $E = \mathbb{R}^2/\sim$ be the quotient space, and let $q : \mathbb{R}^2 \to E$ be the quotient map. For any $[-r, r] \subseteq \mathbb{R}$, the image of the strip $\mathbb{R} \times [-r, r]$ under q is called the **Möbius band**. It is a compact manifold with boundary. It is easy to see that $E \to S^1$ is a smooth vector bundle of rank 1 over the circle, called the **Möbius bundle**, which is nontrivial.

Proposition: **Tangent Bundle as Vector Bundle**

Let M be a smooth manifold of dimension n, with or without boundary, and let TM be its tangent bundle. With the projection map $\pi : TM \to M$ defined by $\pi(v) = p$ for $v \in T_pM$, TM is a smooth vector bundle of rank n over M.

Next, we show how two local trivializations of a vector bundle are related on their overlap.

Lemma 10.1.1: Local Trivialization Overlap

Let $\pi : E \to M$ be a smooth vector bundle of rank k over a smooth manifold M. Suppose $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ and $\Psi : \pi^{-1}(V) \to V \times \mathbb{R}^k$ are two local trivializations of E that $U \cap V \neq \varnothing$. Then there exists a smooth map $\tau : U \cap V \to GL(k, \mathbb{R})$ such that the composition $\Phi \circ \Psi^{-1} : (U \cap V) \times \mathbb{R}^k \to (U \cap V) \times \mathbb{R}^k$ is given by

$$\Phi \circ \Psi^{-1}(p, v) = (p, \tau(p)v) \tag{10.1}$$

Here τ is called the **transition function** between the two local trivializations.

This means that the change of coordinates between two local trivializations is a smoothly varying linear isomorphism along the manifold points.

Lemma 10.1.2: Vector Bundle Chart Lemma

Let M be a smooth manifold, with or without boundary, and suppose for each $p \in M$, we have a vector space E_p of dimension k. Let $E = \bigsqcup_{p \in M} E_p$ and $\pi : E \to M$ be the natural projection map $E_p \mapsto p$. Suppose we are given the following data:

- An open cover $\{U_\alpha\}$ of M.
- For each α, a bijection $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ whose restriction to each E_p is a vector space isomorphism onto $\{p\} \times \mathbb{R}^k \cong \mathbb{R}^k$.
- For each α, β that $U_\alpha \cap U_\beta \neq \varnothing$, a smooth map $\tau_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(k, \mathbb{R})$ such that for all $p \in U_\alpha \cap U_\beta$ and $v \in \mathbb{R}^k$,

$$\Phi_\alpha \circ \Phi_\beta^{-1}(p, v) = (p, \tau_{\alpha\beta}(p)v). \tag{10.2}$$

Then E has a unique topology and smooth structure such that it is a smooth manifold with or without boundary, and also a smooth vector bundle of rank k over M with smooth local trivializations $\{U_\alpha, \Phi_\alpha\}$.

Example: **Whitney Sums**

Given a smooth manifold M and two smooth vector bundles $\pi_1 : E_1 \to M$ and $\pi_2 : E_2 \to M$ of ranks k_1 and k_2 respectively, we can construct a new vector bundle called the **Whitney sum** of E_1 and E_2,:

- The total space is $E = E_1 \oplus E_2 = \bigsqcup_{p \in M} (E_1)_p \oplus (E_2)_p$.
- The projection map $\pi : E \to M$ is defined by $(E_1)_p \oplus (E_2)_p \to p$.
- For a neighborhood U of $p \in M$ with local trivializations $\Phi_1 : \pi_1^{-1}(U) \to U \times \mathbb{R}^{k_1}$ and $\Phi_2 : \pi_2^{-1}(U) \to U \times \mathbb{R}^{k_2}$, we define a local trivialization for E by

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^{k_1+k_2}, \quad \Phi(v_1, v_2) = (p, (\pi_{\mathbb{R}^{k_1}} \circ \Phi_1(v_1), \pi_{\mathbb{R}^{k_2}} \circ \Phi_2(v_2))),$$

Example: **Restricting a Vector Bundle**

Suppose $\pi : E \to M$ is a vector bundle of rank k and $S \subseteq M$ is any subset. Then the restriction of E to S is a vector bundle. The total space is $E|_S = \bigsqcup_{p \in S} E_p$, and the projection map is the restriction $\pi|_{E|_S} : E|_S \to S$. If Φ is a local trivialization of E over an open set U in M, then its restriction is $\Phi|_S : \pi^{-1}(U \cap S) \to (U \cap S) \times \mathbb{R}^k$ is a local trivialization of $E|_S$ over $U \cap S$.

If E is a smooth vector bundle and $S \subseteq M$ is an immersed or embedded submanifold, then $E|_S$ is also a smooth vector bundle over S.
For the tangent bundle, the restriction $TM|_S$ is called the **ambient tangent bundle** of S in M. (Note it is NOT the same as the tangent bundle TS of S because it still maintains full dimension of M in each fiber.)

10.2 Local and Global Sections

Similar to previous definitions about sections.

Definition 10.2.1: Sections of Vector Bundles

Let $\pi : E \to M$ be a vector bundle. A (global) section of E is a section of the projection map π, i.e., a continuous map $\sigma : M \to E$ such that $\pi \circ \sigma = \mathrm{id}_M$. For every $p \in M$, we have $\sigma(p) \in E_p$.
A local section of E over an open subset $U \subseteq M$ is a continuous map $\sigma : U \to E$ such that $\pi \circ \sigma = \mathrm{id}_U$.
If M is a smooth manifold, with or without boundary, and E is a smooth vector bundle over M, then a smooth (global or local) section of E is a smooth map σ as above. A not necessarily continuous section is called a **rough section**. The zero section is a global section $\zeta : M \to E$ defined by $\zeta(p) = 0 \in E_p$ for all $p \in M$. The support of a section σ is defined as

$$\mathrm{supp}(\sigma) = \overline{\{p \in M : \sigma(p) \neq 0\}}.$$

Example: **Sections of Vector Bundles**

Suppose M is a smooth manifold, with or without boundary.

- Sections of TM are vector fields on M.
- Given an immersed submanifold $S \subseteq M$, sections of the ambient tangent bundle $TM|_S$ are called vector fields along S. It is different from a vector field on S because the vectors lie in the tangent spaces of M instead of S.
- If $E = M \times \mathbb{R}^k$ is a trivial vector bundle, then sections of E are exactly (one-to-one correspondence) continuous functions from M to $\mathbb{R}^k$:

  $$F : M \to \mathbb{R}^k, \Leftrightarrow \tilde{F} : M \to M \times \mathbb{R}^k, \quad \tilde{F}(p) = (p, F(p)).$$

  So $C^\infty(M)$ can be identified as the space of smooth sections of the trivial line bundle $M \times \mathbb{R}$.

If $E \to M$ is a smooth vector bundle, then the set of all smooth global sections of E is a vector space over $\mathbb{R}$ under pointwise addition and scalar multiplication, denoted by $\Gamma(E)$.

$$(c_1\sigma_1 + c_2\sigma_2)(p) = c_1\sigma_1(p) + c_2\sigma_2(p), \quad \forall p \in M. \tag{10.3}$$

Then we have $\mathfrak{X}(M) = \Gamma(TM)$.

We can also multiply a section by a smooth function. If $f \in C^\infty(M)$ and $\sigma \in \Gamma(E)$, then we define a new section $f\sigma \in \Gamma(E)$ by

$$(f\sigma)(p) = f(p)\sigma(p), \quad \forall p \in M.$$

Lemma 10.2.1: Extension Lemma for Vector Bundles

Let $\pi : E \to M$ be a smooth vector bundle over a smooth manifold M, with or without boundary. If $A \subseteq M$ is closed and $\sigma : A \to E$ is a section of $E|_A$ that is smooth in a neighborhood of each point of A. For each open neighborhood U of A in M, there exists a smooth global section $\tilde{\sigma} \in \Gamma(E)$ such that $\tilde{\sigma}|_A = \sigma$ and $\operatorname{supp}(\tilde{\sigma}) \subseteq U$.

10.2.1 Local and Global Frames

We can view sections as generalizations of vector fields. Similarly, we generalize the concept of frames.

Definition 10.2.2: Frames of Vector Bundles

Let $\pi : E \to M$ be a vector bundle. If $U \subseteq M$ is an open subset, a k-tuple of local sections $(\sigma_1, \ldots, \sigma_k)$ of E over U is linearly independent if for each $p \in U$, the set $\{\sigma_1(p), \ldots, \sigma_k(p)\}$ is linearly independent in the fiber E_p. It is a local frame of E over U if for each $p \in U$, the set $\{\sigma_1(p), \ldots, \sigma_k(p)\}$ is a basis of the fiber E_p. It is called a **global frame** of E if it is defined over all of M.

Proposition: **Completion of Local Frames for Vector Bundles**

Let $\pi : E \to M$ be a smooth vector bundle of rank k.

- If $(\sigma_1, \ldots, \sigma_m)$ is a linearly independent m-tuple of smooth local sections of E over an open subset $U \subseteq M$, with $m < k$, then for each $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and smooth local sections $\sigma_{m+1}, \ldots, \sigma_k$ of E over V such that $(\sigma_1, \ldots, \sigma_k)$ is a local frame of E over V.
- If $(v_1, \ldots, v_m)$ is a linearly independent m-tuple of vectors in the fiber E_p over some point $p \in M$, with $m \leqslant k$, then there exists a smooth local frame $(\sigma_1, \ldots, \sigma_k)$ of E over an open neighborhood U of p such that $\sigma_i(p) = v_i$ for $1 \leqslant i \leqslant m$.
- If $A \subseteq M$ is a closed set and $(\tau_1, \ldots, \tau_k)$ is a linearly independent k-tuple of sections of $E|_A$ that is smooth in a neighborhood of each point of A, then there exists a smooth global frame $(\sigma_1, \ldots, \sigma_k)$ of E over some open neighborhood U of A such that $\sigma_i|_A = \tau_i$ for $1 \leqslant i \leqslant k$.

Intuitively, local frames of a vector bundle are intimately related to local trivializations. This is indeed the case.

Proposition: **Local Frames and Local trivializations**

Let $\pi : E \to M$ be a smooth vector bundle. If $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ is a local trivialization of E over an open subset $U \subseteq M$, then we can construct a local frame $(\sigma_1, \ldots, \sigma_k)$ of E over U by defining

$$\sigma_i(p) = \Phi^{-1}(p, e_i), \quad \forall p \in U,$$

Conversely, if $(\sigma_1, \ldots, \sigma_k)$ is a smooth local frame of E over an open subset $U \subseteq M$, then we can construct a smooth local trivialization $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ of E over U by defining

$$\Phi(v) = (p, (v^1, \ldots, v^k)), \quad \text{where } v = v^i \sigma_i(p) \in E_p,\ p \in U.$$

Therefore, a smooth vector bundle is smoothly trivial if and only if it admits a smooth global frame.

In a local chart (V, φ) of M with coordinates (x^i), and suppose there is a local frame (σ_i) of E over V. The define $\tilde{\varphi} : \pi^{-1}(V) \to \varphi(V) \times \mathbb{R}^k$ by

$$\tilde{\varphi}(v^i \sigma_i(p)) = (x^1(p), \ldots, x^n(p), v^1, \ldots, v^k).$$

Then $(\pi^{-1}(V), \tilde{\varphi})$ is a smooth chart of the total space E.

Proposition: **Local Frame Criterion for Smoothness**

Let $\pi : E \to M$ be a smooth vector bundle, and let $\tau : M \to E$ be a rough section of E. If (σ_i) is a smooth local frame of E over an open subset $U \subseteq M$, then τ is smooth on U if and only if the component functions $\tau^i : U \to \mathbb{R}$ defined by

$$\tau(p) = \tau^i(p)\sigma_i(p), \quad \forall p \in U$$

are smooth.

Proposition: **Uniqueness of Smooth Structure on TM**

Let M be a smooth n-manifold, with or without boundary. The topology and smooth structure on the tangent bundle TM defined before are the unique ones such that $\pi : TM \to M$ is a smooth vector bundle with the given vector space structure on each fiber by derivations, and such that every coordinate vector field on M is a smooth section of TM.

Proof. Suppose there is another topology and smooth structure on TM satisfying the same conditions. If (U, φ) is a smooth chart of M, then the corresponding frame $\partial/\partial x^i$ of TM over U is a smooth local frame over U. So there is a smooth local trivialization $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n$ associated to this local frame. But this is exactly the same as the local trivialization defined before. $\square$

10.3 Bundle Homomorphisms

Definition 10.3.1: Bundle Homomorphisms

Let $\pi : E \to M$ and $\pi' : E' \to M'$ be vector bundles. A continuous map $F : E \to E'$ is a **bundle homomorphism** from E to E' if there exists a map $f : M \to M'$ such that the following diagram commutes:

$$\begin{tikzcd} E \arrow[r, "F"] \arrow[d, "\pi"] & E' \arrow[d, "\pi'"] \\ M \arrow[r, "f"] & M' \end{tikzcd}$$

with the property that for each $p \in M$, the restriction $F|_{E_p} : E_p \to E'_{f(p)}$ is a linear map between vector spaces. We say that F covers f.
We can deduce that f is actually continuous and if the bundles and F are smooth, then f is also smooth.
A bijection bundle homomorphism whose inverse is also a bundle homomorphism is called a **bundle isomorphism**. If there exists a bundle isomorphism between two vector bundles, then they are said to be **isomorphic**.
If $M' = M$ is the same space, we say that F is a bundle homomorphism over M with base map id_M.

Example: **Bundle Homomorphisms**

- If $F : M \to N$ is a smooth map between smooth manifolds, then its differential $dF : TM \to TN$ is a smooth bundle homomorphism covering F.
- If $\pi : E \to M$ is a smooth vector bundle, and $S \subseteq M$ is an immersed submanifold, with or without boundary, then the inclusion map $i : E|_S \to E$ is a smooth bundle homomorphism covering the inclusion map of S into M.

Definition 10.3.2: Linear over $C^\infty(M)$

Let $\pi : E \to M$ and $\pi' : E' \to M$ be smooth vector bundles over the same smooth manifold M, with or without boundary. A map $\mathcal{F} : \Gamma(E) \to \Gamma(E')$ is said to be **linear over** $C^\infty(M)$ if for all $f, g \in C^\infty(M)$ and $\sigma, \tau \in \Gamma(E)$, we have

$$\mathcal{F}(f\sigma + g\tau) = f\mathcal{F}(\sigma) + g\mathcal{F}(\tau).$$

Now, if $E \to M$ and $E' \to M$ are smooth vector bundles over a smooth manifold M, with or without boundary, then if $F : E \to E'$ is a smooth bundle homomorphism over M, then F induces a map $\tilde{F} : \Gamma(E) \to \Gamma(E')$ defined by

$$\tilde{F}(\sigma)(p) = F(\sigma(p)), \quad \forall p \in M. \tag{10.4}$$

It is easily verified that $\tilde{F}$ is linear over $C^\infty(M)$.

Lemma 10.3.1: Bundle Homomorphism Characterization Lemma

Let $\pi : E \to M$ and $\pi' : E' \to M$ be smooth vector bundles over the same smooth manifold M, with or without boundary. A map $\mathcal{F} : \Gamma(E) \to \Gamma(E')$ is linear over $C^\infty(M)$ if and only if there exists a unique smooth bundle homomorphism $F : E \to E'$ over M such that $\mathcal{F}(\sigma) = F \circ \sigma$ for all $\sigma \in \Gamma(E)$.

Proof. SORRY □

Example: **Bundle Homomorphism over Manifolds**

- Let M be a smooth manifold, and $f \in C^\infty(M)$. Then $\mathfrak{X}(M) \to \mathfrak{X}(M)$ defined by $X \mapsto fX$ is linear over $C^\infty(M)$, and thus defines a smooth bundle homomorphism $TM \to TM$ over M.
- If Z is a smooth vector field in $\mathbb{R}^3$, then the cross product with Z: $X \mapsto X \times Z$ is linear over $C^\infty(\mathbb{R}^3)$, and thus defines a smooth bundle homomorphism $T\mathbb{R}^3 \to T\mathbb{R}^3$ over $\mathbb{R}^3$.
- If Z is a smooth vector field on $\mathbb{R}^n$, then the Euclidean inner product with Z: $X \mapsto \langle X, Z\rangle$ is linear over $C^\infty(\mathbb{R}^n)$, and thus defines a smooth bundle homomorphism $T\mathbb{R}^n \to \mathbb{R} \times \mathbb{R}^n$ over $\mathbb{R}^n$.

Note that many maps the involve differentials are NOT bundle homomorphisms because they do not satisfy the right linearity conditions. For example, the Lie bracket $[X, Y]$ of two vector fields is not linear over $C^\infty(M)$ in either argument.

10.4 Subbundles

Definition 10.4.1: Subbundles

Given a vector bundle $\pi_E : E \to M$, a **subbundle** of E is a vector bundle $\pi_D : D \to M$, where D is a topological subspace of E, and π_D is the restriction of π_E to D, such that for each $p \in M$, the fiber $D_p = D \cap E_p$ is a vector subspace of E_p.
If $E \to M$ is a smooth vector bundle, then a **smooth subbundle** of E is a subbundle $D \to M$ such that D is a smooth vector bundle and an embedded submanifold of E, with or without boundary.

The following proposition gives a useful criterion for determining when a union of subspace $D = \bigsqcup_{p \in M} D_p, D_p \subseteq E_p$ is a smooth subbundle of E.

Lemma 10.4.1: Local Frame Criterion for Subbundles

Let $\pi : E \to M$ be a smooth vector bundle, and for each $p \in M$, we are given an m-dimensional linear subspace $D_p \subseteq E_p$. Let $D = \bigsqcup_{p \in M} D_p \subseteq E$, then D is a smooth subbundle of E if and only if:
Each point $p \in M$ has an open neighborhood $U \subseteq M$ that there exists a smooth local sectrions $\sigma_1, \ldots, \sigma_m : U \to E$ of E over U such that for each $q \in U$, the set $\{\sigma_1(q), \ldots, \sigma_m(q)\}$ is a basis of the subspace $D_q \subseteq E_q$.

Example: **Subbundles**

- If M is a smooth manifold, and V is a nowhere vanishing smooth vector field on M, then $D \subseteq TM$ defined by $D_p = \operatorname{span}\{V(p)\}$ is a smooth subbundle of TM of rank 1.
- Suppose $E \to M$ is any trivial bundle, and let $E_1, \ldots, E_k$ be a smooth global frame of E. For any $0 \leqslant m \leqslant k$, the subspace $D_p = \operatorname{span}\{E_1(p), \ldots, E_m(p)\} \subseteq E_p$ defines a smooth subbundle D of E of rank m.
- Suppose M is a smooth manifold, with or without boundary, and $S \subseteq M$ is an immersed k-dimensional submanifold, with or without boundary. Then TS is a smooth subbundle of the ambient tangent bundle $TM|_S$ of rank k.

Definition 10.4.2: Constant Rank Bundle Homomorphisms

Let $\pi : E \to M$ and $\pi' : E' \to M$ be smooth vector bundles over the same smooth manifold M, with or without boundary. A smooth bundle homomorphism $F : E \to E'$ over M is said to have **constant rank** if for each $p \in M$, the linear map $F_p : E_p \to E'_p$ defined by the restriction of F to the fiber E_p has the same rank.

Theorem 10.4.1: Smooth Subbundles from Constant Rank

Let E, E' be smooth vector bundles over the same smooth manifold M, and let $F : E \to E'$ be a smooth bundle homomorphism over M. Define subsets

$$\ker F = \bigsqcup_{p \in M} \ker F_p \subseteq E, \quad \operatorname{image} F = \bigsqcup_{p \in M} \operatorname{image} F_p \subseteq E'. \tag{10.5}$$

Then both $\ker F$ and $\operatorname{image} F$ are smooth subbundles of E and E' respectively if and only if F has constant rank.

Proof. Quite obvious. □

Lemma 10.4.2: Orthogonal Complement Bundles

Let M be an immersed submanifold of $\mathbb{R}^n$, with or without boundary. Let D be a smooth rank-k subbundle of $T\mathbb{R}^n|_M$. For each $p \in M$, define $D_p^\perp \subseteq T_p\mathbb{R}^n$ to be the orthogonal complement of D_p in $T_p\mathbb{R}^n \cong \mathbb{R}^n$ with respect to the standard Euclidean inner product. Then $D^\perp = \bigsqcup_{p \in M} D_p^\perp$ is a smooth subbundle of $T\mathbb{R}^n|_M$ of rank $n - k$.
If we take $D = TM$ to be the tangent bundle of M, then $D^\perp$ is called the **normal bundle** of M in $\mathbb{R}^n$.

10.5 Fiber Bundles

A vector bundle is a special case of a more general concept called a fiber bundle, where the fibers are not necessarily vector spaces.

Definition 10.5.1: Fiber Bundles

A **fiber bundle** is a triple (E, M, π), where E and M are topological spaces (or smooth manifolds, with or without boundary), and $\pi : E \to M$ is a continuous surjection (or smooth map) satisfying the following conditions:

- For each $p \in M$, the fiber $E_p = \pi^{-1}(p)$ is homeomorphic (or diffeomorphic) to a fixed topological space (or smooth manifold) F, called the **fiber**.
- For each $p \in M$, there exists an open neighborhood U of p in M and a homeomorphism (or diffeomorphism) $\Phi : \pi^{-1}(U) \to U \times F$ called a **local trivialization** such that $\pi_U \circ \Phi = \pi$, where $\pi_U : U \times F \to U$ is the projection onto the first factor.

Example: **Fiber Bundles**

- Every product space $M \times F$ with the projection map $\pi : M \times F \to M$ defined by $\pi(p, x) = p$ is a trivial fiber bundle over M with fiber F.
- Every rank-k vector bundle is a fiber bundle with fiber $\mathbb{R}^k$.
- The Möbius strip is a fiber bundle over S^1 with fiber $[-1, 1]$. It is not a trivial fiber bundle.
- Every covering map $\pi : E \to M$ is a fiber bundle with discrete fiber.

Chapter 11

The Cotangent Bundle

11.1 Covectors

As we know from linear algebra, given a finite-dimensional vector space V, we can form its dual space V^*, the space of all linear functionals from V to $\mathbb{R}$. Elements of V^* are called **covectors**. Now let M be a smooth manifold, with or without boundary. At each point $p \in M$, we have the tangent space T_pM, which is a finite-dimensional vector space. We can then consider the dual space of the tangent space at each point, denoted by $T_p^*M = (T_pM)^*$. Elements of T_p^*M are called **cotangent vectors** or **covectors** at the point p.

Given smooth local coordinates $(x^1, x^2, \ldots, x^n)$ in $U \subseteq M$, the coordinate basis $\partial/\partial x^i|_p$ of T_pM induces a dual basis of T_p^*M, for now denoted by $\lambda^i|_p$. Any $\omega \in T_p^*M$ can be expressed uniquely as $\omega = \omega_i \lambda^i|_p$, where

$$\omega_i = \omega\left(\frac{\partial}{\partial x^i}\bigg|_p\right).$$

Now if $(\tilde{x}^1, \tilde{x}^2, \ldots, \tilde{x}^n)$ is another smooth coordinate whose domain $\tilde{U}$ contains p, then we have

$$\frac{\partial}{\partial x^i}\bigg|_p = \frac{\partial \tilde{x}^j}{\partial x^i}(p)\frac{\partial}{\partial \tilde{x}^j}\bigg|_p,$$

and thus we can write $\omega = \omega_i \lambda^i|_p = \tilde{\omega}_j \tilde{\lambda}^j|_p$, where

$$\omega_i = \omega\left(\frac{\partial}{\partial x^i}\bigg|_p\right) = \tilde{\omega}_j \frac{\partial \tilde{x}^j}{\partial x^i}(p).$$

Remark:
In the early days of differential geometry, tangent vectors were often thought of a n-tuple of reals assigned to a point p in a coordinate chart. Note this definition DEPENDS on the choice of the coordinate chart. A real coordinate-free tangent vector can be thought of as an equivalence class of such n-tuples under the transformation law:

$$\tilde{v}^j = \frac{\partial \tilde{x}^j}{\partial x^i}(p) v^i.$$

for any coordinate change. Similarly, a covector can be thought of as an equivalence class of

n-tuples of reals assigned to p under the transformation law:

$$\omega_i = \tilde{\omega}_j \frac{\partial \tilde{x}^j}{\partial x^i}(p).$$

It has been a custom calling tangent covectors **covariant vectors** and tangent vectors **contravariant vectors** because of the way their components transform under coordinate changes. However, this terminology is somewhat outdated and can be confusing, so it is generally avoided in modern texts.

11.1.1 Covector Fields

Definition 11.1.1: Cotangent Bundle

Let M be a smooth manifold, with or without boundary. The **cotangent bundle** of M is the disjoint union of the cotangent spaces at each point in M:

$$T^*M = \bigsqcup_{p \in M} T_p^*M = \{(p, \omega) : p \in M, \omega \in T_p^*M\}.$$

There is a natural projection map $\pi : T^*M \to M$ defined by $\pi(p, \omega) = p$. For any $U \subseteq M$ with local coordinates $(x^1, x^2, \ldots, x^n)$, the coordinate covectors $\lambda^i|_p$ defines n maps $\lambda^i : U \to T^*M$ called coordinate covector fields.

Proposition: **Cotangent Bundle as Vector Bundle**

Let M be a smooth n-dimensional manifold, with or without boundary. Then with the projection and the natural vector space structure on each fiber, the cotangent bundle T^*M can be uniquely made into a smooth vector bundle of rank n over M, for which all coordinate covector fields λ^i are smooth local sections.

Proof. It is quite easy to find local trivializations, just take any coordinate chart $(U, (x^1, x^2, \ldots, x^n))$ on M, and define

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n, \quad \Phi(\xi_i \lambda^i|_p) = (p, (\xi_1, \xi_2, \ldots, \xi_n)).$$

Next follows straight by vector bundle chart lemma 10.1.2. □

This process can be generalized to any vector bundle. Suppose $E \to M$ is a smooth vector bundle over M. Then we can define the dual bundle $E^* \to M$ by taking the dual space of each fiber.

The projection map produce natural coordinate chart for T^*M. For any chart (U, x^i) on M, the map

$$\pi^{-1}(U) \to \mathbb{R}^{2n}, \quad \xi_i \lambda^i|_p \mapsto (x^1(p), x^2(p), \ldots, x^n(p), \xi_1, \xi_2, \ldots, \xi_n)$$

is a smooth local coordinate chart for T^*M. We call (x^i, ξ_i) the **natural coordinates** on T^*M associated to the coordinate chart (U, x^i) on M.

A (local or global) section of T^*M is called a **covector field** or a **differential 1-form**. For any covector field ω, we denote its value at p by $\omega|_p$ or simply ω_p. If ω is a rough covector field

and X is a rough vector field, then we can define a rough real-valued function $\omega(X) : M \to \mathbb{R}$ by

$$\omega(X)(p) = \omega|_p(X|_p).$$

In component form, if we write $\omega = \omega_i \lambda^i$ and $X = X^i \partial/\partial x^i$, then

$$\omega(X) = \omega_i X^i.$$

The set of smooth covector fields on M is denoted by $\mathfrak{X}^*(M)$.

Proposition: **Smoothness Criterion for Covector Fields**

Let M be a smooth manifold, with or without boundary, and let $\omega : M \to T^*M$ be a rough covector field. The following are equivalent:

- ω is smooth.
- For every smooth chart, the component functions of ω are smooth.
- Each point $p \in M$ has a smooth chart $(U, (x^1, x^2, \ldots, x^n))$ such that the component functions of ω with respect to this chart are smooth on U.
- For every smooth vector field X on M, the function $\omega(X) : M \to \mathbb{R}$ is smooth.
- For every open subset $U \subseteq M$ and every smooth vector field X on U, the function $\omega(X) : U \to \mathbb{R}$ is smooth on U.

11.1.2 Coframes

Definition 11.1.2: Coframe

Let M be a smooth manifold, with or without boundary. Take open subset $U \subseteq M$, a **local coframe** is a collection of n covector fields $(\omega^1, \omega^2, \ldots, \omega^n)$ on U such that at each point $p \in U$, the set $(\omega^1|_p, \omega^2|_p, \ldots, \omega^n|_p)$ forms a basis for the cotangent space T_p^*M. If $U = M$, then we call $(\omega^1, \omega^2, \ldots, \omega^n)$ a **global coframe** on M.

Example: **Coframes**

- The coordinate covector fields $(\lambda^1, \lambda^2, \ldots, \lambda^n)$ associated to any smooth chart $(U, (x^1, x^2, \ldots, x^n))$ form a local coframe on U.
- For any local frame $(E_1, E_2, \ldots, E_n)$ for TM over U, there is a unique local coframe $(\epsilon^1, \epsilon^2, \ldots, \epsilon^n)$ for T^*M over U such that $\epsilon^i(E_j) = \delta^i_j$, called the **dual coframe** to $(E_1, E_2, \ldots, E_n)$. Conversely, given any local coframe, there is a unique dual local frame. And E_i is smooth if and only if ϵ^i is smooth.

Given any local coframe (ϵ^i) over U, any covector field ω over U can be uniquely expressed as $\omega = \omega_i \epsilon^i$ for some rough functions $\omega_i : U \to \mathbb{R}$. If the coframe and ω are smooth, then so are the component functions ω_i.

Proposition: **Coframe Criterion for Smoothness**

Let M be a smooth manifold, with or without boundary, and let ω be a rough covector field on an open subset $U \subseteq M$. Let $(\epsilon^1, \epsilon^2, \ldots, \epsilon^n)$ be a local coframe on U, and write $\omega = \omega_i \epsilon^i$ for some rough functions $\omega_i : U \to \mathbb{R}$. Then ω is smooth if and only if each component function ω_i is smooth.

11.2 The Differential of a Smooth Function

In elementary analysis, the gradient of a smooth real-valued function $f : \mathbb{R}^n \to \mathbb{R}$ is defined as the vector field whose components are the partial derivatives of f. As:

$$\operatorname{grad} f = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}. \tag{11.1}$$

Unfortunately, this definition does not generalize to the coordinate free setting here (the index convention is a hint). However, we find that the partial derivatives $\partial f / \partial x^i$ behave like the components of a covector field rather than a vector field.

Definition 11.2.1: Differential of a Smooth Function

Let M be a smooth manifold, with or without boundary, and let $f : M \to \mathbb{R}$ be a smooth function. The **differential** of f is the covector field $\mathrm{d}f \in \mathfrak{X}^*(M)$ defined by

$$\mathrm{d}f_p(v) = v(f) \tag{11.2}$$

for all $p \in M$ and $v \in T_pM$.

Proposition: **Differential is Smooth Covector Field**

The differential of any smooth function $f : M \to \mathbb{R}$ is a smooth covector field on M.

In coordinate representation, if $(U, (x^1, x^2, \ldots, x^n))$ is a smooth chart on M, then we can write $\mathrm{d}f_p = A_i(p)\lambda^i|_p$ for some functions $A_i : U \to \mathbb{R}$. By definition, for any $p \in U$,

$$A_i(p) = \mathrm{d}f_p \left(\frac{\partial}{\partial x^i}\bigg|_p \right) = \frac{\partial}{\partial x^i}\bigg|_p (f) = \frac{\partial f}{\partial x^i}(p).$$

So we have

$$\mathrm{d}f_p = \frac{\partial f}{\partial x^i}(p)\lambda^i|_p, \tag{11.3}$$

If $f = x^j$, then $\mathrm{d}x^j|_p = \lambda^j|_p$. So the coordinate covector fields can be written as differentials of the coordinate functions! This gives

$$\mathrm{d}f_p = \frac{\partial f}{\partial x^i}(p)\mathrm{d}x^i|_p, \qquad \mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i. \tag{11.4}$$

which is just the classical expression for the differential of f in multivariable calculus.

Proposition: **Properties of Differential**

Let M be a smooth manifold, with or without boundary. For any smooth functions $f, g \in C^\infty(M)$ and any real number $a \in \mathbb{R}$, the following hold:

- Linearity: $\mathrm{d}(af + bg) = a\mathrm{d}f + b\mathrm{d}g$.
- Product Rule: $\mathrm{d}(fg) = f\mathrm{d}g + g\mathrm{d}f$.
- Quotient Rule: $\mathrm{d}\left(\frac{f}{g}\right) = \frac{g\mathrm{d}f - f\mathrm{d}g}{g^2}$, provided g is nowhere zero.
- If $J \subseteq \mathbb{R}$ is an open interval containing the image of f, and $h : J \to \mathbb{R}$ is a smooth function, then $\mathrm{d}(h \circ f) = (h' \circ f)\mathrm{d}f$.
- If f is constant, then $\mathrm{d}f = 0$.

Proposition: **Vanishing Differential**

Let M be a smooth manifold, with or without boundary, and let $f : M \to \mathbb{R}$ be a smooth function. Then $\mathrm{d}f = 0$ if and only if f is constant on each connected component of M.

Proof. If M is connected and $\mathrm{d}f = 0$, then let $p \in M$, let $\mathcal{C} = \{q \in M : f(q) = f(p)\}$. For any $q \in \mathcal{C}$, take a smooth coordinate ball (or half-ball if $q \in \partial M$) U centered at q. From elementary calculus, f is constant on U, so $U \subseteq \mathcal{C}$. This shows that $\mathcal{C}$ is open. On the other hand, it is closed by continuity of f. Since M is connected, we must have $\mathcal{C} = M$, so f is constant. □

Remark:
In elementary calculus, the differential $\mathrm{d}f_p$ at a point p is often thought of as the approximation of the change in f for a small change of x^i. Here the intuition is also valid, just take a local coordinate chart, identify the tangent space to $T_p\mathbb{R}^n$ and the same goes for Taylor's theorem. For any tangent vector $v \in T_p\mathbb{R}^n \cong \mathbb{R}^n$, we have

$$\Delta f = f(p + v) - f(p) \approx \frac{\partial f}{\partial x^i}(p)v^i = \mathrm{d}f_p(v).$$

Proposition: **Derivative along a Curve**

Let M be a smooth manifold, with or without boundary, and let $\gamma : J \to M$ be a smooth curve, and let $f : M \to \mathbb{R}$ be a smooth function. Then the derivative of the composition $f \circ \gamma : J \to \mathbb{R}$ is given by

$$(f \circ \gamma)'(t) = \mathrm{d}f_{\gamma(t)}(\gamma'(t))$$

Remark:
Indeed, there are two ways to interpret $(f \circ \gamma)'(t)$. The first is seeing $f \circ \gamma$ as a smooth function from J to $\mathbb{R}$, and taking its derivative at t. The second is seeing it as a smooth

curve in $\mathbb{R}$, and taking the tangent vector at t. One is an element of $\mathbb{R}$, the other is an element of $T_{f(\gamma(t))}\mathbb{R}$. Obviously, they can be identified via the canonical isomorphism.

11.3 Pullback of Covector Fields

A smooth map yields a linear map on tangent spaces called the differential. Dually, it also yields a linear map on cotangent spaces called the pullback.

Definition 11.3.1: Pullback

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. The differential $\mathrm{d}F_p : T_pM \to T_{F(p)}N$ induces a dual map

$$\mathrm{d}F_p^* : T_{F(p)}^*N \to T_p^*M, \qquad \mathrm{d}F_p^*(\omega)(v) = \omega(\mathrm{d}F_p(v)) \tag{11.5}$$

called the pointwise pullback map.

When we discuss pushforwards of vector fields, we have to impose some conditions on F to ensure the existence of the pushforward vector field: Only when F is a diffeomorphism or a Lie group homomorphism can we guarantee the existence of the pushforward vector field. However, for pullbacks of covector fields, no such conditions are needed. Given a smooth map $F : M \to N$ and a covector field ω on N, we can define a rough covector field $F^*\omega$ on M by

$$(F^*\omega)_p = \mathrm{d}F_p^*(\omega_{F(p)}), \qquad (F^*\omega)_p(v) = \omega_{F(p)}(\mathrm{d}F_p(v)) \tag{11.6}$$

It does not have any problem because the pointwise pullback map is defined for all points in M. (This can be seen as a consequence of the "direction" of the smooth map).

Proposition: **Function Multiplication under Pullback**

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. Let u be a continuous real-valued function on N, and let ω be a rough covector field on N. Then

$$F^*(u\omega) = (u \circ F)F^*\omega. \tag{11.7}$$

If u is smooth, then

$$F^*\mathrm{d}u = \mathrm{d}(u \circ F). \tag{11.8}$$

Proof. From computation

$$(F^*(u\omega))_p(v) = (u\omega)_{F(p)}(\mathrm{d}F_p(v)) = u(F(p))\omega_{F(p)}(\mathrm{d}F_p(v)) = ((u \circ F)F^*\omega)_p(v).$$

$$(F^*\mathrm{d}u)_p(v) = \mathrm{d}u_{F(p)}(\mathrm{d}F_p(v)) = \mathrm{d}(u \circ F)_p(v).$$

would do. □

Proposition: **Smoothness of Pullback**

Let M and N be smooth manifolds, with or without boundary, and let $F : M \to N$ be a smooth map. If ω is a smooth covector field on N, then the pullback $F^*\omega$ is a smooth covector field on M.

Proof. Take any $p \in M$, and choose smooth coordinates (y^i) on an open neighborhood V of $F(p)$ in N, and $U = F^{-1}(V)$. Then let $\omega = \omega_j \mathrm{d}y^j$ on V. So we have

$$F^*\omega = F^*(\omega_j \mathrm{d}y^j) = (\omega_j \circ F)F^*\mathrm{d}y^j = (\omega_j \circ F)\mathrm{d}(y^j \circ F) = (\omega_j \circ F)\mathrm{d}F^j,$$

This also gives a simple expression to compute the components of $F^*\omega$ in local coordinates. This is exactly how we do it in multivariable calculus. □

11.3.1 Restricting Covector Fields to Submanifolds

Suppose M is a smooth manifold, with or without boundary, and $S \subseteq M$ is an immersed submanifold with inclusion map $\iota : S \hookrightarrow M$. Given any smooth covector field ω on M, we can define a smooth covector field on S by restricting ω to S via the pullback $\iota^*\omega$. To see it,

$$\iota^*\omega|_p(v) = \omega_{\iota(p)}(\mathrm{d}\iota_p(v)) = \omega_p(v)$$

which is just the restriction of ω to S. However, ω may vanish on S even if it is nonvanishing on M, when ω annihilates all vectors in T_pS for each $p \in S$.

Remark:
Here is something to clarify. We say ω vanishes along S if $\omega_p = 0$ for all $p \in S$ in the context of being a covector field on M. We say the pullback ω to S vanishes if $\iota^*\omega|_p = 0$ for all $p \in S$ in the context of being a covector field on S. The former implies the latter, but not vice versa.

11.4 Line Integrals

Using covector fields, we can define line integrals on smooth manifolds coordinnate-freely.

We begin with $\mathbb{R}$. Let $[a, b]$ be a closed interval, and ω be a smooth covector field on $[a, b]$ (meaning it is smooth on some open neighborhood of $[a, b]$ in $\mathbb{R}$). Let t be the standard coordinate on $\mathbb{R}$, then we can write $\omega = f(t)\mathrm{d}t$ for some smooth function $f : [a, b] \to \mathbb{R}$. We DEFINE the line integral of ω over $[a, b]$ by

$$\int_{[a,b]} \omega = \int_a^b f(t)\mathrm{d}t. \tag{11.9}$$

Well, the left hand side does not seem to depend on the choice of coordinate t. We shall see that this is indeed the case.

Proposition: **Diffeomorphism Invariance of Line Integrals**

Let ω be a smooth covector field on $[a, b] \subseteq \mathbb{R}$, and $\varphi : [c, d] \to [a, b]$ be an increasing

diffeomorphism. Then

$$\int_{[a,b]} \omega = \int_{[c,d]} \varphi^* \omega. \tag{11.10}$$

If φ is decreasing, then

$$\int_{[a,b]} \omega = -\int_{[c,d]} \varphi^* \omega. \tag{11.11}$$

Proof. By computation, we have $(\varphi^*\omega)_s = \omega_{\varphi(s)}(\varphi'(s)\partial/\partial t|_{\varphi(s)}) = f(\varphi(s))\varphi'(s)\mathrm{d}s$. Thus,

$$\int_{[c,d]} \varphi^* \omega = \int_c^d f(\varphi(s))\varphi'(s)\mathrm{d}s = \int_a^b f(t)\mathrm{d}t = \int_{[a,b]} \omega.$$

□

Now let M be a smooth manifold, with or without boundary. A curve segment in M is a continuous curve $\gamma : [a, b] \to M$.

- A smooth curve segment is a smooth map $\gamma : [a, b] \to M$, where $[a, b]$ is regarded as a smooth manifold with boundary.
- A piecewise smooth curve segment if there is a finite partition $a = t_0 < t_1 < \cdots < t_k = b$ such that each restriction $\gamma|_{[t_{i-1},t_i]}$ is a smooth curve segment.

Proposition: **Connecting Points via Curves**

If M is a connected smooth manifold, with or without boundary, then for any two points $p, q \in M$, there exists a piecewise smooth curve segment $\gamma : [a, b] \to M$ such that $\gamma(a) = p$ and $\gamma(b) = q$.

Proof. Take $p \in M$, let $\mathcal{C} = \{q \in M : \text{there is a piecewise smooth curve segment from } p \text{ to } q\}$. For any $q \in \mathcal{C}$, take a smooth coordinate ball (or half-ball if $q \in \partial M$) U centered at q. For any $r \in U$, we can connect q to r via a straight line in the coordinate chart, so there is a piecewise smooth curve segment from p to r. This shows that $\mathcal{C}$ is open. On the other hand, if $q \in \partial\mathcal{C}$, then any smooth coordinate ball (or half-ball) U centered at q must contain some point $r \in \mathcal{C}$. Again, we can connect r to q via a straight line in the coordinate chart, so there is a piecewise smooth curve segment from p to q. This shows that $\mathcal{C}$ is closed. Since M is connected, we must have $\mathcal{C} = M$, so any two points in M can be connected via a piecewise smooth curve segment. □

If $\gamma : [a, b] \to M$ is a smooth curve segment and ω is a smooth covector field on M, we define the line integral of ω along γ by

$$\int_\gamma \omega = \int_{[a,b]} \gamma^* \omega. \tag{11.12}$$

if γ is piecewise smooth with partition $a = t_0 < t_1 < \cdots < t_k = b$, we define

$$\int_\gamma \omega = \sum_{i=1}^k \int_{\gamma|_{[t_{i-1},t_i]}} \omega. \tag{11.13}$$

Proposition: **Properties of Line Integrals**

Let M be a smooth manifold, with or without boundary, and let $\gamma : [a, b] \to M$ be a piecewise smooth curve segment. For any smooth covector fields $\omega, \omega_1, \omega_2$ on M,

- $\forall c_1, c_2 \in \mathbb{R}, \quad \int_\gamma (c_1\omega_1 + c_2\omega_2) = c_1 \int_\gamma \omega_1 + c_2 \int_\gamma \omega_2.$
- If γ is a constant curve, then $\int_\gamma \omega = 0.$
- If $\gamma_1 = \gamma|_{[a,c]}$ and $\gamma_2 = \gamma|_{[c,b]}$ for some $c \in (a, b)$, then $\int_\gamma \omega = \int_{\gamma_1} \omega + \int_{\gamma_2} \omega.$
- If $F : M \to N$ is a smooth map between smooth manifolds, with or without boundary, and η is a smooth covector field on N, then $\int_\gamma F^*\eta = \int_{F\circ\gamma} \eta.$

Line integrals are independent of reparameterization up to sign. If $\gamma : [a, b] \to M$ and $\tilde{\gamma} : [c, d] \to M$ are two piecewise smooth curve segments, we say that they are **reparameterizations** of each other if there is a diffeomorphism $\varphi : [c, d] \to [a, b]$ such that $\tilde{\gamma} = \gamma \circ \varphi$. If φ is increasing, then it is said to be an **forward reparameterization**; if φ is decreasing, then it is said to be a **backward reparameterization**.

Proposition: **Parameter Independence of Line Integrals**

Let M be a smooth manifold, with or without boundary, and let $\gamma : [a, b] \to M$ and $\tilde{\gamma} : [c, d] \to M$ be two piecewise smooth curve segments that are reparameterizations of each other. For any smooth covector field ω on M,

$$\int_{\tilde{\gamma}} \omega = \begin{cases} \int_\gamma \omega, & \text{if } \tilde{\gamma} \text{ is a forward reparameterization of } \gamma; \\ -\int_\gamma \omega, & \text{if } \tilde{\gamma} \text{ is a backward reparameterization of } \gamma. \end{cases}$$

Proof. Using proposition 11.4, we have, for forward reparameterization,

$$\int_{\tilde{\gamma}} \omega = \int_{[c,d]} \tilde{\gamma}^*\omega = \int_{[c,d]} (\gamma \circ \varphi)^*\omega = \int_{[c,d]} \varphi^*(\gamma^*\omega) = \int_{[a,b]} \gamma^*\omega = \int_\gamma \omega,$$

Same goes for backward reparameterization. □

Also, from the definition, we have (using local coordinates):

$$\gamma^*\omega = \omega_{\gamma(t)}(\gamma'(t))\mathrm{d}t,$$

So we can write the line integral in the classical form:

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\gamma'(t))\mathrm{d}t. \tag{11.14}$$

Theorem 11.4.1: Fundamental Theorem of Line Integrals

Let M be a smooth manifold, with or without boundary, and let $\gamma : [a, b] \to M$ be a piecewise smooth curve segment. For any smooth function $f : M \to \mathbb{R}$,

$$\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a)).$$

Proof. Suppose γ is smooth (then the piecewise smooth case follows by additivity). We have

$$\int_\gamma \mathrm{d}f = \int_a^b (\mathrm{d}f)_{\gamma(t)}(\gamma'(t))\mathrm{d}t = \int_a^b (f \circ \gamma)'(t)\mathrm{d}t = f(\gamma(b)) - f(\gamma(a)).$$

□

11.5 Conservative Covector Fields

Definition 11.5.1: Exact, Conservative and Closed Covector Fields

Let M be a smooth manifold, with or without boundary. A smooth covector field ω on M is said to be **exact** if there exists a smooth function $f : M \to \mathbb{R}$ such that $\omega = \mathrm{d}f$. f is called a **potential function** for ω.
We say that a smooth covector field ω on M is **conservative** if for every piecewise smooth closed curve segment γ in M,

$$\int_\gamma \omega = 0.$$

We say that a smooth covector field ω on M is **closed** if for every smooth chart $(U, (x^1, x^2, \ldots, x^n))$ on M, writing $\omega = \omega_i \mathrm{d}x^i$ on U, we have

$$\frac{\partial \omega_i}{\partial x^j} = \frac{\partial \omega_j}{\partial x^i}$$

Theorem 11.5.1: Conservative Field Theorem

Let M be a smooth manifold, with or without boundary. A smooth covector field ω on M is conservative if and only if it is exact.

Proof. SORRY □

As we did in calculus, there is an easy necessary condition for a covector field to be conservative from the exchange of mixed partial derivatives.

Proposition: **Exact Covector Field is Closed**

Every exact (hence every conservative) covector field on a smooth manifold, with or without boundary, is closed.

Proof. Just take a local chart and use the equality of mixed partial derivatives. □

Well, actually, we do not need to check every chart to see if a covector field is closed.

Proposition: **Criterion for Closed Covector Fields**

Let M be a smooth manifold, with or without boundary, and let ω be a smooth covector field on M. Then the following are equivalent:

- ω is closed.
- For every point $p \in M$, there is a smooth chart $(U, (x^1, x^2, \ldots, x^n))$ containing p such that writing $\omega = \omega_i \mathrm{d}x^i$ on U, we have
$$\frac{\partial \omega_i}{\partial x^j} = \frac{\partial \omega_j}{\partial x^i}$$
- For any open subset $U \subseteq M$ and smooth vector fields $X, Y \in \mathfrak{X}(U)$,
$$X(\omega(Y)) - Y(\omega(X)) = \omega([X, Y]). \tag{11.15}$$

We also know pullbacks of local diffeomorphisms preserve these properties, because it is just like an isomorphism.

Corollary 11.5.1: Pullback by Local Diffeomorphisms

Suppose $F : M \to N$ is a local diffeomorphism between smooth manifolds, with or without boundary. Then the pullback $F^* : \mathfrak{X}^*(N) \to \mathfrak{X}^*(M)$ takes exact (conservative) covector fields on N to exact (conservative) covector fields on M. It also takes closed covector fields on N to closed covector fields on M.

Proof. Exactness follows directly from
$$F^* \mathrm{d}u = \mathrm{d}(u \circ F).$$
Closedness follows from local coordinate representation and change of variables. □

The question of whether a closed covector field is exact is of great importance in differential geometry, leading to the development of de Rham cohomology. For now, we just state the result for a specific type of manifold:

A **Star-shaped domain** in $\mathbb{R}^n$ is a subset $U \subseteq \mathbb{R}^n$ such that there exists a point $c \in U$ such that for every point $x \in U$, the line segment connecting c to x lies entirely in U.

Theorem 11.5.2: Poincaré Lemma for Covector Fields

Let U be a star-shaped open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$. Then every closed covector field on U is exact.

Proof. Assume $c = 0$ by some diffeomorphism. For any $x \in U$, define $\gamma_x : [0, 1] \to U$ by $\gamma_x(t) = tx$. Define a function $f : U \to \mathbb{R}$ by
$$f(x) = \int_{\gamma_x} \omega = \int_0^1 \omega_{\gamma_x(t)}(\gamma_x'(t))\mathrm{d}t = \int_0^1 \omega_i(tx)x^i \mathrm{d}t.$$

As the integrand is smooth in x, f is smooth. Now we have

$$\begin{aligned}\frac{\partial f}{\partial x^j}(x) &= \int_0^1 \left(t\frac{\partial \omega_i}{\partial x^j}(tx)x^i + \omega_j(tx)\right) \mathrm{d}t \\ &= \int_0^1 \left(t\frac{\partial \omega_j}{\partial x^i}(tx)x^i + \omega_j(tx)\right) \mathrm{d}t \\ &= \int_0^1 \frac{\partial}{\partial t}\left(t\omega_j(tx)\right) \mathrm{d}t = \omega_j(x).\end{aligned}$$

from the closedness of ω. Thus, $\mathrm{d}f = \omega$. □

Corollary 11.5.2: Local Exactness of Closed Covector Fields

Let M be a smooth manifold, with or without boundary, $andlet\omega$ be a closed covector field on M. Then for every point $p \in M$, there exists an open neighborhood U of p such that the restriction $\omega|_U$ is exact.

Chapter 12

Tensors

We new generalize the application of linear algebra in differential geometry by introducing tensor fields on smooth manifolds. Tensor fields are multilinear maps that take in multiple vector and covector fields and output smooth functions. They play a crucial role in various areas of differential geometry, including the study of Riemannian metrics, differential forms, and curvature.

12.1 Multilinear Algebra

We have already encountered the concept of multilinear maps in the context of vector spaces in linear algebra. A multilinear map is $F : V_1 \times V_2 \times \cdots \times V_k \to W$ that is linear in each argument when the others are held fixed.

Example: **Multilinear Map**

- The dot product on $\mathbb{R}^n$ is a bilinear map.
- The cross product on $\mathbb{R}^3$ is a bilinear map.
- The determinant function on $n \times n$ matrices is a multilinear map.
- The bracket operation in a Lie algebra is a bilinear map.

Example: **Tensor Product of Covectors**

Let V be a vector space and let $\omega, \eta \in V^*$ be covectors. Define a bilinear map $\omega \otimes \eta : V \times V \to \mathbb{R}$ by

$$\omega \otimes \eta(v_1, v_2) = \omega(v_1)\eta(v_2). \tag{12.1}$$

This can also be generalized to define the tensor product of multilinear maps: if $F : V_1 \times \cdots \times V_k \to \mathbb{R}$ and $G : W_1 \times \cdots \times W_m \to \mathbb{R}$ are multilinear maps, then their tensor product $F \otimes G : (V_1 \times \cdots \times V_k) \times (W_1 \times \cdots \times W_m) \to \mathbb{R}$ is defined by $(F \otimes G)(v_1, \ldots, v_k, w_1, \ldots, w_m) = F(v_1, \ldots, v_k)G(w_1, \ldots, w_m)$.
It is easily verified that the tensor product operation acting on multilinear maps is itself multilinear and also associative.

12.1.1 Tensors Products of Vector Spaces

The vector space of multilinear functions $L(V_1, \ldots, V_k; \mathbb{R})$ can be viewed as all linear combinations of objects of the form $\omega_1 \otimes \cdots \otimes \omega_k$ where each $\omega_i \in V_i^*$. This motivates the following contruction of tensor products of vector spaces.

The intuition is clear: given finite-dimensional vector spaces $V_1, \ldots, V_k$, we want to construct a new vector space $V_1 \otimes \cdots \otimes V_k$ which formally consists of all linear combinations of objects of the form $v_1 \otimes \cdots \otimes v_k$ where each $v_i \in V_i$, depending multilinearly on each v_i.

Definition 12.1.1: Formal Linear Combinations

Let S be a set. A **formal linear combination** of elements of S is a function $f : S \to \mathbb{R}$ such that $f(s) \neq 0$ for only finitely many $s \in S$. The set of all formal linear combinations of elements of S is called the **free vector space** on S, denoted by $\mathcal{F}(S)$.
If we identify $x \subseteq S$ with the formal linear combination δ_x, we can say $S \subseteq \mathcal{F}(S)$ and every element of $\mathcal{F}(S)$ can be written as a finite linear combination of elements of S by $f = \sum_{i=1}^m a_i x_i$, where $a_i = f(x_i)$. Therefore, S is a basis for $\mathcal{F}(S)$.

We can directly see this as construction from a set as basis to form a vector space. So $\mathcal{F}(S)$ is finite-dimensional if and only if S is finite. However, usually the elements of S can be transformed to each other by some linear relations, which we need to quotient out.

Also, as S is thought of as a basis, for any vector space W, any function $A : S \to W$ extends uniquely to a linear map $\tilde{A} : \mathcal{F}(S) \to W$ by defining $\tilde{A}(\sum_{i=1}^m a_i x_i) = \sum_{i=1}^m a_i A(x_i)$.

Now, we need to rigorously form the tensor product of vector spaces.

- We start from the intuitive formal linear combinations of elements of the form $v_1 \otimes \cdots \otimes v_k$ where each $v_i \in V_i$. So we consider the free vector space $\mathcal{F}(V_1 \times \cdots \times V_k)$.

- Next, we need to identify elements that are related by multilinearity. We define an equivalence relation. Let $\mathcal{R}$ be the vector space spanned by all elements of the following forms:

  - $(v_1, \ldots, v_i + v_i', \ldots, v_k) - (v_1, \ldots, v_i, \ldots, v_k) - (v_1, \ldots, v_i', \ldots, v_k)$ for all $v_i, v_i' \in V_i$,
  - $(v_1, \ldots, av_i, \ldots, v_k) - a(v_1, \ldots, v_i, \ldots, v_k)$ for all $v_i \in V_i$ and $a \in \mathbb{R}$.

  for all $1 \leqslant i \leqslant k$. By quotienting out $\mathcal{R}$, we identify elements that are related by multilinearity.

Definition 12.1.2: Tensor Product of Vector Spaces

Let $V_1, \ldots, V_k$ be vector spaces. The **tensor product** of $V_1, \ldots, V_k$, denoted by $V_1 \otimes \cdots \otimes V_k$, is defined as the quotient vector space

$$V_1 \otimes \cdots \otimes V_k = \mathcal{F}(V_1 \times \cdots \times V_k)/\mathcal{R}, \tag{12.2}$$

where $\mathcal{R}$ is the subspace defined above.
The natural projection map is $\Pi : \mathcal{F}(V_1 \times \cdots \times V_k) \to V_1 \otimes \cdots \otimes V_k$. Then for $v_i \in V_i$, we define

$$v_1 \otimes \cdots \otimes v_k = \Pi((v_1, \ldots, v_k)). \tag{12.3}$$

called the tensor product of $v_1, \ldots, v_k$.

From the definition we have multilinearity built in:

$$v_1 \otimes \cdots \otimes (v_i + v_i') \otimes \cdots \otimes v_k = v_1 \otimes \cdots \otimes v_i \otimes \cdots \otimes v_k + v_1 \otimes \cdots \otimes v_i' \otimes \cdots \otimes v_k,$$
$$v_1 \otimes \cdots \otimes (av_i) \otimes \cdots \otimes v_k = a(v_1 \otimes \cdots \otimes v_i \otimes \cdots \otimes v_k).$$

Proposition: **Characterization Property of Tensor Product Space**

Let $V_1, \ldots, V_k$ be finite-dimensional vector spaces and let $A : V_1 \times \cdots \times V_k \to X$ be a multilinear map to a vector space X. Then there exists a unique linear map $\tilde{A} : V_1 \otimes \cdots \otimes V_k \to X$ such that the following diagram commutes:

$$\begin{tikzcd} V_1 \times \cdots \times V_k \arrow[r, "A"] \arrow[d, "\pi"'] & X \\ V_1 \otimes \cdots \otimes V_k \arrow[ur, "\tilde{A}"'] & \end{tikzcd}$$

Proof. Note first that every A extends uniquely to a linear map $\overline{A} : \mathcal{F}(V_1 \times \cdots \times V_k) \to X$ by defining $\overline{A}(\sum_{i=1}^m a_i(v_{1,i}, \ldots, v_{k,i})) = \sum_{i=1}^m a_i A(v_{1,i}, \ldots, v_{k,i})$. So the subspace $\mathcal{R}$ is contained in the kernel of $\overline{A}$ by the multilinearity of A. Therefore, by the first isomorphism theorem, there exists a unique linear map $\tilde{A} : V_1 \otimes \cdots \otimes V_k \to X$ such that $\tilde{A} \circ \Pi = \overline{A}$. Since $\pi = \Pi \circ i$ where $i : V_1 \times \cdots \times V_k \to \mathcal{F}(V_1 \times \cdots \times V_k)$ is the inclusion map, so we have $\tilde{A} \circ \pi = \tilde{A} \circ \Pi \circ i = \overline{A} \circ i = A$.

Uniqueness follows from requiring $\tilde{A}(v_1 \otimes \cdots \otimes v_k) = A(v_1, \ldots, v_k)$ for all $v_i \in V_i$ and linearity. □

We can find a basis for the tensor product space easily.

Proposition: **Basis of Tensor Product Space**

Let $V_1, \ldots, V_k$ be finite-dimensional vector spaces with dimensions $n_1, \ldots, n_k$ respectively. Let $\{E_1^{(j)}, \ldots, E_{n_j}^{(j)}\}$ be a basis for V_j for each $1 \leqslant j \leqslant k$. Then the set

$$\mathcal{C} = \{E_{i_1}^{(1)} \otimes E_{i_2}^{(2)} \otimes \cdots \otimes E_{i_k}^{(k)} \mid 1 \leqslant i_j \leqslant n_j, 1 \leqslant j \leqslant k\}$$

is a basis for $V_1 \otimes \cdots \otimes V_k$. So the dimension of $V_1 \otimes \cdots \otimes V_k$ is $n_1 n_2 \cdots n_k$.

Proposition: **Associativity of Tensor Product**

Let V_1, V_2, V_3 be finite-dimensional vector spaces. Then there is a natural isomorphism

$$(V_1 \otimes V_2) \otimes V_3 \cong V_1 \otimes (V_2 \otimes V_3) \cong V_1 \otimes V_2 \otimes V_3.$$

Proposition: **Concrete Tensor Products**

If $V_1, \ldots, V_k$ are finite-dimensional vector spaces, there is a canonical isomorphism

$$V_1^* \otimes \cdots \otimes V_k^* \cong L(V_1, \ldots, V_k; \mathbb{R}) \tag{12.4}$$

under which the tensor product $\omega_1 \otimes \cdots \otimes \omega_k$ defined abstractly corresponds to the multilinear map defined earlier.

Similarly, there is a canonical isomorphism

$$V_1 \otimes \cdots \otimes V_k \cong L(V_1^*, \ldots, V_k^*; \mathbb{R}) \tag{12.5}$$

from the identification $V \cong V^{**}$.

Proof. Define a multilinear map $\Phi : V_1^* \times \cdots \times V_k^* \to L(V_1, \ldots, V_k; \mathbb{R})$ by $\Phi(\omega_1, \ldots, \omega_k)(v_1, \ldots, v_k) = \omega_1(v_1) \cdots \omega_k(v_k)$. By the characterization property of tensor product space, Φ induces a linear map $\tilde{\Phi} : V_1^* \otimes \cdots \otimes V_k^* \to L(V_1, \ldots, V_k; \mathbb{R})$ such that $\tilde{\Phi}(\omega_1 \otimes \cdots \otimes \omega_k) = \Phi(\omega_1, \ldots, \omega_k)$.

Now $\tilde{\Phi}$ is an isomorphism. We can use a basis above to prove it, but itself is canonical. □

12.1.2 Covariant and Contravariant Tensors

Definition 12.1.3: Covariant and Contravariant Tensors

Let V be a finite-dimensional vector space, $k \in \mathbb{N}$.

- A **covariant k-tensor** on V is an element of $(V^*)^{\otimes k} = V^* \otimes \cdots \otimes V^*$ (k times).
- A **contravariant k-tensor** on V is an element of $V^{\otimes k} = V \otimes \cdots \otimes V$ (k times).

We also denote the vector space of all covariant k-tensors on V by $T^k(V^*)$. Similarly, we denote the vector space of all contravariant k-tensors on V by $T^k(V)$.

- A 1-covariant tensor is just a covector, $T^1(V^*) = V^*$.
- A covariant 2-tensor is an element of $V^* \otimes V^*$, which can be thought of as a bilinear form on V.
- The determinant is a covariant n-tensor on $\mathbb{R}^n$.

Remark:
Even if we can identify $V^{\otimes k}$ with $L((V^*)^k; \mathbb{R})$, we usually just think of contravariant tensors as formal objects instead of multilinear maps on covectors.

More generally, we can mix covariant and contravariant tensors.

Definition 12.1.4: Mixed Tensors

Let V be a finite-dimensional vector space, $k, l \in \mathbb{N}$. A **mixed (k, l)-tensor** on V is an element of $V^{\otimes k} \otimes (V^*)^{\otimes l}$. We denote the vector space of all mixed (k, l)-tensors on V by $T^{(k,l)}(V)$.

12.2 Symmetric and Alternating Tensors

In this section, we describe two classes of tensors that change in a simple way when their arguments are rearranged.

12.2.1 Symmetric Tensors

Let V be a finite-dimensional vector space. A covariant k-tensor α is said to be symmetric on V if for all v_i:

$$\alpha(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = \alpha(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k), \qquad \forall 1 \leqslant i < j \leqslant k \tag{12.6}$$

Or equivalently, the value of α is unchanged when $v_1, \ldots, v_k$ is rearranged in any order.

It is easy to verify that the symmetric covariant k-tensors form a subspace of $T^k(V^*)$, denoted by $\Sigma^k(V^*)$. There is a natural projection from $T^k(V^*)$ to $\Sigma^k(V^*)$: Let S_k be the symmetric group, and $\sigma \in S_k$, denote

$$^{\sigma}\alpha(v_1, \ldots, v_k) = \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$

Note that $^{\tau}(^{\sigma}\alpha) = ^{\tau\sigma}\alpha$ from group multiplication.

Now we define the symmetric projection $\operatorname{Sym} : T^k(V^*) \to \Sigma^k(V^*)$ by

$$\operatorname{Sym}\alpha = \frac{1}{k!}\sum_{\sigma \in S_k} {}^{\sigma}\alpha. \tag{12.7}$$

Proposition: **Properties of Symmetrization**

Let α be a covariant k tensor on a finite finite-dimensional vector space, then

- $\operatorname{Sym}\alpha$ is symmetric.
- $\operatorname{Sym}\alpha = \alpha$ if and only if α is symmetric.

Proof. If $\alpha \in T^k(V^*)$, then for any $\tau \in S_k$, we have

$$\begin{aligned}(\operatorname{Sym}\alpha)(v_{\tau(1)}, \ldots, v_{\tau(k)}) &= \frac{1}{k!}\sum_{\sigma \in S_k} {}^{\sigma}\alpha(v_{\tau(1)}, \ldots, v_{\tau(k)}) = \frac{1}{k!}\sum_{\sigma \in S_k} {}^{\tau\sigma}\alpha(v_1, \ldots, v_k) \\ &= (\operatorname{Sym}\alpha)(v_1, \ldots, v_k)\end{aligned}$$

This shows $\operatorname{Sym}\alpha$ is symmetric. The next part follows easily. □

Definition 12.2.1: Symmetric Product

If α, β are symmetric tensors of rank k, l on V, then the **symmetric product** is defined by a $(k+l)$-symmetric tensor:

$$\alpha\beta = \operatorname{Sym}\alpha \otimes \beta \tag{12.8}$$

Proposition: **Properties of Symmetric Product**

- The symmetric product is symmetric and bilinear: for all symmetric tensors α, β, γ, and for all $a, b \in \mathbb{R}$, we have

$$\alpha\beta = \beta\alpha, \qquad (a\alpha + b\beta)\gamma = a\alpha\gamma + b\beta\gamma.$$

- If $\alpha, \beta \in V^*$, we have
$$\alpha\beta = \frac{1}{2}(\alpha \otimes \beta + \beta \otimes \alpha)$$

12.2.2 Alternating Tensors

Similarly, if V is a finite-dimensional vector space, a covariant k-tensor α on V is said to be **alternating** or skew-symmetric if for all vectors $v_1, \ldots, v_k \in V$ we have

$$\alpha(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -\alpha(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k) \tag{12.9}$$

They are also called **exterior forms**. They also construct a subspace denoted by $\Lambda^k(V^*) \subseteq T^k(V^*)$.

For any permutation $\sigma \in S_k$, we have

$$^{\sigma}\alpha(v_1, \ldots, v_k) = \operatorname{sgn}(\sigma)\alpha(v_1, \ldots, v_k)$$

Specially, any 0-tensor is both symmetric and alternating. Any 1-tensor is both symmetric and alternating. Any 2-tensor α can be decomposed into a sum of a symmetric part and an alternating part by

$$\beta(v, w) = \frac{1}{2}(\alpha(v, w) + \alpha(w, v)), \qquad \gamma(v, w) = \frac{1}{2}(\alpha(v, w) - \alpha(w, v))$$

12.3 Tensors and Tensor Fields on Manifolds

Now we can define tensor fields on smooth manifolds. Let M be a smooth manifold, with or without boundary. Define the **bundle of covariant k-tensors** on M to be

$$T^kT^*M = \bigsqcup_{p \in M} T^k(T_p^*M) \tag{12.10}$$

Similar, the bundle of contravariant k-tensors on M is

$$T^kTM = \bigsqcup_{p \in M} T^k(T_pM) \tag{12.11}$$

And the bundle of mixed (k, l)-tensors on M is

$$T^{(k,l)}TM = \bigsqcup_{p \in M} T^{(k,l)}(T_pM) \tag{12.12}$$

They all have natural smooth vector bundle structures by local trivializations. A section of tensor bundle is called a **tensor field**. The space of all smooth tensor fields, $\Gamma(T^kT^*M)$, $\Gamma(T^kTM)$, $\Gamma(T^{(k,l)}TM)$, are all modules over $C^\infty(M)$, and infinite-dimensional vector spaces over $\mathbb{R}$. We usually denote

$$\mathcal{T}^k(M) = \Gamma(T^kT^*M).$$

Proposition: **Smoothness Criterion for Tensor Fields**

Let M be a smooth manifold, with or without boundary. And let $A : M \to T^kT^*M$ be a rough section. The following are equivalent:

- A is smooth.
- In every smooth coordinate chart (U, φ), the component functions of A with respect to the natural basis of T^kT^*M are smooth.
- Each point $p \in M$ has a smooth coordinate chart (U, φ) such that the component functions of A with respect to the natural basis of T^kT^*M are smooth on U.

Proposition: **Operations on Tensor Fields**

Let M be a smooth manifold, with or without boundary. Let $A \in \mathcal{T}^k(M)$, $B \in \mathcal{T}^l(M)$, and $f \in C^\infty(M)$. Then $fA, A \otimes B$ are also smooth tensor fields on M. In any smooth coordinate chart, the component functions are

$$\begin{aligned}(fA)_{i_1\cdots i_k} &= fA_{i_1\cdots i_k},\\ (A\otimes B)_{i_1\cdots i_{k+l}} &= A_{i_1\cdots i_k}B_{i_{k+1}\cdots i_{k+l}}.\end{aligned}$$

If A is a smooth covariant k-tensor field and $X_1, \dots, X_k$ are smooth vector fields on M, then we can define a smooth function $A(X_1, \dots, X_k)$ on M. Then A induces a map

$$\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M) \to C^\infty(M), \qquad (X_1, \dots, X_k) \mapsto A(X_1, \dots, X_k)$$

We can see it is not only multilinear over $\mathbb{R}$, but also multilinear over $C^\infty(M)$ in the sense that

$$A(X_1, \dots, fX_i + gY_i, \dots, X_k) = fA(X_1, \dots, X_i, \dots, X_k) + gA(X_1, \dots, Y_i, \dots, X_k)$$

for all $f, g \in C^\infty(M)$.

Lemma 12.3.1: Tensor Characterization Lemma

Let M be a smooth manifold, with or without boundary. Let

$$\mathcal{A} : \mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M) \to C^\infty(M)$$

be a map. Then it is multilinear over $C^\infty(M)$ if and only if there exists a unique smooth covariant k-tensor field A such that $\mathcal{A}(X_1, \dots, X_k) = A(X_1, \dots, X_k)$ for all $X_i \in \mathfrak{X}(M)$.

12.3.1 Pullbacks of Tensor Fields

Suppose $F : M \to N$ is a smooth map between smooth manifolds, with or without boundary, $p \in M$, and $\alpha \in \mathcal{T}^k(T^*_{F(p)}N)$ is a covariant k-tensor at $F(p)$. We can define the **pullback** of α by F at p to be the covariant k-tensor at p defined by

$$\mathrm{d}F_p^*(\alpha)(v_1, \dots, v_k) = \alpha(\mathrm{d}F_p(v_1), \dots, \mathrm{d}F_p(v_k)) \tag{12.13}$$

Which just follows our intuition of "doing the same operation on the domain side". If A is a covariant k-tensor field on N, we can define the pullback of A by F to be the covariant k-tensor field on M defined by

$$(F^*A)_p = \mathrm{d}F_p^*(A_{F(p)}), \qquad \forall p \in M \tag{12.14}$$

which is a rough section of T^kT^*M.

Proposition: **Properties of Tensor Pullbacks**

Suppose $F : M \to N$ and $G : N \to P$ are smooth maps between smooth manifolds, with or without boundary. Let A, B are covariant k-tensor fields on N, and let $f \in C^\infty(N)$. Then

- $F^*(fB) = (f \circ F)F^*B$.
- $F^*(A \otimes B) = F^*A \otimes F^*B$.
- $F^*(A + B) = F^*A + F^*B$.
- F^*B is a (continuous) tensor field on M. It is smooth B is smooth.
- $(G \circ F)^*A = F^*(G^*A)$.
- $\mathrm{Id}_N^* B = B$.

Well, if we consistently interpret fB by $f \otimes B$, then it is easy to see.

Corollary 12.3.1: Coordinate Expression of Pullback

Let $F : M \to N$ be a smooth map between smooth manifolds, with or without boundary. Let B be a covariant k-tensor field on N. Let $p \in M$ and let (y^i) be a smooth coordinate chart around $F(p)$ on N. Then we have

$$F^*(B_{i_1\cdots i_k}\mathrm{d}y^{i_1} \otimes \cdots \otimes \mathrm{d}y^{i_k}) = (B_{i_1\cdots i_k} \circ F)\mathrm{d}(y^{i_1} \circ F) \otimes \cdots \otimes \mathrm{d}(y^{i_k} \circ F). \tag{12.15}$$

12.3.2 Lie Derivatives of Tensor Fields

Let M be a smooth manifold, and let V be a smooth vector field on M, with θ be the flow of V. We shall assume $\partial M = \varnothing$ here for simplicity. But the same result generalizes easily to the case where V is tangent to the boundary. Let $p \in M$ and t sufficiently small such that θ_t is a diffeomorphism in a neighborhood.

$$\mathrm{d}(\theta_t)_p^*(A_{\theta_t(p)})(v_1, \ldots, v_k) = A_{\theta_t(p)}(\mathrm{d}(\theta_t)_p(v_1), \ldots, \mathrm{d}(\theta_t)_p(v_k))$$

We define the Lie derivative of A with respect to V at p by

$$(\mathcal{L}_V A)_p = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}(\theta_t^* A)_p = \lim_{t\to 0}\frac{\mathrm{d}(\theta_t)_p^*(A_{\theta_t(p)}) - A_p}{t} \in T^k(T_p^*M) \tag{12.16}$$

Lemma 12.3.2: Existence of Lie Derivative

Let M be a smooth manifold, then the derivative above exists for every $p \in M$ and defines a smooth covariant k-tensor field $\mathcal{L}_V A$ on M.

Proposition: **Properties of Lie Derivative on Tensors**

Let M be a smooth manifold, and let $V \in \mathfrak{X}(M)$. Suppose $f \in C^\infty(M)$, regarded as a covariant 0-tensor field, and A, B are smooth covariant tensor fields on M. Then

- $\mathcal{L}_V f = Vf$.
- $\mathcal{L}_V(fA) = (Vf)A + f\mathcal{L}_V A$.
- $\mathcal{L}_V(A \otimes B) = \mathcal{L}_V A \otimes B + A \otimes \mathcal{L}_V B$.
- If $X_1, \ldots, X_k$ are smooth vector fields and A is a smooth covariant k-tensor field, then

$$\mathcal{L}_V(A(X_1, \ldots, X_k)) = (\mathcal{L}_V A)(X_1, \ldots, X_k) + \sum_{i=1}^{k} A(X_1, \ldots, \mathcal{L}_V X_i, \ldots, X_k).$$

Proof. Easy by coordinate representation. □

So we have, just replacing $\mathcal{L}_V X_i$ by $[V, X_i]$,

$$(\mathcal{L}_V A)(X_1, \ldots, X_k) = V(A(X_1, \ldots, X_k)) - \sum_{i=1}^{k} A(X_1, \ldots, [V, X_i], \ldots, X_k)$$

Also, if $f \in C^\infty(M)$, then we have

$$\mathcal{L}_V(\mathrm{d}f) = \mathrm{d}(\mathcal{L}Vf) = \mathrm{d}(Vf) \tag{12.17}$$

Proof. Direct computation, for any $X \in \mathfrak{X}(M)$, we have

$$\begin{aligned}(\mathcal{L}_V(\mathrm{d}f))(X) &= V(\mathrm{d}f(X)) - \mathrm{d}f([V, X]) = V(Xf) - [V, X]f \\ &= V(Xf) - V(Xf) + X(Vf) = X(Vf) = \mathrm{d}(Vf)(X)\end{aligned}$$

□

We say that A is **invariant** under θ if for each t, we have

$$\mathrm{d}(\theta_t)_p^*(A_{\theta_t(p)}) = A_p, \tag{12.18}$$

for all t, p in the domain of θ. If θ is a global flow, then it is equivalent to saying $\theta_t^* A = A$ for all $t \in \mathbb{R}$.

To compute Lie derivative at other points, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=t_0} (\theta_t^* A)_p = (\theta_{t_0}^*(\mathcal{L}_V A))_p \tag{12.19}$$

Theorem 12.3.1: Invariance of Tensor field under Flow

Let M be a smooth manifold, and let $V \in \mathfrak{X}(M)$ with flow θ. A smooth covariant k-tensor field A on M is invariant under θ if and only if $\mathcal{L}_V A = 0$.

Chapter 13

Riemannian Metrics

We introduce geometry to smooth manifold theory now.

13.1 Riemannian Manifolds

An inner product on a vector space V allows us to measure lengths and angles. We use this idea to define a Riemannian metric on a smooth manifold.

Definition 13.1.1: Riemannian Metric

Let M be a smooth manifold, with or without boundary. A **Riemannian metric** on M is a smooth symmetric covariant 2-tensor field on M that is positive definite at each point. A **Riemannian manifold** is a pair (M, g), where M is a smooth manifold, with or without boundary and g is a Riemannian metric on M.

Note that a Riemannian metric is NOT a metric in the sense of metric spaces. We shall call the metric on metric space by distance function here to avoid confusion.

If g is a Riemannian metric on M, then for each $p \in M$, g_p is an inner product on T_pM. We denote $\langle v, w\rangle_g = g_p(v, w)$ for $v, w \in T_pM$.

In any smooth coordinate chart, a Riemannian metric can be expressed in terms of its components: (g is symmetric)

$$g = g_{ij}dx^i \otimes dx^j = g_{ij}dx^i dx^j,$$

Example: **Riemannian Metrics**

- The Euclidean metric $\overline{g}$ on $\mathbb{R}^n$: $\overline{g} = \delta_{ij}dx^i dx^j$. And we have $\overline{g}_p(v, w) = v \cdot w$ for $v, w \in T_p\mathbb{R}^n \cong \mathbb{R}^n$.
- Product Metric: If (M, g) and $(\tilde{M}, \tilde{g})$ are Riemannian manifolds, then the **product metric** on $M \times \tilde{M}$ is the Riemannian metric $\hat{g} = g \oplus \tilde{g}$ defined by

  $$\hat{g}((v, \tilde{v}), (w, \tilde{w})) = g(v, w) + \tilde{g}(\tilde{v}, \tilde{w})$$

  In the coordinate $(x_1, \ldots, x_m, y_1, \ldots, y_n)$ on $M \times \tilde{M}$, we have

  $$(\hat{g}_{ij}) = \begin{pmatrix} (g_{ij}) & 0 \\ 0 & (\tilde{g}_{ij}) \end{pmatrix}$$

Specially, the Euclidean metric on $\mathbb{R}^{m+n} \cong \mathbb{R}^m \times \mathbb{R}^n$ is the product metric of the Euclidean metrics on $\mathbb{R}^m$ and $\mathbb{R}^n$.

Theorem 13.1.1: Existence of Riemannian Metrics

Every smooth manifold with or without boundary admits a Riemannian metric.

Proof. Let M be a smooth manifold, with or without boundary. Choose a covering of M by smooth coordinate charts $\{(U_\alpha, \phi_\alpha)\}$, and in each chart, there is a Riemannian metric $g_\alpha = \phi_\alpha^* \overline{g}$, whose coordinate expression is $\delta_{ij} dx^i dx^j$.

Let $\{\psi_\alpha\}$ be a smooth partition of unity subordinate to $\{U_\alpha\}$. Then define a Riemannian metric g on M by

$$g = \sum_\alpha \psi_\alpha g_\alpha \tag{13.1}$$

From local finiteness, this defines a smooth tensor field on M. Since each g_α is positive definite, g is also positive definite. Hence g is a Riemannian metric on M. □

There are also many ways to define a Riemannian metric on a smooth manifold. So there is nothing canonical about it.

One useful tool to study Riemannian manifolds is orthonormal frames.

Definition 13.1.2: Orthonormal Frame

Let (M, g) be a Riemannian manifold. An **(local) orthonormal frame** on an open subset $U \subseteq M$ is a collection of smooth vector fields $E_1, \ldots, E_n$ on U such that for each $p \in U$, $\{E_1(p), \ldots, E_n(p)\}$ is an orthonormal basis of T_pM with respect to the inner product g_p.

Proposition: **Existence of Orthonormal Frames**

Let (M, g) be a Riemannian manifold, with or without boundary. Let (X_j) be a smooth local frame for M on an open subset $U \subseteq M$. Then there exists a smooth local orthonormal frame (E_j) on U such that for each $p \in U$ and $j = 1, \ldots, n$, we have $\operatorname{span}\{E_1(p), \ldots, E_j(p)\} = \operatorname{span}\{X_1(p), \ldots, X_j(p)\}$.

As all smooth manifolds admit smooth local frames, every Riemannian manifold admits smooth local orthonormal frames.

13.1.1 Pullback Metrics

Suppose M, N are smooth manifolds, with or without boundary, and g is a Riemannian metric on N. If $F : M \to N$ is a smooth map, the pullback F^*g is a smooth 2-tensor field on M. If it is positive definite at each point, then it is a Riemannian metric on M, called the **pullback metric** of g by F.

Proposition: **Pullback Metric Criterion**

Suppose $F : M \to N$ is a smooth map, and g is a Riemannian metric on N. Then the pullback F^*g is a Riemannian metric on M if and only if F is a smooth immersion.

Proof. If F^*g is a Riemannian metric on M, then for each $p \in M$ and $v \in T_pM \backslash \{0\}$, we have

$$(F^*g)_p(v, v) = g_{F(p)}(dF_p(v), dF_p(v)) > 0$$

Hence $dF_p(v) \neq 0$, which implies that dF_p is injective. So F is a smooth immersion. The converse is the same. □

Definition 13.1.3: Riemannian Isometry

If (M, g) and $(\tilde{M}, \tilde{g})$ are Riemannian manifolds, a **Riemannian isometry** from (M, g) to $(\tilde{M}, \tilde{g})$ is a smooth diffeomorphism $F : M \to \tilde{M}$ such that $F^*\tilde{g} = g$.
Generally, F is called a local Riemannian isometry if for each $p \in M$, there exists an open neighborhood U of p such that $F|_U : U \to F(U)$ is a Riemannian isometry from $(U, g|_U)$ to $(F(U), \tilde{g}|_{F(U)})$.
The study of properties of Riemannian manifolds that are invariant under Riemannian isometries is the main goal of Riemannian geometry.

Definition 13.1.4: Flatness

A Riemannian manifold (M, g) is **flat** if it is locally isometric to the Euclidean space $(\mathbb{R}^n, \overline{g})$.

Flatness is a geometric property of a Riemannian manifold, and it is invariant under Riemannian isometries.

Theorem 13.1.2: Equivalent Characteristics for Flatness

For a Riemannian manifold (M, g), the following are equivalent:

- g is flat.
- $\forall p \in M$, there exists a smooth coordinate chart (U, ϕ) around p such that in the coordinate we have $g = \delta_{ij} dx^i dx^j$.
- $\forall p \in M$, there exists a smooth coordinate chart (U, ϕ) around p such that the coordinate frame $(\partial/\partial x^i)$ is an orthonormal frame on U.
- $\forall p \in M$, there exists a smooth local orthonormal frame (E_i) around p such that $[E_i, E_j] = 0$ for all i, j.

Proof. The $4 \to 1$ is by the canonical form theorem of commuting frames 9.6.3, where we can see that if $[E_i, E_j] = 0$ for all i, j, then there exists a smooth coordinate chart (U, ϕ) around p such that $E_i = \partial/\partial x^i$ for all i. Hence the coordinate frame is an orthonormal frame on U, which implies that g is flat. □

13.1.2 Riemannian Submanifolds

Pullback metrics are useful to construct Riemannian metrics on submanifolds. If (M, g) is a Riemannian manifold, with or without boundary, then every immersed or embedded submanifold $S \subseteq M$, with or without boundary, inherits a Riemannian metric ι^*g from M, where $\iota : S \to M$ is

the inclusion map. We call $\iota^* g$ the **induced metric** on S. We have for all $v, w \in T_pS$,

$$(\iota^* g)_p(v, w) = g_p(\mathrm{d}\iota_p(v), \mathrm{d}\iota_p(w)) = g_p(v, w)$$

Thus $\iota^* g$ is the restriction of g to S. With this metric structure, S is a Riemannian manifold, called a **Riemannian submanifold** of M.

Example: **Round Metric**

The metric $\overset{\circ}{g} = \iota^* \overline{g}$ on the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ is called the **round metric** on S^n.

Usually, it is easy to compute the induced metric on a submanifold by local parametrization. That is, an injective immersion $X : U \to M$ from an open subset $U \subseteq \mathbb{R}^n$ to M such that $X(U)$ is an open subset of S, and whose inverse $X^{-1} : X(U) \to U$ is a smooth coordinate map for S. Since $\iota \circ X = X$, we have the coordinate representation of $\iota^* g$ is $X^*(\iota^* g) = X^* g$.

Example: **Induced Metrics in Graph Coordinates**

Let $U \subseteq \mathbb{R}^n$ be an open subset, and $f : U \to \mathbb{R}$ be a smooth function. Then the graph $S \subseteq \mathbb{R}^{n+1}$ of f is a submanifold of $\mathbb{R}^{n+1}$. The map $X : U \to \mathbb{R}^{n+1}$ defined by $X(u^1, \ldots, u^n) = (u^1, \ldots, u^n, f(u))$ is a smooth global parametrization of S.
The induced metric on S is

$$X^* \overline{g} = X^*((\mathrm{d}x^1)^2 + \cdots + (\mathrm{d}x^{n+1})^2) = (\mathrm{d}u^1)^2 + \cdots + (\mathrm{d}u^n)^2 + (\mathrm{d}f)^2$$

Example: **Induced Metrics of Surface of Revolution**

Let C be an embedded 1-dimensional submanifold of the half plane $\{(r, z) \in \mathbb{R}^2 : r > 0\}$, and S_C be the surface of revolution. Choose any local parametrization $\gamma(t) = (a(t), b(t))$ of C, we have $X(t, \theta) = (a(t)\cos\theta, a(t)\sin\theta, b(t))$ is a local parametrization of S_C. The induced metric on S_C is

$$X^* \overline{g} = (a'(t)^2 + b'(t)^2)\mathrm{d}t^2 + a(t)^2 \mathrm{d}\theta^2$$

Here are some examples:

- The torus is the surface of revolution of a circle. For $(r-2)^2 + z^2 = 1$, we have $a(t) = 2 + \cos t$ and $b(t) = \sin t$. Hence the induced metric on the torus is

$$X^* \overline{g} = \mathrm{d}t^2 + (2 + \cos t)^2 \mathrm{d}\theta^2$$

- The punctured unit sphere $S^2 \backslash \{(0,0,1), (0,0,-1)\}$ is the surface of revolution of the semicircle $r^2 + z^2 = 1$ with $r > 0$. For $a(t) = \sin t$ and $b(t) = \cos t$, we have the induced metric on $S^2 \backslash \{(0,0,1), (0,0,-1)\}$ is

$$X^* \overline{g} = \mathrm{d}t^2 + \sin^2 t \mathrm{d}\theta^2$$

- The unit cylinder $S^1 \times \mathbb{R}$ is the surface of revolution of the vertical line $r = 1$. For $a(t) = 1$ and $b(t) = t$, we have the induced metric on $S^1 \times \mathbb{R}$ is

$$X^* \overline{g} = \mathrm{d}t^2 + \mathrm{d}\theta^2$$

This is a flat metric, which implies that the unit cylinder is a flat Riemannian manifold.

Proposition: **Flatness Criterion for Surfaces of Revolution**

Let $C \subseteq H = \{(r, z) \in \mathbb{R}^2 : r > 0\}$ be a connected embedded 1-dimensional submanifold, and let S_C be the surface of revolution of C. Then S_C is flat if and only if C is a line segment.

Proof. The difficult part is the "only if" part. If S_C is flat, then let $(a(t), b(t))$ be a local parametrization of C. Assume $a'^2 + b'^2 = 1$, then the induced metric on S_C is $\mathrm{d}t^2 + a(t)^2\mathrm{d}\theta^2$. The local frame

$$E_1 = \frac{\partial}{\partial t}, \quad E_2 = \frac{1}{a(t)}\frac{\partial}{\partial \theta}$$

is orthonormal. Any other orthonormal frame $(\tilde{E}_1, \tilde{E}_2)$ can be written as

$$\tilde{E}_1 = uE_1 + vE_2, \quad \tilde{E}_2 = vE_1 - uE_2$$

where $u^2 + v^2 = 1$ and are smooth functions. Because the metric is flat, we can choose $(\tilde{E}_1, \tilde{E}_2)$ such that $[\tilde{E}_1, \tilde{E}_2] = 0$. Hence a simplification yields

$$uv_t - vu_t = 0, \qquad vu_\theta - uv_\theta = a'.$$

For any point p, there is a neighborhood U of p such that either u or v is nonzero on U. Assume $v \neq 0$ on U, then $(u/v)_t = 0$, which implies that $u = f(\theta)v$ for some smooth function f. Hence $u^2 + v^2 = (f^2 + 1)v^2 = 1$, which implies that u, v are functions of θ only. Hence $a' = vu_\theta - uv_\theta$ is a function of θ only, so it is a constant. Therefore we have our conclusion. □

This implies that the torus and the punctured unit sphere are not flat.

13.1.3 The Normal Bundle

Suppose (M, g) is a n-dimensional Riemannian manifold, with or without boundary, and $S \subseteq M$ is a k-dimensional immersed or embedded submanifold, with or without boundary. For each $p \in S$, we say that $v \in T_pM$ is **normal** to S at p if $g_p(v, w) = 0$ for all $w \in T_pS$. The set of all normal vectors to S at p is a linear subspace of T_pM, called the **normal space** to S at p, denoted by N_pS. The **normal bundle** of S in M is the disjoint union of all normal spaces to S:

$$NS = \bigsqcup_{p \in S} N_pS$$

Proposition: **The Normal Bundle to a Riemannian Submanifold**

Let (M, g) be a Riemannian manifold, with or without boundary, and $S \subseteq M$ be a k-dimensional immersed submanifold, with or without boundary. Then the normal bundle NS is a rank $(n-k)$ subbundle of the tangent bundle $TM|_S$ of M restricted to S. Moreover, for each $p \in S$, there is a smooth orthonormal frame for NS defined on an open neighborhood of p in S.

13.2 The Riemannian Distance Function

A Riemannian metric on a smooth manifold allows us to measure lengths of curves, and hence define a distance function on the manifold.

Definition 13.2.1: Length of a Curve

Let (M, g) be a Riemannian manifold, with or without boundary. If $\gamma : [a, b] \to M$ is a piecewise smooth curve, then the **length** of γ is defined by

$$L_\gamma = \int_a^b |\gamma'(t)|_g \mathrm{d}t. \tag{13.2}$$

Because $|\gamma'(t)|_g = \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))}$ is continuous except at finitely many points, and there left and right limits are defined, the length of γ is well-defined.

Lengths are isometric invariants, obviously. If $F : (M, g) \to (\tilde{M}, \tilde{g})$ is a Riemannian isometry, then for any piecewise smooth curve $\gamma : [a, b] \to M$, we have

$$L_{F\circ\gamma} = \int_a^b |(F \circ \gamma)'(t)|_{\tilde{g}} \mathrm{d}t = \int_a^b |\mathrm{d}F_{\gamma(t)}(\gamma'(t))|_{\tilde{g}} \mathrm{d}t = \int_a^b |\gamma'(t)|_g \mathrm{d}t = L_\gamma$$

Proposition: **Parameter Independence of Length**

Let (M, g) be a Riemannian manifold, with or without boundary. If $\gamma : [a, b] \to M$ is a piecewise smooth curve, and $\tilde{\gamma} : [c, d] \to M$ is a piecewise smooth curve such that $\tilde{\gamma} = \gamma \circ \phi$ for some diffeomorphism $\phi : [c, d] \to [a, b]$, then $L_{\tilde{\gamma}} = L_\gamma$. Here $\tilde{\gamma}$ is called a **reparametrization** of γ.

Now, suppose (M, g) is a connected Riemannian manifold, with or without boundary. Any two points $p, q \in M$ can be connected by a piecewise smooth curve from proposition 11.4. Hence we can define a distance function on M by

$$d_g(p, q) = \inf\{L_\gamma : \gamma \text{ is a piecewise smooth curve from } p \text{ to } q\} \tag{13.3}$$

Obviously, d_g is symmetric and satisfies the triangle inequality. Moreover, $d_g(p, q) = 0$ if and only if $p = q$. Hence d_g is a metric on M in the sense of metric spaces, called the **Riemannian distance function** on M. Also, the distance function is an isometric invariant.

In $(\mathbb{R}^n, \overline{g})$, the distance function $d_{\overline{g}}$ is the standard Euclidean distance function.

Lemma 13.2.1: Equivalence of Riemannian Metrics on Compact Sets

Let g be a Riemannian metric on an open subset $U \subseteq \mathbb{R}^n$. Then for each compact subset $K \subseteq U$, there exist positive constants c and C such that for all $x \in K$ and $v \in T_xU \cong \mathbb{R}^n$, we have

$$c|v|_{\overline{g}} \leqslant |v|_g \leqslant C|v|_{\overline{g}} \tag{13.4}$$

Proof. Let $L \subseteq T\mathbb{R}^n$ by

$$L = \{(x, v) \in T\mathbb{R}^n : x \in K, |v|_{\overline{g}} = 1\}.$$

Under the identification $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$, $L = K \times S^{n-1}$, which is compact. Because $|v|_g$ is a continuous function on L and is positive, there exists positive constants c and C such that for all $(x, v) \in L$, we have $c \leqslant |v|_g \leqslant C$. Hence for all $x \in K$ and $v \in T_xU \cong \mathbb{R}^n$, we have

$$|v|_g = |v|_{\overline{g}} \cdot \left| \frac{v}{|v|_{\overline{g}}} \right|_g \geqslant c|v|_{\overline{g}}, \qquad |v|_g = |v|_{\overline{g}} \cdot \left| \frac{v}{|v|_{\overline{g}}} \right|_g \leqslant C|v|_{\overline{g}}$$

□

Theorem 13.2.1: Riemannian Manifolds as Metric Spaces

Let (M, g) be a connected Riemannian manifold, with or without boundary. Then the topology on M induced by the Riemannian distance function d_g is the same as the original topology on M.

Therefore, all the terminology and results about metric spaces can be applied to Riemannian manifolds with the Riemannian distance function.

Corollary 13.2.1: Metrizability of Smooth Manifolds

Every smooth manifold with or without boundary is metrizable.

Proof. Take any Riemannian metric g on M, and let $\{M_i\}$ be the connected components of M. Take $p_i \in M_i$ for each i, and define a distance function on M by

$$d(p, q) = \begin{cases} d_g(p, q), & \text{if } p, q \in M_i \text{ for some } i \\ d_g(p, p_i) + d_g(q, p_j) + 1, & \text{if } p \in M_i, q \in M_j \text{ for some } i \neq j \end{cases}$$

We can see this as building a "bridge" of length 1 between any two different connected components.

If M has nonempty boundary, then just embed M to its double, and note that every subspace of a metrizable space is metrizable. □

13.3 The Tangent-Cotangent Isomorphism

A Riemannian metric on a smooth manifold allows us to identify tangent bundles and cotangent bundles. We define a bundle isomorphism $\hat{g} : TM \to T^*M$ by

$$\hat{g}(v)(w) = g_p(v, w), \qquad \forall p \in M, v, w \in T_pM \tag{13.5}$$

In any smooth coordinate chart (x^i), if $g = g_{ij}dx^i dx^j$, then for smooth vector fields $X = X^i \partial/\partial x^i$ and $Y = Y^i \partial/\partial x^i$, we have

$$\hat{g}(X)(Y) = g(X, Y) = g_{ij}X^iY^j, \qquad \hat{g}(X) = g_{ij}X^i dx^j$$

That is, the matrix representation of $\hat{g}$ with respect to the coordinate frame $(\partial/\partial x^i)$ and the coordinate coframe (dx^i) is (g_{ij}).

It is customary to denote

$$X_j = g_{ij}X^i, \quad \text{so that} \quad X^\flat = \hat{g}(X) = X_j dx^j \tag{13.6}$$

So we say $X^\flat$ lowers the index of X. Similarly, the inverse $\hat{g}^{-1} : T^*M \to TM$ is given in component matrix form by $(g^{ij}) = (g_{ij})^{-1}$, so we have

$$g^{ij} g_{jk} = \delta^i_k$$

For any covector field $\omega = \omega_i dx^i$, we have

$$\omega^\sharp = \hat{g}^{-1}(\omega) = g^{ij}\omega_i \frac{\partial}{\partial x^j} = \omega^j \frac{\partial}{\partial x^j}, \quad \text{where } \omega^j = g^{ij}\omega_i, \tag{13.7}$$

These two isomorphisms are inverses of each other, so we have $X = X^{\flat\sharp}$ and $\omega = \omega^{\sharp\flat}$. They are called the **musical isomorphisms** for obvious reasons.

Now, we can make gradient a vector field. For a smooth function f on M, the **gradient** of f is defined by

$$\operatorname{grad} f = (\mathrm{d}f)^\sharp = \hat{g}^{-1}(\mathrm{d}f) \tag{13.8}$$

For any $X \in \mathfrak{X}(M)$, we have

$$\langle \operatorname{grad} f, X\rangle_g = \hat{g}(\operatorname{grad} f)(X) = \mathrm{d}f(X) = Xf, \qquad \langle \operatorname{grad} f, \cdot\rangle_g = \mathrm{d}f$$

In smooth coordinates, we have

$$\operatorname{grad} f = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}$$

This shows that the gradient of f is smooth, and for the Euclidean metric $\overline{g}$, we have

$$\operatorname{grad} f = \delta^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j} = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}$$

Example: **Polar Coordinate Gradient**

As an example, in the polar coordinate (r, θ) on $\mathbb{R}^2 \backslash \{(0,0)\}$, the Euclidean metric is $\overline{g} = dr^2 + r^2 d\theta^2$. Hence we have its inverse matrix representation

$$\overline{g}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} \end{pmatrix}$$

So for a smooth function f on $\mathbb{R}^2 \backslash \{(0,0)\}$, we have

$$\operatorname{grad} f = \frac{\partial f}{\partial r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial f}{\partial \theta} \frac{\partial}{\partial \theta}$$

13.4 Pseudo-Riemannian Metrics

An important generalization of Riemannian metrics is to relax the positive definiteness condition.

Definition 13.4.1: Pseudo-Riemannian Metric

Let M be a smooth manifold, with or without boundary. A **pseudo-Riemannian metric** on M is a smooth symmetric covariant 2-tensor field on M that is nondegenerate at each point. A **pseudo-Riemannian manifold** is a pair (M, g), where M is a smooth manifold, with or without boundary and g is a pseudo-Riemannian metric on M.

From the Sylvester's law of inertia, every symmetric 2-tensor can be diagonalized at each point to a diagonal matrix with entries $+1$, -1 and 0. And for the nondegenerate condition, there are only $+1$ and -1 entries:

$$(g_{ij}) = \begin{pmatrix} I_r & 0 \\ 0 & -I_s \end{pmatrix}$$

The number r, s are called the **signature** of g, is an isometric invariant of g.

We shall note that the existence of Riemannian metrics on smooth manifolds does NOT carry over to pseudo-Riemannian metrics, for a linear combination of nondegenerate symmetric 2-tensors may be degenerate. Actually, not all smooth manifolds admit pseudo-Riemannian metrics of a given signature.

Remark:
In physics, the most important pseudo-Riemannian metric is the Lorentzian metric, which has signature $(1, n-1)$ or $(n-1, 1)$, which plays a fundamental role in the theory of relativity.

Chapter 14

Differential Forms

14.1 Alternating Tensors

Alternating covariant tensors are also called **exterior forms**, **multicovectors**, or **k-covectors**. They play an important role in differential geometry, especially in the theory of integration on manifolds.

Lemma 14.1.1: Charaterization of Alternating Tensors

Let V be a finite-dimensional vector space and α a covariant k-tensor on V. Then the following are equivalent:

- α is alternating.
- $\alpha(v_1, \ldots, v_k) = 0$ whenever the vectors $v_1, \ldots, v_k$ are linearly dependent.
- $\alpha(v_1, \ldots, v_k) = 0$ whenever $v_i = v_j$ for some $i \neq j$.

As we did for symmetric tensors, we define the **alternation** of a covariant k-tensor α by

$$\mathrm{Alt}(\alpha) = \frac{1}{k!} \sum_{\sigma \in S_k} \mathrm{sgn}(\sigma)({}^{\sigma}\alpha), \tag{14.1}$$

Proposition: **Properties of Alternation**

Let V be a finite-dimensional vector space, and α a covariant k-tensor on V. Then

- $\mathrm{Alt}(\alpha)$ is an alternating tensor.
- $\mathrm{Alt}(\alpha) = \alpha$ if α is alternating.

14.1.1 Elementary Alternating Tensors

Notation 14.1.1: Multi-index

For a positive integer k, and ordered k-tuple $I = (i_1, \dots, i_k)$ of positive integers is called a **multi-index**. If $\sigma \in S_k$, we denite

$$I_\sigma = (i_{\sigma(1)}, \dots, i_{\sigma(k)}).$$

Note that $(I_\sigma)_\tau = I_{\tau\sigma}$ for any $\sigma, \tau \in S_k$.

Now let V be a n-dimensional vector space, and $\{\epsilon^1, \dots, \epsilon^n\}$ be a basis for V^*. We define a collection of alternating covariant k-tensors on V as follows: for each multi-index $I = (i_1, \dots, i_k)$, where $1 \leqslant i_1, \dots i_k \leqslant n$, we define

$$\epsilon^I(v_1, \dots, v_k) = \det \begin{pmatrix} \epsilon^{i_1}(v_1) & \cdots & \epsilon^{i_1}(v_k) \\ \vdots & \ddots & \vdots \\ \epsilon^{i_k}(v_1) & \cdots & \epsilon^{i_k}(v_k) \end{pmatrix} = \det \begin{pmatrix} v_1^{i_1} & \cdots & v_k^{i_1} \\ \vdots & \ddots & \vdots \\ v_1^{i_k} & \cdots & v_k^{i_k} \end{pmatrix}, \tag{14.2}$$

It is obvious that ϵ^I is an alternating covariant k-tensor on V. We call ϵ^I an **elementary alternating tensor** on V.

We extend the Kronecker delta notation to multi-indices as follows. If I, J are multi-indices of length k, we define

$$\delta^I_J = \det \begin{pmatrix} \delta^{i_1}_{j_1} & \cdots & \delta^{i_1}_{j_k} \\ \vdots & \ddots & \vdots \\ \delta^{i_k}_{j_1} & \cdots & \delta^{i_k}_{j_k} \end{pmatrix}. \tag{14.3}$$

Then we have

$$\delta^I_J = \begin{cases} 0 & \text{if } I \text{ and } J \text{ are not permutations of each other, or has repeated indices,} \\ \operatorname{sgn}\sigma & \text{if } I = J_\sigma \text{ for some } \sigma \in S_k, \text{ (or } J = I_\sigma\text{, equivalently)} \end{cases}$$

Lemma 14.1.2: Properties of Elementary Alternating Tensors

Let $(E_i)_{i=1}^n$ be the basis of V, and let $\{\epsilon^i\}_{i=1}^n$ be the dual basis of V^*. Then

- If I has repeated indices, then $\epsilon^I = 0$.
- If $J = I_\sigma$ for some $\sigma \in S_k$, then $\epsilon^I = \operatorname{sgn}\sigma \epsilon^J$.
- $\epsilon^I(E_{j_1}, \dots, E_{j_k}) = \delta^I_J$ for any multi-index $J = (j_1, \dots, j_k)$.

The elementary alternating tensors are a good candidate for a basis of the space of alternating covariant k-tensors $\Lambda^k(V^*)$. Of course, not all of them are linearly independent, some are zero, and some only differs by a permutation so differs by a sign. But the increasing ones do form a basis.

A multi-index $I = (i_1, \dots, i_k)$ is called **increasing** if $i_1 < i_2 < \cdots < i_k$. We denote the increasing sum:

$$\sum_I{}' \alpha_I \epsilon^I = \sum_{I:\text{increasing}} \alpha_I \epsilon^I. \tag{14.4}$$

Proposition: **A Basis for $\Lambda^k(V^*)$**

Let V be an n-dimensional vector space, and $\{\epsilon^1, \dots, \epsilon^n\}$ be a basis for V^*. Then for every $k \leqslant n$, the collection

$$\mathcal{E} = \{\epsilon^I : I \text{ is an increasing multi-index of length } k\} \tag{14.5}$$

is a basis for $\Lambda^k(V^*)$. Therefore, we have

$$\dim \Lambda^k(V^*) = \binom{n}{k}. \tag{14.6}$$

If $k > n$, then $\dim \Lambda^k(V^*) = 0$.

Proof. To shoe $\mathcal{E}$ spans $\Lambda^k(V^*)$, let $\alpha \in \Lambda^k(V^*)$ be arbitrary. Define

$$\alpha_I = \alpha(E_{i_1}, \dots, E_{i_k})$$

As α is alternating, we have $\alpha_I = 0$ if I has repeated indices, and $\alpha_J = \operatorname{sgn} \sigma \alpha_I$ if $J = I_\sigma$ for some $\sigma \in S_k$. Therefore, we have

$$\sum_I{}' \alpha_I \epsilon^I(E_{j_1}, \dots, E_{j_k}) = \sum_I{}' \alpha_I \delta^I_J = \alpha_J.$$

So we have our result. □

Specially, for $\mathbb{R}^n$ with the standard basis, $\epsilon^{1\cdots n}$ is the determinant function. This implies that for any $\omega \in \Lambda^n(\mathbb{R}^n)^*$, we have

$$\omega(Tv_1, \dots, Tv_n) = (\det T)\omega(v_1, \dots, v_n)$$

for any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ and any $v_1, \dots, v_n \in \mathbb{R}^n$. In particular, ω is a scalar multiple of $\epsilon^{1\cdots n}$.

14.1.2 Wedge Product

Similar to the symmetric case, we can define a product on the space of alternating covariant tensors.

Definition 14.1.1: Wedge Product

Let V be a finite-dimensional vector space. For $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^l(V^*)$, we define the **wedge product** of ω and η by

$$\omega \wedge \eta = \frac{(k+l)!}{k!l!} \operatorname{Alt}(\omega \otimes \eta). \tag{14.7}$$

The mysterious coefficient comes from the following lemma. Intuitively, what we do is just removing the "normalizing" factor $\frac{1}{k!}$ in the definition of alternation, and we have $k!$ ways to

permute the first k variables, and $l!$ ways to permute the last l variables, so we need to multiply by $\frac{(k+l)!}{k!l!}$ to get the correct normalizing factor $\frac{1}{(k+l)!}$. To write it more illustratively, we have

$$k!\omega \wedge l!\eta = (k+l)!\,\mathrm{Alt}(\omega \otimes \eta).$$

Lemma 14.1.3: Wedge Product of Elementary Alternating Tensors

Let V be a finite-dimensional vector space, and $\{\epsilon^1, \ldots, \epsilon^n\}$ be a basis for V^*. Then for any multi-indices I, J of length k, l, we have

$$\epsilon^I \wedge \epsilon^J = \frac{(k+l)!}{k!l!}\,\mathrm{Alt}(\epsilon^I \otimes \epsilon^J) = \epsilon^{IJ}, \tag{14.8}$$

where $IJ = (i_1, \ldots, i_k, j_1, \ldots, j_l)$ is the concatenation of I and J.

Proof. By direct verifying

$$\epsilon^I \wedge \epsilon^J(E_{p_1}, \ldots, E_{p_{k+l}}) = \epsilon^{IJ}(E_{p_1}, \ldots, E_{p_{k+l}}).$$

would do. □

Remark:
Geometrically, the wedge product of two alternating tensors ω and η can be thought of as a way to calculate the volume of the parallelepiped spanned by the vectors $v_1, \ldots, v_{k+l}$ in terms of the volumes of the parallelepipeds spanned by the first k vectors and the last l vectors. For example, in $\mathbb{R}^3$, we have $\mathrm{d}x^1 \wedge \mathrm{d}x^2 = \mathrm{d}x^1 \otimes \mathrm{d}x^2 - \mathrm{d}x^2 \otimes \mathrm{d}x^1$, which represents the oriented area element in the x^1x^2-plane. When we evaluate this wedge product on two vectors, it gives us the signed area of the parallelogram spanned by those vectors projected onto the x^1x^2-plane. For an arbitrary alternating 2-tensor

$$\omega = a\mathrm{d}x^1 \wedge \mathrm{d}x^2 + b\mathrm{d}x^2 \wedge \mathrm{d}x^3 + c\mathrm{d}x^3 \wedge \mathrm{d}x^1,$$

It can be interpreted as measuring the oriented area of projections onto the plane with normal vector (a, b, c).

Proposition: **Properties of Wedge Products**

Suppose $\omega, \omega', \eta, \eta', \zeta$ are multicovectors on a finite-dimensional vector space V, then

- Bilinearity: For $a, a' \in \mathbb{R}$, we have

$$(a\omega + a'\omega') \wedge \eta = a(\omega \wedge \eta) + a'(\omega' \wedge \eta),$$
$$\omega \wedge (a\eta + a'\eta') = a(\omega \wedge \eta) + a'(\omega \wedge \eta').$$

- Associativity: We have

$$(\omega \wedge \eta) \wedge \zeta = \omega \wedge (\eta \wedge \zeta).$$

- Anticommutativity: We have for $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^l(V^*)$,

$$\omega \wedge \eta = (-1)^{kl}\eta \wedge \omega.$$

- If (ϵ^i) is any basis for V^*, and $I = (i_1, \dots, i_k)$ is any multi-index, then

$$\epsilon^I = \epsilon^{i_1} \wedge \cdots \wedge \epsilon^{i_k}.$$

- For any covectors $\omega^1, \dots, \omega^k$ and vectors $v_1, \dots, v_k$, we have

$$(\omega^1 \wedge \cdots \wedge \omega^k)(v_1, \dots, v_k) = \det \begin{pmatrix} \omega^1(v_1) & \cdots & \omega^1(v_k) \\ \vdots & \ddots & \vdots \\ \omega^k(v_1) & \cdots & \omega^k(v_k) \end{pmatrix}.$$

A k-covector ω is called **decomposable** if $\omega = \omega^1 \wedge \cdots \wedge \omega^k$ for some covectors $\omega^1, \dots, \omega^k$. Not all k-covectors are decomposable, but all of them are sums of decomposable ones.

The wedge product can be uniquely characterized by the above properties actually.

Proposition: **Characterization of Wedge Product**

Let V be a finite-dimensional vector space, then the wedge product is the unique bilinear, associative, and anticommutative map $\wedge : \Lambda^k(V^*) \times \Lambda^l(V^*) \to \Lambda^{k+l}(V^*)$ such that $\epsilon^I = \epsilon^{i_1} \wedge \cdots \wedge \epsilon^{i_k}$ for any basis (ϵ^i) of V^* and any multi-index $I = (i_1, \dots, i_k)$.

For any n-dimensional vector space V, we define a vector space

$$\Lambda(V^*) = \bigoplus_{k=0}^{n} \Lambda^k(V^*). \tag{14.9}$$

with dimension 2^n. The wedge product turns $\Lambda(V^*)$ into an associative algebra, called the **exterior algebra** or **Grassmann algebra** of V. It is not commutative, but it has a closely related property.

An algebra A is called **graded** if it can be written as a direct sum of vector spaces $A = \bigoplus_{k \in \mathbb{Z}} A^k$ such that $A^k A^l \subseteq A^{k+l}$ for all $k, l \in \mathbb{Z}$. A graded algebra A is called an **anti-commutative graded algebra** if $ab = (-1)^{kl} ba$ for all $a \in A^k$ and $b \in A^l$. The exterior algebra $\Lambda(V^*)$ is an anti-commutative graded algebra with the grading given by $\Lambda(V^*)^k = \Lambda^k(V^*)$.

Some authors define the wedge product without the mysterious coefficient, by

$$\omega \overline{\wedge} \eta = \mathrm{Alt}(\omega \otimes \eta).$$

So we have

$$\epsilon^I \overline{\wedge} \epsilon^J = \frac{k!l!}{(k+l)!} \epsilon^{IJ}, \qquad \omega^1 \overline{\wedge} \cdots \overline{\wedge} \omega^k(v_1, \dots, v_k) = \frac{1}{k!} \det(\omega^i(v_j)).$$

We call our version of wedge product the **determinant convention**, and the other version the **Alt convention**.

14.1.3 Interior Multiplication

We can reduce the degree of an alternating tensor by contracting it with a vector. This operation is called **interior multiplication**. Take $v \in V$ and define $i_v : \Lambda^k(V^*) \to \Lambda^{k-1}(V^*)$ by

$$(i_v \omega)(w_1, \dots, w_{k-1}) = \omega(v, w_1, \dots, w_{k-1}). \tag{14.10}$$

for any $\omega \in \Lambda^k(V^*)$ and $w_1, \ldots, w_{k-1} \in V$. To be consistent, we interpret $i_v\omega = 0$ if $\omega \in \Lambda^0(V^*) = \mathbb{R}$. Another notation is "$v$ into ω", denoted by

$$i_v\omega = v \lrcorner\ \omega \tag{14.11}$$

Proposition: **Properties of Interior Multiplication**

Let V be a finite-dimensional vector space, and $v \in V$. Then

- $i_v \circ i_v = 0$.
- For $\omega \in \Lambda^k(V^*)$ and $\eta \in \Lambda^l(V^*)$, we have

$$i_v(\omega \wedge \eta) = (i_v\omega) \wedge \eta + (-1)^k \omega \wedge (i_v\eta).$$

Proof. First part follows from any alternating tensor vanishes when two of its arguments are the same. For the second part, we only need to verify it for decomposable tensors, generally,

$$v \lrcorner\ (\omega^1 \wedge \cdots \wedge \omega^k) = \sum_{i=1}^{k} (-1)^{i-1} \omega^i(v) \omega^1 \wedge \cdots \wedge \widehat{\omega^i} \wedge \cdots \wedge \omega^k,$$

where $\widehat{\omega^i}$ means we omit ω^i in the wedge product. This can be verified by direct calculating: Take $v_1 = v$ and $v_2, \ldots, v_k$ arbitrary, then equivalently we need to verify

$$(\omega^1 \wedge \cdots \wedge \omega^k)(v_1, \ldots, v_k) = \sum_{i=1}^{k} (-1)^{i-1} \omega^i(v_1)(\omega^1 \wedge \cdots \wedge \widehat{\omega^i} \wedge \cdots \wedge \omega^k)(v_2, \ldots, v_k).$$

The left hand side is the determinant of the matrix $(\omega^i(v_j))$, and the right hand side is the expansion of this determinant along the first column, so they are equal. □

If we use Alt convention for wedge product, the interior multiplication would be defined by

$$\tilde{\iota}_v\omega(w_1, \ldots, w_{k-1}) = k\omega(v, w_1, \ldots, w_{k-1}),$$

The extra factor k is to make proposition 14.1.3 still holds.

14.2 Differential Forms on Manifolds

Consider M a smooth manifold, with or without boundary. Recall T^kT^*M is the vector bundle of covariant k-tensors on M, and denote

$$\Lambda^k T^*M = \bigsqcup_{p \in M} \Lambda^k(T_p^*M) \subseteq T^kT^*M \tag{14.12}$$

be the subbundle of alternating covariant k-tensors on M of rank $\binom{n}{k}$. A section of $\Lambda^k T^*M$ is called a **differential k-form** on M. We denote the vector space of smooth k-forms on M by $\Omega^k(M) = \Gamma(\Lambda^k T^*M)$. The wedge product of alternating tensors induces a wedge product on differential forms, defined pointwisely. If we denote

$$\Omega^*(M) = \bigoplus_{k=0}^{n} \Omega^k(M),$$

then $\Omega^*(M)$ is an anti-commutative graded algebra with the wedge product as multiplication.

In any smooth coordinate chart, a k-form ω can be written as

$$\omega = \sum_I{}' \omega_I \mathrm{d}x^I = \sum_I{}' \omega_I \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k},$$

where ω_I are continuous functions on the coordinate chart. ω is smooth if and only if all ω_I are smooth functions. In this case

$$\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}(\frac{\partial}{\partial x^{j_1}}, \ldots, \frac{\partial}{\partial x^{j_k}}) = \delta^I_J, \qquad \omega_I = \omega(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}).$$

If $F : M \to N$ is a smooth map between smooth manifolds, with or without boundary, and ω is a k-form on N, the pullback $F^*\omega$ is a k-form on M, defined by the usual

$$(F^*\omega)_p(v_1, \ldots, v_k) = \omega_{F(p)}(\mathrm{d}F_p(v_1), \ldots, \mathrm{d}F_p(v_k))$$

Lemma 14.2.1: Properties of Pullback of Forms

Suppose $F : M \to N$ is a smooth map between smooth manifolds, with or without boundary, and ω, η are smooth forms on N, then

- $F^* : \Omega^k(N) \to \Omega^k(M)$ is linear over $\mathbb{R}$.
- $F^*(\omega \wedge \eta) = (F^*\omega) \wedge (F^*\eta)$.
- In any smooth coordinate chart

$$F^*\left(\sum_I{}' \omega_I \mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_k}\right) = \sum_I{}' (\omega_I \circ F)\mathrm{d}(y^{i_1} \circ F) \wedge \cdots \wedge \mathrm{d}(y^{i_k} \circ F).$$

Proposition: **Pullback of Top Degree Forms**

Let $F : M \to N$ be a smooth map between smooth n-dimensional manifolds, with or without boundary, and (x^i), (y^i) be smooth coordinate charts on open subsets $U \subseteq M$ and $V \subseteq N$ respectively, and u be a continuous function on V, then on $F^{-1}(V) \cap U$ we have

$$F^*(u\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n) = (u \circ F)(\det DF)\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n, \tag{14.13}$$

where DF is the Jacobian matrix of F in the coordinates (x^i) and (y^i).

Proof. Well, we have

$$F^*(u\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n) = (u \circ F)\mathrm{d}(y^1 \circ F) \wedge \cdots \wedge \mathrm{d}(y^n \circ F).$$

And

$$\mathrm{d}F^1 \wedge \cdots \wedge \mathrm{d}F^n(\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^n}) = \det\left(\mathrm{d}F^j(\frac{\partial}{\partial x^i})\right) = \det DF,$$

So we have our result. □

The pullback looks exactly like the change of variable formula in multivariable calculus, and it is indeed a generalization of it, as we will see later.

This also gives us a simpler formula of changing the basis:

Proposition: **Change of Basis for Top Degree Forms**

If $(U, (x^i))$ and $(\tilde{U}, (\tilde{x}^i))$ are two smooth coordinate charts on a smooth n-dimensional manifold M, with or without boundary, then on $U \cap \tilde{U}$ we have

$$\mathrm{d}\tilde{x}^1 \wedge \cdots \wedge \mathrm{d}\tilde{x}^n = \det\left(\frac{\partial \tilde{x}^j}{\partial x^i}\right) \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n. \tag{14.14}$$

Proof. Just take F to be the identity map on M in the previous proposition, then we have

$$\mathrm{d}\tilde{x}^1 \wedge \cdots \wedge \mathrm{d}\tilde{x}^n = F^*(\mathrm{d}\tilde{x}^1 \wedge \cdots \wedge \mathrm{d}\tilde{x}^n) = \det\left(\frac{\partial \tilde{x}^j}{\partial x^i}\right) \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

□

Interior multiplication can also be defined for differential forms, if $X \in \mathfrak{X}(M)$ is a vector field and $\omega \in \Omega^k(M)$ is a k-form, we define a $(k-1)$-form $i_X\omega$ by

$$(i_X\omega)_p(v_1, \ldots, v_{k-1}) = \omega_p(X_p, v_1, \ldots, v_{k-1}), \qquad (X \lrcorner\, \omega)_p = X_p \lrcorner\, \omega_p.$$

Proposition: **Properties of Interior Multiplication for Forms**

Let M be a smooth manifold, with or without boundary, and $X \in \mathfrak{X}(M)$ be a vector field on M.

- If ω is a smooth k-form on M, then $i_X\omega$ is a smooth $(k-1)$-form on M.
- $i_X : \Omega^k(M) \to \Omega^{k-1}(M)$ is linear over $C^\infty(M)$, and corresponds to a smooth bundle homomorphism $\Lambda^k T^*M \to \Lambda^{k-1}T^*M$.

14.3 Exterior Derivatives

Exterior derivatives are a way to differentiate differential forms, generalizing the usual differentiation of functions.

To give some motivation, recall that given a smooth 1-form ω, a necessary condition for ω to be the differential of a smooth function (0-form) is that ω is closed, i.e. for any smooth coordinate chart we have

$$\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} = 0.$$

Since this is a coordinate-independent condition, we may think that this has some intrinsic meaning. We may found that it is antisymmetric in index, so can be thought of as a 2-form.

Locally in a coordinate chart, we define a 2-form $\mathrm{d}\omega$ by

$$\mathrm{d}\omega = \sum_{i<j} \left(\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j}\right) \mathrm{d}x^i \wedge \mathrm{d}x^j.$$

Then ω is closed if and only if $\mathrm{d}\omega = 0$ in each chart.

Actually, it turns out that $\mathrm{d}\omega$ is actually well-defined globally, and has a very significant generalization to arbitrary k-forms. We can see that there is a differential operator $\mathrm{d} : \Omega^k(M) \to \Omega^{k+1}(M)$, called the **exterior derivative**, such that $\mathrm{d}(\mathrm{d}\omega) = 0$ for any $\omega \in \Omega^k(M)$. In Euclidean space, the definition of $\mathrm{d}\omega$ is straightforward: If $\omega = \sum_J{}' \omega_J \mathrm{d}x^J$ is a k-form on an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, then we define

$$\mathrm{d}\omega = \mathrm{d}\left(\sum_J{}' \omega_J \mathrm{d}x^J\right) = \sum_J{}' \mathrm{d}\omega_J \wedge \mathrm{d}x^J = \sum_J{}' \sum_{i=1}^n \frac{\partial \omega_J}{\partial x^i} \mathrm{d}x^i \wedge \mathrm{d}x^J,$$

where $\mathrm{d}\omega_J$ is the usual differential of the function ω_J. Then when ω is a 1-form, we have

$$\mathrm{d}(\omega_j \mathrm{d}x^j) = \sum_j{}' \sum_{i=1}^n \frac{\partial \omega_j}{\partial x^i} \mathrm{d}x^i \wedge \mathrm{d}x^j = \sum_{i,j} \frac{\partial \omega_j}{\partial x^i} \mathrm{d}x^i \wedge \mathrm{d}x^j = \sum_{i<j} \left(\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j}\right) \mathrm{d}x^i \wedge \mathrm{d}x^j.$$

For a 0-form, we have

$$\mathrm{d}f = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \mathrm{d}x^i,$$

So it is consistent with the usual definition of differential of a function.

Now, we need to transfer this to smooth manifolds.

Proposition: **Properties of Exterior Derivative on $\mathbb{R}^n$**

- d is linear over $\mathbb{R}$.
- If ω is a smooth k-form and η is a smooth l-form, then
$$\mathrm{d}(\omega \wedge \eta) = \mathrm{d}\omega \wedge \eta + (-1)^k \omega \wedge \mathrm{d}\eta.$$
- $\mathrm{d} \circ \mathrm{d} = 0$.
- d commutes with pullbacks: If U is an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, and V is an open subset of $\mathbb{R}^m$ or $\mathbb{H}^m$, and $F : U \to V$ is a smooth map, and $\omega \in \Omega^k(V)$ is a smooth k-form on V, then
$$F^*(\mathrm{d}\omega) = \mathrm{d}(F^*\omega).$$

Proof.

- Linearity is straightforward.
- From linearity, it suffices to assume $\omega = u\mathrm{d}x^I \in \Omega^k(U)$ and $\eta = v\mathrm{d}x^J \in \Omega^l(U)$ are decomposable, then we have
$$\begin{aligned}\mathrm{d}(\omega \wedge \eta) &= \mathrm{d}(uv\mathrm{d}x^I \wedge \mathrm{d}x^J) = (v\mathrm{d}u + u\mathrm{d}v) \wedge \mathrm{d}x^I \wedge \mathrm{d}x^J \\ &= \mathrm{d}u \wedge \mathrm{d}x^I \wedge v\mathrm{d}x^J + u\mathrm{d}x^I \wedge \mathrm{d}v \wedge \mathrm{d}x^J = \mathrm{d}\omega \wedge \eta + (-1)^k \omega \wedge \mathrm{d}\eta.\end{aligned}$$
- If ω is a 0-form, then $\omega = u$ for some smooth function u,
$$\mathrm{d}(\mathrm{d}u) = \mathrm{d}\left(\frac{\partial u}{\partial x^j}\mathrm{d}x^j\right) = \frac{\partial^2 u}{\partial x^i \partial x^j}\mathrm{d}x^i \wedge \mathrm{d}x^j = \sum_{i<j}\left(\frac{\partial^2 u}{\partial x^i \partial x^j} - \frac{\partial^2 u}{\partial x^j \partial x^i}\right)\mathrm{d}x^i \wedge \mathrm{d}x^j = 0.$$

For a general case, we have

$$\begin{aligned}
\mathrm{d}(\mathrm{d}\omega) &= \mathrm{d}\left(\sum_J{}' \mathrm{d}\omega_J \wedge \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_k}\right) \\
&= \sum_J{}' \mathrm{d}(\mathrm{d}\omega_J) \wedge \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_k} \\
&\quad + \sum_J{}' \sum_{i=1}^{k} (-1)^i \mathrm{d}\omega_J \wedge \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}(\mathrm{d}x^{j_i}) \wedge \cdots \wedge \mathrm{d}x^{j_k} = 0.
\end{aligned}$$

- From linearity, it suffices to assume $\omega = u\mathrm{d}y^I$. Then

$$F^*(\mathrm{d}\omega) = \mathrm{d}(u \circ F) \wedge \mathrm{d}(y^{i_1} \circ F) \wedge \cdots \wedge \mathrm{d}(y^{i_k} \circ F).$$

$$\mathrm{d}(F^*\omega) = \mathrm{d}((u \circ F)\mathrm{d}(y^{i_1} \circ F) \wedge \cdots \wedge \mathrm{d}(y^{i_k} \circ F)) = \mathrm{d}(u \circ F) \wedge \mathrm{d}(y^{i_1} \circ F) \wedge \cdots \wedge \mathrm{d}(y^{i_k} \circ F).$$

So we have our result.

□

This proposition leads to the general definition of exterior derivatives on smooth manifolds.

Theorem 14.3.1: Existence and Uniqueness of Exterior Differentiation

Let M be a smooth manifold, with or without boundary. Then there exist unique operators $\mathrm{d} : \Omega^k(M) \to \Omega^{k+1}(M)$ for all $k \geqslant 0$ such that

- d is linear over $\mathbb{R}$.
- If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$, then

$$\mathrm{d}(\omega \wedge \eta) = \mathrm{d}\omega \wedge \eta + (-1)^k \omega \wedge \mathrm{d}\eta.$$

- $\mathrm{d} \circ \mathrm{d} = 0$
- For $f \in C^\infty(M) = \Omega^0(M)$, $\mathrm{d}f$ is the usual differential of f.

In any smooth coordinate chart $(U, (x^i))$, if $\omega = \sum_J{}' \omega_J \mathrm{d}x^J$ is a smooth k-form on U, then

$$\mathrm{d}\omega = \sum_J{}' \mathrm{d}\omega_J \wedge \mathrm{d}x^J \tag{14.15}$$

Proof. First we prove existence. Suppose $\omega \in \Omega^k(M)$, then we wish to DEFINE $\mathrm{d}\omega$ by the coordinate formula. For each smooth coordinate chart (U, φ), we define

$$\mathrm{d}\omega_U = \varphi^* \mathrm{d}((\varphi^{-1})^* \omega)$$

To see this is well-defined, suppose (V, ψ) is another smooth coordinate chart, then on $U \cap V$ we have $\varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$ is a diffeomorphism between open subsets of $\mathbb{R}^n$ or $\mathbb{H}^n$, and we have

$$(\varphi \circ \psi^{-1})^* \mathrm{d}((\varphi^{-1})^* \omega) = \mathrm{d}((\varphi \circ \psi^{-1})^* (\varphi^{-1})^* \omega) = \mathrm{d}(\psi^{-1})^* \omega,$$

Therefore we have

$$\mathrm{d}\omega_U = \varphi^*\mathrm{d}(\varphi^{-1})^*\omega = \psi^*(\varphi\circ\psi^{-1})^*\mathrm{d}((\varphi^{-1})^*\omega) = \psi^*\mathrm{d}(\psi^{-1})^*\omega = \mathrm{d}\omega_V,$$

So $\mathrm{d}\omega$ is well-defined. For uniqueness, suppose d satisfies the properties in the theorem, we shall show that $\mathrm{d}\omega$ is determined locally: if $\omega_1 = \omega_2$ are two smooth k-forms on $U \subseteq M$, then $\mathrm{d}\omega_1 = \mathrm{d}\omega_2$ on U. First let $p \in U$, and $\eta = \omega_1 - \omega_2$, and $\psi \in C^\infty(M)$ be a bump function that is identically 1 on some neighborhood of p and supported in U, then $\psi\eta = 0$, so

$$\mathrm{d}(\psi\eta) = \mathrm{d}\psi \wedge \eta + \psi\mathrm{d}\eta = 0,$$

Evaluating at p, by $\psi(p) = 1$ and $\mathrm{d}\psi_p = 0$, we have $\mathrm{d}\eta_p = 0$. Since p is arbitrary, we have $\mathrm{d}\omega_1 = \mathrm{d}\omega_2$ on U.

Now let $\omega \in \Omega^k(M)$ be any smooth k-form, and (U, φ) be any smooth coordinate chart, we can write $\omega = \sum_I{}'\omega_I \mathrm{d}x^I$ on U, then for any $p \in U$, using a bump function we can constrict smooth global functions $\tilde{\omega}_I$ and $\tilde{x}^i$ that agree with ω_I and x^i on some neighborhood of p, then we have

$$\omega = \tilde{\omega} = \sum_I{}'\tilde{\omega}_I \mathrm{d}\tilde{x}^I$$

at some neighborhood of p, so from the preceding part we have

$$\mathrm{d}\omega = \mathrm{d}\tilde{\omega} = \sum_I{}'\mathrm{d}\tilde{\omega}_I \wedge \mathrm{d}\tilde{x}^I = \sum_I{}'\mathrm{d}\omega_I \wedge \mathrm{d}x^I$$

on some neighborhood of p. So we have our result, since p is arbitrary. □

Remark:
Note that we need the construction $\tilde{\omega}$ because we only require the properties holds for smooth forms defined on the whole manifold, the coordinate formula only holds for a subset of M do we cannot directly use the properties on it. This is also why we first need to prove the locality of $\mathrm{d}\omega$ before handling the coordinate formula.

If $A = \bigoplus_k A^k$ is a graded algebra, a linear map $T : A \to A$ is called a **map of degree** m if $T(A^k) \subseteq A^{k+m}$ for all k. It is said to be a **anti-derivation** if $T(xy) = T(x)y + (-1)^k xT(y)$ for all $x \in A^k$ and $y \in A^l$. The theorem above says that the differential of functions uniquely extends to an anti-derivation of $\Omega^*(M)$ of degree 1 that squares to zero.

Proposition: **Structure of Interior Multiplication**

The interior multiplication $i_X : \Omega^*(M) \to \Omega^*(M)$ is an anti-derivation of degree -1 whose square is zero.

Proposition: **Naturality of Exterior Derivative**

Let $F : M \to N$ be a smooth map between smooth manifolds, with or without boundary,

and ω be a smooth k-form on N, then

$$F^*(\mathrm{d}\omega) = \mathrm{d}(F^*\omega).$$

Proof. Let (U, φ) be a smooth coordinate chart on M, and (V, ψ) be a smooth coordinate chart on N. On $F^{-1}(V) \cap U$, we apply the coordinate formula for the coordinate expression $\psi \circ F \circ \varphi^{-1}$ of F to get

$$\begin{aligned} F^*(\mathrm{d}\omega) &= F^*\psi^*\mathrm{d}((\psi^{-1})^*\omega) = \varphi^* \circ (\psi \circ F \circ \varphi^{-1})^*\mathrm{d}((\psi^{-1})^*\omega) \\ &= \varphi^*\mathrm{d}((\psi \circ F \circ \varphi^{-1})^*(\psi^{-1})^*\omega) = \varphi^*\mathrm{d}((\varphi^{-1})^*F^*\omega) = \mathrm{d}(F^*\omega). \end{aligned}$$

□

Definition 14.3.1: Closed and Exact Forms

Let M be a smooth manifold, with or without boundary, and $\omega \in \Omega^k(M)$ be a smooth k-form on M. We say ω is **closed** if $\mathrm{d}\omega = 0$, and ω is **exact** if there exists a smooth $(k-1)$-form η such that $\omega = \mathrm{d}\eta$.

Then as $\mathrm{d} \circ \mathrm{d} = 0$, we have any exact form is closed. The converse is not true in general, and the obstruction to the converse is measured by the de Rham cohomology, which we will discuss later.

14.3.1 Vector Calculus on $\mathbb{R}^3$

The exterior derivative on $\mathbb{R}^3$ can be used to unify the gradient, curl and divergence operators in vector calculus.

Any smooth 1-form ω on $\mathbb{R}^3$ can be written as

$$\omega = P\mathrm{d}x + Q\mathrm{d}y + R\mathrm{d}z,$$

where P, Q, R are smooth functions on $\mathbb{R}^3$. Then we have

$$\mathrm{d}\omega = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathrm{d}y \wedge \mathrm{d}z + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathrm{d}z \wedge \mathrm{d}x + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}x \wedge \mathrm{d}y,$$

Any smooth 2-form η on $\mathbb{R}^3$ can be written as

$$\eta = u\mathrm{d}y \wedge \mathrm{d}z + v\mathrm{d}z \wedge \mathrm{d}x + w\mathrm{d}x \wedge \mathrm{d}y,$$

where u, v, w are smooth functions on $\mathbb{R}^3$. Then we have

$$\mathrm{d}\eta = \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z,$$

Recall in vector calculus, the gradient and divergence in $\mathbb{R}^n$ are defined by

$$\operatorname{grad} f = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\frac{\partial}{\partial x^i}, \qquad \operatorname{div} X = \sum_{i=1}^{n} \frac{\partial X^i}{\partial x^i},$$

where $f \in C^\infty(\mathbb{R}^n)$ is a smooth function and $X \in \mathfrak{X}(\mathbb{R}^n)$ is a smooth vector field on $\mathbb{R}^n$. The curl in $\mathbb{R}^3$ is defined by

$$\operatorname{curl} X = \left(\frac{\partial X^3}{\partial y} - \frac{\partial X^2}{\partial z}\right)\frac{\partial}{\partial x} + \left(\frac{\partial X^1}{\partial z} - \frac{\partial X^3}{\partial x}\right)\frac{\partial}{\partial y} + \left(\frac{\partial X^2}{\partial x} - \frac{\partial X^1}{\partial y}\right)\frac{\partial}{\partial z},$$

Note the similarity between the formulas!

The Euclidean metric on $\mathbb{R}^3$ induces an index lowering isomorphism $\flat : \mathfrak{X}(\mathbb{R}^3) \to \Omega^1(\mathbb{R}^3)$, interior multiplication induces another map $\beta : \mathfrak{X}(\mathbb{R}^3) \to \Omega^2(\mathbb{R}^3)$ defined by

$$\beta(X) = X \lrcorner\, (\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z) = X^1 \mathrm{d}y \wedge \mathrm{d}z + X^2 \mathrm{d}z \wedge \mathrm{d}x + X^3 \mathrm{d}x \wedge \mathrm{d}y,$$

It is easily verified that β is an isomorphism. Similarly, we define a smooth isomorphism $* : C^\infty(\mathbb{R}^3) \to \Omega^3(\mathbb{R}^3)$ by

$$*(f) = f \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z,$$

So we have the commutative diagram

$$\begin{tikzcd}
C^\infty(\mathbb{R}^3) \arrow[r, "\mathrm{grad}"] \arrow[d, "*"] & \mathfrak{X}(\mathbb{R}^3) \arrow[r, "\mathrm{curl}"] \arrow[d, "\beta"] & \mathfrak{X}(\mathbb{R}^3) \arrow[r, "\mathrm{div}"] \arrow[d, "\flat"] & C^\infty(\mathbb{R}^3) \arrow[d, "*"] \\
\Omega^3(\mathbb{R}^3) \arrow[r, "\mathrm{d}"] & \Omega^2(\mathbb{R}^3) \arrow[r, "\mathrm{d}"] & \Omega^1(\mathbb{R}^3) \arrow[r, "\mathrm{d}"] & \Omega^0(\mathbb{R}^3)
\end{tikzcd} \tag{14.16}$$

14.3.2 An Invariant Formulation of Exterior Derivative

There is a more invariant formula for the exterior derivative, which does not involve any coordinate chart.

We begin with 1-forms.

Proposition: **Invariant Formula for Exterior Derivative of 1-Forms**

Let M be a smooth manifold, with or without boundary, and $\omega \in \Omega^1(M)$ be a smooth 1-form on M, then for any $X, Y \in \mathfrak{X}(M)$ we have

$$\mathrm{d}\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]).$$

Proof. Suppose $\omega = u\mathrm{d}v$ and compute. □

Proposition: **Exterior Derivative and Lie Bracket**

Let M be a smooth n-dimensional manifold, with or without boundary, and (E_i) be a smooth local frame for M. Let (ϵ^i) be the dual coframe, then let

$$\mathrm{d}\epsilon^i = \sum_{j<k} b^i_{jk} \epsilon^j \wedge \epsilon^k, \qquad [E_j, E_k] = c^i_{jk} E_i, \tag{14.17}$$

then we have $b^i_{jk} = -c^i_{jk}$ for all i, j, k.

The generalization to higher degree forms is more complicated.

Proposition: **Invariant Formula for Exterior Derivative**

Let M be a smooth manifold, with or without boundary, and $\omega \in \Omega^k(M)$. For any $X_1, \ldots, X_{k+1} \in \mathfrak{X}(M)$, we have

$$\begin{aligned} &\mathrm{d}\omega(X_1, \ldots, X_{k+1}) \\ =&\sum_{i=1}^{k+1}(-1)^{i-1}X_i(\omega(X_1, \ldots, \hat{X}_i, \ldots, X_{k+1})) \\ &+\sum_{1\leqslant i<j\leqslant k+1}(-1)^{i+j}\omega([X_i, X_j], X_1, \ldots, \hat{X}_i, \ldots, \hat{X}_j, \ldots, X_{k+1}), \end{aligned} \tag{14.18}$$

where $\hat{X}_i$ means we omit X_i.

14.3.3 Lie Derivative of Differential Forms

We restrict ourselves to manifolds without boundary in this subsection, as we did in Chapter 12.

Proposition: **Lie Derivative of Wedge Product**

Let M be a smooth manifold without boundary, and $V \in \mathfrak{X}(M)$ and $\omega, \eta \in \Omega^*(M)$, then

$$\mathcal{L}_V(\omega \wedge \eta) = \mathcal{L}_V\omega \wedge \eta + \omega \wedge \mathcal{L}_V\eta.$$

Proof. From the tensor product formula for Lie derivatives, obviously. □

The following is a very important formula, called the Cartan's magic formula.

Theorem 14.3.2: Cartan's Magic Formula

Let M be a smooth manifold without boundary, and $V \in \mathfrak{X}(M)$ and $\omega \in \Omega^*(M)$, then

$$\mathcal{L}_V\omega = V \lrcorner\, \mathrm{d}\omega + \mathrm{d}(V \lrcorner\, \omega).$$

Proof. We prove it by induction on k. If ω is a 0-form f, then

$$V \lrcorner\, \mathrm{d}f + \mathrm{d}(V \lrcorner\, f) = \mathrm{d}f(V) + \mathrm{d}0 = V(f) = \mathcal{L}_V f.$$

Now suppose ω is a k-form for some $k \geqslant 1$, and the formula holds for all forms of degree less than k. In some coordinate chart, we can write $\omega = \sum_I{}'\omega_I \mathrm{d}x^I$, then take $u = x^{i_1}, \beta = \omega_I \mathrm{d}x^{i_2} \wedge \cdots \wedge \mathrm{d}x^{i_k}$, all we need to do is verify the formula for the form $\mathrm{d}u \wedge \beta$, where u is a smooth function and β is a smooth $(k-1)$-form. We have $\mathcal{L}_V(\mathrm{d}u) = \mathrm{d}(V(u))$ by the 0-form case,

$$\begin{aligned} \mathcal{L}_V(\mathrm{d}u \wedge \beta) &= \mathcal{L}_V(\mathrm{d}u) \wedge \beta + \mathrm{d}u \wedge \mathcal{L}_V\beta \\ &= \mathrm{d}(V(u)) \wedge \beta + \mathrm{d}u \wedge (V \lrcorner\, \mathrm{d}\beta + \mathrm{d}(V \lrcorner\, \beta)) \end{aligned}$$

on the other hand,

$$\begin{aligned}
&V \lrcorner\, \mathrm{d}(\mathrm{d}u \wedge \beta) + \mathrm{d}(V \lrcorner\, (\mathrm{d}u \wedge \beta)) \\
&= V \lrcorner\, (\mathrm{d}(\mathrm{d}u) \wedge \beta - \mathrm{d}u \wedge \mathrm{d}\beta) + \mathrm{d}(V(u)\beta - \mathrm{d}u \wedge (V \lrcorner\, \beta)) \\
&= -V(u)\mathrm{d}u \wedge \beta + V(u)\mathrm{d}u \wedge \beta + V(u)\mathrm{d}u \wedge (V \lrcorner\, \beta) - \mathrm{d}u \wedge (V \lrcorner\, \mathrm{d}\beta) + \mathrm{d}u \wedge \mathrm{d}(V \lrcorner\, \beta) \\
&= \mathrm{d}(V(u)) \wedge \beta + \mathrm{d}u \wedge (V \lrcorner\, \mathrm{d}\beta + \mathrm{d}(V \lrcorner\, \beta)),
\end{aligned}$$

So we have our result. □

Corollary 14.3.1: Lie Derivative Commutes with Exterior Derivative

Let M be a smooth manifold without boundary, and $V \in \mathfrak{X}(M)$ and $\omega \in \Omega^*(M)$, then

$$\mathcal{L}_V(\mathrm{d}\omega) = \mathrm{d}(\mathcal{L}_V\omega).$$

Proof. From Cartan's magic formula, we have

$$\begin{aligned}
\mathcal{L}_V(\mathrm{d}\omega) &= V \lrcorner\, \mathrm{d}(\mathrm{d}\omega) + \mathrm{d}(V \lrcorner\, \mathrm{d}\omega) = \mathrm{d}(V \lrcorner\, \mathrm{d}\omega), \\
\mathrm{d}(\mathcal{L}_V\omega) &= \mathrm{d}(V \lrcorner\, \mathrm{d}\omega + \mathrm{d}(V \lrcorner\, \omega)) = \mathrm{d}(V \lrcorner\, \mathrm{d}\omega),
\end{aligned}$$

So we have our result. □

Chapter 15

Orientations

Simply speaking, the orientation of a line or curve is a choice of direction along it, but we can generalize this idea to higher-dimensional manifolds.

15.1 Orientation of Vector Spaces

Definition 15.1.1: Orientation of a Vector Space

Let V be a n-dimensional real vector space. We say that two basis $(E_1, \ldots, E_n)$ and $(\tilde{E}_1, \ldots, \tilde{E}_n)$ of V are **consistently oriented** if the transition matrix B defined by

$$E_i = B_i^j \tilde{E}_j$$

has positive determinant. If $n = 0$, we define orientations by a choice of $+1$ or -1 specially. It is easy to see that this is an equivalence relation, and thus we can define an **orientation** of V to be one of the two equivalence classes, denoted by $[E_1, \ldots, E_n]$ and its opposite $-[E_1, \ldots, E_n]$. A vector space with a choice of orientation is called an **oriented vector space**, and the chosen orientation is called **oriented** or **positively oriented**, while the opposite orientation is called **negatively oriented**.

Example: **Orientated Vector Spaces**

The orientation $[e_1, \ldots, e_n]$ of $\mathbb{R}^n$ is called the **standard orientation** of $\mathbb{R}^n$. An oriented basis for $\mathbb{R}^3$ is called a **right-handed** basis.

There is a connection between orientations and alternating tensors.

Proposition: **Orientation and Alternating Tensors**

Let V be a n-dimensional real vector space. For any nonzero $\omega \in \Lambda^n(V^*)$, we can define an orientation $\mathcal{O}_\omega$ of V by:

- If $n \geqslant 1$, then $[E_1, \ldots, E_n] = \mathcal{O}_\omega$ if and only if $\omega(E_1, \ldots, E_n) > 0$.

- If $n = 0$, then $\mathcal{O}_\omega = +1$ if $\omega > 0$ and $\mathcal{O}_\omega = -1$ if $\omega < 0$.

Proof. We assume $n \geqslant 1$, and (E_i) and $(\tilde{E}_i)$ are two bases of V and let B be the linear map sending E_i to $\tilde{E}_i$, or $\tilde{E}_j = B^i_j E_i$. Then we have

$$\omega(\tilde{E}_1, \dots, \tilde{E}_n) = \omega(B^i_1 E_i, \dots, B^i_n E_i) = \det(B)\omega(E_1, \dots, E_n).$$

So we have $\omega(\tilde{E}_1, \dots, \tilde{E}_n) > 0$ if and only if $\det(B) > 0$ and $\omega(E_1, \dots, E_n) > 0$, which means that the two bases are consistently oriented. □

If ω here determines the positive orientation, then we say that ω is a **positive orientated n-covector** of V. As $\Lambda^n(V^*)$ is one-dimensional, so choosing orientation for V is equivalent to choosing a component of $\Lambda^n(V^*)\backslash\{0\}$. This formula also shows why we define the 0-dimensional case like this.

15.2 Orientation of Manifolds

Definition 15.2.1: Orientation of Manifolds

Let M be a smooth manifold, with or without boundary. We define a **pointwise orientation** of M to be a choice of orientation for each tangent space T_pM at each point $p \in M$.
To connect different points, we give the following definitions.

- If (E_i) is a local frame for TM, we say that (E_i) is **positively oriented** if $(E_i|_p)$ is positively oriented for each $p \in U \subseteq M$. Same for negatively oriented.
- A pointwise orientation of M is called **continuous** if for each $p \in M$, there exists a local frame (E_i) defined on a neighborhood of p such that (E_i) is positively oriented.

An **orientation** of M is a continuous pointwise orientation. We say M is **orientable** if it admits an orientation, and **non-orientable** otherwise. An oriented manifold is a pair $(M, \mathcal{O})$ where M is a manifold and $\mathcal{O}$ is an orientation of M. An oriented manifold with boundary is defined similarly.

Specially, if M is 0-dimensional (discrete), then an orientation of M is a choice of $+1$ or -1 for each point in M, and the continuity condition is vacuous. Every 0-dimensional manifold is orientable.

If $n \geqslant 1$, then every connected domain local frame is either positively oriented or negatively oriented.

We can also define orientation by n-forms.

Proposition: **Orientation by n-forms**

Let M be a smooth n-dimensional manifold, with or without boundary. For any nonvanishing $\omega \in \Omega^n(M)$, we can define an orientation $\mathcal{O}_\omega$ of M by being positively oriented at each point p. Conversely, if M is given an orientation $\mathcal{O}$, then there exists a nonvanishing $\omega \in \Omega^n(M)$ such that $\mathcal{O} = \mathcal{O}_\omega$.

Proof. Assume ω bes a nonvanishing n-form on M, then ω defines a pointwise orientation $\mathcal{O}_\omega$ of M by being positively oriented at each point. We need to show that $\mathcal{O}_\omega$ is continuous. It is trivial when $n = 0$, so we assume $n \geqslant 1$. For each $p \in M$, we can choose a local frame (E_i) defined on a connected neighborhood U of p and let (ϵ^i) be the dual coframe. Then we can write $\omega = f\epsilon^1 \wedge \cdots \wedge \epsilon^n$ for some smooth function f on U. Since ω is nonvanishing, we have

$$\omega(E_1, \ldots, E_n) = f \neq 0$$

at all points in U. So from connectedness of U, we have $f > 0$ or $f < 0$ at all points in U. So (E_i) is positively oriented or negatively oriented at all points in U, which means that $\mathcal{O}_\omega$ is continuous.

Conversely, assume $\mathcal{O}$ is an orientation of M. Define $\Lambda^n_+ T^*M \subseteq \Lambda^n T^*M$ to be the open subset consisting of all positive orientated n-covectors at each point of M. At each $p \in M$, the intersection of $\Lambda^n_+ T^*M$ with the fiber $\Lambda^n T^*_p M$ is a half line, and therefore convex. So locally in some trivialization, we can find a smooth section of $\Lambda^n_+ T^*M$ over some neighborhood of p.

So we can choose a smooth local section ω_α of $\Lambda^n_+ T^*M$ over each element U_α of an open cover of M. Then we can choose a partition of unity $\{\rho_\alpha\}$ subordinate to $\{U_\alpha\}$, and define

$$\omega = \sum_\alpha \rho_\alpha \omega_\alpha.$$

Then ω is a smooth section of $\Lambda^n_+ T^*M$, and thus a nonvanishing n-form on M. It is easy to check that $\mathcal{O} = \mathcal{O}_\omega$. □

A smooth coordinate chart (U, φ) of M is called **positively oriented** if the local coordinate frame $(\partial/\partial x^i)$ is positively oriented. A smooth atlas $\{(U_\alpha, \varphi_\alpha)\}$ of M is called **consistently oriented** if the transition maps $\varphi_\beta \circ \varphi_\alpha^{-1}$ have positive Jacobian determinant at each point of their domains.

Proposition: **Orientation by Coordinate Atlas**

Let M be a smooth manifold with positive dimension, with or without boundary. Given any consistently oriented smooth atlas $\{(U_\alpha, \varphi_\alpha)\}$ of M, there is a unique orientation $\mathcal{O}$ of M such that each chart in the atlas is positively oriented.
Conversely, if M is given an orientation $\mathcal{O}$, and either $\partial M = \varnothing$ or $\dim M \geqslant 2$, then the collection of all positively oriented charts forms a consistently oriented smooth atlas of M.

Proof. All is very obvious. But we shall note that for smooth 1-dimensional manifolds with boundary, the problem exists because we require the last coordinate of a boundary chart to be nonnegative, which means that if we get a negatively oriented chart at the boundary, then we cannot change it to a positively oriented one by changing the sign. For the rest cases, we can always change the sign of x^1 to get a positively oriented chart. □

Proposition: **Product Orientation**

Let $M_1, \ldots, M_k$ be orientable smooth manifolds. There is a unique orientation on the product manifold $M_1 \times \cdots \times M_k$ such that for each ω_i that are orientation form for M_i, the form $\pi_1^*\omega_1 \wedge \cdots \wedge \pi_k^*\omega_k$ is an orientation form for $M_1 \times \cdots \times M_k$, where $\pi_i : M_1 \times \cdots \times M_k \to M_i$ is the projection map.

Proof. Obvious. □

Proposition: **Orientations of Orientable Manifolds**

Let M be a connected orientable smooth manifold, with or without boundary. Then M has exactly two orientations. If two orientations $\mathcal{O}$ and $\tilde{\mathcal{O}}$ of M agree at some point $p \in M$, then $\mathcal{O} = \tilde{\mathcal{O}}$.

Proposition: **Orientation of Codimension 0 Submanifolds**

Let M be a smooth manifold, with or without boundary, and $N \subseteq M$ be a codimension 0 submanifold. If M is orientable, then N is orientable. Moreover, if $\mathcal{O}$ is an orientation of M, then the restriction $\mathcal{O}|_N$ is an orientation of N. If ω is an orientation form for M, then $\iota_N^*\omega$ is an orientation form for N, where $\iota_N : N \to M$ is the inclusion map.

Let M, N be oriented smooth manifolds, with or without boundary. Let $F : M \to N$ be a local diffeomorphism. We say that F is **orientation-preserving** if for each $p \in M$, the differential $dF_p : T_pM \to T_{F(p)}N$ is an orientation-preserving linear isomorphism, i.e. dF_p sends positively oriented bases of T_pM to positively oriented bases of $T_{F(p)}N$. The same goes for **orientation-reversing**.

Proposition: **Equivalent Criterion for Orientation-Preserving Local Diffeomorphisms**

Let M, N be oriented smooth manifolds with positive dimension, with or without boundary. Let $F : M \to N$ be a local diffeomorphism. Then the following are equivalent:

- F is orientation-preserving.
- For any oriented smooth charts for M and N, the Jacobian of F has positive determinant.
- For any positively oriented form ω for N, the pullback $F^*\omega$ is a positively oriented form for M.

We can also construct orientations by pullbacks.

Proposition: **The Pullback Orientation**

Suppose M, N are smooth manifolds, with or without boundary, and $F : M \to N$ is a local diffeomorphism. If N is oriented by $\mathcal{O}$, then M has a unique orientation $F^*\mathcal{O}$ called the **pullback orientation** of M by F such that F is orientation-preserving.

Proof. For each $p \in M$, there is a unique orientation of T_pM such that $dF_p : T_pM \to T_{F(p)}N$ is orientation-preserving. This defines a pointwise orientation of M. To show that it is continuous, we choose an orientation form ω for N. Then $F^*\omega$ is a nonvanishing n-form on M, and thus defines a continuous orientation of M. □

For this we have the usual pullback composition formula:

$$(G \circ F)^* \mathcal{O} = F^*(G^* \mathcal{O}) \tag{15.1}$$

Recall that a smooth manifold M is called **parallelizable** if it admits a smooth global frame for.

Proposition: **Parallelizable Manifolds are Orientable**

Every parallelizable smooth manifold is orientable.

Proof. Just define the orientation by the global frame. □

So $\mathbb{R}^n, T^n, S^1, S^3, S^7$ and there products are orientable, and any codimension 0 submanifolds of them are orientable. Also, every Lie group is orientable because it is parallelizable. We say that an orientation of a Lie group G is **left-invariant** if every left translation $L_g : G \to G$ is orientation-preserving.

Proposition: **Orientations of Lie Groups**

Every Lie group has exactly two left-invariant orientations, corresponding to the two orientations of its Lie algebra.

Proof. Quite natural, just define the orientation at $\mathrm{Lie}(G) \cong T_e G$ and then extend it to the whole G by left translations. □

15.2.1 Orientation of Hypersurfaces

An orientation of some smooth manifold M do not generally restrict to a submanifold easily. A directly restriction of a full rank form to a lower dimensional submanifold is just zero. As it turns out, submanifolds of oriented manifolds may not actually be orientable, take the Möbius strip as an example.

Instead, here we focus on the case of hypersurfaces, which are codimension 1 submanifolds, with one extra information: A vector field that is nowhere tangent to the hypersurface.

Recall that a vector field along $S \subseteq M$ is a section of the ambient tangent bundle $TM|_S$, which is a continuous map $N : S \to TM$ such that $N_p \in T_p M$ for each $p \in S$.

Theorem 15.2.1: Orientation of Hypersurfaces

Let M be an oriented smooth manifold, with or without boundary, and $S \subseteq M$ be an immersed hypersurface with or without boundary, and N is a vector field along S that is nowhere tangent to S. Then there is a unique orientation $\mathcal{O}$ of S such that for each $p \in S$, a basis $(E_1, \ldots, E_{n-1})$ of $T_p S$ is positively oriented if and only if the basis $(N_p, E_1, \ldots, E_{n-1})$ of $T_p M$ is positively oriented.
If ω is an orientation form for M, then $\iota_S^*(N \lrcorner\, \omega)$ is an orientation form for S with respect to $\mathcal{O}$, where $\iota_S : S \hookrightarrow M$ is the inclusion map.

For $n = 1$ case, since S is 0-dimensional, we interpret the condition as follows: $+1$ is the positive orientation of $T_p S$ if and only if (N_p) is a positively oriented basis of $T_p M$.

Proof. We already knows that $\sigma = \iota_S^*(N \lrcorner\, \omega)$ is an $(n-1)$-form on S. We need to show that σ is nonvanishing: Take any $p \in S$ and a basis $(E_1, \ldots, E_{n-1})$ of T_pS. Then $(N_p, E_1, \ldots, E_{n-1})$ is a basis of T_pM because N_p is not tangent to S. So we have

$$\sigma_p(E_1, \ldots, E_{n-1}) = \omega_p(N_p, E_1, \ldots, E_{n-1}) \neq 0.$$

So σ is nonvanishing, and thus defines an orientation $\mathcal{O}$ of S. □

However, not all hypersurfaces have a nowhere tangent vector field, however for most cases we can. For example, $S^n \subseteq \mathbb{R}^{n+1}$ has a nowhere tangent vector field given by $N = x^i \partial/\partial x^i$, which defines the standard orientation of S^n.

Proposition: **Regular Level Sets are Orientable**

Let M be an oriented smooth manifold, with or without boundary, and $f : M \to \mathbb{R}$ be a smooth function. If $S \subseteq M$ is a regular level set of f, then S is orientable.

Proof. Choose any Riemannian metric on M, and let $N = \operatorname{grad} f|_S$ would do. □

15.2.2 Boundary Orientation

If M is a smooth manifold with boundary, then ∂M is an embedded hypersurface of M. And we know from lemma 9.4.1

Proposition: **Induced Orientation on the Boundary**

Let M be an oriented smooth n-dimensional manifold with boundary, $n \geqslant 1$. Then ∂M is orientable, and all outward-pointing vector fields along ∂M define the same orientation of ∂M, called the **induced orientation** or **Stokes orientation** of ∂M.

Example: **Induced Orientation on the Boundary**

First, consider S^n to be the boundary of the unit ball in $\mathbb{R}^{n+1}$, then the induced orientation of S^n is the standard orientation.
Next, consider the upper half space $\mathbb{H}^n$. Identify $\partial\mathbb{H}^n \subseteq \mathbb{R}^n$ by $\{(x^1, \ldots, x^{n-1}, 0) : x^i \in \mathbb{R}\}$, then the vector field $-\partial/\partial x^n$ is outward-pointing along $\partial\mathbb{H}^n$, and in $\mathbb{R}^n$, we have

$$\left[-\frac{\partial}{\partial x^n}, \frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^{n-1}}\right] = (-1)^n [e_1, \ldots, e_n].$$

Thus, the induced orientation of $\partial\mathbb{H}^n$ is equal to the standard orientation of $\mathbb{R}^{n-1}$ if n is even, and the opposite of the standard orientation if n is odd.

In many settings, we use parametrization to describe submanifolds.

Lemma 15.2.1: Local Parametrization of Boundary and Orientation

Let M be an oriented smooth n-dimensional manifold with boundary. Suppose $U \subseteq \mathbb{R}^{n-1}$ is an open subset, and $a, b \in \mathbb{R}, a < b$, and $F : (a, b] \times U \to M$ is a smooth embedding that restricts to a smooth embedding of $\{b\} \times U$ into ∂M. Then the parametrization $f : U \to \partial M, f(x) = F(b, x)$ is orientation-preserving for ∂M if and only if F is orientation-preserving for M.

Proof. Quite natural, note that a same-dimension embedding is a diffeomorphism onto its image, so we can identify U with $\{b\} \times U$ and then the condition is just the same as the definition of orientation-preserving for F. □

For example, consider the smooth local parametrization of $S^2 = \partial \overline{B}^3$, let $U = (0, \pi) \times (0, 2\pi)$ and $X : U \to S^2 \subseteq \mathbb{R}^3$ be given by

$$X(\theta, \varphi) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta).$$

Then it is a restriction of a smooth embedding $F : (0, 1] \times U \to \overline{B}^3$ given by

$$F(\rho, \theta, \varphi) = (\rho\sin\theta\cos\varphi, \rho\sin\theta\sin\varphi, \rho\cos\theta), \qquad F(1, \theta, \varphi) = X(\theta, \varphi).$$

So, as $\det J_F = \rho^2 \sin\theta > 0$, we have X is orientation-preserving for S^2.

15.3 The Riemannian Volume Form

Let (M, g) be an oriented Riemannian manifold of positive dimension. Then there is a smooth orthonormal local frame $(E_1, \ldots, E_n)$ in a neighborhood of each point in M. Assume it is positively oriented (else we can change the sign of E_1 to make it positively oriented), then we can find an oriented orthonormal frame in a neighborhood of each point in M.

Proposition: **Riemannian Volume Form**

Suppose (M, g) is an oriented Riemannian n-dimensional manifold, with or without boundary, and $n \geqslant 1$. Then there is a unique orientation form $\omega_g \subseteq \Omega^n(M)$ called the **Riemannian volume form**, such that

$$\omega_g(E_1, \ldots, E_n) = 1. \tag{15.2}$$

for every local oriented orthonormal frame $(E_1, \ldots, E_n)$ for M.
In local smooth oriented coordinates (U, x^i), we have

$$\omega_g = \sqrt{\det(g_{ij})}dx^1 \wedge \cdots \wedge dx^n, \tag{15.3}$$

Proof. Existence and uniqueness is easy. To see the coordinate form, let (E_i) be the local oriented orthonormal frame, then let

$$\frac{\partial}{\partial x^i} = A_i^j E_j, \qquad \omega_g = f dx^1 \wedge \cdots \wedge dx^n.$$

Then we have

$$f = \omega_g\left(\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^n}\right) = \omega_g(A_1^j E_j, \ldots, A_n^j E_j) = \det(A).$$

$$g_{ij} = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) = g(A_i^k E_k, A_j^l E_l) = A_i^k A_j^l g(E_k, E_l) = \sum_{k=1}^{n} A_i^k A_j^k,$$

So $\det(g_{ij}) = \det(A)^2$, and thus $f = \sqrt{\det(g_{ij})}$, since both frames are positively oriented. (If (x^i) is a negatively oriented coordinate chart, then we have $f = -\sqrt{\det(g_{ij})}$, which is consistent with the definition of ω_g.) □

This form is also isometric invariant. Suppose (M, g) and $(\tilde{M}, \tilde{g})$ are positive-dimensional oriented Riemannian manifolds, with or without boundary, and $F : M \to \tilde{M}$ is a local isometry, then $F^*\omega_{\tilde{g}} = \omega_g$.

15.3.1 Hypersurfaces in Riemannian Manifolds

Let (M, g) be an oriented Riemannian manifold, with or without boundary, and $S \subseteq M$ be an immersed hypersurface with or without boundary. Any unit normal vector field N along S is nowhere tangent to S, so it defines an orientation of S by previous discussions.

Proposition: **Volume Form on Submanifolds**

Let (M, g) be an oriented Riemannian manifold, with or without boundary, and $S \subseteq M$ be an immersed hypersurface with or without boundary. Let $\tilde{g}$ be the induced Riemannian metric on S, and suppose N is a unit normal vector field along S. Then with respect to the orientation of S defined by N, the Riemannian volume form of $(S, \tilde{g})$ is given by

$$\omega_{\tilde{g}} = \iota_S^*(N \lrcorner\, \omega_g), \tag{15.4}$$

Proposition: **Outward Pointing Normal Field**

Let (M, g) be a Riemannian manifold with boundary. There is a unique smooth outward-pointing unit normal vector field N along ∂M.

Proof. Existence: Let f be a boundary defining function for M, then

$$N = -\frac{\operatorname{grad} f}{|\operatorname{grad} f|_g}\bigg|_{\partial M}$$

is a smooth outward-pointing unit normal vector field along ∂M. Uniqueness is obvious. □

Corollary 15.3.1: Structure of the Boundary Volume Form

Let (M, g) be an oriented Riemannian manifold with boundary, and $\tilde{g}$ be the induced Riemannian metric on ∂M. Then the Riemannian volume form of $(\partial M, \tilde{g})$ is

$$\omega_{\tilde{g}} = \iota_{\partial M}^*(N \lrcorner\, \omega_g),$$

where N is the unique smooth outward-pointing unit normal vector field along ∂M.

15.4 Orientations and Covering Maps

Proving that a manifold is non-orientable is usually more difficult than proving that it is orientable. One useful tool for this is the following result about covering maps.

First here is a simply special case of the pullback orientation:

Proposition: **Orientation of Covering Spaces**

If $\pi : E \to M$ is a smooth covering map, and M is orientable, then E is orientable.

Proof. Covering maps are local diffeomorphisms, so it follows from proposition 15.2. □

If G is a Lie group acting smoothly on a smooth manifold E, we say that the action is **orientation-preserving** if for each $g \in G$, the diffeomorphism $E \to E, p \mapsto g \cdot p$ is orientation-preserving.

A covering map $\pi : E \to M$ is called **normal** (also **regular** or **Galois**) if the group G of deck transformations of π acts transitively on each fiber of π. In this case, we have $M \cong E/G$. Here G is denoted by $\mathrm{Aut}_\pi(E)$, and is called the **automorphism group** of π.

Theorem 15.4.1: Orientability and Normal Covering Maps

Suppose E is a connected, oriented smooth manifold, with or without boundary, and $\pi : E \to M$ is a normal smooth covering map. Then M is orientable if and only if the action of $\mathrm{Aut}_\pi(E)$ on E is orientation-preserving.

Proof. Let $\mathcal{O}_E$ be the orientation of E. First suppose M is orientable, and $q \in E$. Because M is connected, it has exactly two orientations, and one of them has the property $\mathrm{d}\pi_q : T_qE \to T_{\pi(q)}M$ is orientation-preserving (because local diffeomorphisms are orientation-preserving for some orientation). Denote that orientation by $\mathcal{O}_M$. Then $\pi^*\mathcal{O}_M = \mathcal{O}_E$ by the uniqueness of pullback orientation, so for $\varphi \in \mathrm{Aut}_\pi(E)$, we have $\pi \circ \varphi = \pi$, and thus

$$\varphi^*\mathcal{O}_E = \varphi^*(\pi^*\mathcal{O}_M) = (\pi \circ \varphi)^*\mathcal{O}_M = \pi^*\mathcal{O}_M = \mathcal{O}_E,$$

which means that φ is orientation-preserving.

Conversely, suppose the action of $\mathrm{Aut}_\pi(E)$ on E is orientation-preserving, and let $p \in M$. If $U \subseteq M$ is any evenly covered neighborhood of p, then there is a smooth section $\sigma : U \to E$ that induces an orientation $\sigma^*\mathcal{O}_E$ of U. Suppose $\sigma' : U \to E$ is another smooth section, then there is a covering automorphism $\varphi \in \mathrm{Aut}_\pi(E)$ such that $\sigma' = \varphi \circ \sigma$. So we have

$$\sigma'^*\mathcal{O}_E = \sigma^*(\varphi^*\mathcal{O}_E) = \sigma^*\mathcal{O}_E,$$

because φ is orientation-preserving. So the orientation induced by any local section is the same, and thus defines a continuous orientation of M via pullback, which means that M is orientable. □

Remark:
This is how we say that the manifold "twists" in the Möbius strip, because the nontrivial deck transformation (like walking along the strip and come back to the same point) is orientation-reversing (getting to the back of the strip).

Example: **Orientation by Covering Maps**

- Consider the real projective space $\mathbb{R}P^n = S^n/\{\pm 1\}$, where the only nontrivial covering automorphism is given by $\alpha(x) = -x$. We will see that α is orientation-preserving if and only if n is odd, so $\mathbb{R}P^n$ is orientable if and only if n is odd.

  Proof. Consider $F : \overline{B}^{n+1} \to \overline{B}^{n+1}$ given by $F(x) = -x$, and take the Riemannian volume form ω_g of $\overline{B}^{n+1}$, then the induced orientation of S^n is the standard orientation by $\omega_{\tilde{g}} = \iota_{S^n}^*(N \lrcorner\, \omega_g)$, where N is the outward-pointing normal vector field along S^n. Consider the pullback $\alpha^*\omega_{\tilde{g}}$, we have $\iota_{S^n} \circ \alpha = F \circ \iota_{S^n}$, so

$$\begin{aligned}\alpha^*\omega_{\tilde{g}} &= \alpha^*(\iota_{S^n}^*(N \lrcorner\, \omega_g)) = (\iota_{S^n} \circ \alpha)^*(N \lrcorner\, \omega_g) \\ &= (F \circ \iota_{S^n})^*(N \lrcorner\, \omega_g) = \iota_{S^n}^*(F^*(N \lrcorner\, \omega_g)) = \iota_{S^n}^*((dF^{-1}N) \lrcorner\, F^*\omega_g) \\ &= \iota_{S^n}^*(N \lrcorner\, (-1)^{n+1}\omega_g) = (-1)^{n+1}\iota_{S^n}^*(N \lrcorner\, \omega_g) = (-1)^{n+1}\omega_{\tilde{g}}.\end{aligned}$$

  where we used

$$(\mathrm{d}F^{-1}N)_x(f) = N_{F^{-1}(x)}(f \circ F^{-1}) = N_{-x}(f \circ F^{-1}) = N_x(f)$$

  so $\alpha^*\omega_{\tilde{g}} = \omega_{\tilde{g}}$ if n is odd, and $\alpha^*\omega_{\tilde{g}} = -\omega_{\tilde{g}}$ if n is even. □

- Now consider the Möbius band. Let E be the total space of Möbius bundle. The quotient map $q : \mathbb{R}^2 \to E$. Then $\mathrm{Aut}_q(\mathbb{R}^2) \cong \mathbb{Z}$ by $n \cdot (x, y) = (x + n, (-1)^n y)$. So for odd n, the pullback gives $\mathrm{d}x \wedge \mathrm{d}y \mapsto -\mathrm{d}x \wedge \mathrm{d}y$, which is orientation-reversing, so E is non-orientable.

15.4.1 The Orientation Covering

Now we shall see that every non-orientable manifold has a two-sheeted covering manifold. This gives intuition that every non-orientable manifold is like a twisted version of an orientable manifold.

Recall that a topological covering map $\pi : E \to M$ between **connected and locally path connected topological spaces** is defined as a surjective continuous map such that for each $p \in M$, there is an open neighborhood U of p such that $\pi^{-1}(U)$ is a disjoint union of open subsets of E, each of which is homeomorphic to U by π. A smooth covering map is a topological covering map that is also a smooth map.

We define a **generalized covering map** to be covering map except that we do not require the total space E to be connected. Well, we can easily see that its restriction to each connected component of E is a covering map in the usual sense.

Definition 15.4.1: Orientation Covering

Let M be a connected smooth positive-dimensional manifold, with or without boundary. Let $\hat{M}$ be the set of all orientations of all tangent spaces of M, i.e.

$$\hat{M} = \{(p, \mathcal{O}_p) : p \in M, \mathcal{O}_p \text{ is an orientation of } T_pM\}.$$

Define the projection map $\hat{\pi} : \hat{M} \to M$ by $\hat{\pi}(p, \mathcal{O}_p) = p$. Then $\hat{\pi}$ is called the **orientation**

covering of M, each fiber $\hat{\pi}^{-1}(p)$ consists of exactly two points.

Proposition: **Properties of the Orientation Covering**

Let M be a connected smooth positive-dimensional manifold, with or without boundary, and $\hat{\pi} : \hat{M} \to M$ be the orientation covering of M. Then $\hat{M}$ can be given s structure of smooth oriented manifold with or without boundary such that

- $\hat{\pi}$ is a generalized smooth covering map.
- A connected open subset $U \subseteq M$ is evenly covered by $\hat{\pi}$ if and only if U is orientable. In this case, every orientation of U is the pullback orientation induced by a local section of $\hat{\pi}$ over U.

Remark:
We can view $\hat{M}$ as the two sides of M, geometrically, imaging a smooth surface, and moving its two sides apart, then the resulting manifold is just $\hat{M}$, and the projection map $\hat{\pi}$ just glues the two sides together. For orientable manifolds like S^2, the two sides are completely separated, so $\hat{M}$ is just two disjoint copies of M. For non-orientable manifolds like the Möbius strip, the two sides are twisted together, but $\hat{M}$ is actually orientable because it needs two rounds to get back to the same point.

Proof.

- First, we give $\hat{M}$ a topology: For each orientable open subset $U \subseteq M$ and each orientation $\mathcal{O}$ of U, we define

$$\hat{U}_{\mathcal{O}} = \{(p, \mathcal{O}_p) : p \in U, \mathcal{O}_p = \mathcal{O}|_p\} \subseteq \hat{M}.$$

  Then all $\hat{U}_{\mathcal{O}}$ forms a basis of a topology on $\hat{M}$, and $\hat{\pi}|_{\hat{U}_{\mathcal{O}}} : \hat{U}_{\mathcal{O}} \to U$ is a homeomorphism, so $\hat{\pi}$ is a local homeomorphism. Now, as every orientable connected open subset $U \subseteq M$ has $\hat{\pi}^{-1}(U) = \hat{U}_{\mathcal{O}} \sqcup \hat{U}_{-\mathcal{O}}$, so $\hat{\pi}$ is a generalized covering map. By the preceding discussion, $\hat{\pi}$ restricted to each connected component of $\hat{M}$ is a covering map in the usual sense, so there is a unique smooth structure on $\hat{M}$ such that $\hat{\pi}$ is a smooth covering map, from theorem 4.4.2.

- To give M an orientation, for $\hat{p} = (p, \mathcal{O}_p) \in \hat{M}$, we define $\mathcal{O}_{\hat{p}} = \mathcal{O}_p$. Then it gives a continuous orientation $\mathcal{O}$ of $\hat{M}$.

□

Theorem 15.4.2: Orientation Covering Theorem

Suppose M is a connected smooth manifold, with or without boundary, and $\hat{\pi} : \hat{M} \to M$ is the orientation covering of M.

- If M is orientable, then $\hat{M}$ has exactly two connected components, each of which is diffeomorphic to M by $\hat{\pi}$.
- If M is non-orientable, then $\hat{M}$ is connected, and $\hat{\pi} : \hat{M} \to M$ is a two-sheeted smooth covering map.

Theorem 15.4.3: Uniqueness of the Orientation Covering

Let M be a non-orientable connected smooth manifold, with or without boundary and $\pi : \hat{M} \to M$ is its orientation covering. If $\tilde{M}$ is an oriented smooth manifold, with or without boundary that exists a two-sheeted smooth covering map $\tilde{\pi} : \tilde{M} \to M$, then there is a unique orientation-preserving diffeomorphism $\varphi : \tilde{M} \to \hat{M}$ such that $\pi \circ \varphi = \tilde{\pi}$.

Theorem 15.4.4: Orientatability and Fundamental Group

Let M be a connected smooth manifold, with or without boundary, and the fundamental group of M has no subgroup of index 2, then M is orientable.
In particular, if M is simply connected, then M is orientable.

Chapter 16

Integration on Manifolds

We have already introduced line integrals of covector fields, by pulling a covector field back to a segment of the real line. We will now introduce the more general notion of integrating a differential form over a manifold, which will allow us to state and prove the general Stokes' theorem.

16.1 The Geometry of Volume Measurement

How can we make sense of coordinate independent muultiple integrals. First, there is no way to integrate a real-valued function on a manifold without further structure like Riemannian metric.

When we integrate covector fields along the curve, at each point, the covector puts out a number for each tangent vector, and thus we can see it as a "length ruler" along some direction, coordinate independently. Similarly, a k-form can be seen as a "volume ruler" that measures the volume of k-dimensional parallelepipeds in the tangent space.

Actually, we cam deduce that what we expected to be a volume measurement must by an alternating tensor. To be consistent with our intuition of volume, ω should be a map $V^k \to \mathbb{R}$ that

- Be multilinear, so that scaling and adding would work.
- If the vectors are linearly dependent, the volume should be zero.

The above two conditions uniquely determine ω to be an alternating tensor, which is exactly a k-form, by lemma 14.1.1.

16.2 Integration of Differential Forms

We begin by defining integrals of n-forms over some suitable subsets of $\mathbb{R}^n$. A **domain of integration** in $\mathbb{R}^n$ is a bounded subset $D \subset \mathbb{R}^n$ such that ∂D has measure zero. Let ω be a continuous n-form on $\overline{D}$, then we can write $\omega = f dx^1 \wedge \cdots \wedge dx^n$ for some continuous function $f : \overline{D} \to \mathbb{R}$. We define the integral of ω over D by

$$\int_D \omega = \int_D f dx^1 \cdots dx^n, \tag{16.1}$$

More generally, let U be an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, and ω be a compactly supported n-form on U, then we can define the integral of ω by

$$\int_U \omega = \int_D \omega,$$

where D is any domain of integration containing $\operatorname{supp}\omega$, and ω is extended by zero outside $\operatorname{supp}\omega$.

Proposition: **Invariance under Diffeomorphisms**

Let D, E are open domains of integration in $\mathbb{R}^n$ or $\mathbb{H}^n$, and $G : \overline{D} \to \overline{E}$ is a smooth map that restricts to an orientation-preserving or orientation-reversing diffeomorphism from D to E. Then for any continuous n-form ω on $\overline{E}$, we have

$$\int_D G^*\omega = \begin{cases} \displaystyle\int_E \omega & \text{if } G \text{ is orientation-preserving,} \\ \displaystyle -\int_E \omega & \text{if } G \text{ is orientation-reversing.} \end{cases} \tag{16.2}$$

If U, V are open subsets of $\mathbb{R}^n$ or $\mathbb{H}^n$, and $G : U \to V$ is an orientation-preserving or orientation-reversing diffeomorphism, then for any compactly supported n-form ω on V, we have

$$\int_U G^*\omega = \begin{cases} \displaystyle\int_V \omega & \text{if } G \text{ is orientation-preserving,} \\ \displaystyle -\int_V \omega & \text{if } G \text{ is orientation-reversing.} \end{cases} \tag{16.3}$$

Proof. Direct computation, let (y^i) be the standard coordinates on E and (x^i) be the standard coordinates on D, then we can write $\omega = f dy^1 \wedge \cdots \wedge dy^n$ for some continuous function $f : \overline{E} \to \mathbb{R}$. Then if G is orientation-preserving, we have

$$\int_E \omega = \int_E f dy^1 \cdots dy^n = \int_D (f \circ G)|\det DG| dx^1 \cdots dx^n = \int_D G^*\omega,$$

and if G is orientation-reversing, we have a negative sign when removing the absolute value. □

We would like to extend this to compactly supported n-forms on any open subsets.

Lemma 16.2.1: Integration Domain from Open Subset

Let U be an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, and $K \subseteq U$ is compact, then there exists an open domain of integration D such that $K \subseteq D \subseteq \overline{D} \subseteq U$.

Proof. Just take balls around each point in K and finite many covers K, then take the union of these balls as D. □

So for the second part of the proposition, we can just take a domain of integration D that $\operatorname{supp}\omega \subseteq D \subseteq \overline{D} \subseteq V$, then as diffeomorphisms are homeomorphisms, taking interiors to interiors, boundaries to boundaries, and zero measure sets to zero measure sets, so it takes domain of integration to domain of integration, thus $G^{-1}(D)$ is a domain of integration containing $\operatorname{supp} G^*\omega$, and we can apply the first part of the proposition to get the desired result.

16.2.1 Integration on Manifolds

Let M be an oriented smooth n-manifold, with or without boundary, and ω be an n-form on M. Suppose first that ω is compactly supported in the domain of a single coordinate chart (U, φ), that

is either positively or negatively oriented. Then we can define the integral of ω by

$$\int_M \omega = \pm \int_{\varphi(U)} (\varphi^{-1})^* \omega, \tag{16.4}$$

Proposition: **Chart Independence of Integration**

The above definition of integration does not depend on the choice of the chart.

Proof. Suppose (U, φ) and $(\tilde{U}, \tilde{\varphi})$ are two charts that $\operatorname{supp} \omega \subseteq U \cap \tilde{U}$, if both charts are positively oriented or both charts are negatively oriented, then $\tilde{\varphi} \circ \varphi^{-1}$ is an orientation-preserving diffeomorphism on $\varphi(U \cap \tilde{U})$, thus by proposition 16.2, we have

$$\begin{aligned}\int_{\tilde{\varphi}(\tilde{U})} (\tilde{\varphi}^{-1})^* \omega &= \int_{\tilde{\varphi}(U \cap \tilde{U})} (\tilde{\varphi}^{-1})^* \omega = \int_{\varphi(U \cap \tilde{U})} (\tilde{\varphi} \circ \varphi^{-1})^* (\varphi^{-1})^* \omega \\ &= \int_{\varphi(U \cap \tilde{U})} (\varphi^{-1})^* \tilde{\varphi}^* (\tilde{\varphi}^{-1})^* \omega = \int_{\varphi(U)} (\varphi^{-1})^* \omega.\end{aligned}$$

If one chart is positively oriented and the other chart is negatively oriented, then $\tilde{\varphi} \circ \varphi^{-1}$ is an orientation-reversing diffeomorphism on $\varphi(U \cap \tilde{U})$, thus a negative sign appears when applying proposition 16.2, and we still get the same result. □

To integrate over the entire manifold, we can use a partition of unity to break the integral into integrals over small pieces that are contained in single charts.

Definition 16.2.1: Integration of n-forms on Manifolds

Let M be an oriented smooth n-manifold, with or without boundary, and ω be a compactly supported n-form on M. Let $\{U_i\}$ be a **finite** open cover of $\operatorname{supp} \omega$ by smooth coordinate charts, (each chart is either positively or negatively oriented), and let $\{\psi_i\}$ be a smooth partition of unity subordinate to $\{U_i\}$, then define the integral of ω by

$$\int_M \omega = \sum_i \int_M \psi_i \omega, \tag{16.5}$$

For a 0-manifold, which is just a discrete set of points, we can define a compactly supported 0-form (a function f with only finitely many nonzero points) to be integrable, and define the integral of f by

$$\int_M f = \sum_{p \in M} \pm f(p), \tag{16.6}$$

where the sign is positive if p is positively oriented, and negative if p is negatively oriented.

Remark:
We allow to negatively orient charts because it may not be possible to find a positive oriented chart on a 1-manifold with boundary at the boundary points.

Proposition: **Independence of Integration on Manifolds**

The above definition of integration does not depend on the choice of the open cover and the partition of unity.

Proof. Suppose $\{U_i\}$ and $\{\tilde{U}_j\}$ are two finite open covers of $\operatorname{supp}\omega$ by smooth coordinate charts, and $\{\psi_i\}$ and $\{\tilde{\psi}_j\}$ are two smooth partitions of unity subordinate to $\{U_i\}$ and $\{\tilde{U}_j\}$ respectively, then we have

$$\int_M \psi_i\omega = \int_M \left(\sum_j \tilde{\psi}_j\right)\psi_i\omega = \sum_j \int_M \tilde{\psi}_j\psi_i\omega,$$

Summing over i on both sides, we get

$$\sum_i \int_M \psi_i\omega = \sum_i\sum_j \int_M \tilde{\psi}_j\psi_i\omega = \sum_j \int_M \tilde{\psi}_j\omega,$$

Both sums give the same value, thus the definition does not depend on the choice of the open cover and the partition of unity. □

Definition 16.2.2: Integration on Submanifolds

Let M be an oriented smooth n-manifold, with or without boundary, and $S \subseteq M$ be an immersed oriented smooth k-submanifold, with or without boundary, and ω be a k-form on M whose restriction to S has compact support, we intepret

$$\int_S \omega = \int_S \iota_S^*\omega, \tag{16.7}$$

where $\iota_S : S \hookrightarrow M$ is the inclusion map, the same goes for boundary, we always take the induced orientation on the boundary unless otherwise specified.

Proposition: **Properties of Integrals of Forms**

Let M, N be oriented smooth n-manifolds, with or without boundary, and ω, η be compactly supported n-forms on M, then we have,

- Linearity: $\forall a, b \in \mathbb{R}$, $\displaystyle\int_M (a\omega + b\eta) = a\int_M \omega + b\int_M \eta$.
- Orientation Reversal: If $-M$ is the manifold M with the opposite orientation, then $\displaystyle\int_{-M} \omega = -\int_M \omega$.
- Positivity: If ω is a positively oriented orientation form on M, then $\displaystyle\int_M \omega > 0$.
- Diffeomorphism Invariance: If $F : N \to M$ is an orientation-preserving or orientation-

reversing diffeomorphism, then

$$\int_M \omega = \begin{cases} \displaystyle\int_N F^*\omega & \text{if } F \text{ is orientation-preserving,} \\ \displaystyle -\int_N F^*\omega & \text{if } F \text{ is orientation-reversing.} \end{cases}$$

Although this definition by partition of unity is of great convenience in theoretical proofs, it is not very useful for actual computations. In practice, we usually break the integral into integrals over small pieces that are contained in single charts, and then sum them up.

Proposition: **Integration over Parametrizations**

Let M be an oriented smooth n-manifold, with or without boundary, and ω be a compactly supported n-form on M. Suppose $D_1, \ldots, D_k$ are open domains of integration in $\mathbb{R}^n$ and for each i, we are given smooth maps $F_i : \overline{D_i} \to M$ saitisfying

- $F_i|_{D_i}$ is an orientation-preserving diffeomorphism from D_i to its image $W_i \subseteq M$.
- $W_i \cap W_j = \varnothing$ for $i \neq j$.
- $\operatorname{supp} \omega \subseteq \bigcup_i \overline{W_i}$.

Then we have

$$\int_M \omega = \sum_i \int_{D_i} F_i^*\omega. \tag{16.8}$$

Proof. It suffices to assume ω is supported in a single oriented precompact chart (U, φ), then $Y = \varphi(U)$ is a domain of integration in $\mathbb{R}^n$ or $\mathbb{H}^n$, and φ extends to a diffeomorphism from $\overline{U}$ to $\overline{Y}$. (Just think U has its closure in a larger chart would do).

Then we now have two coordinate route: one with D_i as local parametrization, the other with Y as local structure of M. What we actually care about is the correspondence between D_i and partition of Y. Define

$$A_i = F_i^{-1}(U \cap W_i), \quad B_i = U \cap W_i, \quad C_i = \varphi(B_i),$$

Since $\overline{D_i}$ is compact, so $\partial W_i \subseteq F_i(\partial D_i)$, so ∂W_i has measure zero in M, and $\partial C_i = \varphi(\partial B_i)$ has measure zero in $\mathbb{R}^n$.

Now, in the coordinate chart (U, φ), we have $\operatorname{supp}(\varphi^{-1})^*\omega \subseteq \overline{C_1} \cup \cdots \cup \overline{C_k}$, and any two of C_i's intersect in a measure zero set, so we have

$$\int_M \omega = \int_Y (\varphi^{-1})^*\omega = \sum_i \int_{C_i} (\varphi^{-1})^*\omega = \sum_i \int_{A_i} (\varphi \circ F_i)^*(\varphi^{-1})^*\omega = \sum_i \int_{A_i} F_i^*\omega = \sum_i \int_{D_i} F_i^*\omega,$$

so we get the desired result. □

Remark:

Sometimes, the condition can be relaxed somewhat. For example, F_i is required to be smooth

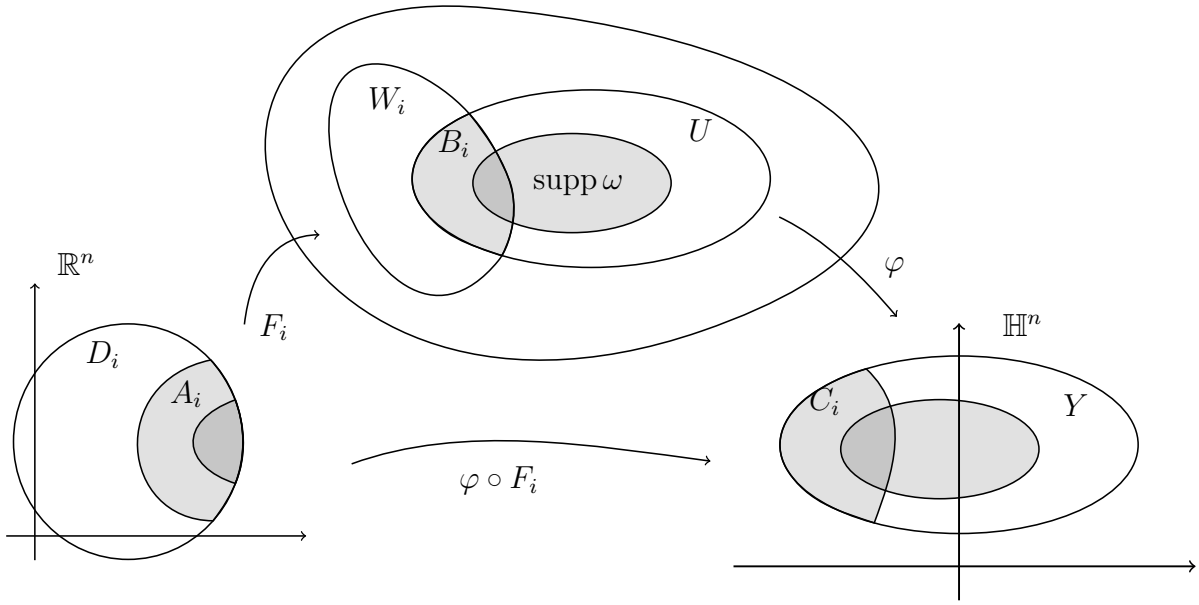

Figure 16.1: Integrating over Parametrizations

on $\overline{D_i}$ to ensure that ∂W_i has measure zero and the pullback $F_i^*\omega$ is continuous on $\overline{D_i}$. If we are given W_i filling up M except a measure zero set, we can allow F_i that do not extend smoothly to the boundary by interpreting the resulting integral as an improper integral or Lebesgue integral.
For example, the hemispheres can be parametrized by

$$F(u,v) = (u, v, \pm\sqrt{1-u^2-v^2}), \quad u^2+v^2<1,$$

which does not extend smoothly to the boundary, but the result still holds.

16.2.2 Integration on Lie Groups

Let G be a Lie group, a covariant tensor field A on G is called **left-invariant** if for each $g \in G$, we have $L_g^*A = A$.

Theorem 16.2.1: Haar Volume Form on Lie Groups

Let G be a compact Lie group with a left-invariant orientation. Then there exists a unique left-invariant n-form ω_G on G such that

$$\int_G \omega_G = 1.$$

Proof. If $\dim G = 0$, then we just let $\omega_G = 1/k$, where k is the number of points in G. If $\dim G > 0$, then let $E_1, \ldots, E_n$ be a left-invariant oriented global frame on G. Let $\epsilon^1, \ldots, \epsilon^n$ be

the dual coframe, then left-invariance of the frame implies that

$$(L_g^*\epsilon^i)(E_j) = \epsilon^i(L_{g*}E_j) = \epsilon^i(E_j) = \delta^i_j,$$

so ϵ^i is left-invariant for each i.

Let $\omega_G = \epsilon^1 \wedge \cdots \wedge \epsilon^n$, then ω_G is a left-invariant n-form on G. Then all the left-invariant orientation forms on G are of the form $c\omega_G$ for some constant $c > 0$.

If G is compact, then $\int_G \omega_G$ is a positive real number, so we can normalize ω_G to make the integral equal to 1. □

Remark:
Actually, every Lie group admits a left-invariant orientation from up to a positive scalar multiple, only in the compact case we can normalize it to make the integral equal to 1.

16.3 Stokes' Theorem

This is the ultimate summit of the theory of calculus on manifolds.

Theorem 16.3.1: Stokes' Theorem

Let M be an oriented smooth n-manifold with boundary, and ω be a compactly supported smooth $(n-1)$-form on M, then we have

$$\int_M \mathrm{d}\omega = \int_{\partial M} \iota^*_{\partial M}\omega, \tag{16.9}$$

where $\iota_{\partial M} : \partial M \hookrightarrow M$ is the inclusion map, and ∂M is given the induced orientation.

Proof. For $n = 1$, assume ω is supported in a single oriented chart (U, φ), then we have $\varphi(U) = (a, b)$ or $[0, a)$ or $(-a, 0]$ for some $a, b > 0$, and $\omega = f dx$ for some compactly supported smooth function $f : \varphi(U) \to \mathbb{R}$ that $\operatorname{supp} f \subseteq \varphi(U)$. Then,

- If $\varphi(U) = (a, b)$, then $\partial M \cap U = \varnothing$, so $\int_{\partial M} \iota^*_{\partial M}\omega = 0$, and $\int_M \mathrm{d}\omega = \int_a^b f'(x)dx = 0$.
- If $\varphi(U) = [0, a)$, then $\partial M \cap U = \varphi^{-1}(0)$, so $\int_{\partial M} \iota^*_{\partial M}\omega = f(0)$, and $\int_M \mathrm{d}\omega = \int_0^a f'(x)dx = f(a) - f(0) = -f(0)$. Same for $\varphi(U) = (-a, 0]$.

So the result holds for $n = 1$ by using a partition of unity to break the integral into integrals over small pieces that are contained in single charts.

For $n > 1$, we precede by a special case. Suppose $M = \mathbb{H}^n$, then as ω is compactly supported, we can assume that $\operatorname{supp}\omega \subseteq A = [-R, R]^{n-1} \times [0, R]$ for some $R > 0$. Then let

$$\omega = \sum_{i=1}^n \omega_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n,$$

where $\widehat{\mathrm{d}x^i}$ means that $\mathrm{d}x^i$ is omitted, and therefore, we have

$$\begin{aligned}\mathrm{d}\omega &= \sum_{i=1}^n \mathrm{d}\omega_i \wedge \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{\partial \omega_i}{\partial x^j} \mathrm{d}x^j \wedge \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n \\ &= \sum_{i=1}^n (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i} \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n,\end{aligned}$$

so we have

$$\begin{aligned}\int_{\mathbb{H}^n} \mathrm{d}\omega &= \sum_{i=1}^n (-1)^{i-1} \int_A \frac{\partial \omega_i}{\partial x^i} \mathrm{d}x^1 \cdots \mathrm{d}x^n \\ &= \sum_{i=1}^{n-1} (-1)^{i-1} \int_{[-R,R]^{n-2}\times[0,R]} \left(\int_{-R}^R \frac{\partial \omega_i}{\partial x^i} \mathrm{d}x^i \right) \mathrm{d}x^1 \cdots \widehat{\mathrm{d}x^i} \cdots \mathrm{d}x^n \\ &\quad + (-1)^{n-1} \int_{[-R,R]^{n-1}} \left(\int_0^R \frac{\partial \omega_n}{\partial x^n} \mathrm{d}x^n \right) \mathrm{d}x^1 \cdots \mathrm{d}x^{n-1} \\ &= (-1)^n \int_{[-R,R]^{n-1}} \omega_n(x^1, \ldots, x^{n-1}, 0) \mathrm{d}x^1 \cdots \mathrm{d}x^{n-1}\end{aligned}$$

on the other hand, we have

$$\begin{aligned}\int_{\partial\mathbb{H}^n} \iota^*_{\partial\mathbb{H}^n}\omega &= \sum_{i=1}^{n-1} \int_{A\cap\partial\mathbb{H}^n} \omega_i(x^1, \ldots, x^{n-1}, 0) \iota^*_{\partial\mathbb{H}^n} \left(\mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n \right) \\ &\quad + \int_{A\cap\partial\mathbb{H}^n} \omega_n(x^1, \ldots, x^{n-1}, 0) \iota^*_{\partial\mathbb{H}^n} (\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1}),\end{aligned}$$

As $\iota^*_{\partial\mathbb{H}^n}(\mathrm{d}x^n) = 0$, so the first sum vanishes, and as $\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1}$ is a positively oriented orientation form on $\partial\mathbb{H}^n$ when n is even and negatively oriented when n is odd, we have

$$\int_{\partial\mathbb{H}^n} \iota^*_{\partial\mathbb{H}^n}\omega = (-1)^n \int_{[-R,R]^{n-1}} \omega_n(x^1, \ldots, x^{n-1}, 0) \mathrm{d}x^1 \cdots \mathrm{d}x^{n-1},$$

If $M = \mathbb{R}^n$, then the same computation as above shows that both sides of the equation are zero, so the result holds.

For the general case when M is an oriented smooth n-manifold, with or without boundary, suppose ω is compactly supported in a single oriented chart (U, φ), then pulling back to the chart, we have the same situation as the above two special cases, so the result holds. For the general case, we can use a partition of unity ψ_i subordinate to a finite open cover U_i of $\operatorname{supp}\omega$ by oriented charts, then we have

$$\begin{aligned}\int_{\partial M} \iota^*_{\partial M}\omega &= \sum_i \int_{\partial M} \iota^*_{\partial M}(\psi_i\omega) = \sum_i \int_M \mathrm{d}(\psi_i\omega) = \sum_i \int_M \mathrm{d}\psi_i \wedge \omega + \psi_i \mathrm{d}\omega \\ &= \int_M \left(\sum_i \mathrm{d}\psi_i \right) \wedge \omega + \left(\sum_i \psi_i \right) \mathrm{d}\omega = \int_M \mathrm{d}\omega,\end{aligned}$$

Thus completing the proof. □

The Stokes' theorem has many important special cases and corollaries.

Proposition: **Integral of Exact Forms**

Let M be a compact oriented smooth n-manifold without boundary, and ω be an $(n-1)$-form on M, then we have

$$\int_M \mathrm{d}\omega = 0. \tag{16.10}$$

Proposition: **Integral of Closed Forms**

Let M be a compact oriented smooth n-manifold with boundary, and ω be a closed form on M, then we have

$$\int_{\partial M} \iota_{\partial M}^* \omega = 0, \qquad \text{if } \mathrm{d}\omega = 0 \text{ on } M \tag{16.11}$$

Usually, we just simply denote $\int_{\partial M} \omega$ instead of $\int_{\partial M} \iota_{\partial M}^* \omega$ when there is no confusion.

Corollary 16.3.1: Inverses of Stokes' Theorem

Suppose M is a smooth manifold, with or without boundary, $S \subseteq M$ is an oriented compact smooth k-dimensional submanifold without boundary, and ω is a closed k-form on M, then if $\int_S \omega \neq 0$, then

- ω is NOT exact on M.
- S is NOT the boundary of some oriented compact smooth submanifold without boundary in M.

Proof. The first is just definition, and the second is because if $S = \partial W$ for some oriented compact smooth $(k+1)$-dimensional submanifold without boundary W, then by Stokes' theorem, we have $\int_S \omega = \int_W \mathrm{d}\omega = 0$, which contradicts the assumption. □

Example: **Inverse of Stokes' Theorem**

Let $M = \mathbb{R}^2 \backslash \{0\}$, and $\omega = (x\mathrm{d}y - y\mathrm{d}x)/(x^2 + y^2)$, then ω is a closed 1-form on M. Let S be the unit circle in $\mathbb{R}^2$, then we have

$$\int_S \omega = \int_0^{2\pi} \frac{\cos^2 t + \sin^2 t}{\cos^2 t + \sin^2 t} dt = 2\pi \neq 0,$$

So ω is not exact on M, and S is not the boundary of some oriented compact smooth submanifold without boundary in M.

16.4 Manifolds with Corners

The Stokes' theorem can be further generalized to manifolds with corners, like squares, cubes, simplices, etc. They are topologically manifolds with boundary, but not smoothly manifolds with

boundary, as the boundary does not resemble $\mathbb{H}^n$ at the corner points.

Denote the space

$$\overline{\mathbb{R}}^n_+ = \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^i \geqslant 0 \text{ for each } i\},$$

all coordinates are nonnegative.

Definition 16.4.1: Smooth Manifolds with Corners

Suppose M is a topological n-dimensional manifold with boundary, a **chart with corners** for M is a pair (U, φ), where $U \subseteq M$ is open and φ is a hoemomorphism from U to an open subset $\hat{U} \subseteq \overline{\mathbb{R}}^n_+$ (relative to the subspace topology). Two charts with corners (U, φ) and (V, ψ) are **smoothly compatible** if $\psi \circ \varphi^{-1}$ is a smooth map.
A **smooth structure with corners** on M is a maximal collection of smoothly compatible interior charts and charts with corners whose domains cover M. A **smooth manifold with corners** is a topological manifold with boundary together with a smooth structure with corners.

In topology, there is no difference between manifolds with corners and manifolds with boundary, because $\overline{\mathbb{R}}^n_+$ is homeomorphic to $\mathbb{H}^n$ (while $\mathbb{H}^n$ is not homeomorphic to $\mathbb{R}^n$). The difference lies in the smooth structure, as as cannot transform a neighborhood of a corner point to a neighborhood of a boundary point smoothly. For $n = 1$ case, there is no difference as $\overline{\mathbb{R}}^1_+ = \mathbb{H}^1$.

The points of $\overline{\mathbb{R}}^n_+$ that more than one coordinate is zero are called **corner points**.

Proposition: **Invariance of Corner Points**

Let M be a smooth manifold with corners, $n \geqslant 2$. Let $p \in M$, and if $\varphi(p)$ is a corner point in some smooth chart with corners (U, φ), then every chart whose domain contains p is a chart with corners, and $\varphi(p)$ is a corner point in every such chart.

If M is a smooth manifold with corners, a point $p \in M$ is called a **corner point** if $\varphi(p)$ is a corner point in some (and therefore every) smooth chart with corners whose domain contains p. Similarly, p is called a **boundary point** if $\varphi(p) \in \partial\overline{\mathbb{R}}^n_+$ with respect to some (and therefore every) smooth chart with corners whose domain contains p.

However, unlike a smooth manifold with boundary, the boundary of a smooth manifold with corners is generally not a smooth manifold with corners, take the cube $[0, 1]^3$ as an example. However, to get Stokes' theorem working, we need to define the integration of a form on the boundary of a smooth manifold with corners:

Let M be an oriented smooth n-dimensional manifold with corners, and ω be an $(n-1)$-form on ∂M that is compactly supported in the domain of a single oriented smooth chart with corners (U, φ) in M. Locally looking, the boundary like $\partial\overline{\mathbb{R}}^n_+$ is a union of n hyperplanes,

$$\partial\overline{\mathbb{R}}^n_+ = \bigcup_{i=1}^n H_i, \qquad H_i = \{(x^1, \dots, x^n) \in \overline{\mathbb{R}}^n_+ : x^i = 0\},$$

Each H_i can be seen as a "boundary" of M, thus can be given the induced orientation, and we can define the integral of ω by

$$\int_{\partial M} \omega = \sum_{i=1}^n \int_{H_i} (\varphi^{-1})^* \omega, \tag{16.12}$$

Then for any compactly supported $(n-1)$-form ω on M, we can use a partition of unity to break the integral into integrals over small pieces that are contained in single charts with corners, and then sum them up.

The patching of parametrizations still work here, from proposition 16.2.1,

Proposition: **Integration over Parametrizations for Manifolds with Corners**

Let M be an oriented smooth $(n+1)$-manifold with corners, and ω be a compactly supported n-form on ∂M. Suppose $D_1, \ldots, D_k$ are open domains of integration in $\mathbb{R}^n$ and for each i, we are given smooth maps $F_i : \overline{D_i} \to M$ saitisfying

- $F_i|_{D_i}$ is an orientation-preserving diffeomorphism from D_i to its image $W_i \subseteq \partial M$.
- $W_i \cap W_j = \varnothing$ for $i \neq j$.
- $\operatorname{supp} \omega \subseteq \bigcup_i \overline{W_i}$.

Then we have

$$\int_{\partial M} \omega = \sum_i \int_{D_i} F_i^* \omega. \tag{16.13}$$

Theorem 16.4.1: Stokes' Theorem for Manifolds with Corners

Let M be an oriented smooth n-manifold with corners, and ω be a compactly supported smooth $(n-1)$-form on M, then we have

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega, \tag{16.14}$$

where $\iota_{\partial M} : \partial M \hookrightarrow M$ is the inclusion map, and ∂M is given the induced orientation.

Theorem 16.4.2: Path Homotopic Integration

Let M be a smooth manifold and $\gamma_0, \gamma_1 : [0,1] \to M$ be two path homotopic piecewise smooth curve segments. For any closed 1-form ω on M, we have

$$\int_{\gamma_0} \omega = \int_{\gamma_1} \omega. \tag{16.15}$$

Specially, on a simply connected smooth manifold, every closed 1-form is exact.

Proof. First suppose γ_0, γ_1 are smooth, then just using a smooth homotopy $H : [0,1] \times [0,1] \to M$ between γ_0 and γ_1, then we have

$$\int_{\partial(I \times I)} H^* \omega = \int_{I \times I} \mathrm{d}(H^* \omega) = \int_{I \times I} H^*(\mathrm{d}\omega) = 0,$$

Then expanding the left hand side, we get our result. The general case is easily generated by mere induction on the number of pieces. □

16.5 Integration on Riemannian Manifolds

16.5.1 Integration of Functions on Riemannian Manifolds

Definition 16.5.1: Integration of Functions on Riemannian Manifolds

Let (M, g) be an oriented Riemannian n-manifold, with or without boundary, denote ω_g be its Riemannian volume form. If f is a compactly supported continuous function on M, then $f\omega_g$ is a compactly supported n-form on M, so we can define the integral of f denoted by

$$\int_M f\omega_g = \int_M f \mathrm{d}V_g, \tag{16.16}$$

If M is compact, then we can define the volume of M by

$$\mathrm{Vol}(M) = \int_M \omega_g = \int_M \mathrm{d}V_g. \tag{16.17}$$

WARNING: The notation $\mathrm{d}V_g$ is just a common notation for ω_g, it does not mean that $\mathrm{d}V_g$ is the exterior derivative of some $(n-1)$-form V_g.

Proposition: **Positivity of Integrals of Functions on Riemannian Manifolds**

Let (M, g) be a nonempty oriented Riemannian n-manifold, with or without boundary, and suppose f is a compactly supported continuous function on M that $f \geqslant 0$ on M, then $\int_M f \mathrm{d}V_g \geqslant 0$, and equality holds if and only if f is identically zero on M.

Proof. Use partition of unity to break down. If f is compactly supported in a chart (U, φ), then we have

$$\int_M f \mathrm{d}V_g = \int_{\varphi(U)} f(x)\sqrt{\det(g_{ij}(x))}\mathrm{d}x^1 \cdots \mathrm{d}x^n \geqslant 0.$$

□

Proposition: **The Triangle Inequality for Integrals of Functions**

Let (M, g) be a nonempty oriented Riemannian n-manifold, with or without boundary, and suppose f is a compactly supported continuous function on M, then we have

$$\left|\int_M f \mathrm{d}V_g\right| \leqslant \int_M |f| \mathrm{d}V_g. \tag{16.18}$$

16.5.2 The Hodge Star Operator

We have already given an operator $* : C^\infty(M) \to \Omega^n(M)$ in a Riemannian manifold before, now here is an important generalization.

Let (M, g) be an oriented Riemannian n-manifold, with or without boundary. For each $k = 0, \ldots, n$, g determines a unique inner product $\langle \cdot, \cdot \rangle_g$ on the space $\Lambda^k(T_p^*M)$ of k-forms satisfying

$$\langle \omega^1 \wedge \cdots \wedge \omega^k, \eta^1 \wedge \cdots \wedge \eta^k \rangle_g = \det(\langle (\omega^i)^\sharp, (\eta^j)^\sharp \rangle_g).$$

when $\omega^1, \ldots, \omega^k$ and $\eta^1, \ldots, \eta^k$ are covectors at p.

Specially, when $k = 0$, $\Lambda^0(T_p^*M) = \mathbb{R}$, so interpret the inner product to be real multiplication.

Remark:
For $k = 1$, the inner product $\langle \cdot, \cdot \rangle_g$ on T_p^*M is just the dual inner product of the inner product $\langle \cdot, \cdot \rangle_g$ on T_pM.
In general case, we can visualize the inner product as the inner product of the two linear subspaces of T_pM that the two k-forms are dual to, that is, the product of the volumes of the two subspaces and the cosine of the angle between them. To prove this, just verify the formula for the inner product on the basis $\epsilon^{i_1} \wedge \cdots \wedge \epsilon^{i_k}$.

Proposition: **Normalization of Full Rank Forms**

Let (M, g) be an oriented Riemannian n-manifold, with or without boundary, then the Riemannian volume form ω_g (or $\mathrm{d}V_g$) is the unique n-form on M such that $\langle \omega_g, \omega_g \rangle_g = 1$.

Proof. Quite straightforward as there is only one dimension. □

Theorem 16.5.1: The Hodge Star Operator

Let (M, g) be an oriented Riemannian n-manifold, with or without boundary, then for each $k = 0, \ldots, n$, there exists a unique smooth bundle homomorphism $* : \Lambda^k T^*M \to \Lambda^{n-k} T^*M$ such that

$$\omega \wedge *\eta = \langle \omega, \eta \rangle_g \mathrm{d}V_g, \tag{16.19}$$

for any $\omega, \eta \in \Lambda^k T_p^*M$ and $p \in M$.

Proof. First, we show uniqueness. Suppose $*$ and $*'$ are two bundle homomorphisms satisfying

$$\omega \wedge *\eta = \langle \omega, \eta \rangle_g \, \mathrm{d}V_g = \omega \wedge *'\eta$$

for all $\omega, \eta \in \Lambda^k T_p^*M$. Then $\omega \wedge (*\eta - *'\eta) = 0$ for every ω. Fix η and consider the linear map $L : \Lambda^k T_p^*M \to \Lambda^n T_p^*M$ given by $L(\omega) = \omega \wedge (*\eta - *'\eta)$. Since this vanishes for all ω, and the wedge product pairing $\Lambda^k \times \Lambda^{n-k} \to \Lambda^n$ is nondegenerate (if $\alpha \in \Lambda^{n-k}$ satisfies $\omega \wedge \alpha = 0$ for all ω, then $\alpha = 0$), we conclude $*\eta - *'\eta = 0$. Thus $* = *'$ pointwise, and hence globally.

For existence, we construct $*$ locally. Let $p \in M$ and choose an oriented orthonormal basis $(e_1, \ldots, e_n)$ of T_pM with dual coframe $(\varepsilon^1, \ldots, \varepsilon^n)$. Then the Riemannian volume form is $\mathrm{d}V_g = \varepsilon^1 \wedge \cdots \wedge \varepsilon^n$. For any increasing multi-index $I = (i_1 < \cdots < i_k)$, set $\varepsilon^I = \varepsilon^{i_1} \wedge \cdots \wedge \varepsilon^{i_k}$. Let $J = (j_1 < \cdots < j_{n-k})$ be the complementary increasing multi-index. Define

$$*(\varepsilon^I) = \varepsilon^J \quad \text{if the permutation } (i_1, \ldots, i_k, j_1, \ldots, j_{n-k}) \text{ is even, and } -\varepsilon^J \text{ if odd.}$$

Equivalently, the sign is chosen so that $\varepsilon^I \wedge *\varepsilon^I = \mathrm{d}V_g$. Extend $*$ linearly to all of $\Lambda^k T_p^* M$. Then for any two basis forms ε^I and ε^K, we have

$$\varepsilon^I \wedge *\varepsilon^K = \begin{cases} \mathrm{d}V_g & \text{if } I = K, \\ 0 & \text{otherwise.} \end{cases}$$

On the other hand, because the ε^i are orthonormal, the inner product on forms satisfies $\langle \varepsilon^I, \varepsilon^K \rangle_g = \delta_{IK}$. Hence

$$\varepsilon^I \wedge *\varepsilon^K = \langle \varepsilon^I, \varepsilon^K \rangle_g \, \mathrm{d}V_g.$$

By bilinearity, for any $\omega, \eta \in \Lambda^k T_p^* M$ we obtain $\omega \wedge *\eta = \langle \omega, \eta \rangle_g \, \mathrm{d}V_g$.

This construction at each point is independent of the choice of oriented orthonormal basis because the inner product and orientation are fixed; a change of basis preserves the definition (the sign is consistent). Moreover, by using a smooth local orthonormal frame (which exists via Gram–Schmidt), we obtain a smooth bundle homomorphism. Thus $*$ exists globally. □

Proposition: **Properties of the Hodge Star Operator**

- $* : \Lambda^0 T^* M \to \Lambda^n T^* M$ is given by $*(f) = f \mathrm{d}V_g$ for any $f \in C^\infty(M)$.
- If ω is a k-form, then $* * \omega = (-1)^{k(n-k)} \omega$.

On Euclidean spaces, we can use the Hodge star operator to rewrite the classical vector calculus operators in terms of differential forms.

16.5.3 The Divergence Theorem

Let (M, g) be an oriented Riemannian n-manifold with or without boundary. In previous chapters, we have defined smooth bundle homomorphisms

$$* : C^\infty(M) \to \Omega^n(M), \qquad *f = f \mathrm{d}V_g,$$

and also

$$\beta : \mathfrak{X}(M) \to \Omega^{n-1}(M), \qquad \beta(X) = X \lrcorner \, \mathrm{d}V_g,$$

writing by Hodge star operator, we have $\beta(X) = *(X^\flat)$, where $X^\flat$ is the 1-form dual to X under the Riemannian metric.

Lemma 16.5.1: Pullback over Immersed Hypersurfaces

Let (M, g) be an oriented Riemannian n-manifold with or without boundary, and $S \subseteq M$ is an immersed hypersurface with the induced orientation by a unit normal vector field N, and $\tilde{g}$ is the induced Riemannian metric on S, then for any vector field X along S, we have

$$\iota_S^* \beta(X) = \langle X, N \rangle_g \mathrm{d}V_{\tilde{g}}, \tag{16.20}$$

where $\iota_S : S \hookrightarrow M$ is the inclusion map.

Geometrically, $\beta(X)$ is the tangent hyperplane flow of X, and $\iota_S^* \beta(X)$ is the projection of the tangent hyperplane flow of X onto S, which is just the normal component of X times the volume form on S.

Definition 16.5.2: The Divergence of a Vector Field

Let (M, g) be an oriented Riemannian n-manifold with or without boundary, and $X \in \mathfrak{X}(M)$ be a smooth vector field on M, then the **divergence** of X is defined as $\operatorname{div} : \mathfrak{X}(M) \to C^\infty(M)$ given by

$$\operatorname{div} X = *^{-1} \mathrm{d}\beta(X) = * \mathrm{d} * (X^\flat). \tag{16.21}$$

Or equivalently,

$$\mathrm{d}\beta(X) = \mathrm{d} * (X^\flat) = (\operatorname{div} X) \mathrm{d}V_g = *(\operatorname{div} X). \tag{16.22}$$

Actually, $* = *^{-1}$ for 0-forms and n-forms, so the first formula holds.

Theorem 16.5.2: The Divergence Theorem

Let (M, g) be an oriented Riemannian n-manifold with boundary, and $X \in \mathfrak{X}(M)$ be a compactly supported smooth vector field on M, then we have

$$\int_M \operatorname{div} X \mathrm{d}V_g = \int_{\partial M} \langle X, N \rangle_g \mathrm{d}V_{\tilde{g}}, \tag{16.23}$$

where $\tilde{g}$ is the induced Riemannian metric on ∂M, and N is the outward unit normal vector field along ∂M.

Proof. This is equivalent to

$$\int_M \mathrm{d} * (X^\flat) = \int_{\partial M} \iota_{\partial M}^* * (X^\flat),$$

which is just Stokes' theorem for the $(n-1)$-form $*(X^\flat)$. □

The name divergence comes from the following geometric interpretation. A smooth flow θ on M is called **volume-preserving** if for every compact regular domain D, we have $\operatorname{Vol}(\theta_t(D)) = \operatorname{Vol}(D)$ whenever θ_t is defined on D.

Proposition: **Geometric Interpretation of Divergence**

Let (M, g) be an oriented Riemannian n-manifold, and let θ be the smooth flow generated by a smooth vector field $X \in \mathfrak{X}(M)$, then θ is

- volume-preserving iff $\operatorname{div} X = 0$ on M.
- volume-increasing iff $\operatorname{div} X > 0$ on M.
- volume-decreasing iff $\operatorname{div} X < 0$ on M.

Proof. We have θ_t being an orientation-preserving diffeomorphism, so we have

$$\operatorname{Vol}(\theta_t(D)) = \int_{\theta_t(D)} \mathrm{d}V_g = \int_D \theta_t^*(\mathrm{d}V_g).$$

As the right hand integrand depends smoothly on t, p, we can differentiate under the integral sign (using a partition of unity to rigorously justify this), and we have by Cartan's magic formula,

$$\mathcal{L}_X \omega_g = X \lrcorner\, \omega_g + \mathrm{d}(X \lrcorner\, \omega_g) = \operatorname{div} X\, \omega_g = \operatorname{div} X \mathrm{d}V_g,$$

so we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=t_0}\mathrm{Vol}(\theta_t(D)) &= \int_D \frac{\partial}{\partial t}\Big|_{t=t_0}\theta_t^*(\mathrm{d}V_g) = \int_D \theta_{t_0}^*(\mathcal{L}_X \mathrm{d}V_g) = \int_D \theta_{t_0}^*(\operatorname{div} X \mathrm{d}V_g)\\ &= \int_{\theta_{t_0}(D)} \operatorname{div} X \mathrm{d}V_g,\end{aligned}$$

Thus we have our result. □

16.5.4 Surface Integrals

Now, we come to the original curl operator in vector calculus.

Definition 16.5.3: The Curl Operator

Let (M, g) be an oriented Riemannian 3-manifold, with or without boundary, and $X \in \mathfrak{X}(M)$ be a smooth vector field on M, then the **curl** of X is defined as $\operatorname{curl} : \mathfrak{X}(M) \to \mathfrak{X}(M)$ given by

$$\operatorname{curl} X = \beta^{-1}\mathrm{d}(X^\flat) = (*\,\mathrm{d}X^\flat)^\sharp, \tag{16.24}$$

or equivalently,

$$(\operatorname{curl} X)\lrcorner\, \mathrm{d}V_g = \mathrm{d}(X^\flat).$$

Now, if $S \subseteq M$ is a compact 2-dimensional submanifold with or without boundary, and N is a smooth unit normal vector field along S, then denote $\mathrm{d}A$ to be the Riemannian volume form on S induced by the Riemannian metric g and the orientation given by N, that is,

$$\mathrm{d}A = \iota_S^*(N\lrcorner\, \mathrm{d}V_g)$$

Then the surface integral of a smooth vector field X on M over S is defined by

$$\int_S \langle X, N\rangle_g \mathrm{d}A \tag{16.25}$$

Theorem 16.5.3: Stokes' Surface Integral Theorem

Let (M, g) be an oriented Riemannian 3-manifold with or without boundary, and $S \subseteq M$ be a compact oriented smooth 2-dimensional submanifold with boundary, and N be the unit normal vector field along S that induces the orientation on S, and T be the unit tangent vector field along ∂S that is positively oriented with respect to the orientation on ∂S, then we have

$$\int_S \langle \operatorname{curl} X, N\rangle_g \mathrm{d}A = \int_{\partial S} \langle X, T\rangle_g \mathrm{d}s, \tag{16.26}$$

Proof. Three identities:

$$\int_S \mathrm{d}X^\flat = \int_{\partial S} \iota_{\partial S}^* X^\flat, \qquad \iota_S^*\mathrm{d}X^\flat = \langle \operatorname{curl} X, N\rangle_g \mathrm{d}A, \qquad \iota_{\partial S}^* X^\flat = \langle X, T\rangle_g \mathrm{d}s,$$

where the first one is just Stokes' theorem, the second one is by the definition of curl and the pullback over immersed hypersurfaces 16.5.1, and the third one is proved by assuming $\iota_{\partial S}^* X^\flat = f\mathrm{d}s$ for some function f, then we have

$$f = f\mathrm{d}s(T) = X^\flat(T) = \langle X, T\rangle_g,$$

thus we have our result. □

16.6 Densities

There is a way to define integration on nonorientable manifolds as well. The problem with differential forms is that the coordinate transformations of a differential form involves the Jacobian determinant of the transformation map, while the transformation law for integrals involves the absolute value of the Jacobian determinant, so we need to restrict the transition maps to be orientation-preserving to get a well-defined integration of forms (invariant under orientation-preserving diffeomorphisms).

Now, we may define objects that transform under the absolute value of the Jacobian determinant, for which orientation is not a problem.

Definition 16.6.1: Densities

Let V be an n-dimensional real vector space. A density on V is a function

$$\nu : V^n \to \mathbb{R} \tag{16.27}$$

satisfying that if $T : V \to V$ is a linear transformation, then

$$\mu(Tv_1, \ldots, Tv_n) = |\det T|\mu(v_1, \ldots, v_n), \tag{16.28}$$

Note that densities are not linear, so they are not elements of the tensor algebra of V, but they have certain relations. Denote $\mathcal{D}(V)$ to be the space of densities on V.

Proposition: **Properties of Densities**

Let V be an $n \geqslant 1$-dimensional real vector space.

- $\mathcal{D}(V)$ is a real vector space under pointwise addition and scalar multiplication:

$$(c_1\mu_1 + c_2\mu_2)(v_1, \ldots, v_n) = c_1\mu_1(v_1, \ldots, v_n) + c_2\mu_2(v_1, \ldots, v_n).$$

- If $\mu_1, \mu_2 \in \mathcal{D}(V)$ and $\mu_1(E_1, \ldots, E_n) = \mu_2(E_1, \ldots, E_n)$ for some basis $E_1, \ldots, E_n$ of V, then $\mu_1 = \mu_2$.
- If $\omega \in \Lambda^n V^*$ is a nonzero n-form, then $|\omega| : V^n \to \mathbb{R}$ given by $|\omega|(v_1, \ldots, v_n) = |\omega(v_1, \ldots, v_n)|$ is a density on V.
- $\mathcal{D}(V)$ is a 1-dimensional real vector space, spanned by any nonzero $|\omega|$ for $\omega \in \Lambda^n V^*$.

A **positive density** is a density μ such that $\mu(v_1, \ldots, v_n) > 0$ for any linear independent $v_1, \ldots, v_n$. A negative density is defined similarly. It is easy to see that a density is either positive, negative, or identically zero. For nonzero $\omega \in \Lambda^n V^*$, $|\omega|$ is a positive density, and the positive densities are exactly the positive scalar multiples of $|\omega|$, thus a convex subset of $\mathcal{D}(V)$, (a half line).

Now, if M is a smooth manifold, with or without boundary, then the set

$$\mathcal{D}M = \bigsqcup_{p \in M} \mathcal{D}(T_pM) \tag{16.29}$$

is called the **density bundle** of M.

Proposition: **Structure of Density Bundles**

Let M be a smooth manifold, with or without boundary, then $\mathcal{D}M$ is a smooth line bundle over M, and the transition functions of $\mathcal{D}M$ are given by the absolute value of the Jacobian determinant of the transition maps of M.

If M is a smooth n-manifold, with or without boundary, then a **density (field)** on M is a smooth section of the density bundle $\mathcal{D}M$.

Proposition: **Existence of Positive Densities**

Let M be a smooth manifold, with or without boundary, then there exists a positive density on M.

Proof. Because the set of positive densities of M in $\mathcal{D}M$ is an open subset whose intersection with each fiber is a half line (convex), so we can use a partition of unity to construct a global positive density from local positive densities. □

Remark:
Unlike forms whose positivity requires an orientation, the positivity of densities is invariant under all diffeomorphisms.

Under smooth maps, densities pull back just like forms. If $F : M \to N$ is a smooth map between smooth n-manifolds, with or without boundary, and μ is a density on N, then the pullback $F^*\mu$ is defined by

$$(F^*\mu)_p(v_1, \ldots, v_n) = \mu_{F(p)}(\mathrm{d}F_p(v_1), \ldots, \mathrm{d}F_p(v_n)),$$

Proposition: **Properties of Pullback of Densities**

Let $G : P \to M$ and $F : M \to N$ be smooth maps between smooth n-manifolds, with or without boundary, and μ be a density on N, then we have

- For any $f \in C^\infty(N)$, we have $F^*(f\mu) = (f \circ F)F^*\mu$.
- If ω is an n-form on N, then $F^*|\omega| = |F^*\omega|$.
- If μ is smooth, then $F^*\mu$ is smooth.
- $(F \circ G)^*\mu = G^*(F^*\mu)$.

Proposition: **Pullback of Densities in Coordinates**

Suppose $F : M \to N$ is a smooth map between smooth n-manifolds, with or without boundary, and (x^i) and (y^i) are local smooth coordinates on open sets $U \subseteq M$ and $V \subseteq N$ respectively, and u is a continuous function on V then on $F^{-1}(V) \cap U$, we have

$$F^*(u|\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n|) = (u \circ F)|\det(\partial F^i/\partial x^j)||\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n|. \tag{16.30}$$

Now for integration. First consider the case when $M = \mathbb{R}^n$. If D is a domain of integration and μ is a density on $\overline{D}$, then we can write $\mu = f|\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n|$ for some continuous function $f : \overline{D} \to \mathbb{R}$, and we can define the integral of μ by

$$\int_D \mu = \int_D f \mathrm{d}x^1 \cdots \mathrm{d}x^n, \tag{16.31}$$

Similarly, if $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$ is an open set and μ is compactly supported on U, then we can define

$$\int_U \mu = \int_D \mu, \tag{16.32}$$

where D is any domain of integration containing $\operatorname{supp} \mu$. This definition is diffeomorphism invariant.

Proposition: **Diffeomorphism Invariance of Integration of Densities**

Let U, V be open sets in $\mathbb{R}^n$ or $\mathbb{H}^n$, and $G : U \to V$ be a diffeomorphism. If μ is a compactly supported density on V, then we have

$$\int_V \mu = \int_U G^* \mu.$$

Now let M be a smooth n-manifold, with or without boundary. If μ is a density on M that is compactly supported in a chart (U, φ), then we can define

$$\int_M \mu = \int_{\varphi(U)} \varphi^* \mu,$$

For arbitrary compactly supported density μ on M, we can use a partition of unity to break down μ into a sum of compactly supported densities that are supported in charts, and define the integral of μ by

$$\int_M \mu = \sum_i \int_M \psi_i \mu$$

where $\{\psi_i\}$ is a partition of unity subordinate to an open cover of M by smooth charts.

Proposition: **Properties of Integration of Densities**

Let M, N be smooth n-manifolds, with or without boundary, and μ, η be compactly supported densities on M, then

- Linearity: $\int_M (c_1\mu + c_2\eta) = c_1 \int_M \mu + c_2 \int_M \eta$ for any $c_1, c_2 \in \mathbb{R}$.
- Positivity: If μ is a positive density on M, then $\int_M \mu > 0$.
- Diffeomorphism invariance: If $F : M \to N$ is a diffeomorphism, then $\int_N \mu = \int_M F^*\mu$.

We can also calculate the integral of a density in coordinates by cutting it into pieces like proposition 16.2.1 and 16.4.

Proposition: **Integration of Densities over Parametrizations**

Let M be a smooth n-manifold, with or without boundary, and μ be a compactly supported density on M. Suppose $D_1, \ldots, D_k$ are open domains of integration in $\mathbb{R}^n$ and for each i, we are given smooth maps $F_i : \overline{D_i} \to M$ saitisfying

- $F_i|_{D_i}$ is a diffeomorphism from D_i to its image $W_i \subseteq M$.
- $W_i \cap W_j = \varnothing$ for $i \neq j$.
- $\operatorname{supp} \mu \subseteq \bigcup_i \overline{W_i}$.

Then we have

$$\int_M \mu = \sum_i \int_{D_i} F_i^* \mu. \tag{16.33}$$

16.6.1 The Riemannian Density

Densities are particularly useful in Riemannian geometry.

Proposition: **Riemannian Density**

Let (M, g) be a smooth Riemannian n-manifold, with or without boundary, then there is a unique smooth positive density μ_g on M called the **Riemannian density** of g such that for any local orthonormal frame $(E_1, \ldots, E_n)$ of TM, we have

$$\mu_g(E_1, \ldots, E_n) = 1.$$

Actually, $\mu_g = |\omega_g|$, where ω_g is the Riemannian volume form of g.

The Riemannian density is invariant under isometries: Suppose $F : (M, g) \to (\tilde{M}, \tilde{g})$ is an isometry between Riemannian manifolds with or without boundary, then $F^*\mu_{\tilde{g}} = \mu_g$.

Sometimes we simply denote $\mathrm{d}V_g$ to be the Riemannian density μ_g. If $f : M \to \mathbb{R}$ is a compactly supported continuous function, then we can define the integral of f by

$$\int_M f \mathrm{d}V_g = \int_M f\mu_g,$$

When M is oriented, it does not matter whether we use the Riemannian volume form ω_g or the Riemannian density μ_g to define the integral of f, because even when the orientation is reversed, both the integral and ω_g change sign, so there is no change for the result. When M is nonorientable, we can only use the Riemannian density μ_g to define the integral of f, and this definition is invariant under all diffeomorphisms.

It is fairly surprising that we actually can generalize the divergence theorem to nonorientable Riemannian manifolds.

Theorem 16.6.1: Divergence Theorem for Nonorientable Riemannian Manifolds

Let (M, g) be a nonorientable Riemannian n-manifold with boundary, and $X \in \mathfrak{X}(M)$ be a compactly supported smooth vector field on M, then we have

$$\int_M \operatorname{div} X \mu_g = \int_{\partial M} \langle X, N \rangle_g \mu_{\tilde{g}}, \tag{16.34}$$

where N is an outward unit normal vector field along ∂M, and $\tilde{g}$ is the induced Riemannian metric on ∂M.

Proof. Let $\hat{\pi} : \hat{M} \to M$ be the orientation double cover of M, then it restricts to a generalized smooth covering map $\hat{\pi} : \partial \hat{M} \to \partial M$. Let $\square$

Chapter 17

Symplectic Manifolds

A symplectic structure is a new kind of geometric structure that can be put on a smooth manifold.

17.1 Symplectic Tensors

A 2-covector ω on on a finite dimensional vector space V defines a linear map $\tilde{\omega} : V \to V^*$ by $\tilde{\omega}(v)(w) = \omega(v, w)$. If ω is nondegenerate, then $\tilde{\omega}$ is invertible (and vice versa), which is also equivalent of saying that ω has nonsingular matrix representation in some and hence every basis.

Definition 17.1.1: Symplectic Tensor

A **symplectic tensor** on a finite dimensional vector space V is a nondegenerate 2-covector.

Example: **Symplectic Tensor**

Let V be a $2n$-dimensional vector space with basis denoted by $(A_1, B_1, \ldots, A_n, B_n)$ and its dual $(\alpha^1, \beta^1, \ldots, \alpha^n, \beta^n)$. Define a 2-covector ω by

$$\omega = \sum_{i=1}^{n} \alpha^i \wedge \beta^i. \tag{17.1}$$

Then $\omega(A_i, B_j) = -\omega(B_j, A_i) = \delta_{ij}$ and $\omega(A_i, A_j) = \omega(B_i, B_j) = 0$ for all i, j. If for some $u \in V$, we have $\omega(u, v) = 0$ for all $v \in V$, then $v = a^i A_i + b^i B_i$ and $\omega(v, A_j) = -b^j, \omega(v, B_j) = a^j$ for all j. Hence $a^i = b^i = 0$ for all i, which means $u = 0$. Therefore ω is nondegenerate and hence a symplectic tensor.
when $n = 1, \dim V = 2$, then every nonzero 2-covector is a symplectic tensor.

Definition 17.1.2: Symplectic Complement

Let (V, ω) be a symplectic vector space and $S \subseteq V$ be a linear subspace. The **symplectic complement** of S is the linear subspace

$$S^\perp = \{v \in V : \omega(v, w) = 0 \text{ for all } w \in S\}. \tag{17.2}$$

Lemma 17.1.1: Dimension Theorem for Symplectic Complement

Let (V, ω) be a symplectic vector space and $S \subseteq V$ be a linear subspace. Then $\dim S + \dim S^\perp = \dim V$. And also $S^{\perp\perp} = S$.

Proof. Define a linear map $\Phi : V \to S^*$ by $\Phi(v)(w) = \omega(v, w)$ for all $v \in V$ and $w \in S$. Then $\ker \Phi = S^\perp$. For any $\varphi \in S^*$, we can extend φ to a linear functional $\tilde{\varphi}$ on V. Since $\tilde{\omega} : V \to V^*$ is an isomorphism, there exists a unique $v \in V$ such that $\tilde{\omega}(v) = v \lrcorner\, \omega = \tilde{\varphi}$. Then Φ has full rank and hence $\dim S^* = \dim V - \dim S^\perp$. Since $\dim S^* = \dim S$, we have $\dim S + \dim S^\perp = \dim V$. □

Remark:
There is a more intuitive visulization. Writing ω in matrix form, left taking a row vector v and right taking a column vector w, then $v\omega w$ is the value of $\omega(v, w)$, and the (i, j)-entry of ω is $\omega(e_i, e_j)$ where (e_i) is the standard basis. Then ω is invertible anti-symmetric. For the lemma, take w to be a basis for S, then ωv has the same dimension as S and $S^\perp$ is spanned by the row vectors v such that $v(\omega w) = 0$, which is just the orthogonal complement of the column space of ωw.
We can see that this is quite different from the case of inner product spaces, where the inner product is represented by a symmetric matrix.

A major difference between symplectic vector spaces and inner product spaces is that $S \cap S^\perp$ is not necessarily $\{0\}$ for a symplectic vector space, while it is always $\{0\}$ for an inner product space. In fact, we have the following classification of subspaces of a symplectic vector space.

Definition 17.1.3: Subspace Classification of Symplectic Vector Space

Let (V, ω) be a symplectic vector space and $S \subseteq V$ be a linear subspace. Then

- S is **symplectic** if $S \cap S^\perp = \{0\}$, which is equivalent to saying that $\omega|_S$ is nondegenerate.
- S is **isotropic** if $S \subseteq S^\perp$, which is equivalent to saying that $\omega|_S = 0$.
- S is **coisotropic** if $S^\perp \subseteq S$, which is equivalent to saying that $S^\perp$ is isotropic.
- S is **Lagrangian** if $S = S^\perp$, which is equivalent to saying that S is both isotropic and coisotropic, or equivalently $\omega|_S = 0$ and $\dim S = \frac{1}{2} \dim V$.

In matrix form, we assume S is spanned by the first k standard basis vectors. Then ω can be written in block form as

$$\omega = \begin{pmatrix} A & B \\ -B^T & C \end{pmatrix}, \qquad \omega(v, w) = v_1^T A w_1 + v_1^T B w_2 - v_2^T B^T w_1 + v_2^T C w_2 \tag{17.3}$$

where A, C are anti-symmetric. When $w \in S$, we have $w_2 = 0$ and hence $\omega(v, w) = v_1^T A w_1 - v_2^T B^T w_1$. Then:

$$S^\perp = \{(v_1, v_2) : A v_1 + B v_2 = 0\}.$$

- S is symplectic if and only if A is invertible.
- S is isotropic if and only if $A = 0$.

- S is coisotropic if and only if B is injective and $\operatorname{range} A \cap \operatorname{range} B = \{0\}$.
- S is Lagrangian if and only if $A = 0$ and B is injective.

Proof.
- S is symplectic if and only if $(v_1, 0)$ does not satisfy $Av_1 + B0 = Av_1 = 0$ for any nonzero v_1, which is equivalent to saying that A is invertible.
- S is isotropic if and only if all $(v_1, 0)$ satisfy $Av_1 + B0 = Av_1 = 0$, which is equivalent to saying that $A = 0$.
- S is coisotropic if and only if all (v_1, v_2) satisfying $Av_1 + Bv_2 = 0$ would imply $v_2 = 0$. So if B is not injective, then take $v_1 = 0$ would give a nonzero $(0, v_2)$ in $S^\perp$ but not in S, so B must be injective. And also, this means that $\operatorname{range} A \cap \operatorname{range} B = \{0\}$. Conversely, it is easy to see.
- S is Lagrangian if and only if $A = 0$ and B is injective and $\operatorname{range} A \cap \operatorname{range} B = \{0\}$, which is equivalent to saying that $A = 0$ and B is injective.

□

Example: **Symplectic Subspace Classification**

Let (V, ω) be a symplectic vector space of dimension $2n$ defined in the previous example. Then for all $k = 0, 1, \ldots, n$,

- the subspace $\operatorname{span}\{A_1, B_1, \ldots, A_k, B_k\}$ is symplectic;
- the subspace $\operatorname{span}\{A_1, A_2, \ldots, A_k\}$ is isotropic;
- the subspace $\operatorname{span}\{A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_k\}$ is coisotropic;
- the subspace $\operatorname{span}\{A_1, A_2, \ldots, A_n\}$ is Lagrangian.

Proof. The matrix representation of ω with respect to the standard basis $(A_1, B_1, \ldots, A_n, B_n)$ is

$$\omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

□

Proposition: **The Canonical Form of Symplectic Tensors**

Let ω be a symplectic tensor on an m-dimensional vector space V. Then V has even dimension $2n$ and there exists a basis of V with respect to which ω has the form

$$\omega = \sum_{i=1}^{n} \alpha^i \wedge \beta^i \tag{17.4}$$

Proof. Writing in matrix form, let $\omega(u, w) = u^T \Omega w$ for some anti-symmetric matrix Ω. From real Schur decomposition, there exists an orthogonal matrix P such that $P^T \Omega P$ is block diagonal with 1×1 blocks and 2×2 blocks of the form $\begin{pmatrix} 0 & a_i \\ -a_i & 0 \end{pmatrix}$. Since Ω is antisymmetric, the 1×1 blocks

must be zero. Since Ω is invertible, there are no zero blocks. Hence Ω has even dimension and is block diagonal with 2×2 blocks of the form above. By rescaling the basis, we can make $a_i = 1$ for all i. Then the new basis is the desired one.

There is also a more geometric proof by induction. We induct on $m = \dim V$. If $m = 0$, then the statement is trivial. If $m > 0$, then suppose the statement holds for all dimensions less than m. Take any $A_1 \in V$, then there exists $B_1 \in V$ such that $\omega(A_1, B_1) = 1$. Since ω is alternating, A_1, B_1 are linearly independent. So $\dim V \geqslant 2$.

Take $S = \operatorname{span}\{A_1, B_1\}$, then S is symplectic, then $S^\perp$ is also symplectic and $V = S \oplus S^\perp$. By induction hypothesis, there exists a basis of $S^\perp$ with respect to which $\omega|_{S^\perp}$ has the desired form. Then the union of this basis and $\{A_1, B_1\}$ is the desired basis for V. □

The basis of V with respect to which ω has the form above is called a **symplectic basis** of V.

Proposition: **Criterion for Nondegeneracy of 2-Covectors**

Let ω be a 2-covector on an $2n$-dimensional vector space V. Then ω is nondegenerate if and only if $\omega^n = \omega \wedge \cdots \wedge \omega \neq 0$.

Proof. If ω is a symplectic tensor, then write in canonical form, we have $\omega = \sum_{i=1}^n \alpha^i \wedge \beta^i$. Then

$$\omega^n = \sum_I \alpha^{i_1} \wedge \beta^{i_1} \wedge \cdots \wedge \alpha^{i_n} \wedge \beta^{i_n} = n!\alpha^1 \wedge \beta^1 \wedge \cdots \wedge \alpha^n \wedge \beta^n.$$

where $I = (i_1, \ldots, i_n)$ runs through all permutations of $(1, \ldots, n)$.

Conversely, if ω is degenerate, then there exists a nonzero $v \in V$ such that $\omega(v, w) = 0$ for all $w \in V$. Then for any $v_2, \ldots, v_{2n} \in V$, we have $\omega^n(v, v_2, \ldots, v_{2n}) = 0$. Hence $\omega^n = 0$. □

17.2 Symplectic Structures on Manifolds

Let M be a smooth manifold. A nondegenerate 2-form on M is a 2-form ω such that ω_p is a nondegenerate 2-covector on T_pM for all $p \in M$.

Definition 17.2.1: Symplectic Form and Symplectic Manifold

A **symplectic form** ω on a smooth manifold M is a closed nondegenerate 2-form. A **symplectic manifold** is a pair (M, ω) where M is a smooth manifold and ω is a symplectic form on M.

Proposition 17.1 implies that a symplectic manifold must be even-dimensional. However, not all even-dimensional manifolds admit symplectic structures. For example, proposition 17.1 implies that ω^n is a nonvanishing $2n$-form on M, which means that M is orientable.

If (M_1, ω_1) and (M_2, ω_2) are symplectic manifolds, then a diffeomorphism $F : M_1 \to M_2$ is called a **symplectomorphism** if $F^*\omega_2 = \omega_1$. The study of properties of symplectic manifolds that are invariant under symplectomorphisms is called **symplectic geometry**.

Example: **Symplectic Manifolds**

- The standard symplectic structure on $\mathbb{R}^{2n}$ with coordinates $(x_1, y_1, \ldots, x_n, y_n)$ is given by
$$\omega = \sum_{i=1}^{n} \mathrm{d}x^i \wedge \mathrm{d}y^i.$$
- Suppose Σ is any orientable smooth 2-manifold and ω is a nonvanishing 2-form on Σ. Then $\mathrm{d}\omega = 0$ so ω is closed. Also ω is nondegenerate since in two dimensions, a 2-covector is nondegenerate if and only if it is nonvanishing. Hence (Σ, ω) is a symplectic manifold.

Suppose (M, ω) is a symplectic manifold. An immersed or embedded submanifold $S \subseteq M$ is called **symplectic**, **isotropic**, **coisotropic** or **Lagrangian** if T_pS has the corresponding property as a subspace of the symplectic vector space (T_pM, ω_p) for all $p \in S$. Generally, a smooth immersion or embedding $F : S \to M$ is called these properties if $\mathrm{d}F_p(T_pS)$ has the corresponding property as a subspace of $(T_{F(p)}M$ for all $p \in S$.

17.2.1 The Canonical Symplectic Form on Cotangent Bundles

There is a natural 1-form on the total space of the cotangent bundle T^*M of a smooth manifold M, which is called the **tautological 1-form**.

Definition 17.2.2: Tautological 1-Form on Cotangent Bundle

Let $(q, \varphi) \in T^*M$ where $q \in M$ and $\varphi \in T_q^*M$. Then the natural projection $\pi : T^*M \to M$ is given by $\pi(q, \varphi) = q$. It induces a pullback map $\pi^*_{(q,\varphi)} : T_q^*M \to T^*_{(q,\varphi)}T^*M$. The **tautological 1-form** $\tau \in \Omega^1(T^*M)$ is defined by
$$\tau_{(q,\varphi)} = \pi^*_{(q,\varphi)}\varphi, \qquad \tau_{(q,\varphi)}(v) = \varphi(\mathrm{d}\pi_{(q,\varphi)}v) \tag{17.5}$$

Remark:
We can visualize this as taking a vector at the (q, φ)-plane. And then projecting it down to the q-plane and applying φ to it.

In coordinates, in a local trivialization of T^*M with coordinates (x^i, ξ_i) where x^i are coordinates on M and ξ_i are the corresponding fiber coordinates, we have $\pi(x, \xi) = x$ and $\mathrm{d}\pi_{(x,\xi)}(\mathrm{d}x^i) = \mathrm{d}x^i$ so we have
$$\tau_{(x,\xi)} = \mathrm{d}\pi^*_{(x,\xi)}(\xi_i \mathrm{d}x^i) = \xi_i \mathrm{d}x^i. \tag{17.6}$$

Proposition: **Symplectic Form from Tautological 1-Form**

Let M be a smooth manifold and τ be the tautological 1-form. Then τ is smooth and $\omega = -\mathrm{d}\tau$ is a symplectic form on T^*M, called the **canonical symplectic form** on T^*M.

Proof. It follows clearly from the coordinate expression that τ is smooth. Also, we have
$$\omega = -\mathrm{d}\tau = -\mathrm{d}(\xi_i \mathrm{d}x^i) = \mathrm{d}x^i \wedge \mathrm{d}\xi_i, \tag{17.7}$$

□

Proposition: **1-Form as Embedding**

Let M be a smooth manifold and $\sigma : M \to T^*M$ be a 1-form on M. Then σ is a smooth embedding and is closed iff $\sigma(M)$ is a Lagrangian submanifold of (T^*M, ω).

Proof. Taking $\omega = \sigma_i \mathrm{d}x^i$ in local coordinates, we have σ as a map from M to T^*M given by

$$\sigma(x) = (x, \sigma_i(x)).$$

Then σ is a smooth embedding for it is a proper map.

Because $\sigma(M)$ is n-dimensional, it is Lagrangian if and only if it is isotropic, which is equivalent to saying that $\sigma^*\omega = 0$. Since

$$\sigma^*\tau = \sigma^*(\xi_i \mathrm{d}x^i) = \sigma_i \mathrm{d}x^i = \sigma,$$

we have

$$\sigma^*\omega = -\sigma^*\mathrm{d}\tau = -\mathrm{d}\sigma^*\tau = -\mathrm{d}\sigma.$$

Thus we have our conclusion. □

17.3 The Darboux Theorem

Unlike Riemannian geometry, there is no local obstruction to a symplectic structure being locally equivalent to the standard flat model.

Theorem 17.3.1: Darboux Theorem

Let (M, ω) be a $2n$-dimensional symplectic manifold. For any point $p \in M$, there exists smooth coordinates $(x_1, y_1, \ldots, x_n, y_n)$ defined on a neighborhood of p such that

$$\omega = \sum_{i=1}^{n} \mathrm{d}x^i \wedge \mathrm{d}y^i. \tag{17.8}$$

17.4 Hamiltonian Vector Fields

There is a symplectic analogue of the gradient. Suppose (M, ω) is a symplectic manifold and for any smooth function $f \in C^\infty(M)$, we define the **Hamiltonian vector field** X_f of f by

$$X_f = \hat{\omega}^{-1}(\mathrm{d}f), \qquad \text{or} \qquad X_f \lrcorner\, \omega = \mathrm{d}f. \tag{17.9}$$

where $\hat{\omega} : TM \to T^*M$ is the isomorphism defined by $\hat{\omega}(v) = v \lrcorner\, \omega$ for all $v \in TM$.

In any Darboux coordinates, we write

$$X_f = \sum_{i=1}^{n} \left(a_i \frac{\partial}{\partial x^i} + b_i \frac{\partial}{\partial y^i} \right).$$

Then we have

$$X_f \lrcorner\, \omega = \sum_{i=1}^{n}(a_i \mathrm{d}y^i - b_i \mathrm{d}x^i) = \mathrm{d}f = \sum_{i=1}^{n}\left(\frac{\partial f}{\partial x^i}\mathrm{d}x^i + \frac{\partial f}{\partial y^i}\mathrm{d}y^i\right).$$

Thus we have

$$X_f = \sum_{i=1}^{n}\left(\frac{\partial f}{\partial y^i}\frac{\partial}{\partial x^i} - \frac{\partial f}{\partial x^i}\frac{\partial}{\partial y^i}\right). \tag{17.10}$$

Proposition: **Properties of Hamiltonian Vector Fields**

Let (M, ω) be a symplectic manifold and $f \in C^\infty(M)$, then

- f is constant along the flow of X_f.
- At each regular point of f, the vector field X_f is tangent to the level set of f.

Proof. We have

$$X_f(f) = \mathrm{d}f(X_f) = (X_f \lrcorner\, \omega)(X_f) = 0.$$

Hence f is constant along the flow of X_f. Also, at each regular point of f, we have $\mathrm{d}f \neq 0$, so $X_f = \hat{\omega}^{-1}(\mathrm{d}f) \neq 0$. Since $\hat{\omega}$ is an isomorphism, we have $\hat{\omega}(X_f) = X_f \lrcorner\, \omega = \mathrm{d}f$, which means that X_f is tangent to the level set of f. □

A smooth vector field X on a symplectic manifold (M, ω) is called **symplectic** if ω is invariant under the flow of X, which is equivalent to saying that $\mathcal{L}_X\omega = 0$. It is said to be **Hamiltonian** if there exists a smooth function f such that $X = X_f$, and locally Hamiltonian if for each point $p \in M$, there exists a neighborhood U of p and a smooth function f defined on U such that $X|_U = X_f$.

Proposition: **Hamiltonian and Symplectic Vector Fields**

Let (M, ω) be a symplectic manifold and X be a smooth vector field on M. Then X is symplectic if and only if X is locally Hamiltonian. Every locally Hamiltonian vector field is globally Hamiltonian if and only if the first de Rham cohomology group $H^1_{\mathrm{dR}}(M)$ is trivial.

Proof. Using Carten's magic formula, we have

$$\mathcal{L}_X\omega = \mathrm{d}(X \lrcorner\, \omega) + X \lrcorner\, \mathrm{d}\omega = \mathrm{d}(X \lrcorner\, \omega).$$

Then X is symplectic if and only if $\mathrm{d}(X \lrcorner\, \omega) = 0$. If X is locally Hamiltonian, then in a neighborhood of each point there is a smooth function f such that $X = X_f$, so $X \lrcorner\, \omega = X_f \lrcorner\, \omega = \mathrm{d}f$ is closed. Conversely, if X is symplectic, then by Poincaré lemma, in a neighborhood of each point there is a smooth function f such that $X \lrcorner\, \omega = \mathrm{d}f$, so $X = X_f$ is locally Hamiltonian.

The second part follows from the fact that a closed 1-form is exact if and only if its cohomology class is zero. □

In analytical mechanics, a Hamiltonian system is a symplectic manifold (M, ω) together with a smooth function $H \in C^\infty(M)$ called the Hamiltonian. The flow of the Hamiltonian vector field X_H is called the **Hamiltonian flow** of H. The integral curves of X_H are called the **trajectories**

of the Hamiltonian system. In Darboux coordinates, the Hamiltonian flow of H is given by the system of ordinary differential equations

$$\begin{aligned} \frac{\mathrm{d}x^i}{\mathrm{d}t} &= \frac{\partial H}{\partial y^i}(x(t), y(t)), \\ \frac{\mathrm{d}y^i}{\mathrm{d}t} &= -\frac{\partial H}{\partial x^i}(x(t), y(t)). \end{aligned} \tag{17.11}$$

called the **Hamilton's equations**.

17.4.1 Poisson Bracket

In Hamiltonian system, each smooth function f on M is related to a vector field X_f on M.

Definition 17.4.1: Poisson Bracket

Let (M, ω) be a symplectic manifold and $f, g \in C^\infty(M)$ be two smooth functions. The **Poisson bracket** of f and g is the smooth function $\{f, g\} \in C^\infty(M)$ defined by

$$\{f, g\} = \omega(X_f, X_g) = X_g(f) = \mathrm{d}f(X_g) \tag{17.12}$$

In Darboux coordinates, we have

$$\{f, g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x^i} \frac{\partial g}{\partial y^i} - \frac{\partial f}{\partial y^i} \frac{\partial g}{\partial x^i} \right). \tag{17.13}$$

Remark:
Geometrically, the Poisson bracket $\{f, g\}$ is the rate of change of f along the flow of X_g.

Proposition: **Properties of Poisson Bracket**

Let (M, ω) be a symplectic manifold and $f, g, h \in C^\infty(M)$. Then

- Bilinearity: $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$ and $\{h, af + bg\} = a\{h, f\} + b\{h, g\}$ for all $a, b \in \mathbb{R}$.
- Antisymmetry: $\{f, g\} = -\{g, f\}$.
- Jacobi identity: $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$.
- $X_{\{f,g\}} = -[X_f, X_g]$.

Proof. The first two properties are clear from $X_f = \hat{\omega}^{-1}(\mathrm{d}f)$ depending linearly on f and ω being antisymmetric. For the Lie bracket, it suffices to show that for any vector field Y, we have

$$\omega(X_{\{f,g\}}, Y) + \omega([X_f, X_g], Y) = 0.$$

Firstly, we have

$$\omega(X_{\{f,g\}}, Y) = \mathrm{d}\{f, g\}(Y) = Y(\{f, g\}) = Y(X_g(f))$$

Also, as Hamiltonian vector fields are symplectic, we have

$$\begin{aligned}
0 &= (\mathcal{L}_{X_g}\omega)(X_f, Y) = X_g(\omega(X_f, Y)) - \omega([X_g, X_f], Y) - \omega(X_f, [X_g, Y]) \\
&= X_g(Y(f)) - \omega([X_g, X_f], Y) - \mathrm{d}f([X_g, Y]) \\
&= X_g(Y(f)) - \omega([X_g, X_f], Y) - X_g(Y(f)) + Y(X_g(f)) \\
&= -\omega([X_g, X_f], Y) + Y(X_g(f)) = \omega([X_f, X_g], Y) + Y(\{f, g\}).
\end{aligned}$$

The Jacobi identity follows from the Lie bracket property and antisymmetry:

$$\begin{aligned}
\{f, \{g, h\}\} &= X_{\{g,h\}}(f) = -[X_g, X_h](f) = -X_g(X_h(f)) + X_h(X_g(f)) \\
&= -X_g(\{f, h\}) + X_h(\{f, g\}) = -\{f, \{g, h\}\} + \{f, \{h, g\}\}.
\end{aligned}$$

Thus we have our conclusion. □

Corollary 17.4.1: Lie Algebra Structure on Smooth Functions

Let (M, ω) be a symplectic manifold. Then $C^\infty(M)$ is a Lie algebra under the Poisson bracket.

If (M, ω, H) is a Hamiltonian system, then any function $f \in C^\infty(M)$ that is constant along every integral curve of X_H is called a **conserved quantity**. A smooth vector field V on M is called an **infinitesimal symmetry** of the Hamiltonian system if both ω and H are invariant under the flow of V.

Proposition: **Properties of Conserved Quantities and Infinitesimal Symmetries**

Let (M, ω, H) be a Hamiltonian system, $f \in C^\infty(M)$ be a smooth function and V be a smooth vector field on M. Then

- f is a conserved quantity if and only if $\{f, H\} = 0$.
- V is an infinitesimal symmetry if and only if $VH = 0$.
- If θ is the flow of an infinitesimal symmetry V and γ is a trajectory of the Hamiltonian system, then for all $s \in \mathbb{R}$ we have $\theta_s \circ \gamma$ is also a trajectory on its domain.

Now we shall see that there is a one-to-one correspondence between conserved quantities and infinitesimal symmetries of a Hamiltonian system.

Theorem 17.4.1: Noether's Theorem

Let (M, ω, H) be a Hamiltonian system. Then if f is a conserved quantity, then X_f is an infinitesimal symmetry. Conversely, if $H^1_{\mathrm{dR}}(M) = 0$ then each infinitesimal symmetry V is a Hamiltonian vector field X_f for some conserved quantity f, unique up to a constant on each connected component of M.